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Edition

Laravel Up & Running

A Framework for
Building Modern PHP Apps



Matt Stauffer

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THIRD EDITION

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Beijing • Boston • Farnham • Sebastopol • Tokyo

Laravel: Up & Running

by Matt Stauffer

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[LSI]

Dedication

This book is dedicated to my families and my communities—my birth family, my new nuclear family, my chosen family, the Tighten crew and alumni, the Laravel community, Dexter, Ann Arbor, Gainesville, Chicago, Decatur, and Atlanta. All y'all.

Preface

The story of how I got started with Laravel is a common one: I had written PHP for years, but I was on my way out the door, pursuing the power of Rails and other modern web frameworks. Rails in particular had a lively community, a perfect combination of opinionated defaults and flexibility, and the power of Ruby-Gems to leverage prepackaged common code.

Something kept me from jumping ship, and I was glad for that when I found Laravel. It offered everything I was drawn to in Rails, but it wasn't just a Rails clone; this was an innovative framework with incredible documentation, a welcoming community, and clear influences from many languages and frameworks.

Since that day I've been able to share my journey of learning Laravel through blogging, podcasting, and speaking at conferences; I've written dozens of apps in Laravel for work and side projects; and I've met thousands of Laravel developers online and in person. I have plenty of tools in my development toolkit, but I am honestly happiest when I sit down in front of a command line and type `laravel new projectName`.

What This Book Is About

This is not the first book about Laravel, and it won't be the last. I don't intend for this to be a book that covers every line of code or every implementation pattern. I don't want this to be the sort of book that goes out of date when a new version of Laravel is released. Instead, its primary purpose is to provide developers with a high-level overview and concrete examples to learn what they need to work in any Laravel codebase with any and every Laravel feature and subsystem. Rather than mirroring the docs, I want to help you understand the foundational concepts behind Laravel.

Laravel is a powerful and flexible PHP framework. It has a thriving community and a wide ecosystem of tools, and as a result, it's growing in appeal and reach. This book is for developers who already know how to make websites and applications and want to learn how to do so well in Laravel.

Laravel's documentation is thorough and excellent. If you find that I don't cover any particular topic deeply enough for your liking, I encourage you to visit the [online documentation](#) and dig deeper into that particular topic.

I think you will find the book a comfortable balance between high-level introduction and concrete application, and by the end you should feel comfortable writing an entire application in Laravel from scratch. And, if I did my job well, you'll be excited to try.

Who This Book Is For

This book assumes knowledge of basic object-oriented programming practices, PHP (or at least the general syntax of C-family languages), and the basic concepts of the Model–View–Controller (MVC) pattern and templating. If you’ve never made a website before, you may find yourself in over your head. But as long as you have some programming experience, you don’t have to know anything about Laravel before you read this book—we’ll cover everything you need to know, from the simplest “Hello, world!”

Laravel can run on any operating system, but there will be some bash (shell) commands in the book that are easiest to run on Linux/macOS. Windows users may have a harder time with these commands and with modern PHP development, but if you follow the instructions to get Homestead (a Linux virtual machine) running, you’ll be able to run all of the commands from there.

How This Book Is Structured

This book is structured in what I imagine to be a chronological order: if you're building your first web app with Laravel, the early chapters cover the foundational components you'll need to get started, and the later chapters cover less foundational or more esoteric features.

Each section of the book can be read on its own, but for someone new to the framework, I've tried to structure the chapters so that it's actually very reasonable to start from the beginning and read until the end.

Where applicable, each chapter will end with two sections: "Testing" and "TL;DR." If you're not familiar, "TL;DR" means "too long; didn't read." These final sections will show you how to write tests for the features covered in each chapter and will give a high-level overview of what was covered.

The book is written for Laravel 10.

About the Third Edition

The first edition of *Laravel: Up & Running* came out in December 2016 and covered Laravel versions 5.1 to 5.3. The second edition came out in April 2019 and added coverage for 5.4 to 5.8 and Laravel Dusk and Horizon, and added an 18th chapter about community resources and other noncore Laravel packages that weren't covered in the first 17 chapters. This third edition catches the book up to Laravel 10 and adds Breeze, Jetstream, Fortify, Vite, and much more.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

Constant width

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

Constant width bold

Shows commands or other text that should be typed literally by the user.

Constant width italic

Shows code text that should be replaced with user-supplied values or by values determined by context.

{Italic in braces}

Shows file names or file pathways that should be replaced with user-supplied values or by values determined by context.

TIP

This element signifies a tip or suggestion.

NOTE

This element signifies a general note.

WARNING

This element indicates a warning or caution.

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NOTE

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There are so many people in the Laravel community who deserve thanks at this point that I couldn't even begin to include them all. So, to every one of you who's put in so much love, devotion, care, craft...thank you. Thank you for the ways you've helped this community be an incredible place; thank you for so many of you who've reached out to encourage me through parenting, divorce, pandemic, depression, and whatever else. Y'all are amazing.

Taylor Otwell deserves thanks and honor for creating Laravel—and therefore creating so many jobs and helping so many developers love our lives that much more. He deserves appreciation for how he's focused on developer happiness and how hard he's worked to have empathy for developers and to build a positive and encouraging community. But I also want to thank him for being a kind, encouraging, and challenging friend. Taylor, you're a boss.

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Chapter 1. Why Laravel?

In the early days of the dynamic web, writing a web application looked a lot different than it does today. Developers then were responsible for writing the code for not just the unique business logic of our applications but also each of the components that are so common across sites—user authentication, input validation, database access, templating, and more.

Today, programmers have dozens of application development frameworks and thousands of components and libraries easily accessible. It’s a common refrain among programmers that, by the time you learn one framework, three newer (and purportedly better) frameworks have popped up intending to replace it.

“Just because it’s there” might be a valid justification for climbing a mountain, but there are better reasons to choose to use a specific framework—or to use a framework at all. It’s worth asking the question, why frameworks? More specifically, why Laravel?

Why Use a Framework?

It’s easy to see why it’s beneficial to use the individual components, or packages, that are available to PHP developers. With packages, someone else is responsible for developing and maintaining an isolated piece of code that has a well-defined job, and, in theory, that person has a deeper understanding of this single component than you have time to have.

Frameworks like Laravel—and Symfony, Lumen, and Slim—prepackage a collection of third-party components together with custom framework “glue” like configuration files, service providers, prescribed directory structures, and application bootstraps. So, the benefit of using a framework in general is that someone has made decisions not just about individual components for you but also about *how those components should fit together*.

“I’ll Just Build It Myself”

Let's say you start a new web app without the benefit of a framework. Where do you begin? Well, it should probably route HTTP requests, so you now need to evaluate all of the HTTP request and response libraries available and pick one. Then you'll have to pick a router. Oh, and you'll probably need to set up some form of routes configuration file. What syntax should it use? Where should it go? What about controllers? Where do they live, and how are they loaded? Well, you probably need a dependency injection container to resolve the controllers and their dependencies. But which one?

Furthermore, if you do take the time to answer all those questions and successfully create your application, what's the impact on the next developer? What about when you have four such custom framework-based applications, or a dozen, and you have to remember where the controllers live in each or what the routing syntax is?

Consistency and Flexibility

Frameworks address this issue by providing a carefully considered answer to the question "Which component should we use here?" and ensuring that the particular components chosen work well together. Additionally, frameworks provide conventions that reduce the amount of code a developer new to the project has to understand—if you understand how routing works in one Laravel project, for example, you understand how it works in all Laravel projects.

When someone prescribes rolling your own framework for each new project, what they're really advocating is the ability to *control* what does and doesn't go into your application's foundation. That means the best frameworks will not only provide you with a solid foundation but also give you the freedom to customize to your heart's content. And this, as I'll show you in the rest of this book, is part of what makes Laravel so special.

A Short History of Web and PHP Frameworks

An important part of being able to answer the question "Why Laravel?" is understanding Laravel's history—and understanding what came before it. Prior to Laravel's rise in popularity, there were a variety of frameworks and other movements in PHP and other web development spaces.

Ruby on Rails

David Heinemeier Hansson released the first version of Ruby on Rails in 2004, and it's been hard to find a web application framework since then that hasn't been influenced by Rails in some way.

Rails popularized MVC, RESTful JSON APIs, convention over configuration, ActiveRecord, and many more tools and conventions that had a profound influence on the way web developers approached their applications—especially with regard to rapid application development.

The Influx of PHP Frameworks

It was clear to most developers that Rails and similar web application frameworks were the wave of the future, and PHP frameworks, including those admittedly imitating Rails, started popping up quickly.

CakePHP was the first in 2005, and it was soon followed by Symfony, CodeIgniter, Zend Framework, and Kohana (a CodeIgniter fork). Yii arrived in 2008, and Aura and Slim in 2010. The year 2011 brought FuelPHP and Laravel, both of which were not quite CodeIgniter offshoots, but instead proposed as alternatives.

Some of these frameworks were more Rails-y, focusing on database object-relational mappers (ORMs), MVC structures, and other tools targeting rapid development. Others, like Symfony and Zend, focused more on enterprise design patterns and ecommerce.

The Good and the Bad of CodeIgniter

CakePHP and CodeIgniter were the two early PHP frameworks that were most open about how much their inspiration was drawn from Rails. CodeIgniter quickly rose to fame and by 2010 was arguably the most popular of the independent PHP frameworks.

CodeIgniter was simple, easy to use, and boasted amazing documentation and a strong community. But its use of modern technology and patterns advanced slowly; and as the framework world grew and PHP's tooling advanced, CodeIgniter started falling behind in terms of both technological advances and

out-of-the-box features. Unlike many other frameworks, CodeIgniter was managed by a company, and it was slow to catch up with PHP 5.3's newer features, like namespaces and the moves to GitHub and later Composer. It was in 2010 that Taylor Otwell, Laravel's creator, became dissatisfied enough with CodeIgniter that he set off to write his own framework.

Laravel 1, 2, and 3

The first beta of Laravel 1 was released in June 2011, and it was written completely from scratch. It featured a custom ORM (Eloquent); closure-based routing (inspired by Ruby Sinatra); a module system for extension; and helpers for forms, validation, authentication, and more.

Early Laravel development moved quickly, and Laravel 2 and 3 were released in November 2011 and February 2012, respectively. They introduced controllers, unit testing, a command-line tool, an inversion of control (IoC) container, Eloquent relationships, and migrations.

Laravel 4

With Laravel 4, Taylor rewrote the entire framework from the ground up. By this point Composer, PHP's now-ubiquitous package manager, was showing signs of becoming an industry standard, and Taylor saw the value of rewriting the framework as a collection of components, distributed and bundled together by Composer.

Taylor developed a set of components under the code name *Illuminate* and, in May 2013, released Laravel 4 with an entirely new structure. Instead of bundling the majority of its code as a download, Laravel now pulled in the majority of its components from Symfony (another framework that released its components for use by others) and the Illuminate components through Composer.

Laravel 4 also introduced queues, a mail component, facades, and database seeding. And because Laravel was now relying on Symfony components, it was announced that Laravel would be mirroring (not exactly, but soon after) the six-month release schedule Symfony follows.

Laravel 5

Laravel 4.3 was scheduled to be released in November 2014, but as development progressed it became clear that the significance of its changes merited a major release, and Laravel 5 was released in February 2015.

Laravel 5 featured a revamped directory structure, removal of the form and HTML helpers, the introduction of the contract interfaces, a spate of new views, Socialite for social media authentication, Elixir for asset compilation, Scheduler to simplify cron, dotenv for simplified environment management, form requests, and a brand-new REPL (read–evaluate–print loop). Since then, it's grown in features and maturity, but there have been no major changes like in previous versions.

Laravel 6

In September 2019, Laravel 6 was introduced with two primary changes: first, removal of the string and array global helpers Laravel offers (in favor of facades); and second, moving to SemVer (semantic versioning) for version numbering. The practical effect of this change means that, for all versions of Laravel after 5, both major (6, 7, etc.) and minor (6.1, 6.2, etc.) versions are released much more frequently.

Laravel Versions in the New SemVer World (6+)

Starting with version 6, Laravel releases are less monumental than in the past because of the new SemVer release schedule. So going forward, releases will be more about how much time has passed and less about very specific big new features.

What's So Special About Laravel?

So, what is it that sets Laravel apart? Why is it worth having more than one PHP framework at any time? They all use components from Symfony anyway, right? Let's talk a bit about what makes Laravel “tick.”

The Philosophy of Laravel

You only need to read through the Laravel marketing materials and READMEs

to start seeing its values. Taylor uses light-related words like “Illuminate” and “Spark.” And then there are these: “Artisans.” “Elegant.” Also, these: “Breath of fresh air.” “Fresh start.” And finally: “Rapid.” “Warp speed.”

The two most strongly communicated values of the framework are to increase developer speed and developer happiness. Taylor has described the “Artisan” language as intentionally contrasting against more utilitarian values. You can see the genesis of this sort of thinking in [his 2011 question on StackExchange](#) in which he stated, “Sometimes I spend ridiculous amounts of time (hours) agonizing over making code ‘look pretty’” just for the sake of a better experience of looking at the code itself. And he’s often talked about the value of making it easier and quicker for developers to take their ideas to fruition, getting rid of unnecessary barriers to creating great products.

Laravel is, at its core, about equipping and enabling developers. Its goal is to provide clear, simple, and beautiful code and features that help developers quickly learn, start, develop, and write code that’s simple, clear, and lasting.

The concept of targeting developers is clear across Laravel materials. “Happy developers make the best code” is written in the documentation. “Developer happiness from download to deploy” was the unofficial slogan for a while. Of course, any tool or framework will say it wants developers to be happy. But having developer happiness as a *primary* concern, rather than secondary, has had a huge impact on Laravel’s style and decision-making progress. Where other frameworks may target architectural purity as their primary goal, or compatibility with the goals and values of enterprise development teams, Laravel’s primary focus is on serving the individual developer. That doesn’t mean you can’t write architecturally pure or enterprise-ready applications in Laravel, but it won’t have to be at the expense of the readability and comprehensibility of your codebase.

How Laravel Achieves Developer Happiness

Just saying you want to make developers happy is one thing. Doing it is another, and it requires you to question what in a framework is most likely to make developers unhappy and what is most likely to make them happy. There are a number of ways Laravel tries to make developers’ lives easier.

First, Laravel is a rapid application development framework. This means it focuses on a shallow (easy) learning curve and on minimizing the steps between starting a new app and publishing it. All of the most common tasks in building web applications, from database interactions to authentication to queues to email to caching, are made simpler by the components Laravel provides. But Laravel's components aren't just great on their own; they provide a consistent API and predictable structures across the entire framework. That means that, when you're trying something new in Laravel, you're more than likely going to end up saying, "...and it just works."

This doesn't end with the framework itself, either. Laravel provides an entire ecosystem of tools for building and launching applications. You have Sail and Valet and Homestead for local development, Forge for server management, and Envoyer and Vapor for advanced deployment. And there's a suite of add-on packages: Cashier for payments and subscriptions, Echo for WebSockets, Scout for search, Sanctum and Passport for API authentication, Dusk for frontend testing, Socialite for social login, Horizon for monitoring queues, Nova for building admin panels, and Spark to bootstrap your SaaS. Laravel is trying to take the repetitive work out of developers' jobs so they can do something unique.

Next, Laravel focuses on "convention over configuration"—meaning that if you're willing to use Laravel's defaults, you'll have to do much less work than with other frameworks that require you to declare all of your settings even if you're using the recommended configuration. Projects built on Laravel take less time than those built on most other PHP frameworks.

Laravel also focuses deeply on simplicity. It's possible to use dependency injection and mocking and the Data Mapper pattern and repositories and command query responsibility segregation and all sorts of other more complex architectural patterns with Laravel, if you want. But while other frameworks might suggest using those tools and structures on every project, Laravel and its documentation and community lean toward starting with the simplest possible implementation—a global function here, a facade there, ActiveRecord over there. This allows developers to create the simplest possible application to solve for their needs, without limiting its usefulness in complex environments.

An interesting source of how Laravel is different from other PHP frameworks is that its creator and its community are more connected to and inspired by Ruby

and Rails and functional programming languages than by Java. There's a strong current in modern PHP to lean toward verbosity and complexity, embracing the more Java-esque aspects of PHP. But Laravel tends to be on the other side, embracing expressive, dynamic, and simple coding practices and language features.

The Laravel Community

If this book is your first exposure to the Laravel community, you have something special to look forward to. One of the distinguishing elements of Laravel, which has contributed to its growth and success, is the welcoming, teaching community that surrounds it. From Jeffrey Way's [Laracasts video tutorials](#) to [Laravel News](#) to Slack and IRC and Discord channels, from Twitter friends to bloggers to podcasts to the Laracon conferences, Laravel has a rich and vibrant community full of folks who've been around since day one and folks who are just starting their own "day one." And this isn't an accident:

From the very beginning of Laravel, I've had this idea that all people want to feel like they are part of something. It's a natural human instinct to want to belong and be accepted into a group of other like-minded people. So, by injecting personality into a web framework and being really active with the community, that type of feeling can grow in the community.

—Taylor Otwell, Product and Support interview

Taylor understood from the early days of Laravel that a successful open source project needed two things: good documentation and a welcoming community. And those two things are now hallmarks of Laravel.

How It Works

Up until now, everything I've shared here has been entirely abstract. What about the code, you ask? Let's dig into a simple application ([Example 1-1](#)) so you can see what working with Laravel day-to-day is actually like.

Example 1-1. "Hello, World" in routes/web.php

```
<?php
```

```
Route::get('/', function () {
```

```
    return 'Hello, World!';
});
```

The simplest possible action you can take in a Laravel application is to define a route and return a result any time someone visits that route. If you initialize a brand-new Laravel application on your machine, define the route in [Example 1-1](#), and then serve the site from the *public* directory, you’ll have a fully functioning “Hello, World” example (see [Figure 1-1](#)).

Hello, World!

Figure 1-1. Returning “Hello, World!” with Laravel

It looks very similar with controllers, as you can see in [Example 1-2](#) (which, if you’d like to test right away, requires you to run `php artisan make:controller WelcomeController` first to create the controller).

Example 1-2. “Hello, World” with controllers

```
// File: routes/web.php
<?php

use App\Http\Controllers\WelcomeController;

Route::get('/', [WelcomeController::class, 'index']);
// File: app/Http/Controllers/WelcomeController.php
<?php

namespace App\Http\Controllers;

class WelcomeController extends Controller
{
    public function index()
    {
        return 'Hello, World!';
    }
}
```

And if you’re storing your greetings in a database, it’ll also look pretty similar (see [Example 1-3](#)).

Example 1-3. Multigreeting “Hello, World” with database access

```
// File: routes/web.php
```

```

<?php

use App\Greeting;

Route::get('create-greeting', function () {
    $greeting = new Greeting;
    $greeting->body = 'Hello, World!';
    $greeting->save();
});

Route::get('first-greeting', function () {
    return Greeting::first()->body;
});
// File: app/Models/Greeting.php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Greeting extends Model
{
    use HasFactory;
}
// File: database/migrations/2023_03_12_192110_create_greetings_table.php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('greetings', function (Blueprint $table) {
            $table->id();
            $table->string('body');
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */

```

```
public function down(): void
{
    Schema::dropIfExists('greetings');
}
};
```

Example 1-3 might be a bit overwhelming, and if so, just skip over it. You'll learn about everything that's happening here in later chapters, but you can already see that with just a few lines of code, you can set up database migrations and models and pull records out. It's just that simple.

Why Laravel?

So—why Laravel?

Because Laravel helps you bring your ideas to reality with no wasted code, using modern coding standards, surrounded by a vibrant community, with an empowering ecosystem of tools.

And because you, dear developer, deserve to be happy.

Chapter 2. Setting Up a Laravel Development Environment

Part of PHP's success has been because it's hard to find a web server that *can't* serve PHP. However, modern PHP tools have stricter requirements than those of the past. The best way to develop for Laravel is to ensure a consistent local and remote server environment for your code, and, thankfully, the Laravel ecosystem has a few tools for this.

System Requirements

Everything we'll cover in this chapter is possible with Windows machines, but you'll need dozens of pages of custom instructions and caveats. I'll leave those instructions and caveats to actual Windows users, so the examples here and in the rest of the book will focus on Unix/Linux/macOS developers.

Whether you choose to serve your website by installing PHP and other tools on your local machine, serve your development environment from a virtual machine via Vagrant or Docker, or rely on a tool like MAMP/WAMP/XAMPP, your development environment will need to have all of the following installed in order to serve Laravel sites:

- PHP $\geq$ 8.1
- OpenSSL PHP extension
- PDO PHP extension
- Mbstring PHP extension
- Tokenizer PHP extension
- XML PHP extension
- Ctype PHP extension

- JSON PHP extension
- BCMath PHP extension

Composer

Whatever machine you're developing on will need to have **Composer** installed globally. *Composer* is a tool that's at the foundation of most modern PHP development. It's a dependency manager for PHP, much like NPM (Node Package Manager) for Node or RubyGems for Ruby. But like NPM, Composer is also the foundation of much of our testing, local script loading, installation scripts, and much more. You'll need Composer to install Laravel, update Laravel, and bring in external dependencies.

Local Development Environments

For many projects, hosting your development environment using a simpler toolset will be enough. If you already have MAMP or WAMP or XAMPP installed on your system, that will likely be fine to run Laravel.

You can also just run Laravel with PHP's built-in web server. Run `php -S localhost:8000 -t public` from your Laravel site's root folder, and PHP's built-in web server will serve your site at <http://localhost:8000/>.

However, if you want a little more power in your development environment (different local domains for each project, management of dependencies like MySQL, etc.), you'll want to reach for a more powerful tool than just PHP's built-in server.

Laravel offers five tools for local development: `Artisan serve`, `Sail`, `Valet`, `Herd`, and `Homestead`. We'll cover each briefly. If you're unsure of which to use, I'm personally a fan of `Valet` for Mac users and `Sail` for everyone else.

Artisan Serve

If you run `php artisan serve` after setting up your Laravel application, it'll serve it at <http://localhost:8000>, just like we set it up earlier using PHP's built-in

web server. You're not getting anything else for free here, so its only meaningful benefit is that it's easier to remember.

Laravel Sail

Sail is the simplest way to get started with local Laravel development in a way that's the same regardless of your operating system. It comes with a PHP web server, databases, and a host of other niceties that make it very easy to run a single Laravel installation that's consistent for every developer on your project regardless of the project's dependencies or your developers' work environments.

Why don't I use Sail? It uses Docker to accomplish the above, and Docker on macOS is just slow enough that I prefer Valet. But if you're new to Laravel, especially if you don't use a Mac, Sail is intentionally created as the simplest way to get started building your Laravel apps.

Laravel Valet

If you're a macOS user (there are also unofficial forks for Windows and Linux), Laravel Valet makes it easy to serve every one of your local Laravel apps (and most other static and PHP-based apps) on different local domains.

You'll need to install a few tools using Homebrew, which the documentation will walk you through, but there are very few steps from initial installation to serving your apps.

Install Valet—see the [Valet docs](#) for the latest installation instructions—and point it at one or more directories where your sites will live. I ran `valet park` from my `~/Sites` directory, which is where I put all of my under-development apps. Now, you can just add `.test` to the end of the directory name and visit it in your browser.

Valet makes it easy to serve Laravel apps; we can use `valet park` to serve all subfolders in a given folder as `{foldername}.test`, `valet link` to serve just a single folder, `valet open` to open a browser to show the Valet-served domain for a folder, `valet secure` to serve the Valet site with HTTPS, and `valet share` to open an ngrok or Expose tunnel so you can share your site with others.

Laravel Herd

Herd is a native macOS app that bundles Valet and all its dependencies as a single installer. While Herd is not quite as customizable as Valet CLI is, it removes the need to work with Homebrew, Docker, or any other dependency managers, and it allows you to interact with Valet's core features through a nice graphical interface.

Laravel Homestead

Homestead is another tool you might want to use to set up your local development environment. It's a configuration tool that sits on top of Vagrant (which is a tool for managing virtual machines) and provides a preconfigured virtual machine image that is perfectly set up for Laravel development *and* mirrors the most common production environment that many Laravel sites run on.

The [Homestead docs](#) are robust and kept constantly up to date, so I'll just refer you to them if you want to learn how it works and how to get it set up.

Creating a New Laravel Project

There are two ways to create a new Laravel project, both of which are run from the command line. The first option is to globally install the Laravel installer tool (using Composer); the second is to use Composer's `create-project` feature.

You can learn about both options in greater detail on the [Installation documentation page](#), but I'd recommend the Laravel installer tool.

Installing Laravel with the Laravel Installer Tool

If you have Composer installed globally, installing the Laravel installer tool is as simple as running the following command:

```
composer global require "laravel/installer"
```

Once you have the Laravel installer tool installed, spinning up a new Laravel project is simple. Just run this command from your command line:

```
laravel new projectName
```

This will create a new subdirectory of your current directory named *{projectName}* and install a bare Laravel project in it.

Installing Laravel with Composer's create-project Feature

Composer also offers a feature called `create-project` for creating new projects with a particular skeleton. To use this tool to create a new Laravel project, issue the following command:

```
composer create-project laravel/laravel projectName
```

Just like the installer tool, this will create a subdirectory of your current directory named *{projectName}* that contains a skeleton Laravel install, ready for you to develop.

Installing Laravel with Sail

If you plan to work with Laravel Sail, you can install a Laravel app and begin its Sail installation process at the same time. Ensure you have Docker installed on your computer, and then issue the following command, replacing *example-app* with the name of your app:

```
curl -s "https://laravel.build/example-app" | bash
```

This will install Laravel into the *example-app* folder underneath your current folder and then begin the Sail installation process.

Once that installation process is complete, change to your new directory and spin up Sail:

```
cd example-app  
./vendor/bin/sail up
```

NOTE

The first time you run `sail up`, it'll take quite a bit longer than other installation processes, as

it needs to build up the initial Docker image.

Laravel's Directory Structure

When you open up a directory that contains a skeleton Laravel application, you'll see the following files and directories:

```
app/  
bootstrap/  
config/  
database/  
public/  
resources/  
routes/  
storage/  
tests/  
vendor/  
.editorconfig  
.env  
.env.example  
.gitattributes  
.gitignore  
artisan  
composer.json  
composer.lock  
package.json  
phpunit.xml  
readme.md  
vite.config.js
```

Let's walk through them one by one to get familiar.

The Folders

The root directory contains the following folders by default:

app

Where the bulk of your actual application will go. Models, controllers, commands, and your PHP domain code all go in here.

bootstrap

Contains the files that the Laravel framework uses to boot every time it runs.

config

Where all the configuration files live.

database

Where database migrations, seeds, and factories live.

public

The directory the server points to when it's serving the website. This contains *index.php*, which is the front controller that kicks off the bootstrapping process and routes all requests appropriately. It's also where any public-facing files like images, stylesheets, scripts, or downloads go.

resources

Where files that are needed for other scripts live. Views, and (optionally) source CSS and source JavaScript files live here.

routes

Where all of the route definitions live, both for HTTP routes and “console routes,” or Artisan commands.

storage

Where caches, logs, and compiled system files live.

tests

Where unit and integration tests live.

vendor

Where Composer installs its dependencies. It's Git-ignored (marked to be excluded from your version control system) because Composer is expected to run as a part of your deploy process on any remote servers.

The Loose Files

The root directory also contains the following files:

.editorconfig

Gives your IDE/text editor instructions about Laravel's coding standards (e.g., the size of indents, the charset, and whether to trim trailing whitespace).

.env and .env.example

Dictate the environment variables (variables that are expected to be different in each environment and are therefore not committed to version control).

.env.example is a template that each environment should duplicate to create its own *.env* file, which is Git-ignored.

.gitignore and .gitattributes

Git configuration files.

artisan

Allows you to run Artisan commands (see [Chapter 8](#)) from the command line.

composer.json and composer.lock

Configuration files for Composer; *composer.json* is user-editable and *composer.lock* is not. These files share some basic information about the project and also define its PHP dependencies.

package.json

Like *composer.json*, but for frontend assets and dependencies of the build system; it instructs NPM about which JavaScript-based dependencies to pull in.

phpunit.xml

A configuration file for PHPUnit, the tool Laravel uses for testing out of the box.

readme.md

A Markdown file giving a basic introduction to Laravel. You won't see this file if you use the Laravel installer.

vite.config.js

The (optional) configuration file for Vite. This file instructs your build system about how to compile and process your frontend assets.

Configuration

The core settings of your Laravel application—database connection settings, queue and mail settings, etc.—live in files in the *config* folder. Each of these files returns a PHP array, and each value in the array is accessible by a config key that is comprised of the filename and all descendant keys, separated by dots (.).

So, if you create a file at *config/services.php* that looks like this:

```
// config/services.php
<?php
return [
    'sparkpost' => [
        'secret' => 'abcdefg',
    ],
];
```

You can access that config variable using `config('services.sparkpost.secret')`.

Any configuration variables that should be distinct for each environment (and therefore not committed to source control) will instead live in your *.env* files. Let's say you want to use a different Bugsnag API key for each environment. You'd set the config file to pull it from *.env*:

```
// config/services.php
<?php
return [
    'bugsnag' => [
        'api_key' => env('BUGSNAG_API_KEY'),
    ],
];
```

This `env()` helper function pulls a value from your `.env` file with that same key. So now, add that key to your `.env` (settings for this environment) and `.env.example` (template for all environments) files:

```
# In .env
BUGSNAG_API_KEY=oinfp9813410942
```

```
# In .env.example
BUGSNAG_API_KEY=
```

Your `.env` file will already contain quite a few environment-specific variables needed by the framework, like which mail driver you'll be using and what your basic database settings are.

USING ENV() OUTSIDE OF CONFIG FILES

Certain features in Laravel, including some caching and optimization features, aren't available if you use `env()` calls anywhere outside of config files.

The best way to pull in environment variables is to set up config items for anything you want to be environment-specific. Have those config items read the environment variables, and then reference the config variables anywhere within your app:

```
// config/services.php
return [
    'bugsnag' => [
        'key' => env('BUGSNAG_API_KEY'),
    ],
];

// In controller, or whatever
$bugsnag = new Bugsnag(config('services.bugsnag.key'));
```

The .env File

Let's take a quick look at the default contents of the `.env` file. The exact keys will vary depending on which version of Laravel you're using, but take a look at [Example 2-1](#) to see what they look like.

Example 2-1. The default environment variables in Laravel

```
APP_NAME=Laravel
APP_ENV=local
APP_KEY=
APP_DEBUG=true
APP_URL=http://localhost

LOG_CHANNEL=stack
LOG_DEPRECATIONS_CHANNEL=null
LOG_LEVEL=debug

DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=laravel
DB_USERNAME=root
DB_PASSWORD=

BROADCAST_DRIVER=log
CACHE_DRIVER=file
FILESYSTEM_DISK=local
QUEUE_CONNECTION=sync
SESSION_DRIVER=file
SESSION_LIFETIME=120

MEMCACHED_HOST=127.0.0.1

REDIS_HOST=127.0.0.1
REDIS_PASSWORD=null
REDIS_PORT=6379

MAIL_MAILER=smtp
MAIL_HOST=mailpit
MAIL_PORT=1025
MAIL_USERNAME=null
MAIL_PASSWORD=null
MAIL_ENCRYPTION=null
MAIL_FROM_ADDRESS="hello@example.com"
MAIL_FROM_NAME="${APP_NAME}"

AWS_ACCESS_KEY_ID=
AWS_SECRET_ACCESS_KEY=
AWS_DEFAULT_REGION=us-east-1
AWS_BUCKET=
AWS_USE_PATH_STYLE_ENDPOINT=false

PUSHER_APP_ID=
PUSHER_APP_KEY=
PUSHER_APP_SECRET=
PUSHER_HOST=
```

```
PUSHER_PORT=443
PUSHER_SCHEME=https
PUSHER_APP_CLUSTER=mt1

VITE_PUSHER_APP_KEY="${PUSHER_APP_KEY}"
VITE_PUSHER_HOST="${PUSHER_HOST}"
VITE_PUSHER_PORT="${PUSHER_PORT}"
VITE_PUSHER_SCHEME="${PUSHER_SCHEME}"
VITE_PUSHER_APP_CLUSTER="${PUSHER_APP_CLUSTER}"
```

I won't go into all of them, because quite a few are just groups of authentication information for various services (Pusher, Redis, DB, Mail). Here are two important environment variables you should know about, though:

APP_KEY

A randomly generated string that's used to encrypt data. If this is ever empty, you may run into the error "No application encryption key has been specified." In that case, just run `php artisan key:generate`, and Laravel will generate one for you.

APP_DEBUG

A Boolean determining whether the users of this instance of your application should see debug errors—great for local and staging environments, terrible for production.

The rest of the nonauthentication settings (`BROADCAST_DRIVER`, `QUEUE_CONNECTION`, etc.) are given default values that work with as little reliance on external services as possible, which is perfect for when you're getting started.

When you start your first Laravel app, the only change you'll likely want to make for most projects is to the database configuration settings. I use Laravel Valet, so I change `DB_DATABASE` to the name of my project, `DB_USERNAME` to `root`, and `DB_PASSWORD` to an empty string:

```
DB_DATABASE=myProject
DB_USERNAME=root
DB_PASSWORD=
```

Then, I create a database with the same name as my project in my favorite MySQL client, and I'm ready to go.

Up and Running

You're now up and running with a bare Laravel install. Run `git init`, commit the bare files with `git add .` and `git commit`, and you're ready to start coding. That's it! And if you're using Valet, you can run the following commands and instantly see your site live in your browser:

```
laravel new myProject && cd myProject && valet open
```

Every time I start a new project, these are the steps I take:

```
laravel new myProject
cd myProject
git init
git add .
git commit -m "Initial commit"
```

I keep all of my sites in a `~/Sites` folder, which I have set up as my primary Valet directory, so in this case I'd instantly have *myProject.test* accessible in my browser with no added work. I can edit `.env` and point it to a particular database, add that database in my MySQL app, and I'm ready to start coding.

Testing

In every chapter after this, the “Testing” section at the end of the chapter will show you how to write tests for the feature or features that were covered. Since this chapter doesn't cover a testable feature, let's talk tests quickly. (To learn more about writing and running tests in Laravel, head over to [Chapter 12](#).)

Out of the box, Laravel brings in PHPUnit as a dependency and is configured to run the tests in any file in the *tests* directory whose name ends with *Test.php* (for example, *tests/UserTest.php*).

So, the simplest way to write tests is to create a file in the *tests* directory with a name that ends with *Test.php*. And the easiest way to run them is to run

`./vendor/bin/phpunit` from the command line (in the project root).

If any tests require database access, be sure to run your tests from the machine where your database is hosted—so if you’re hosting your database in Vagrant, make sure to `ssh` into your Vagrant box to run your tests from there. Again, you can learn about this and much more in [Chapter 12](#).

Also, some of the testing sections will use testing syntax and features that you will not be familiar with yet if you’re reading the book for the first time. If code in any of the testing sections is confusing, just skip it and come back to it after you’ve had a chance to read the testing chapter.

TL;DR

Since Laravel is a PHP framework, it’s very simple to serve it locally. Laravel also provides three tools for managing your local development: Sail, a Docker setup; Valet, a simpler macOS-based tool; and Homestead, a preconfigured Vagrant setup. Laravel relies on and can be installed by Composer and comes out of the box with a series of folders and files that reflect both its conventions and its relationship with other open source tools.

Chapter 3. Routing and Controllers

The essential function of any web application framework is to take requests from a user and deliver responses, usually via HTTP(S). This means defining an application's routes is the first and most important project to tackle when learning a web framework; without routes, you have little to no ability to interact with the end user.

In this chapter we will examine routes in Laravel; you'll see how to define them, how to point them to the code they should execute, and how to use Laravel's routing tools to handle a diverse array of routing needs.

A Quick Introduction to MVC, the HTTP Verbs, and REST

Most of what we'll talk about in this chapter references how Model–View–Controller (MVC) applications are structured, and many of the examples we'll be looking at use REST-ish route names and verbs, so let's take a quick look at both.

What Is MVC?

In MVC, you have three primary concepts:

model

Represents an individual database table (or a record from that table)—think “Company” or “Dog.”

view

Represents the template that outputs your data to the end user—think “the login page template with this given set of HTML and CSS and JavaScript.”

controller

Like a traffic cop, takes HTTP requests from the browser, gets the right data out of the database and other storage mechanisms, validates user input, and eventually sends a response back to the user.

In [Figure 3-1](#), you can see that the end user will first interact with the controller by sending an HTTP request using their browser. The controller, in response to that request, may write data to and/or pull data from the model (database). The controller will then likely send data to a view, and then the view will be returned to the end user to display in their browser.

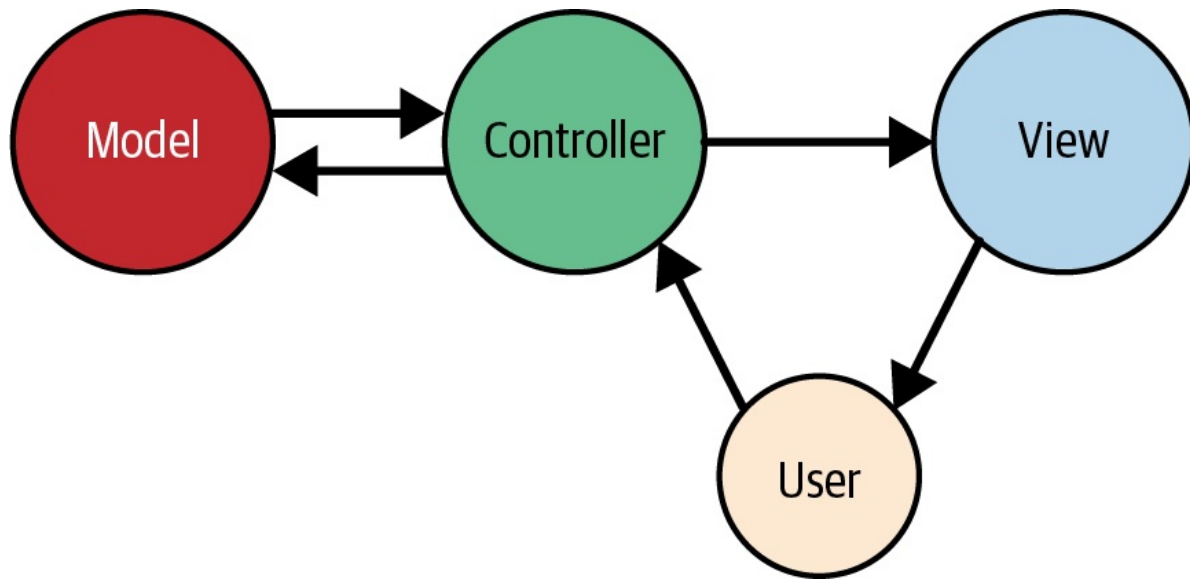


Figure 3-1. A basic illustration of MVC

We'll cover some use cases for Laravel that don't fit this relatively simplistic way of looking at application architecture, so don't get hung up on MVC, but this will at least get you ready to approach the rest of this chapter as we talk about views and controllers.

The HTTP Verbs

the most common HTTP verbs are GET and POST, followed by PUT and DELETE. There are also HEAD, OPTIONS, and PATCH, and two others that are pretty much never used in normal web development, TRACE and CONNECT.

Here's a quick rundown:

GET

Request a resource (or a list of resources).

HEAD

Ask for a headers-only version of the GET response.

POST

Create a resource.

PUT

Overwrite a resource.

PATCH

Modify a resource.

DELETE

Delete a resource.

OPTIONS

Ask the server which verbs are allowed at this URL.

Table 3-1 shows the actions available on a resource controller (more on these in “**Resource Controllers**”). Each action expects you to call a specific URL pattern using a specific verb, so you can get a sense of what each verb is used for.

Table 3-1. The methods of Laravel's resource controllers

Verb	URL	Controller method	Name	Description
GET	tasks	index()	tasks.index	Show all t

GET	tasks/create	create()	tasks.create	Show the create task form
POST	tasks	store()	tasks.store	Accept for submission from the create task form
GET	tasks/{task}	show()	tasks.show	Show one
GET	tasks/{task}/edit	edit()	tasks.edit	Edit one task
PUT/PATCH	tasks/{task}	update()	tasks.update	Accept for submission from the edit task form
DELETE	tasks/{task}	destroy()	tasks.destroy	Delete one task

What Is REST?

We'll cover REST in greater detail in [“The Basics of REST-Like JSON APIs”](#), but as a brief introduction, it's an architectural style for building APIs. When we talk about REST in this book, we'll mainly be referencing a few characteristics, such as:

- Being structured around one primary resource at a time (e.g., tasks)
- Consisting of interactions with predictable URL structures using HTTP verbs (as seen in [Table 3-1](#))
- Returning JSON and often being requested with JSON

There's more to it, but usually "RESTful" as it'll be used in this book will mean "patterned after these URL-based structures so we can make predictable calls like GET /tasks/14/edit for the edit page." This is relevant (even when not building APIs) because Laravel's routing structures are based around a REST-like structure, as you can see in [Table 3-1](#).

REST-based APIs follow mainly this same structure, except they don't have a *create* route or an *edit* route, since APIs just represent actions, not pages that prep for the actions.

Route Definitions

In a Laravel application, you will define your web routes in *routes/web.php* and your API routes in *routes/api.php*. *Web routes* are those that will be visited by your end users; *API routes* are those for your API, if you have one. For now, we'll primarily focus on the routes in *routes/web.php*.

The simplest way to define a route is to match a path (e.g., /) with a closure, as seen in [Example 3-1](#).

Example 3-1. Basic route definition

```
// routes/web.php
Route::get('/', function () {
    return 'Hello, World!';
});
```

WHAT'S A CLOSURE?

Closures are PHP's version of anonymous functions. A closure is a function that you can pass around as an object, assign to a variable, pass as a parameter to other functions and methods, or even serialize.

You've now defined that if anyone visits / (the root of your domain), Laravel's router should run the closure defined there and return the result. Note that we return our content and don't echo or print it.

A QUICK INTRODUCTION TO MIDDLEWARE

You might be wondering, “Why am I returning ‘Hello, World!’ instead of echoing it?”

There are quite a few answers, but the simplest is that there are a lot of wrappers around Laravel’s request and response cycle, including something called *middleware*. When your route closure or controller method is done, it’s not time to send the output to the browser yet; returning the content allows it to continue flowing through the response stack and the middleware before it is returned to the user.

Many simple websites could be defined entirely within the web routes file. With a few simple GET routes combined with some templates, as illustrated in **Example 3-2**, you can serve a classic website easily.

Example 3-2. Sample website

```
Route::get('/', function () {  
    return view('welcome');  
});  
  
Route::get('about', function () {  
    return view('about');  
});  
  
Route::get('products', function () {  
    return view('products');  
});  
  
Route::get('services', function () {  
    return view('services');  
});
```

STATIC CALLS

If you have much experience developing with PHP, you might be surprised to see static calls on the Route class. This is not actually a static method per se, but rather a service location using Laravel’s facades, which we’ll cover in **Chapter 11**.

If you prefer to avoid facades, you can accomplish these same definitions like this:

```
$router->get('/', function () {  
    return 'Hello, World!';  
});
```

Route Verbs

You might've noticed that we've been using `Route::get()` in our route definitions. This means we're telling Laravel to only match for these routes when the HTTP request uses the GET action. But what if it's a form POST, or maybe some JavaScript sending PUT or DELETE requests? There are a few other options for methods to call on a route definition, as illustrated in [Example 3-3](#).

Example 3-3. Route verbs

```
Route::get('/', function () {
    return 'Hello, World!';
});

Route::post('/', function () {
    // Handle someone sending a POST request to this route
});

Route::put('/', function () {
    // Handle someone sending a PUT request to this route
});

Route::delete('/', function () {
    // Handle someone sending a DELETE request to this route
});

Route::any('/', function () {
    // Handle any verb request to this route
});

Route::match(['get', 'post'], '/', function () {
    // Handle GET or POST requests to this route
});
```

Route Handling

As you've probably guessed, passing a closure to the route definition is not the only way to teach it how to resolve a route. Closures are quick and simple, but the larger your application gets, the clumsier it becomes to put all of your routing logic in one file. Additionally, applications using route closures can't take advantage of Laravel's route caching (more on that later), which can shave up to hundreds of milliseconds off of each request.

The other common option is to pass a controller name and method as a string in place of the closure, as in [Example 3-4](#).

Example 3-4. Routes calling controller methods

```
use App\Http\Controllers\WelcomeController;

Route::get('/', [WelcomeController::class, 'index']);
```

This is telling Laravel to pass requests to that path to the `index()` method of the `App\Http\Controllers\WelcomeController` controller. This method will be passed the same parameters and treated the same way as a closure you might've alternatively put in its place.

LARAVEL'S CONTROLLER/METHOD REFERENCE SYNTAX

Laravel has a convention for how to refer to a particular method in a given controller: `[ControllerName::class, methodName]`, known as *tuple syntax* or *callable array syntax*. Sometimes this is just a casual communication convention, but it's also used in real bindings, like in [Example 3-4](#). The first item on the array identifies the controller and the second the method.

Laravel also supports an older “string” syntax (`Route::get('/', 'WelcomeController@index')`), and this is still a common way to describe a method in written communication.

Route Parameters

If the route you're defining has parameters—segments in the URL structure that are variable—it's simple to define them in your route and pass them to your closure (see [Example 3-5](#)).

Example 3-5. Route parameters

```
Route::get('users/{id}/friends', function ($id) {
    //
});
```

You can also make your route parameters optional by including a question mark (?) after the parameter name, as illustrated in [Example 3-6](#). In this case, you should also provide a default value for the route's corresponding variable.

Example 3-6. Optional route parameters

```
Route::get('users/{id?}', function ($id = 'fallbackId') {
    //
});
```

And you can use regular expressions (regexes) to define that a route should only match if a parameter meets particular requirements, as in [Example 3-7](#).

Example 3-7. Regular expression route constraints

```
Route::get('users/{id}', function ($id) {
    //
})->where('id', '[0-9]+');

Route::get('users/{username}', function ($username) {
    //
})->where('username', '[A-Za-z]+');

Route::get('posts/{id}/{slug}', function ($id, $slug) {
    //
})->where(['id' => '[0-9]+', 'slug' => '[A-Za-z]+']);
```

As you’ve probably guessed, if you visit a path that matches a route string but the regex doesn’t match the parameter, it won’t be matched. Since routes are matched top to bottom, `users/abc` would skip the first closure in [Example 3-7](#), but it would be matched by the second closure, so it would get routed there. On the other hand, `posts/abc/123` wouldn’t match any of the closures, so it would return a 404 (Not Found) error.

Laravel also offers convenience methods for common regular expression matching patterns, as you can see in [Example 3-8](#).

Example 3-8. Regular expression route constraint helpers

```
Route::get('users/{id}/friends/{friendname}', function ($id, $friendname) {
    //
})->whereNumber('id')->whereAlpha('friendname');

Route::get('users/{name}', function ($name) {
    //
})->whereAlphaNumeric('name');

Route::get('users/{id}', function ($id) {
    //
})->whereUuid('id');

Route::get('users/{id}', function ($id) {
    //
})->whereUlid('id');
```

```
Route::get('friends/types/{type}', function ($type) {  
    //  
})->whereIn('type', ['acquaintance', 'bestie', 'frenemy']);
```

THE NAMING RELATIONSHIP BETWEEN ROUTE PARAMETERS AND CLOSURE/CONTROLLER METHOD PARAMETERS

As you can see in [Example 3-5](#), it's most common to use the same names for your route parameters (`{id}`) and the method parameters they inject into your route definition (`function ($id)`). But is this necessary?

Unless you're using route model binding, discussed later in this chapter, no. The only thing that defines which route parameter matches with which method parameter is their order (left to right), as you can see here:

```
Route::get('users/{userId}/comments/{commentId}', function (  
    $thisIsActuallyTheUserId,  
    $thisIsReallyTheCommentId  
) {  
    //  
});
```

That having been said, just because you *can* make them different doesn't mean you *should*. I recommend keeping them the same for the sake of future developers, who could get tripped up by inconsistent naming.

Route Names

The simplest way to refer to these routes elsewhere in your application is just by their path. There's a `url()` global helper to simplify that linking in your views, if you need it; see [Example 3-9](#) for an example. The helper will prefix your route with the full domain of your site.

Example 3-9. The `url()` helper

```
<a href="<?php echo url('/'); ?>">  
// Outputs <a href="http://myapp.com/">
```

However, Laravel also allows you to name each route, which enables you to

refer to it without explicitly referencing the URL. This is helpful because it means you can give simple nicknames to complex routes, and also because linking them by name means you don't have to rewrite your frontend links if the paths change (see [Example 3-10](#)).

Example 3-10. Defining route names

```
// Defining a route with name() in routes/web.php:
Route::get('members/{id}', [\App\Http\Controller\MemberController::class, 'show'])
    ->name('members.show');

// Linking the route in a view using the route() helper:
<a href="<?php echo route('members.show', ['id' => 14]); ?>">
```

This example illustrates a few new concepts. First, we're using fluent route definition to add the name, by chaining the `name()` method after the `get()` method. This method allows us to name the route, giving it a short alias to make it easier to reference elsewhere.

In our example, we've named this route `members.show`; *resourcePlural.action* is a common convention within Laravel for route and view names.

ROUTE NAMING CONVENTIONS

You can name your route anything you'd like, but the common convention is to use the plural of the resource name, then a period, then the action. So, here are the routes most common for a resource named `photo`:

```
photos.index
photos.create
photos.store
photos.show
photos.edit
photos.update
photos.destroy
```

To learn more about these conventions, see [“Resource Controllers”](#).

This example also introduced the `route()` helper. Just like `url()`, it's intended to be used in views to simplify linking to a named route. If the route has no

parameters, you can simply pass the route name (`route('members.index')`) and receive a route string (`"http://myapp.com/members"`). If it has parameters, pass them as an array in the second parameter like we did in [Example 3-10](#).

In general, I recommend using route names instead of paths to refer to your routes, and therefore using the `route()` helper instead of the `url()` helper. Sometimes it can get a bit clumsy—for example, if you’re working with multiple subdomains—but it provides an incredible level of flexibility to later change the application’s routing structure without major penalty.

PASSING ROUTE PARAMETERS TO THE ROUTE() HELPER

When your route has parameters (e.g., `users/id`), you need to define those parameters when you’re using the `route()` helper to generate a link to the route.

There are a few different ways to pass these parameters. Let’s imagine a route defined as `users/userId/comments/commentId`. If the user ID is 1 and the comment ID is 2, let’s look at a few options we have available to us:

Option 1:

```
route('users.comments.show', [1, 2])  
// http://myapp.com/users/1/comments/2
```

Option 2:

```
route('users.comments.show', ['userId' => 1, 'commentId' => 2])  
// http://myapp.com/users/1/comments/2
```

Option 3:

```
route('users.comments.show', ['commentId' => 2, 'userId' => 1])  
// http://myapp.com/users/1/comments/2
```

Option 4:

```
route('users.comments.show', ['userId' => 1, 'commentId' => 2, 'opt' => 'a'])  
// http://myapp.com/users/1/comments/2?opt=a
```

As you can see, nonkeyed array values are assigned in order; keyed array values are matched with the route parameters matching their keys, and anything left over is added as a query parameter.

Route Groups

Often a group of routes shares a particular characteristic—a certain authentication requirement, a path prefix, or perhaps a controller namespace. Defining these shared characteristics again and again on each route not only seems tedious but also can muddy up the shape of your routes file and obscure some of the structures of your application.

Route groups allow you to reduce this duplication by grouping several routes together and applying any shared configuration settings once to the entire group. Additionally, route groups are visual cues to future developers (and to your own brain) that these routes are grouped together.

To group two or more routes together, you “surround” the route definitions with a route group, as shown in [Example 3-11](#). In reality, you’re actually passing a closure to the group definition and defining the grouped routes within that closure.

Example 3-11. Defining a route group

```
Route::group(function () {  
    Route::get('hello', function () {  
        return 'Hello';  
    });  
    Route::get('world', function () {  
        return 'World';  
    });  
});
```

By default, a route group doesn’t actually do anything. There’s no difference between using the group in [Example 3-11](#) and separating a segment of your routes with code comments.

Middleware

Probably the most common use for route groups is to apply middleware to a

group of routes. You'll learn more about middleware in [Chapter 10](#), but, among other things, they're what Laravel uses for authenticating users and restricting guest users from using certain parts of a site.

In [Example 3-12](#), we're creating a route group around the dashboard and account views and applying the `auth` middleware to both. In this example, this means users have to be logged in to the application to view the dashboard or the account page.

Example 3-12. Restricting a group of routes to logged-in users only

```
Route::middleware('auth')->group(function() {  
    Route::get('dashboard', function () {  
        return view('dashboard');  
    });  
    Route::get('account', function () {  
        return view('account');  
    });  
});
```

Often, it's clearer and more direct to attach middleware to your routes in the controller instead of at the route definition. You can do this by calling the `middleware()` method in the constructor of your controller. The string you pass to the `middleware()` method is the name of the middleware, and you can optionally chain modifier methods (`only()` and `except()`) to define which methods will receive that middleware:

```
class DashboardController extends Controller  
{  
    public function __construct()  
    {  
        $this->middleware('auth');  
  
        $this->middleware('admin-auth')  
            ->only('editUsers');  
  
        $this->middleware('team-member')  
            ->except('editUsers');  
    }  
}
```

Note that if you're doing a lot of “only” and “except” customizations, that's often a sign that you should break out a new controller for the exceptional

routes.

A BRIEF INTRODUCTION TO ELOQUENT

We'll be covering Eloquent, database access, and Laravel's query builder in depth in [Chapter 5](#), but there will be a few references between now and then that will make a basic understanding useful.

Eloquent is Laravel's ActiveRecord database object-relational mapper (ORM), which makes it easy to relate a `Post` class (model) to the `posts` database table and get all records with a call like `Post::all()`.

The query builder is the tool that makes it possible to make calls like `Post::where('active', true)->get()` or even `DB::table('users')->all()`. You're *building* a query by chaining methods one after another.

Path Prefixes

If you have a group of routes that share a segment of their path—for example, if your site's dashboard is prefixed with `/dashboard`—you can use route groups to simplify this structure (see [Example 3-13](#)).

Example 3-13. Prefixing a group of routes

```
Route::prefix('dashboard')->group(function () {
    Route::get('/', function () {
        // Handles the path /dashboard
    });
    Route::get('users', function () {
        // Handles the path /dashboard/users
    });
});
```

Note that each prefixed group also has a `/` route that represents the root of the prefix—in [Example 3-13](#) that's `/dashboard`.

Subdomain Routing

Subdomain routing is the same as route prefixing, but it's scoped by subdomain instead of route prefix. There are two primary uses for this. First, you may want

to present different sections of the application (or entirely different applications) to different subdomains. [Example 3-14](#) shows how you can achieve this.

Example 3-14. Subdomain routing

```
Route::domain('api.myapp.com')->group(function () {  
    Route::get('/', function () {  
        //  
    });  
});
```

Second, you might want to set part of the subdomain as a parameter, as illustrated in [Example 3-15](#). This is most often done in cases of multitenancy (think Slack or Harvest, where each company gets its own subdomain, like *tighten.slack.co*).

Example 3-15. Parameterized subdomain routing

```
Route::domain('{account}.myapp.com')->group(function () {  
    Route::get('/', function ($account) {  
        //  
    });  
    Route::get('users/{id}', function ($account, $id) {  
        //  
    });  
});
```

Note that any parameters for the group get passed into the grouped routes' methods as the first parameter(s).

Name Prefixes

It's common that route names will reflect the inheritance chain of path elements, so `users/comments/5` will be served by a route named `users.comments.show`. In this case, it's common to use a route group around all of the routes that are beneath the `users.comments` resource.

Just like we can prefix URL segments, we can also prefix strings to the route name. With route group name prefixes, we can define that every route within this group should have a given string prefixed to its name. In this context, we're prefixing `"users."` to each route name, then `"comments."` (see [Example 3-16](#)).

Example 3-16. Route group name prefixes

```
Route::name('users.')->prefix('users')->group(function () {
```

```
Route::name('comments.')->prefix('comments')->group(function () {
    Route::get('{id}', function () {
        // ...
    }->name('show')); // Route named 'users.comments.show'

    Route::destroy('{id}', function () {}->name('destroy'));
});
```

Route Group Controllers

When you’re grouping routes that are served by the same controller, such as when we’re showing, editing, and deleting users, for example, we can use the `route group controller()` method, as shown in [Example 3-17](#), to avoid having to define the full tuple for every route.

Example 3-17. Route group controllers

```
use App\Http\Controllers\UserController;

Route::controller(UserController::class)->group(function () {
    Route::get('/', 'index');
    Route::get('{id}', 'show');
});
```

Fallback Routes

In Laravel you can define a “fallback route” (which you need to define at the end of your routes file) to catch all unmatched requests:

```
Route::fallback(function () {
    //
});
```

Signed Routes

Many applications regularly send notifications about one-off actions (resetting a password, accepting an invitation, etc.) and provide simple links to take those actions. Let’s imagine sending an email confirming the recipient was willing to be added to a mailing list.

There are three ways to send that link:

- Make that URL public and hope no one else discovers the approval URL or modifies their own approval URL to approve someone else.
- Put the action behind authentication, link to the action, and require the user to log in if they're not logged in yet (which, in this case, may be impossible, as many mailing list recipients likely won't be users with accounts).
- “Sign” the link so that it uniquely proves that the user received the link from your email, without them having to log in—something like `http://myapp.com/invitations/5816/yes?signature=030ab0ef6a8237bd86a8b8`.

One simple way to accomplish the last option is to use a feature called *signed URLs*, which makes it easy to build a signature authentication system for sending out authenticated links. These links are composed of the normal route link with a “signature” appended that proves that the URL has not been changed since it was sent (and therefore that no one has modified the URL to access someone else's information).

Signing a Route

To build a signed URL to access a given route, the route must have a name:

```
Route::get('invitations/{invitation}/{answer}', InvitationController::class)
->name('invitations');
```

To generate a normal link to this route you would use the `route()` helper, as we've already covered, but you could also use the URL facade to do the same thing: `URL::route('invitations', ['invitation' => 12345, 'answer' => 'yes'])`. To generate a *signed* link to this route, simply use the `signedRoute()` method instead. And if you want to generate a signed route with an expiration, use `temporarySignedRoute()`:

```
// Generate a normal link
URL::route('invitations', ['invitation' => 12345, 'answer' => 'yes']);

// Generate a signed link
URL::signedRoute('invitations', ['invitation' => 12345, 'answer' => 'yes']);
```



```
// Generate an expiring (temporary) signed link
URL::temporarySignedRoute(
    'invitations',
    now()->addHours(4),
    ['invitation' => 12345, 'answer' => 'yes']
);
```

USING THE NOW() HELPER

Laravel offers a `now()` helper that's the equivalent of `Carbon::now()`; it returns a Carbon object representative of today, right at this second.

Carbon is a datetime library that's included with Laravel.

Modifying Routes to Allow Signed Links

Now that you've generated a link to your signed route, you need to protect against any unsigned access. The easiest option is to apply the signed middleware:

```
Route::get('invitations/{invitation}/{answer}', InvitationController::class)
    ->name('invitations')
    ->middleware('signed');
```

If you'd prefer, you can manually validate using the `hasValidSignature()` method on the Request object instead of using the signed middleware:

```
class InvitationController
{
    public function __invoke(Invitation $invitation, $answer, Request $request)
    {
        if (! $request->hasValidSignature()) {
            abort(403);
        }

        //
    }
}
```

Views

In a few of the route closures we've looked at so far, we've seen something along the lines of `return view('account')`. What's going on here?

In the MVC pattern ([Figure 3-1](#)), views (or templates) are files that describe what some particular output should look like. You might have views that output JSON or XML or email, but the most common views in a web framework output HTML.

In Laravel, there are two view formats you can use out of the box: plain PHP and Blade templates (see [Chapter 4](#)). The difference is in the filename: *about.php* will be rendered with the PHP engine, and *about.blade.php* will be rendered with the Blade engine.

THREE WAYS TO LOAD A VIEW

There are three ways to return a view. For now, just concern yourself with `view()`, but if you ever see `View::make()`, it's the same thing, or you could inject `Illuminate\View\ViewFactory` if you prefer.

Once you've "loaded" a view with the `view()` helper, you have the option to simply return it (as in [Example 3-18](#)), which will work fine if the view doesn't rely on any variables from the controller.

Example 3-18. Simple `view()` usage

```
Route::get('/', function () {  
    return view('home');  
});
```

This code looks for a view in *resources/views/home.blade.php* or *resources/views/home.php* and loads its contents and parses any inline PHP or control structures until you have just the view's output. Once you return it, it's passed on to the rest of the response stack and eventually returned to the user.

But what if you need to pass in variables? Take a look at [Example 3-19](#).

Example 3-19. Passing variables to views

```
Route::get('tasks', function () {  
    return view('tasks.index')  
        ->with('tasks', Task::all());  
});
```

This closure loads the *resources/views/tasks/index.blade.php* or *resources/views/tasks/index.php* view and passes it a single variable named *tasks*, which contains the result of the `Task::all()` method. `Task::all()` is an Eloquent database query you'll learn about in [Chapter 5](#).

Returning Simple Routes Directly with `Route::view()`

Because it's so common for a route to just return a view with no custom data, Laravel allows you to define a route as a “view” route without even passing the route definition a closure or a controller/method reference, as you can see in [Example 3-20](#).

Example 3-20. `Route::view()`

```
// Returns resources/views/welcome.blade.php
Route::view('/', 'welcome');

// Passing simple data to Route::view()
Route::view('/', 'welcome', ['User' => 'Michael']);
```

Using View Composers to Share Variables with Every View

Sometimes it can become a hassle to pass the same variables over and over. There may be a variable that you want accessible to every view in the site or to a certain class of views or a certain included subview—for example, all views related to tasks or the header partial.

It's possible to share certain variables with every template or just certain templates, like in the following code:

```
view()->share('variableName', 'variableValue');
```

To learn more, check out [“View Composers and Service Injection”](#).

Controllers

I've mentioned controllers a few times, but until now, most of the examples have shown route closures. In the MVC pattern, controllers are essentially classes that organize the logic of one or more routes together in one place. Controllers tend

to group similar routes together, especially if your application is structured in a traditionally CRUD-like format; in this case, a controller might handle all the actions that can be performed on a particular resource.

WHAT IS CRUD?

CRUD stands for *create*, *read*, *update*, *delete*, which are the four primary operations that web applications most commonly provide on a resource. For example, you can create a new blog post, you can read that post, you can update it, or you can delete it.

It may be tempting to cram all of the application’s logic into the controllers, but it’s better to think of controllers as the traffic cops that route HTTP requests around your application. Since there are other ways requests can come into your application—cron jobs, Artisan command-line calls, queue jobs, etc.—it’s wise to not rely on controllers for much behavior. This means a controller’s primary job is to capture the intent of an HTTP request and pass it on to the rest of the application.

So, let’s create a controller. One easy way to do this is with an Artisan command, so from the command line, run the following:

```
php artisan make:controller TaskController
```

ARTISAN AND ARTISAN GENERATORS

Laravel comes bundled with a command-line tool called Artisan. Artisan can be used to run migrations, create users and other database records manually, and perform many other manual, one-time tasks.

Under the `make` namespace, Artisan provides tools for generating skeleton files for a variety of system files. That’s what allows us to run `php artisan make:controller`.

To learn more about this and other Artisan features, see [Chapter 8](#).

This will create a new file named *TaskController.php* in *app/Http/Controllers*, with the contents shown in [Example 3-21](#).

Example 3-21. Default generated controller

```
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;

class TaskController extends Controller
{
    //
}
```

Modify this file as shown in [Example 3-22](#), creating a new public method called `index()`. We'll just return some text there.

Example 3-22. Simple controller example

```
<?php

namespace App\Http\Controllers;

class TaskController extends Controller
{
    public function index()
    {
        return 'Hello, World!';
    }
}
```

Then, like we learned before, we'll hook up a route to it, as shown in [Example 3-23](#).

Example 3-23. Route for the simple controller

```
// routes/web.php
<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\TaskController;

Route::get('/', [TaskController::class, 'index']);
```

That's it. Visit the `/` route and you'll see the words "Hello, World!"

The most common use of a controller method, then, will be something like [Example 3-24](#), which provides the same functionality as our route closure in [Example 3-19](#).

Example 3-24. Common controller method example

```
// TaskController.php
...
public function index()
{
    return view('tasks.index')
        ->with('tasks', Task::all());
}
```

This controller method loads the *resources/views/tasks/index.blade.php* or *resources/views/tasks/index.php* view and passes it a single variable named *tasks*, which contains the result of the `Task::all()` Eloquent method.

GENERATING RESOURCE CONTROLLERS

If you want to create a resource controller with autogenerated methods for all the basic resource routes like `create()` and `update()`, you can pass the `--resource` flag when using `php artisan make:controller`:

```
php artisan make:controller TaskController --resource
```

Getting User Input

The second most common action to perform in a controller method is to take input from the user and act on it. That introduces a few new concepts, so let's take a look at a bit of sample code and walk through the new pieces.

First, let's bind our route; see [Example 3-25](#).

Example 3-25. Binding basic form actions

```
// routes/web.php
Route::get('tasks/create', [TaskController::class, 'create']);
Route::post('tasks', [TaskController::class, 'store']);
```

Notice that we're binding the GET action of `tasks/create` (which shows a form for creating a new task) and the POST action of `tasks` (which is where our form will POST to when we're creating a new task). We can assume the `create()` method in our controller just shows a form, so let's look at the `store()` method in [Example 3-26](#).

Example 3-26. Common form input controller method

```
// TaskController.php
...
public function store()
{
    Task::create(request()->only(['title', 'description']));

    return redirect('tasks');
}
```

This example makes use of Eloquent models and the `redirect()` functionality, and we'll talk about them more later, but for now let's talk quickly about how we're getting our data here.

We're using the `request()` helper to represent the HTTP request (more on that later) and using its `only()` method to pull just the `title` and `description` fields the user submitted.

We're then passing that data into the `create()` method of our `Task` model, which creates a new instance of the `Task` with `title` set to the passed-in title and `description` set to the passed-in description. Finally, we redirect back to the page that shows all tasks.

There are a few layers of abstraction at work here, which we'll cover in a second, but know that the data coming from the `only()` method comes from the same pool of data all common methods used on the `Request` object draw from, including `all()` and `get()`. The set of data each of these methods is pulling from represents all user-provided data, whether from query parameters or POST values. So, our user filled out two fields on the “add task” page: “title” and “description.”

To break down the abstraction a bit, `request()->only()` takes an associative array of input names and returns them:

```
request()->only(['title', 'description']);
// returns:
[
    'title' => 'Whatever title the user typed on the previous page',
    'description' => 'Whatever description the user typed on the previous page',
]
```

And `Task::create()` takes an associative array and creates a new task from it:

```
Task::create([
    'title' => 'Buy milk',
    'description' => 'Remember to check the expiration date this time, Norbert!',
]);
```

Combining them together creates a task with just the user-provided “title” and “description” fields.

Injecting Dependencies into Controllers

Laravel’s facades and global helpers present a simple interface to the most useful classes in Laravel’s codebase. You can get information about the current request and user input, the session, caches, and much more.

But if you prefer to inject your dependencies, or if you want to use a service that doesn’t have a facade or a helper, you’ll need to find some way to bring instances of these classes into your controller.

This is our first exposure to Laravel’s service container. For now, if this is unfamiliar, you can think about it as a little bit of Laravel magic; or, if you want to know more about how it’s actually functioning, you can skip ahead to [Chapter 11](#).

All controller methods (including the constructors) are resolved out of Laravel’s container, which means anything you typehint that the container knows how to resolve will be automatically injected.

TYPEHINTS IN PHP

Typehinting in PHP means putting the name of a class or interface in front of a variable in a method signature:

```
public function __construct(Logger $logger) {}
```

This typehint is telling PHP that whatever is passed into the method *must* be of type `Logger`, which could be either an interface or a class.

As a nice example, what if you’d prefer having an instance of the `Request` object instead of using the global helper? Just typehint

`\Illuminate\Http\Request` in your method parameters, like in [Example 3-27](#).

Example 3-27. Controller method injection via typehinting

```
// TaskController.php
...
public function store(\Illuminate\Http\Request $request)
{
    Task::create($request->only(['title', 'description']));

    return redirect('tasks');
}
```

So, you’ve defined a parameter that must be passed into the `store()` method. And since you typehinted it, and since Laravel knows how to resolve that class name, you’re going to have the `Request` object ready for you to use in your method with no work on your part. No explicit binding, no anything else—it’s just there as the `$request` variable.

And, as you can tell from comparing [Examples 3-26](#) and [3-27](#), the `request()` helper and the `Request` object behave exactly the same.

Resource Controllers

Sometimes naming the methods in your controllers can be the hardest part of writing a controller. Thankfully, Laravel has some conventions for all of the routes of a traditional REST/CRUD controller (called a *resource controller* in Laravel); additionally, it comes with a generator out of the box and a convenience route definition that allows you to bind an entire resource controller at once.

To see the methods that Laravel expects for a resource controller, let’s generate a new controller from the command line:

```
php artisan make:controller MySampleResourceController --resource
```

Now open `app/Http/Controllers/MySampleResourceController.php`. You’ll see it comes prefilled with quite a few methods. Let’s walk through what each represents. We’ll use a `Task` as an example.

The methods of Laravel’s resource controllers

Remember the table from earlier? **Table 3-1** shows the HTTP verb, the URL, the controller method name, and the name for each of these default methods that are generated in Laravel’s resource controllers.

Binding a resource controller

So, we’ve seen that these are the conventional route names to use in Laravel, and also that it’s easy to generate a resource controller with methods for each of these default routes. Thankfully, you don’t have to generate routes for each of these controller methods by hand, if you don’t want to. There’s a trick for that, called *resource controller binding*. Take a look at **Example 3-28**.

Example 3-28. Resource controller binding

```
// routes/web.php
Route::resource('tasks', TaskController::class);
```

This will automatically bind all of the routes listed in **Table 3-1** for this resource to the appropriate method names on the specified controller. It’ll also name these routes appropriately; for example, the `index()` method on the `tasks` resource controller will be named `tasks.index`.

ARTISAN ROUTE:LIST

If you ever find yourself in a situation where you’re wondering what routes your current application has available, there’s a tool for that: from the command line, run `php artisan route:list` and you’ll get a listing of all of the available routes. I prefer `php artisan route:list --exclude-vendor` so I don’t see all the weird routes my dependencies register in order for them to operate (see **Figure 3-2**).

```
mattstauffer at LaunchpdMcQuack in ~/RealSites/book-up-and-running
○ php artisan route:list --except-vendor
```

```
GET|HEAD / .....
GET|HEAD api/user .....
GET|HEAD dogs ..... DogsController@index
GET|HEAD dogs/create ..... DogsController@create
POST dogs/store ..... DogsController@store
GET|HEAD dogs/{dog} ..... DogsController@show
PUT dogs/{dog} ..... DogsController@update
DELETE dogs/{dog} ..... DogsController@destroy
GET|HEAD dogs/{dog}/edit ..... DogsController@edit
```

Showing [9] routes

Figure 3-2. `artisan route:list`

API Resource Controllers

When you’re working with RESTful APIs, the list of potential actions on a resource is not the same as it is with an HTML resource controller. For example, you can send a POST request to an API to create a resource, but you can’t really “show a create form” in an API.

To generate an *API resource controller*, which is a controller that has the same structure as a resource controller except it excludes the *create* and *edit* actions, pass the `--api` flag when creating a controller:

```
php artisan make:controller MySampleResourceController --api
```

To bind an API resource controller, use the `apiResource()` method instead of the `resource()` method, as shown in [Example 3-29](#).

Example 3-29. API resource controller binding

```
// routes/web.php
Route::apiResource('tasks', TaskController::class);
```

Single Action Controllers

There will be times in your applications when a controller should only service a single route. You may find yourself wondering how to name the controller method for that route. Thankfully, you can point a single route at a single controller without concerning yourself with naming the one method.

As you may already know, the `__invoke()` method is a PHP magic method that allows you to “invoke” an instance of a class, treating it like a function and calling it. This is the tool Laravel’s *single action controllers* use to allow you to point a route to a single controller, as you can see in [Example 3-30](#).

Example 3-30. Using the `__invoke()` method

```
// \App\Http\Controllers\UpdateUserAvatar.php
public function __invoke(User $user)
{
    // Update the user's avatar image
}

// routes/web.php
Route::post('users/{user}/update-avatar', UpdateUserAvatar::class);
```

Route Model Binding

One of the most common routing patterns is that the first line of any controller method tries to find the resource with the given ID, like in [Example 3-31](#).

Example 3-31. Getting a resource for each route

```
Route::get('conferences/{id}', function ($id) {
    $conference = Conference::findOrFail($id);
});
```

Laravel provides a feature that simplifies this pattern called *route model binding*. This allows you to define that a particular parameter name (e.g., `{conference}`) will indicate to the route resolver that it should look up an Eloquent database record with that ID and then pass it in as the parameter *instead* of just passing the ID.

There are two kinds of route model binding: implicit and custom (or explicit).

Implicit Route Model Binding

The simplest way to use route model binding is to name your route parameter something unique to that model (e.g., name it `$conference` instead of `$id`), then typehint that parameter in the closure/controller method and use the same variable name there. It’s easier to show than to describe, so take a look at [Example 3-32](#).

Example 3-32. Using an implicit route model binding

```
Route::get('conferences/{conference}', function (Conference $conference) {  
    return view('conferences.show')->with('conference', $conference);  
});
```

Because the route parameter (`{conference}`) is the same as the method parameter (`$conference`), and the method parameter is typehinted with a `Conference` model (`Conference $conference`), Laravel sees this as a route model binding. Every time this route is visited, the application will assume that whatever is passed into the URL in place of `{conference}` is an ID that should be used to look up a `Conference`, and then that resulting model instance will be passed in to your closure or controller method.

CUSTOMIZING THE ROUTE KEY FOR AN ELOQUENT MODEL

Any time an Eloquent model is looked up via a URL segment (usually because of route model binding), the default column Eloquent will look it up by is its primary key (ID).

To change the column your Eloquent model uses for URL lookups in all your routes, add a method to your model named `getRouteKeyName()`:

```
public function getRouteKeyName()  
{  
    return 'slug';  
}
```

Now, a URL like `conferences/{conference}` will expect to get an entry from the `slug` column instead of the `ID` and will perform its lookups accordingly.

CUSTOMIZING THE ROUTE KEY IN A SPECIFIC ROUTE

In Laravel, you can also change the route key on a specific route instead of globally by appending a colon and the column name in the route definition:

```
Route::get(  
    'conferences/{conference:slug}',  
    function (Conference $conference) {  
        return view('conferences.show')  
            ->with('conference', $conference);  
    });
```

If you have two dynamic segments in your URL (for example: `organizers/{organizer}/conferences/{conference:slug}`), Laravel will automatically attempt to scope the second model's queries to only those related to the first. So it'll check the `Organizer` model for a `conferences` relationship, and if it exists, only return `Conferences` that are associated with the `Organizer` found by the first segment.

```
use App\Models\Conference;
use App\Models\Organizer;

Route::get(
    'organizers/{organizer}/conferences/{conference:slug}',
    function (Organizer $organizer, Conference $conference) {
        return $conference;
    });
```

Custom Route Model Binding

To manually configure route model bindings, add a line like the one in [Example 3-33](#) to the `boot()` method in `App\Providers\RouteServiceProvider`.

Example 3-33. Adding a route model binding

```
public function boot()
{
    // Perform the binding
    Route::model('event', Conference::class);
}
```

You've now specified that whenever a route has a parameter in its definition named `{event}`, as demonstrated in [Example 3-34](#), the route resolver will return an instance of the `Conference` class with the ID of that URL parameter.

Example 3-34. Using an explicit route model binding

```
Route::get('events/{event}', function (Conference $event) {
    return view('events.show')->with('event', $event);
});
```

Route Caching

If you're looking to squeeze every millisecond out of your load time, you may want to take a look at *route caching*. One of the pieces of Laravel's bootstrap that can take anywhere from a few dozen to a few hundred milliseconds is parsing the *routes/** files, and route caching speeds up this process dramatically.

To cache your routes file, you need to be using all controller, redirect, view, and resource routes (no route closures). If your app isn't using any route closures, you can run `php artisan route:cache` and Laravel will serialize the results of your *routes/** files. If you want to delete the cache, run `php artisan route:clear`.

Here's the drawback: Laravel will now match routes against that cached file instead of your actual *routes/** files. You can make endless changes to your routes files, and they won't take effect until you run `route:cache` again. This means you'll have to recache every time you make a change, which introduces a lot of potential for confusion.

Here's what I would recommend instead: since Git ignores the route cache file by default anyway, consider only using route caching on your production server, and run the `php artisan route:cache` command every time you deploy new code (whether via a Git post-deploy hook, a Forge deploy command, or as a part of whatever other deployment system you use). This way, you won't have confusing local development issues, but your remote environment will still benefit from route caching.

Form Method Spoofing

Sometimes you need to manually define which HTTP verb a form should send as. HTML forms only allow for GET or POST, so if you want any other sort of verb, you'll need to specify that yourself.

HTTP Verbs in Laravel

As we've seen already, you can define which verbs a route will match in the route definition using `Route::get()`, `Route::post()`, `Route::any()`, or

`Route::match()`. You can also match with `Route::patch()`, `Route::put()`, and `Route::delete()`.

But how does one send a request other than GET with a web browser? First, the `method` attribute in an HTML form determines its HTTP verb: if your form has a `method` of "GET", it will submit via query parameters and a GET method; if the form has a `method` of "POST", it will submit via the post body and a POST method.

JavaScript frameworks make it easy to send other requests, like DELETE and PATCH. But if you find yourself needing to submit HTML forms in Laravel with verbs other than GET or POST, you'll need to use *form method spoofing*, which means spoofing the HTTP method in an HTML form.

HTTP Method Spoofing in HTML Forms

To inform Laravel that the form you're currently submitting should be treated as something other than a POST, add a hidden variable named `_method` with the value of "PUT", "PATCH", or "DELETE", and Laravel will match and route that form submission as if it were actually a request with that verb.

The form in [Example 3-35](#), since it's passing Laravel the method of "DELETE", will match routes defined with `Route::delete()`, but not those with `Route::post()`.

Example 3-35. Form method spoofing

```
<form action="/tasks/5" method="POST">
  <input type="hidden" name="_method" value="DELETE">
  <!-- or: -->
  @method('DELETE')
</form>
```

CSRF Protection

If you've tried to submit a form in a Laravel application already, including the one in [Example 3-35](#), you've likely run into the dreaded `TokenMismatchException`.

By default, all routes in Laravel except “read-only” routes (those using GET, HEAD, or OPTIONS) are protected against cross-site request forgery (CSRF) attacks by requiring a token, in the form of an input named `_token`, to be passed along with each request. This token is generated at the start of every session, and every non-read-only route compares the submitted `_token` against the session token.

WHAT IS CSRF?

A *cross-site request forgery* is when one website pretends to be another. The goal is for someone to hijack your users’ access to your website by submitting forms from *their* website to *your* website via the logged-in user’s browser.

The best way around CSRF attacks is to protect all inbound routes—POST, DELETE, etc.—with a token, which Laravel does out of the box.

You have two options for getting around this CSRF error. The first, and preferred, method is to add the `_token` input to each of your submissions. In HTML forms, there’s a simple way to do it, as you can see in [Example 3-36](#).

Example 3-36. CSRF tokens

```
<form action="/tasks/5" method="POST">
  @csrf
</form>
```

In JavaScript applications, it takes a bit more work, but not much. The most common solution for sites using JavaScript frameworks is to store the token on every page in a `<meta>` tag like this one:

```
<meta name="csrf-token" content="<?php echo csrf_token(); ?>">
```

Storing the token in a `<meta>` tag makes it easy to bind it to the correct HTTP header, which you can do once globally for all requests from your JavaScript framework, like in [Example 3-37](#).

Example 3-37. Globally binding a header for CSRF

```
// In jQuery:
$.ajaxSetup({
  headers: {
```

```
        'X-CSRF-TOKEN': $('meta[name="csrf-token"]').attr('content')
    }
});
```

// With Axios: it automatically retrieves it from a cookie. Nothing to do!

Laravel will check the X-CSRF-TOKEN (and X-XSRF-TOKEN, which Axios and other JavaScript frameworks like Angular use) on every request, and valid tokens passed there will mark the CSRF protection as satisfied.

BINDING CSRF TOKENS WITH VUE RESOURCE

Bootstrapping the CSRF token into Vue Resource looks a bit different than it does for Laravel; see the [Vue Resource docs](#) for examples.

Redirects

So far, the only things we've explicitly talked about returning from a controller method or route definition have been views. But there are a few other structures we can return to give the browser instructions on how to behave.

First, let's cover the *redirect*. You've already seen a few of these in other examples. There are two common ways to generate a redirect; we'll use the `redirect()` global helper here, but you may prefer the facade. Both create an instance of `Illuminate\Http\RedirectResponse`, perform some convenience methods on it, and then return it. You can also do this manually, but you'll have to do a little more work yourself. Take a look at [Example 3-38](#) to see a few ways you can return a redirect.

Example 3-38. Different ways to return a redirect

```
// Using the global helper to generate a redirect response
Route::get('redirect-with-helper', function () {
    return redirect()->to('login');
});

// Using the global helper shortcut
Route::get('redirect-with-helper-shortcut', function () {
    return redirect('login');
});
```

```
// Using the facade to generate a redirect response
Route::get('redirect-with-facade', function () {
    return Redirect::to('login');
});

// Using the Route::redirect shortcut
Route::redirect('redirect-by-route', 'login');
```

Note that the `redirect()` helper exposes the same methods as the `Redirect` facade, but it also has a shortcut; if you pass parameters directly to the helper instead of chaining methods after it, it's a shortcut to the `to()` redirect method.

Also note that the (optional) third parameter for the `Route::redirect()` route helper can be the status code (e.g., 302) for your redirect.

redirect()->to()

The method signature for the `to()` method for redirects looks like this:

```
function to($to = null, $status = 302, $headers = [], $secure = null)
```

`$to` is a valid internal path, `$status` is the HTTP status (defaulting to 302), `$headers` allows you to define which HTTP headers to send along with your redirect, and `$secure` allows you to override the default choice of http versus https (which is normally set based on your current request URL). [Example 3-39](#) shows an example of its use.

Example 3-39. redirect()->to()

```
Route::get('redirect', function () {
    return redirect()->to('home');

    // Or same, using the shortcut:

    return redirect('home');
});
```

redirect()->route()

The `route()` method is the same as the `to()` method, but rather than pointing to a particular path, it points to a particular route name (see [Example 3-40](#)).

Example 3-40. `redirect()->route()`

```
Route::get('redirect', function () {  
    return redirect()->route('conferences.index');  
});
```

Note that since some route names require parameters, its parameter order is a little different. `route()` has an optional second parameter for the route parameters:

```
function route($to = null, $parameters = [], $status = 302, $headers = [])
```

So, using it might look a little like [Example 3-41](#).

Example 3-41. `redirect()->route()` with parameters

```
Route::get('redirect', function () {  
    return to_route('conferences.show', [  
        'conference' => 99,  
    ];  
});
```

REDIRECT TO_ROUTE() HELPER

You can use the `to_route()` helper as an alias for the `redirect()->route()` method. The signature for both is the same:

```
Route::get('redirect', function () {  
    return to_route('conferences.show', ['conference' => 99]);  
});
```

`redirect()->back()`

Because of some of the built-in conveniences of Laravel's session implementation, your application will always have knowledge of what the user's previously visited page was. That opens up the opportunity for a `redirect()->back()` redirect, which simply redirects the user to whatever page they came from. There's also a global shortcut for this: `back()`.

Other Redirect Methods

The redirect service provides other methods that are less commonly used, but still available:

`refresh()`

Redirects to the same page the user is currently on.

`away()`

Allows for redirecting to an external URL without the default URL validation.

`secure()`

Like `to()` with the `secure` parameter set to `"true"`.

`action()`

Allows you to link to a controller and method in one of two ways: as a string (`redirect()->action('MyController@myMethod')`) or as a tuple (`redirect()\->action([MyController::class, 'myMethod'])`).

`guest()`

Used internally by the authentication system (discussed in [Chapter 9](#)); when a user visits a route they're not authenticated for, this captures the "intended" route and then redirects the user (usually to a login page).

`intended()`

Also used internally by the authentication system; after a successful authentication, this grabs the "intended" URL stored by the `guest()` method and redirects the user there.

`redirect()->with()`

While it is structured similarly to the other methods you can call on `redirect()`, `with()` is different in that it doesn't define where you're redirecting to, but what data you're passing along with the redirect. When you're redirecting users to different pages, you often want to pass certain data along with them. You could

manually flash the data to the session, but Laravel has some convenience methods to help you with that.

Most commonly, you can pass along either an array of keys and values or a single key and value using `with()`, like in [Example 3-42](#). This saves your `with()` data to the session just for the next page load.

Example 3-42. Redirect with data

```
Route::get('redirect-with-key-value', function () {
    return redirect('dashboard')
        ->with('error', true);
});

Route::get('redirect-with-array', function () {
    return redirect('dashboard')
        ->with(['error' => true, 'message' => 'Whoops!']);
});
```

CHAINING METHODS ON REDIRECTS

As with many other facades, most calls to the `Redirect` facade can accept fluent method chains, like the `with()` calls in [Example 3-42](#). You'll learn more about fluency in [“What Is a Fluent Interface?”](#).

You can also use `withInput()`, as in [Example 3-43](#), to redirect with the user's form input flashed; this is most common in the case of a validation error, where you want to send the user back to the form they just came from.

Example 3-43. Redirect with form input

```
Route::get('form', function () {
    return view('form');
});

Route::post('form', function () {
    return redirect('form')
        ->withInput()
        ->with(['error' => true, 'message' => 'Whoops!']);
});
```

The easiest way to get the flashed input that was passed with `withInput()` is using the `old()` helper, which can be used to get all old input (`old()`) or just the value for a particular key, as shown in the following example, with the second

parameter as the default if there is no old value. You’ll commonly see this in views, which allows this HTML to be used both on the “create” and the “edit” view for this form:

```
<input name="username" value="<?=
    old('username', 'Default username instructions here');
?>">
```

Speaking of validation, there is also a useful method for passing errors along with a redirect response: `withErrors()`. You can pass it any “provider” of errors, which may be an error string, an array of errors, or, most commonly, an instance of the `Illuminate Validator`, which we’ll cover in [Chapter 10](#).

[Example 3-44](#) shows an example of its use.

Example 3-44. Redirect with errors

```
Route::post('form', function (Illuminate\Http\Request $request) {
    $validator = Validator::make($request->all(), $this->validationRules);

    if ($validator->fails()) {
        return back()
            ->withErrors($validator)
            ->withInput();
    }
});
```

`withErrors()` automatically shares an `$errors` variable with the views of the page it’s redirecting to, for you to handle however you’d like.

THE VALIDATE() METHOD ON REQUESTS

Don’t like how [Example 3-44](#) looks? There’s a simple and powerful tool that will make it easy for you to clean up that code. Read more in [“validate\(\) on the Request Object”](#).

Aborting the Request

Aside from returning views and redirects, the most common way to exit a route is to abort. There are a few globally available methods (`abort()`, `abort_if()`, and `abort_unless()`), which optionally take HTTP status codes, a message, and a headers array as parameters.

As [Example 3-45](#) shows, `abort_if()` and `abort_unless()` take a first parameter that is evaluated for its truthiness and perform the abort depending on the result.

Example 3-45. 403 Forbidden aborts

```
Route::post('something-you-cant-do', function (Illuminate\Http\Request $request) {  
    abort(403, 'You cannot do that!');  
    abort_unless($request->has('magicToken'), 403);  
    abort_if($request->user()->isBanned, 403);  
});
```

Custom Responses

There are a few other options available for us to return, so let's go over the most common responses after views, redirects, and aborts. Just like with redirects, you can run these methods on either the `response()` helper or the `Response` facade.

`response()->make()`

If you want to create an HTTP response manually, just pass your data into the first parameter of `response()->make()`: for example, return `response()->make(Hello, World!)`. Once again, the second parameter is the HTTP status code and the third is your headers.

`response()->json()` and `->jsonp()`

To create a JSON-encoded HTTP response manually, pass your JSON-able content (arrays, collections, or whatever else) to the `json()` method: for example return `response()->json(User::all())`. It's just like `make()`, except it `json_`encodes your content and sets the appropriate headers.

`response()->download()`, `->streamDownload()`, and `->file()`

To send a file for the end user to download, pass either an `SplFileInfo` instance or a string filename to `download()`, with an optional second parameter of the download filename: for example, return `response()->download('file501751.pdf', 'myFile.pdf')`, which would send a file

that's at *file501751.pdf* and rename it, as it's sent, to *myFile.pdf*.

To display the same file in the browser (if it's a PDF or an image or something else the browser can handle), use `response()->file()` instead, which takes the same parameters as `response()->download()`.

If you want to make some content from an external service available as a download without having to write it directly to your server's disk, you can stream the download using `response()->streamDownload()`. This method expects as parameters a closure that echoes a string, a filename, and optionally an array of headers; see [Example 3-46](#).

Example 3-46. Streaming downloads from external servers

```
return response()->streamDownload(function () {  
    echo DocumentService::file('myFile')->getContent();  
}, 'myFile.pdf');
```

Testing

In some other communities the idea of unit-testing controller methods is common, but within Laravel (and most of the PHP community) it's typical to rely on *application testing* to test the functionality of routes.

For example, to verify that a POST route works correctly, we can write a test like [Example 3-47](#).

Example 3-47. Writing a simple POST route test

```
// tests/Feature/AssignmentTest.php  
public function test_post_creates_new_assignment()  
{  
    $this->post('/assignments', [  
        'title' => 'My great assignment',  
    ]);  
  
    $this->assertDatabaseHas('assignments', [  
        'title' => 'My great assignment',  
    ]);  
}
```

Did we directly call the controller methods? No. But we ensured that the goal of this route—to receive a POST and save its important information to the database—was met.

You can also use similar syntax to visit a route and verify that certain text shows up on the page, or that clicking certain buttons does certain things (see [Example 3-48](#)).

Example 3-48. Writing a simple GET route test

```
// AssignmentTest.php
public function test_list_page_shows_all_assignments()
{
    $assignment = Assignment::create([
        'title' => 'My great assignment',
    ]);

    $this->get('/assignments')
        ->assertSee('My great assignment');
}
```

TL;DR

Laravel's routes are defined in *routes/web.php* and *routes/api.php*. You can define the expected path for each route, which segments are static and which are parameters, which HTTP verbs can access the route, and how to resolve it. You can also attach middleware to routes, group them, and give them names.

What is returned from the route closure or controller method dictates how Laravel responds to the user. If it's a string or a view, it's presented to the user; if it's other sorts of data, it's converted to JSON and presented to the user; and if it's a redirect, it forces a redirect.

Laravel provides a series of tools and conveniences to simplify common routing-related tasks and structures. These include resource controllers, route model binding, and form method spoofing.

Chapter 4. Blade Templating

Compared to most other backend languages, PHP actually functions relatively well as a templating language. But it has its shortcomings, and it's also just ugly to be using `<?php` inline all over the place, so you can expect most modern frameworks to offer a templating language.

Laravel offers a custom templating engine called *Blade*, which is inspired by .NET's Razor engine. It boasts a concise syntax, a shallow learning curve, a powerful and intuitive inheritance model, and easy extensibility.

For a quick look at what writing Blade looks like, check out [Example 4-1](#).

Example 4-1. Blade samples

```
<h1>{{ $group->title }}</h1>
{!! $group->heroImageHtml() !!}

@forelse ($users as $user)
    • {{ $user->first_name }} {{ $user->last_name }}<br>
@empty
    No users in this group.
@endforelse
```

As you can see, Blade uses curly braces for its “echo” and introduces a convention in which its custom tags, called “directives,” are prefixed with an `@`. You'll use directives for all of your control structures and also for inheritance and any custom functionality you want to add.

Blade's syntax is clean and concise, so at its core it's just more pleasant and tidy to work with than the alternatives. But the moment you need anything of any complexity in your templates—nested inheritance, complex conditionals, or recursion—Blade starts to really shine. Just like the best Laravel components, it takes complex application requirements and makes them easy and accessible.

Additionally, since all Blade syntax is compiled into normal PHP code and then cached, it's fast and it allows you to use native PHP in your Blade files if you want. However, I'd recommend avoiding using PHP if at all possible—usually if you need to do anything that you can't do with Blade or a custom Blade

directive, it doesn't belong in the template.

USING TWIG WITH LARAVEL

Unlike many other Symfony-based frameworks, Laravel doesn't use Twig by default. But if you're just in love with Twig, there's a **TwigBridge package** that makes it easy to use Twig instead of Blade.

Echoing Data

As you can see in **Example 4-1**, `{{` and `}}` are used to wrap sections of PHP that you'd like to echo. `{{ $variable }}` is similar to `<?= $variable ?>` in plain PHP.

It's different in one way, however, and you might've guessed this already: Blade escapes all echoes by default using PHP's `htmlspecialchars()` to protect your users from malicious script insertion. That means `{{ $variable }}` is functionally equivalent to `<?= htmlspecialchars($variable) ?>`. If you want to echo without the escaping, use `{!!` and `!!}}` instead.

{{ AND }} WHEN USING A FRONTEND TEMPLATING FRAMEWORK

You might've noticed that the echo syntax for Blade (`{{ }}`) is similar to the echo syntax for many frontend frameworks. So how does Laravel know when you're writing Blade versus Handlebars?

Blade will ignore any `{{` that's prefaced with an `@`. So, it will parse the first of the following examples, but the second will be echoed out directly:

```
// Parsed as Blade; the value of $bladeVariable is echoed to the view
{{ $bladeVariable }}
```

```
// @ is removed and "{{ handlebarsVariable }}" echoed to the view directly
@{{ handlebarsVariable }}
```

You can also wrap any large sections of script content with the **@verbatim**

directive.

Control Structures

Most of the control structures in Blade will be very familiar. Many directly echo the name and structure of the same tag in PHP.

There are a few convenience helpers, but in general, the control structures just look cleaner than they would in PHP.

Conditionals

First, let's take a look at the control structures that allow for logic.

@if

Blade's `@if ($condition)` compiles to `<?php if ($condition): ?>`. `@else`, `@elseif`, and `@endif` also compile to the exact same style of syntax in PHP.

Take a look at [Example 4-2](#) for some examples.

Example 4-2. @if, @else, @elseif, and @endif

```
@if (count($talks) === 1)
    There is one talk at this time period.
@elseif (count($talks) === 0)
    There are no talks at this time period.
@else
    There are {{ count($talks) }} talks at this time period.
@endif
```

Just like with the native PHP conditionals, you can mix and match these how you want. They don't have any special logic; there's literally a parser looking for something with the shape of `@if ($condition)` and replacing it with the appropriate PHP code.

@unless and @endunless

`@unless`, on the other hand, is a new syntax that doesn't have a direct equivalent in PHP. It's the direct inverse of `@if`. `@unless ($condition)` is the same as `<?php if (! $condition)`. You can see it in use in [Example 4-3](#).

Example 4-3. @unless and @endunless

```
@unless ($user->hasPaid())  
    You can complete your payment by switching to the payment tab.  
@endunless
```

Loops

Next, let's take a look at the loops.

@for, @foreach, and @while

@for, @foreach, and @while work the same in Blade as they do in PHP; see Examples 4-4, 4-5, and 4-6.

Example 4-4. @for and @endfor

```
@for ($i = 0; $i < $talk->slotsCount(); $i++)  
    The number is {{ $i }}<br>  
@endfor
```

Example 4-5. @foreach and @endforeach

```
@foreach ($talks as $talk)  
    • {{ $talk->title }} ({{ $talk->length }} minutes)<br>  
@endforeach
```

Example 4-6. @while and @endwhile

```
@while ($item = array_pop($items))  
    {{ $item->orSomething() }}<br>  
@endwhile
```

@forelse and @endforelse

@forelse is a @foreach that also allows you to program in a fallback if the object you're iterating over is empty. We saw it in action at the start of this chapter; Example 4-7 shows another example.

Example 4-7. @forelse

```
@forelse ($talks as $talk)  
    • {{ $talk->title }} ({{ $talk->length }} minutes)<br>  
@empty  
    No talks this day.  
@endforelse
```

\$LOOP WITHIN @FOREACH AND @FORELSE

The @foreach and @forelse directives add one feature that's not available in PHP foreach loops: the \$loop variable. When used within a @foreach or @forelse loop, this variable will return a stdClass object with these properties:

index

The 0-based index of the current item in the loop; 0 would mean “first item.”

iteration

The 1-based index of the current item in the loop; 1 would mean “first item.”

remaining

How many items remain in the loop.

count

The count of items in the loop.

first *and* last

Booleans indicating whether this is the first or last item in the loop.

even *and* odd

Booleans indicating whether this is an even or odd iteration.

depth

How many “levels” deep this loop is: 1 for a loop, 2 for a loop within a loop, etc.

parent

A reference to the \$loop variable for the parent loop item if this loop is

within another @foreach loop; otherwise, null.

Here's an example of how to use it:

```
<ul>
@foreach ($pages as $page)
  <li>{{ $loop->iteration }}: {{ $page->title }}
    @if ($page->hasChildren())
      <ul>
        @foreach ($page->children() as $child)
          <li>{{ $loop->parent->iteration }}
            .{{ $loop->iteration }}:
            {{ $child->title }}</li>
        @endforeach
      </ul>
    @endif
  </li>
@endforeach
</ul>
```

Template Inheritance

Blade provides a structure for template inheritance that allows views to extend, modify, and include other views.

Let's take a look at how inheritance is structured with Blade.

Defining Sections with @section/@show and @yield

Let's start with a top-level Blade layout, like in [Example 4-8](#). This is the definition of a generic page wrapper that we'll later place page-specific content into.

Example 4-8. Blade layout

```
<!-- resources/views/layouts/master.blade.php -->
<html>
  <head>
    <title>My Site | @yield('title', 'Home Page')</title>
  </head>
  <body>
    <div class="container">
```



```

        @yield('content')
    </div>
    @section('footerScripts')
        <script src="app.js"></script>
    @show
</body>
</html>

```

This looks a bit like a normal HTML page, but you can see we’ve *yielded* in two places (title and content) and we’ve defined a *section* in a third (footerScripts). We have three Blade directives here: `@yield('content')` alone, `@yield('title', 'Home Page')` with a defined default, and `@section/@show` with actual content in it.

Although they each look a little different, *all three function essentially the same*. All three are defining that there’s a section with a given name (the first parameter) that can be extended later, and all three are defining what to do if the section isn’t extended. They do this either by providing a string fallback ('Home Page'), no fallback (which will just not show anything if it’s not extended), or an entire block fallback (in this case, `<script src="app.js"></script>`).

What’s different? Well, clearly, `@yield('content')` has no default content. But additionally, the default content in `@yield('title')` will *only* be shown if it’s never extended. If it is extended, its child sections will not have programmatic access to the default value. `@section/@show`, on the other hand, is both defining a default *and* doing so in such a way that its default contents will be available to its children, through `@parent`.

Once you have a parent layout like this, you can extend it in a new template file like in [Example 4-9](#).

Example 4-9. Extending a Blade layout

```

<!-- resources/views/dashboard.blade.php -->
@extends('layouts.master')

@section('title', 'Dashboard')

@section('content')
    Welcome to your application dashboard!
@endsection

@section('footerScripts')

```

```
@parent
<script src="dashboard.js"></script>
@endsection
```

@SHOW VERSUS @ENDSECTION

You may have noticed that [Example 4-8](#) uses `@section/@show`, but [Example 4-9](#) uses `@section/@endsection`. What's the difference?

Use `@show` when you're defining the place for a section, in the parent template. Use `@endsection` when you're defining the content for a template in a child template.

This child view allows us to cover a few new concepts in Blade inheritance.

@extends

In [Example 4-9](#), with `@extends('layouts.master')`, we define that this view should not be rendered on its own but that it instead *extends* another view. That means its role is to define the content of various sections, but not to stand alone. It's almost more like a series of buckets of content, rather than an HTML page. This line also defines that the view it's extending lives at *resources/views/layouts/master.blade.php*.

Each file should only extend one other file, and the `@extends` call should be the first line of the file.

@section and @endsection

With `@section('title', 'Dashboard')`, we provide our content for the first section, `title`. Since the content is so short, instead of using `@section` and `@endsection`, we're just using a shortcut. This allows us to pass the content in as the second parameter of `@section` and then move on. If it's a bit disconcerting to see `@section` without `@endsection`, you could just use the normal syntax.

With `@section('content')` and following, we use the normal syntax to define the contents of the `content` section. We'll just throw a little greeting in for now. Note, however, that when you're using `@section` in a child view, you end it with `@endsection` (or its alias `@stop`), instead of `@show`, which is reserved for

defining sections in parent views.

@parent

Finally, with `@section('footerScripts')` and on, we use the normal syntax to define the contents of the `footerScripts` section.

But remember, we actually defined that content (or, at least, its “default”) already in the master layout. So this time, we have two options: we can either *overwrite* the content from the parent view, or we can *add* to it.

You can see that we have the option to include the content from the parent by using the `@parent` directive within the section. If we didn’t, the content of this section would entirely overwrite anything defined in the parent for this section.

Including View Partial

Now that we’ve established the basics of inheritance, there are a few more tricks we can perform.

@include

What if we’re in a view and want to pull in another view? Maybe we have a call-to-action “Sign up” button that we want to reuse around the site. And maybe we want to customize the button text every time we use it. Take a look at

Example 4-10.

Example 4-10. Including view partials with @include

```
<!-- resources/views/home.blade.php -->
<div class="content" data-page-name="{{ $pageName }}">
    <p>Here's why you should sign up for our app: <strong>It's Great.</strong></p>

    @include('sign-up-button', ['text' => 'See just how great it is'])
</div>

<!-- resources/views/sign-up-button.blade.php -->
<a class="button button--callout" data-page-name="{{ $pageName }}">
    <i class="exclamation-icon"></i> {{ $text }}
</a>
```

`@include` pulls in the partial and, optionally, passes data into it. Note that not only can you *explicitly* pass data to an include via the second parameter of

`@include`, but you can also reference any variables within the included file that are available to the including view (`$pageName`, in this example). Once again, you can do whatever you want, but I would recommend you consider always explicitly passing every variable that you intend to use, just for clarity.

You also use the `@includeIf`, `@includeWhen`, and `@includeFirst` directives, as shown in [Example 4-11](#).

Example 4-11. Conditionally including views

```
{{!-- Include a view if it exists --}}
@includeIf('sidebars.admin', ['some' => 'data'])

{{!-- Include a view if a passed variable is truth-y --}}
@includeWhen($user->isAdmin(), 'sidebars.admin', ['some' => 'data'])

{{!-- Include the first view that exists from a given array of views --}}
@includeFirst(['customs.header', 'header'], ['some' => 'data'])
```

@each

You can probably imagine some circumstances in which you'd need to loop over an array or collection and `@include` a partial for each item. There's a directive for that: `@each`.

Let's say we have a sidebar composed of modules, and we want to include multiple modules, each with a different title. Take a look at [Example 4-12](#).

Example 4-12. Using view partials in a loop with @each

```
<!-- resources/views/sidebar.blade.php -->
<div class="sidebar">
    @each('partials.module', $modules, 'module', 'partials.empty-module')
</div>

<!-- resources/views/partials/module.blade.php -->
<div class="sidebar-module">
    <h1>{{ $module->title }}</h1>
</div>

<!-- resources/views/partials/empty-module.blade.php -->
<div class="sidebar-module">
    No modules :(
</div>
```

Consider that `@each` syntax. The first parameter is the name of the view partial.

The second is the array or collection to iterate over. The third is the variable name that each item (in this case, each element in the `$modules` array) will be passed as to the view. And the optional fourth parameter is the view to show if the array or collection is empty (or, optionally, you can pass a string in here that will be used as your template).

Using Components

Laravel offers another pattern for including content between views: *components*. Components make the most sense in contexts where you find yourself using view partials and passing large chunks of content into them as variables. Take a look at [Example 4-13](#) for an example of a modal, or popover, that might alert the user in response to an error or other action.

Example 4-13. A modal as an awkward view partial

```
<!-- resources/views/partials/modal.blade.php -->
<div class="modal">
    <h2>{{ $title }}</h2>
    <div>{!! $content !!}</div>
    <div class="close button etc">...</div>
</div>

<!-- in another template -->
@include('partials.modal', [
    'title' => 'Insecure password',
    'content' => '<p>The password you have provided is not valid. Here are the rules
    for valid passwords: [...]</p><p><a href="#">...</a></p>'
])
```

This is too much for these poor variables, and it's the perfect fit for a component.

Laravel's components are another way of structuring view partials that looks much closer to how components work in frontend frameworks like Vue. They may be more familiar to frontend developers, but they also have a few significant benefits compared to view partials, including that it's much easier to pass large sections of template code into them.

Take a look at [Example 4-14](#) to see how to refactor [Example 4-13](#) with components.

Example 4-14. A modal as a more appropriate component

```
<!-- resources/views/components/modal.blade.php -->
```

```

<div class="modal">
    <h2>{{ $title }}</h2>
    <div>{{ $slot }}</div>
    <div class="close button etc">...</div>
</div>

<!-- in another template -->
<x-modal title="Insecure password">
    <p>The password you have provided is not valid.
    Here are the rules for valid passwords: [...]</p>

    <p><a href="#">...</a></p>
</x-modal>

```

As you can see in [Example 4-14](#), components allow us to pull our HTML out of a cramped variable string and back into the template space.

Let's dig into more of the features of components, how they're structured, and how we write them.

Creating components

Components can exist either as purely Blade templates (*anonymous components*), or as Blade templates backed by a PHP class that injects data and functionality (*class-based components*).

If you only need a template, you can generate your component with the `--view` flag:

```
php artisan make:component modal --view
```

If you also want to generate the PHP class, exclude that flag:

```
php artisan make:component modal
```

If you'd like to group your components under folders, you can use the `.` separator:

```

# To create it:
php artisan make:component modals.cancellation

// To use it:
<x-modals.cancellation />

```

Passing data into components

There are four ways to pass data into components: string attributes, PHP attributes, the default slot, and named slots.

Passing data into components via attributes

Let's start with attributes. You can pass strings directly into components by passing attributes with no prefix, or you can pass PHP variables and expressions with a colon prefix, as you can see in [Example 4-15](#).

Example 4-15. Passing data to components via attributes

```
<!-- Passing the data in -->
<x-modal title="Title here yay" :width="$width" />
<!-- Accessing the data in the template -->
<div style="width: {{ $width }}">
    <h1>{{ $title }}</h1>
</div>
```

For class-based components, you'll need to define every attribute in the PHP class and set it as a public property on the class, as in [Example 4-16](#).

Example 4-16. Defining attributes as public on component classes

```
class Modal extends Component
{
    public function __construct(
        public string $title,
        public string $width,
    ) {}
}
```

For anonymous components, you'll need to define the attributes in a props array at the top of your template:

```
@props([
    'width',
    'title',
])

<div style="width: {{ $width }}">
    <h1>{{ $title }}</h1>
</div>
```

Passing data into components via slots

In [Example 4-14](#) you may have noticed that the contents of the modal were referred to as a variable, `$slot`. But where did this come from?

By default, every component that has an opening and a closing tag when it's referenced has a `$slot` variable, and it's filled with all the HTML between those two tags. In [Example 4-14](#), the `$slot` variable contains the two `<p>` tags and everything inside (and between) them.

But what if you need two or more slots? You can add more than just the default slot, giving each slot its own name and variable. Let's rework [Example 4-14](#) assuming we want to define the title in a slot; take a look at [Example 4-17](#).

Example 4-17. Defining multiple slots

```
<x-modal>
  <x-slot:title>
    <h2 class="uppercase">Password requirements not met</h2>
  </x-slot>

  <p>The password you have provided is not valid.
  Here are the rules for valid passwords: [...]</p>

  <p><a href="#">...</a></p>
</x-modal>
```

The contents of this new `$slot` variable will be accessible to the component template as a `$title` variable, just like the attribute was before.

Component methods

At times it can be helpful to have a helper method in a component that performs some logic. One common pattern is to use these methods for complex logic checks you'd rather keep out of your templates.

Components allow you to call any public method on their associated PHP class in the template by prefixing the method name with `$`, as you can see in [Example 4-18](#).

Example 4-18. Defining and calling component methods

```
// in the component definition
public function isPromoted($item)
{
    return $item->promoted_at !== null && ! $item->promoted_at->isPast();
}
```



```

<!-- in the template -->
<div>
    @if ($isPromoted($item))
        <!-- show promoted badge -->
    @endif
    <!-- ... -->
</div>

```

Attributes grab bag

Most of the attributes we'll pass into our components will be named, specific, and similar to passing parameters to a PHP function.

But sometimes there are just loose HTML attributes we want to pass in, almost always so they can be assigned to the root element of our template.

With components, you can grab all of those attributes at once, using the `$attributes` variable. This variable captures all the attributes not defined as properties and allows you to echo them out (by treating it like a string) or interact with some of its methods for grabbing or inspecting data.

Take a look at [the docs](#) to learn about all the ways you can interact with the `$attributes` object, but here is one very useful trick:

```

<!-- Merge default classes with passed-in classes -->
<!-- Definition -->
<div {{ $attributes->merge(['class' => 'p-4 m-4']) }}>
    {{ $message }}
</div>

<!-- Usage -->
<x-notice class="text-blue-200">
    Message here
</x-notice>

<!-- Outputs: -->
<div class="p-4 m-4 text-blue-200">
    Message here
</div>

```

Using Stacks

One common pattern that can be difficult to manage using basic Blade includes is when each view in a Blade includes hierarchy needs to add something to a

certain section—almost like adding an entry to an array.

The most common situation for this is when certain pages (and sometimes, more broadly, certain sections of a website) have specific, unique CSS and JavaScript files they need to load. Imagine you have a site-wide “global” CSS file, a “jobs section” CSS file, and an “apply for a job” page CSS file.

Blade’s *stacks* are built for exactly this situation. In your parent template, define a stack, which is just a placeholder. Then, in each child template you can “push” entries onto that stack with `@push/@endpush`, which adds them to the bottom of the stack in the final render. You can also use `@prepend/@endprepend` to add them to the top of the stack. [Example 4-19](#) illustrates.

Example 4-19. Using Blade stacks

```
<!-- resources/views/layouts/app.blade.php -->
<html>
<head>
    <link href="/css/global.css">
    <!-- the placeholder where stack content will be placed -->
    @stack('styles')
</head>
<body>
    <!-- // -->
</body>
</html>

<!-- resources/views/jobs.blade.php -->
@extends('layouts.app')

@push('styles')
    <!-- push something to the bottom of the stack -->
    <link href="/css/jobs.css">
@endpush

<!-- resources/views/jobs/apply.blade.php -->
@extends('jobs')

@prepend('styles')
    <!-- push something to the top of the stack -->
    <link href="/css/jobs--apply.css">
@endprepend
```

These generate the following result:

```
<html>
```

```

<head>
  <link href="/css/global.css">
  <!-- the placeholder where stack content will be placed -->
  <!-- push something to the top of the stack -->
  <link href="/css/jobs--apply.css">
  <!-- push something to the bottom of the stack -->
  <link href="/css/jobs.css">
</head>
<body>
  <!-- // -->
</body>
</html>

```

View Composers and Service Injection

As we covered in [Chapter 3](#), it's simple to pass data to our views from the route definition (see [Example 4-20](#)).

Example 4-20. Reminder of how to pass data to views

```

Route::get('passing-data-to-views', function () {
    return view('dashboard')
        ->with('key', 'value');
});

```

There may be times, however, when you find yourself passing the same data over and over to multiple views. Or you might find yourself using a header partial or something similar that requires some data; will you have to pass that data in from every route definition that might ever load that header partial?

Binding Data to Views Using View Composers

Thankfully, there's a simpler way. The solution is called a *view composer*, and it allows you to define that *any time a particular view loads, it should have certain data passed to it*—without the route definition having to pass that data in explicitly.

Let's say you have a sidebar on every page, which is defined in a partial named `partials.sidebar` (*resources/views/partials/sidebar.blade.php*) and then included on every page. This sidebar shows a list of the last seven posts that were published on your site. If it's on every page, every route definition would normally have to grab that list and pass it in, like in [Example 4-21](#).

Example 4-21. Passing sidebar data in from every route

```
Route::get('home', function () {
    return view('home')
        ->with('posts', Post::recent());
});

Route::get('about', function () {
    return view('about')
        ->with('posts', Post::recent());
});
```

That could get annoying quickly. Instead, we’re going to use view composers to “share” that variable with a prescribed set of views. We can do this a few ways, so let’s start simple and move up.

Sharing a variable globally

First, the simplest option: just globally “share” a variable with every view in your application, like in [Example 4-22](#).

Example 4-22. Sharing a variable globally

```
// Some service provider
public function boot()
{
    ...
    view()->share('recentPosts', Post::recent());
}
```

If you want to use `view()->share()`, the best place would be the `boot()` method of a service provider so that the binding runs on every page load. You can create a custom `ViewComposerServiceProvider` (see “[Service Providers](#)” for more detail), but for now just put it in

`App\Providers\AppServiceProvider` in the `boot()` method.

Using `view()->share()` makes the variable accessible to every view in the entire application, however, so it might be overkill.

View-scoped view composers with closures

The next option is to use a closure-based view composer to share variables with a single view, like in [Example 4-23](#).

Example 4-23. Creating a closure-based view composer

```
view()->composer('partials.sidebar', function ($view) {  
    $view->with('recentPosts', Post::recent());  
});
```

As you can see, we've defined the name of the view we want it shared with in the first parameter (`partials.sidebar`) and then passed a closure to the second parameter; in the closure we've used `$view->with()` to share a variable, but only with a specific view.

VIEW COMPOSERS FOR MULTIPLE VIEWS

Anywhere a view composer is binding to a particular view (like in [Example 4-23](#), which binds to `partials.sidebar`), you can pass an array of view names instead to bind to multiple views.

You can also use an asterisk in the view path, as in `partials.*` or `tasks.*`:

```
view()->composer(  
    ['partials.header', 'partials.footer'],  
    function ($view) {  
        $view->with('recentPosts', Post::recent());  
    }  
);  
  
view()->composer('partials.*', function ($view) {  
    $view->with('recentPosts', Post::recent());  
});
```

View-scoped view composers with classes

Finally, the most flexible, but also the most complex, option is to create a dedicated class for your view composer.

First, let's create the view composer class. There's no formally defined place for view composers to live, but the docs recommend `App\Http\ViewComposers`. So, let's create `App\Http\ViewComposers\RecentPostsComposer` like in [Example 4-24](#).

Example 4-24. A view composer

```
<?php
```

```

namespace App\Http\ViewComposers;

use App\Post;
use Illuminate\Contracts\View\View;

class RecentPostsComposer
{
    public function compose(View $view)
    {
        $view->with('recentPosts', Post::recent());
    }
}

```

As you can see, when this composer is called, it runs the `compose()` method, in which we bind the `recentPosts` variable to the result of running the `Post` model's `recent()` method.

Like the other methods of sharing variables, this view composer needs to have a binding somewhere. Again, you'd likely create a custom `ViewComposerServiceProvider`, but for now, as seen in [Example 4-25](#), we'll just put it in the `boot()` method of `App\Providers\AppServiceProvider`.

Example 4-25. Registering a view composer in `AppServiceProvider`

```

public function boot(): void
{
    view()->composer(
        'partials.sidebar',
        \App\Http\ViewComposers\RecentPostsComposer::class
    );
}

```

Note that this binding is the same as a closure-based view composer, but instead of passing a closure, we're passing the class name of our view composer. Now, every time Blade renders the `partials.sidebar` view, it'll automatically run our provider and pass the view a `recentPosts` variable set to the results of the `recent()` method on our `Post` model.

Blade Service Injection

There are three primary types of data we're most likely to inject into a view: collections of data to iterate over, single objects that we're displaying on the page, and services that generate data or views.

With a service, the pattern will most likely look like [Example 4-26](#), where we inject an instance of our analytics service into the route definition by typehinting it in the route’s method signature, and then pass it into the view.

Example 4-26. Injecting services into a view via the route definition constructor

```
Route::get('backend/sales', function (AnalyticsService $analytics) {
    return view('backend.sales-graphs')
        ->with('analytics', $analytics);
});
```

Just as with view composers, Blade’s service injection offers a convenient shortcut to reduce duplication in your route definitions. Normally, the content of a view using our analytics service might look like [Example 4-27](#).

Example 4-27. Using an injected navigation service in a view

```
<div class="finances-display">
    {{ $analytics->getBalance() }} / {{ $analytics->getBudget() }}
</div>
```

Blade service injection makes it easy to inject an instance of a class from the container directly into the view, like in [Example 4-28](#).

Example 4-28. Injecting a service directly into a view

```
@inject('analytics', 'App\Services\Analytics')

<div class="finances-display">
    {{ $analytics->getBalance() }} / {{ $analytics->getBudget() }}
</div>
```

As you can see, this `@inject` directive has actually made an `$analytics` variable available, which we’re using later in our view.

The first parameter of `@inject` is the name of the variable you’re injecting, and the second parameter is the class or interface that you want to inject an instance of. This is resolved just like when you typehint a dependency in a constructor elsewhere in Laravel; if you’re unfamiliar with how that works, check out [Chapter 11](#) to learn more.

Just like view composers, Blade service injection makes it easy to make certain data or functionality available to every instance of a view, without having to inject it via the route definition every time.

Custom Blade Directives

All of the built-in syntax of Blade that we've covered so far—`@if`, `@unless`, and so on—are called *directives*. Each Blade directive is a mapping between a pattern (e.g., `@if ($condition)`) and a PHP output (e.g., `<?php if ($condition): ?>`).

Directives aren't just for the core; you can actually create your own. You might think directives are good for making little shortcuts to bigger pieces of code—for example, using `@button('buttonName')` and having it expand to a larger set of button HTML. This isn't a *terrible* idea, but for simple code expansion like this you might be better off including a view partial.

Custom directives tend to be most useful when they simplify some form of repeated logic. Say we're tired of having to wrap our code with `@if (auth()->guest())` (to check if a user is logged in or not) and we want a custom `@ifGuest` directive. As with view composers, it might be worth having a custom service provider to register these, but for now let's just put it in the `boot()` method of `App\Providers\AppServiceProvider`. Take a look at [Example 4-29](#) to see what this binding will look like.

Example 4-29. Binding a custom Blade directive in a service provider

```
public function boot(): void
{
    Blade::directive('ifGuest', function () {
        return "<?php if (auth()->guest()): ?>";
    });
}
```

We've now registered a custom directive, `@ifGuest`, which will be replaced with the PHP code `<?php if (auth()->guest()): ?>`.

This might feel strange. You're writing a *string* that will be returned and then executed as PHP. But what this means is that you can now take the complex, ugly, unclear, or repetitive aspects of your PHP templating code and hide them behind clear, simple, and expressive syntax.

CUSTOM DIRECTIVE RESULT CACHING

You might be tempted to do some logic to make your custom directive faster by performing an operation *in* the binding and then embedding the result within the returned string:

```
Blade::directive('ifGuest', function () {
    // Antipattern! Do not copy.
    $ifGuest = auth()->guest();
    return "<?php if ({ $ifGuest }): ?>";
});
```

The problem with this idea is that it assumes this directive will be re-created on every page load. However, Blade caches aggressively, so you're going to find yourself in a bad spot if you try this.

Parameters in Custom Blade Directives

What if you want to accept parameters in your custom logic? Check out [Example 4-30](#).

Example 4-30. Creating a Blade directive with parameters

```
// Binding
Blade::directive('newlinesToBr', function ($expression) {
    return "<?php echo nl2br({ $expression }); ?>";
});

// In use
<p>@newlinesToBr($message->body)</p>
```

The `$expression` parameter received by the closure represents whatever's within the parentheses. As you can see, we then generate a valid PHP code snippet and return it.

If you find yourself constantly writing the same conditional logic over and over, you should consider a Blade directive.

Example: Using Custom Blade Directives for a Multitenant App

Let's imagine we're building an application that supports *multitenancy*, which means users might be visiting the site from *www.myapp.com*, *client1.myapp.com*, *client2.myapp.com*, or elsewhere.

Suppose we have written a class to encapsulate some of our multitenancy logic

and named it `Context`. This class will capture information and logic about the context of the current visit, such as who the authenticated user is and whether the user is visiting the public website or a client subdomain.

We'll probably frequently resolve that `Context` class in our views and perform conditionals on it, like in [Example 4-31](#). `app('context')` is a shortcut to get an instance of a class from the container, which we'll learn more about in [Chapter 11](#).

Example 4-31. Conditionals on context without a custom Blade directive

```
@if (app('context')->isPublic())
    &copy; Copyright MyApp LLC
@else
    &copy; Copyright {{ app('context')->client->name }}
@endif
```

What if we could simplify `@if (app('context')->isPublic())` to just `@ifPublic`? Let's do it. Check out [Example 4-32](#).

Example 4-32. Conditionals on context with a custom Blade directive

```
// Binding
Blade::directive('ifPublic', function () {
    return "<?php if (app('context')->isPublic()): ?>";
});

// In use
@ifPublic
    &copy; Copyright MyApp LLC
@else
    &copy; Copyright {{ app('context')->client->name }}
@endif
```

Since this resolves to a simple `if` statement, we can still rely on the native `@else` and `@endif` conditionals. But if we wanted, we could also create a custom `@elseifClient` directive, or a separate `@ifClient` directive, or really whatever else we want.

Easier Custom Directives for “if” Statements

Although custom Blade directives are powerful, the most common use for them is `if` statements. So there's a simpler way to create custom “if” directives: `Blade::if()`. [Example 4-33](#) shows how we could refactor [Example 4-32](#) using

the `Blade::if()` method:

Example 4-33. Defining a custom “if” Blade directive

```
// Binding
Blade::if('ifPublic', function () {
    return (app('context'))->isPublic();
});
```

You’ll use the directives exactly the same way, but as you can see, defining them is a bit simpler. Instead of having to manually type out PHP braces, you can just write a closure that returns a Boolean.

Testing

The most common method of testing views is through application testing, meaning that you’re actually calling the route that displays the views and ensuring the views have certain content (see [Example 4-34](#)). You can also click buttons or submit forms and ensure that you are redirected to a certain page or that you see a certain error. (You’ll learn more about testing in [Chapter 12](#).)

Example 4-34. Testing that a view displays certain content

```
// EventsTest.php
public function test_list_page_shows_all_events()
{
    $event1 = Event::factory()->create();
    $event2 = Event::factory()->create();

    $this->get('events')
        ->assertSee($event1->title)
        ->assertSee($event2->title);
}
```

You can also test that a certain view has been passed a particular set of data, which, if it accomplishes your testing goals, is less fragile than checking for certain text on the page. [Example 4-35](#) demonstrates this approach.

Example 4-35. Testing that a view was passed certain content

```
// EventsTest.php
public function test_list_page_shows_all_events()
{
    $event1 = Event::factory()->create();
    $event2 = Event::factory()->create();
```

```

$response = $this->get('events');

$response->assertViewHas('events', Event::all());
$response->assertViewHasAll([
    'events' => Event::all(),
    'title' => 'Events Page',
]);
$response->assertViewMissing('dogs');
}

```

With `assertViewHas()` we can pass in a closure, meaning we can customize how we want to check more complex data structures. [Example 4-36](#) illustrates how we might use this.

Example 4-36. Passing a closure to `assertViewHas()`

```

// EventsTest.php
public function test_list_page_shows_all_events()
{
    $event1 = Event::factory()->create();

    $response = $this->get("events/{ $event1->id }");

    $response->assertViewHas('event', function ($event) use ($event1) {
        return $event->id === $event1->id;
    });
}

```

TL;DR

Blade is Laravel’s templating engine. Its primary focus is a clear, concise, and expressive syntax with powerful inheritance and extensibility. Its “safe echo” brackets are `{{` and `}}`, its unprotected echo brackets are `{!!` and `!!`, and it has a series of custom tags called “directives” that all begin with `@` (`@if` and `@unless`, for example).

You can define a parent template and leave “holes” in it for content using `@yield` and `@section/@show`. You can then teach its child views to extend the parent using `@extends('parent.view')` and define their sections using `@section/@endsection`. You use `@parent` to reference the content of the block’s parent.

View composers make it easy to define that every time a particular view or

subview loads, it should have certain information available to it. And service injection allows the view itself to request data straight from the application container.

Chapter 5. Databases and Eloquent

Laravel provides a suite of tools for interacting with your application's databases, the most notable of which is Eloquent, Laravel's ActiveRecord ORM.

Eloquent is one of Laravel's most popular and influential features. It's a great example of how Laravel is different from the majority of PHP frameworks; in a world of DataMapper ORMs that are powerful but complex, Eloquent stands out for its simplicity. There's one class per table, which is responsible for retrieving, representing, and persisting data in that table.

Whether or not you choose to use Eloquent, you'll still get a ton of benefit from the other database tools Laravel provides. So before we dig into Eloquent, we'll start by covering the basics of Laravel's database functionality: migrations, seeders, and the query builder.

Then we'll cover Eloquent: defining your models; inserting, updating, and deleting; customizing your responses with accessors, mutators, and attribute casting; and, finally, relationships. There's a lot going on here, and it's easy to get overwhelmed, but if we take it one step at a time, we'll make it through.

Configuration

Before we get into how to use Laravel's database tools, let's pause for a second and go over how to configure your database credentials and connections.

The configuration for database access lives in *config/database.php* and *.env*. Like many other configuration areas in Laravel, you can define multiple "connections" and then decide which the code will use by default.

Database Connections

By default, there's one connection for each of the drivers, as you can see in [Example 5-1](#).

Example 5-1. The default database connections list

```
'connections' => [

    'sqlite' => [
        'driver' => 'sqlite',
        'url' => env('DATABASE_URL'),
        'database' => env('DB_DATABASE', database_path('database.sqlite')),
        'prefix' => '',
        'foreign_key_constraints' => env('DB_FOREIGN_KEYS', true),
    ],

    'mysql' => [
        'driver' => 'mysql',
        'url' => env('DATABASE_URL'),
        'host' => env('DB_HOST', '127.0.0.1'),
        'port' => env('DB_PORT', '3306'),
        'database' => env('DB_DATABASE', 'forge'),
        'username' => env('DB_USERNAME', 'forge'),
        'password' => env('DB_PASSWORD', ''),
        'unix_socket' => env('DB_SOCKET', ''),
        'charset' => 'utf8mb4',
        'collation' => 'utf8mb4_unicode_ci',
        'prefix' => '',
        'prefix_indexes' => true,
        'strict' => true,
        'engine' => null,
        'options' => extension_loaded('pdo_mysql') ? array_filter([
            PDO::MYSQL_ATTR_SSL_CA => env('MYSQL_ATTR_SSL_CA'),
        ]) : [],
    ],

    'pgsql' => [
        'driver' => 'pgsql',
        'url' => env('DATABASE_URL'),
        'host' => env('DB_HOST', '127.0.0.1'),
        'port' => env('DB_PORT', '5432'),
        'database' => env('DB_DATABASE', 'forge'),
        'username' => env('DB_USERNAME', 'forge'),
        'password' => env('DB_PASSWORD', ''),
        'charset' => 'utf8',
        'prefix' => '',
        'prefix_indexes' => true,
        'search_path' => 'public',
        'sslmode' => 'prefer',
    ],

    'sqlsrv' => [
        'driver' => 'sqlsrv',
        'url' => env('DATABASE_URL'),
```

```

    'host' => env('DB_HOST', 'localhost'),
    'port' => env('DB_PORT', '1433'),
    'database' => env('DB_DATABASE', 'forge'),
    'username' => env('DB_USERNAME', 'forge'),
    'password' => env('DB_PASSWORD', ''),
    'charset' => 'utf8',
    'prefix' => '',
    'prefix_indexes' => true,
    // 'encrypt' => env('DB_ENCRYPT', 'yes'),
    // 'trust_server_certificate' => env('DB_TRUST_SERVER_CERTIFICATE', 'false'),
  ],

```

]

Nothing is stopping you from deleting or modifying these named connections or creating your own. You can create new named connections, and you’ll be able to set the drivers (MySQL, PostgreSQL, etc.) in them. So although there’s one connection per driver by default, that’s not a constraint; you could have five different connections, all with the `mysql` driver, if you wanted.

Each connection allows you to define the properties necessary for connecting to and customizing each connection type.

There are a few reasons for the idea of multiple drivers. To start with, the “connections” section as it comes out of the box is a simple template that makes it easy to start apps that use any of the supported database connection types. In many apps, you can pick the database connection you’ll be using, fill out its information, and even delete the others if you’d like. I usually just keep them all there, in case I might eventually use them.

But there are also some cases where you might need multiple connections within the same application. For example, you might use different database connections for two different types of data, or you might read from one and write to another. Support for multiple connections makes this possible.

URL Configurations

Often services like Heroku will provide an environment variable with a URL that contains all of the information you need to connect to the database. It’ll look something like this:

```
mysql://root:password@127.0.0.1/forge?charset=UTF-8
```


You don't have to write code to parse this URL out; instead, pass it in as the `DATABASE_URL` environment variable, and Laravel will understand it.

Other Database Configuration Options

The `config/database.php` configuration section has quite a few other configuration settings. You can configure Redis access, customize the table name used for migrations, determine the default connection, and toggle whether non-Eloquent calls return `stdClass` or array instances.

With any service in Laravel that allows connections from multiple sources—sessions can be backed by the database or file storage, the cache can use Redis or Memcached, databases can use MySQL or PostgreSQL—you can define multiple connections and also choose that a particular connection will be the “default,” meaning it will be used any time you don't explicitly ask for a particular connection. Here's how you ask for a specific connection, if you want to:

```
$users = DB::connection('secondary')->select('select * from users');
```

Migrations

Modern frameworks like Laravel make it easy to define your database structure with code-driven migrations. Every new table, column, index, and key can be defined in code, and any new environment can be brought from bare database to your app's perfect schema in seconds.

Defining Migrations

A migration is a single file that defines two things: the modifications desired when running this migration *up* and, optionally, the modifications desired when running this migration *down*.

“UP” AND “DOWN” IN MIGRATIONS

Migrations are always run in order by date. Every migration file is named something like this: `2018_10_12_000000_create_users_table.php`. When a

new system is migrated, the system grabs each migration, starting at the earliest date, and runs its `up()` method—you’re migrating it “up” at this point. But the migration system also allows you to “roll back” your most recent set of migrations. It’ll grab each of them and run its `down()` method, which should undo whatever changes the up migration made.

So, the `up()` method of a migration should “do” its migration, and the `down()` method should “undo” it.

Example 5-2 shows what the default “create users table” migration that comes with Laravel looks like.

Example 5-2. Laravel’s default “create users table” migration

```
<?php
```

```
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     *
     * @return void
     */
    public function up(): void
    {
        Schema::create('users', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->string('email')->unique();
            $table->timestamp('email_verified_at')->nullable();
            $table->string('password');
            $table->rememberToken();
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     *
     * @return void
     */
    public function down(): void
```

```
{  
    Schema::dropIfExists('users');  
}  
};
```

EMAIL VERIFICATION

The `email_verified_at` column stores a timestamp indicating when the user verified their email address.

As you can see, we have an `up()` method and a `down()` method. `up()` tells the migration to create a new table named `users` with a few fields, and `down()` tells it to drop the `users` table.

Creating a migration

As you will see in [Chapter 8](#), Laravel provides a series of command-line tools you can use to interact with your app and generate boilerplate files. One of these commands allows you to create a migration file. You can run it using `php artisan make:migration`, and it has a single parameter, which is the name of the migration. For example, to create the table we just covered, you would run `php artisan make:migration create_users_table`.

There are two flags you can optionally pass to this command. `--create=table_name` prefills the migration with code designed to create a table named `table_name`, and `--table=table_name` just prefills the migration for modifications to an existing table.

```
php artisan make:migration create_users_table  
php artisan make:migration add_votes_to_users_table --table=users  
php artisan make:migration create_users_table --create=users
```

Creating tables

We already saw in the default `create_users_table` migration that our migrations depend on the `Schema` facade and its methods. Everything we can do in these migrations will rely on the methods of `Schema`.

To create a new table in a migration, use the `create()` method—the first parameter is the table name, and the second is a closure that defines its columns:

```
Schema::create('users', function (Blueprint $table) {  
    // Create columns here  
});
```

Creating columns

To create new columns in a table, whether in a create table call or a modify table call, use the instance of `Blueprint` that's passed into your closure:

```
Schema::create('users', function (Blueprint $table) {  
    $table->string('name');  
});
```

Let's look at the various methods available on `Blueprint` instances for creating columns. I'll describe how they work in MySQL, but if you're using another database, Laravel will just use the closest equivalent.

The following are the simple field `Blueprint` methods:

`id()`

An alias for `$table->bigIncrements('id')`

`integer(colName)`, `tinyInteger(colName)`, `smallInteger(colName)`,
`mediumInteger(colName)`, `bigInteger(colName)`,
`unsignedTinyInteger(colName)`, `unsignedSmallInteger(colName)`,
`unsignedMediumInteger(colName)`,
`unsignedBigInteger(colName)`

Adds an `INTEGER` type column, or one of its many variations

`string(colName, length)`

Adds a `VARCHAR` type column with an optional length

`binary(colName)`

Adds a BLOB type column

`boolean(colName)`

Adds a BOOLEAN type column (a TINYINT(1) in MySQL)

`char(colName, length)`

Adds a CHAR column with an optional length

`date(colName)`, `datetime(colName)`, `dateTimeTz(colName)`

Adds a DATE or DATETIME column; if time zone awareness is needed, use the `dateTimeTz()` method to create a DATETIME column with time zone

`decimal(colName, precision, scale)`,

`unsignedDecimal(colName, precision, scale)`

Adds a DECIMAL column, with precision and scale—for example, `decimal('amount', 5, 2)` specifies a precision of 5 and a scale of 2; for an unsigned column, use the `unsignedDecimal` method

`double(colName, total digits, digits after decimal)`

Adds a DOUBLE column—for example, `double('tolerance', 12, 8)` specifies 12 digits long, with 8 of those digits to the right of the decimal place, as in 7204.05691739

`enum(colName, [choiceOne, choiceTwo])`

Adds an ENUM column, with provided choices

`float(colName, precision, scale)`

Adds a FLOAT column (same as `double` in MySQL)

`foreignId(colName)`, `foreignUuid(colName)`

Adds an UNSIGNED BIGINT or a UUID column, with provided choices

`foreignIdFor(colName)`

Adds an UNSIGNED BIG INT column with the name *colName*

`geometry(colName), geometryCollection(colName)`

Adds a GEOMETRY or a GEOMETRYCOLLECTION column

`ipAddress(colName)`

Adds a VARCHAR column

`json(colName), jsonb(colName)`

Adds a JSON or JSONB column

`lineString(colName), multiLineString(colName)`

Adds a LINESTRING or MULTILINESTRING column with the given *colName*

`text(colName), tinyText(colName), mediumText(colName),
longText(colName)`

Adds a TEXT column (or its various sizes)

`macAddress(colName)`

Adds a MACADDRESS column in the databases that support it (like PostgreSQL); on other database systems, it creates a string equivalent

`multiPoint(colName), multiPolygon(colName), polygon(colName),
point(colName)`

Adds columns of the types MULTIPOINT, MULTIPOLYGON, POLYGON, and POINT, respectively

`set(colName, membersArray)`

Creates a SET column with the *colName* name and *membersArray* as members

`time(colName, precision), timeTz(colName, precision)`

Adds a TIME column with *colName* name; for time zone awareness, use the `timeTz()` method

`timestamp(colName, precision),`
`timestampTz(colName, precision)`

Adds a TIMESTAMP column; for time zone awareness, use the `timestampTz()` method

`uuid(colName)`

Adds a UUID column (CHAR(36) in MySQL)

`year()`

Adds a YEAR column

And these are the special (joined) Blueprint methods:

`increments(colName), tinyIncrements(colName),`
`smallIncrements(colName), mediumIncrements(colName),`
`bigIncrements(colName)`

Adds an unsigned incrementing INTEGER primary key ID, or one of its many variations

`timestamps(precision), nullableTimestamps(precision),`
`timestampsTz(precision)`

Adds `created_at` and `updated_at` timestamp columns with optional precision, nullable, and time zone-aware variations

`rememberToken()`

Adds a `remember_token` column (VARCHAR(100)) for user “remember me” tokens

`softDeletes(colName, precision), softDeletesTz(colName, precision)`

Adds a `deleted_at` timestamp for use with soft deletes with optional

precision and time zone–aware variations

```
morphs(colName), nullableMorphs(colName),  
uuidMorphs(relationshipName),  
nullableUuidMorphs(relationshipName)
```

For a provided *colName*, adds an integer *colName_id* and a string *colName_type* (e.g., `morphs(tag)` adds integer *tag_id* and string *tag_type*); for use in polymorphic relationships, using IDs or UUIDs, and can be set as nullable as per method name

Building extra properties fluently

Most of the properties of a field definition—its length, for example—are set as the second parameter of the field creation method, as we saw in the previous section. But there are a few other properties that we’ll set by chaining more method calls after the creation of the column. For example, this `email` field is nullable and will be placed (in MySQL) right after the `last_name` field:

```
Schema::table('users', function (Blueprint $table) {  
    $table->string('email')->nullable()->after('last_name');  
});
```

The following methods are some of those used to set additional properties of a field; look to the [migrations docs](#) for an exhaustive list.

`nullable()`

Allows NULL values to be inserted into this column

`default('default content')`

Specifies the default content for this column if no value is provided

`unsigned()`

Marks integer columns as unsigned (not negative or positive, but just an integer)

`first()` (*MySQL only*)

Places the column first in the column order

`after(colName)` (*MySQL only*)

Places the column after another column in the column order

`charset(charset)` (*MySQL only*)

Sets the charset for a column

`collation(collation)`

Sets the collation for a column

`invisible()` (*MySQL only*)

Makes the column invisible to SELECT queries

`useCurrent()`

Used on TIMESTAMP columns to use CURRENT_TIMESTAMP as the default value

`isGeometry()` (*PostgreSQL only*)

Sets a column type to GEOMETRY (the default is GEOGRAPHY)

`unique()`

Adds a UNIQUE index

`primary()`

Adds a primary key index

`index()`

Adds a basic index

Note that `unique()`, `primary()`, and `index()` can also be used outside of the fluent column building context, which we'll cover later.

Dropping tables

If you want to drop a table, use the `dropIfExists()` method on `Schema`, which takes one parameter, the table name:

```
Schema::dropIfExists('contacts');
```

Modifying columns

To modify a column, just write the code you would write to create the column as if it were new, and then append a call to the `change()` method after it.

REQUIRED DEPENDENCY BEFORE MODIFYING COLUMNS

If you are not using a database that natively supports renaming and dropping columns (the latest versions of the most common databases support these operations), before you can modify any columns, you'll need to run `composer require doctrine/dbal`.

So, if we have a string column named `name` that has a length of 255 and we want to change its length to 100, this is how we would write it:

```
Schema::table('users', function (Blueprint $table) {  
    $table->string('name', 100)->change();  
});
```

The same is true if we want to adjust any of its properties that aren't defined in the method name. To make a field nullable, we do this:

```
Schema::table('contacts', function (Blueprint $table) {  
    $table->string('deleted_at')->nullable()->change();  
});
```

Here's how we rename a column:

```
Schema::table('contacts', function (Blueprint $table)  
{  
    $table->renameColumn('promoted', 'is_promoted');  
});
```

And this is how we drop a column:

```
Schema::table('contacts', function (Blueprint $table)
{
    $table->dropColumn('votes');
});
```

MODIFYING MULTIPLE COLUMNS AT ONCE IN SQLITE

If you try to drop or modify multiple columns within a single migration closure and you are using SQLite, you'll run into errors.

In [Chapter 12](#) I recommend that you use SQLite for your testing database, so even if you're using a more traditional database, you may want to consider this a limitation for testing purposes.

However, you don't have to create a new migration for each. Instead, just create multiple calls to `Schema::table()` within the `up()` method of your migration:

```
public function up(): void
{
    Schema::table('contacts', function (Blueprint $table)
    {
        $table->dropColumn('is_promoted');
    });

    Schema::table('contacts', function (Blueprint $table)
    {
        $table->dropColumn('alternate_email');
    });
}
```

Squashing migrations

If you have too many migrations to reason with, you can merge them all into a single SQL file that Laravel will run before it runs any future migrations. This is called “squashing” your migrations.

```
// Squash the schema but keep your existing migrations
php artisan schema:dump
```

```
// Dump the current database schema and delete all existing migrations
php artisan schema:dump --prune
```

Laravel only runs these dumps if it detects no migrations have been run so far. That means you can squash your migrations and it won't break your already-deployed applications.

WARNING

If you use schema dumps, you can't use in-memory SQLite; it only works on MySQL, PostgreSQL, and local file SQLite.

Indexes and foreign keys

We've covered how to create, modify, and delete columns. Let's move on to indexing and relating them.

If you're not familiar with indexes, your databases can survive if you just never use them, but they're pretty important for performance optimization and for some data integrity controls with regard to related tables. I'd recommend reading up on them, but if you absolutely must, you can skip this section for now.

Adding indexes

Check out [Example 5-3](#) for examples of how to add indexes to your column.

Example 5-3. Adding column indexes in migrations

```
// After columns are created...
$table->primary('primary_id'); // Primary key; unnecessary if used increments()
$table->primary(['first_name', 'last_name']); // Composite keys
$table->unique('email'); // Unique index
$table->unique('email', 'optional_custom_index_name'); // Unique index
$table->index('amount'); // Basic index
$table->index('amount', 'optional_custom_index_name'); // Basic index
```

Note that the first example, `primary()`, is not necessary if you're using the `increments()` or `bigIncrements()` methods to create your index; this will automatically add a primary key index for you.

Removing indexes

We can remove indexes as shown in [Example 5-4](#).

Example 5-4. Removing column indexes in migrations

```
$table->dropPrimary('contacts_id_primary');
$table->dropUnique('contacts_email_unique');
$table->dropIndex('optional_custom_index_name');

// If you pass an array of column names to dropIndex, it will
// guess the index names for you based on the generation rules
$table->dropIndex(['email', 'amount']);
```

Adding and removing foreign keys

To add a foreign key that defines that a particular column references a column on another table, Laravel's syntax is simple and clear:

```
$table->foreign('user_id')->references('id')->on('users');
```

Here we're adding a foreign index on the `user_id` column, showing that it references the `id` column on the `users` table. Couldn't get much simpler.

If we want to specify foreign key constraints, we can do that too, with `cascadeOnUpdate()`, `restrictOnUpdate()`, `cascadeOnDelete()`, `restrictOnDelete()`, and `nullOnDelete()`. For example:

```
$table->foreign('user_id')
    ->references('id')
    ->on('users')
    ->cascadeOnDelete();
```

There is also an alias for creating foreign key constraints. Using it, the above example can be written like so:

```
$table->foreignId('user_id')->constrained()->cascadeOnDelete();
```

To drop a foreign key, we can either delete it by referencing its index name (which is automatically generated by combining the names of the columns and tables being referenced):

```
$table->dropForeign('contacts_user_id_foreign');
```

or by passing it an array of the fields that it's referencing in the local table:

```
$table->dropForeign(['user_id']);
```

Running Migrations

Once you have your migrations defined, how do you run them? There's an Artisan command for that:

```
php artisan migrate
```

This command runs all “outstanding” migrations (by running the `up()` method on each). Laravel keeps track of which migrations you have run and which you haven't. Every time you run this command, it checks whether you've run all available migrations, and if you haven't, it'll run any that remain.

There are a few options in this namespace that you can work with. First, you can run your migrations *and* your seeds (which we'll cover next):

```
php artisan migrate --seed
```

You can also run any of the following commands:

`migrate:install`

Creates the database table that keeps track of which migrations you have and haven't run; this is run automatically when you run your migrations, so you can basically ignore it.

`migrate:reset`

Rolls back every database migration you've run on this instance.

`migrate:refresh`

Rolls back every database migration you've run on this instance, and then runs every migration available. It's the same as running `migrate:reset` followed by `migrate`.

`migrate:fresh`

Drops all of your tables and runs every migration again. It's the same as `refresh` but doesn't bother with the “down” migrations—it just deletes the tables and then runs the “up” migrations again.

`migrate:rollback`

Rolls back *just* the migrations that ran the last time you ran `migrate`, or, with the added option `--step=n`, rolls back the number of migrations you specify.

`migrate:status`

Shows a table listing every migration, with a Y or N next to each indicating whether or not it has run yet in this environment.

MIGRATING WITH HOMESTEAD/VAGRANT

If you're running migrations on your local machine and your `.env` file points to a database in a Vagrant box, your migrations will fail. You'll need to `ssh` into your Vagrant box and then run the migrations from there. The same is true for seeds and any other Artisan commands that affect or read from the database.

Inspecting Your Database

If you want to dig into the status or definition of your database, its tables, and its models, there are a few Artisan commands for exactly that purpose:

`db:show`

Shows a table overview of your entire database, including the connection details, tables, size, and open connections

`db:table {tableName}`

Passed a table name, shows the size and lists the columns

`db:monitor`

List, the number of open connections to the database

Seeding

Seeding with Laravel is so simple, it has gained widespread adoption as a part of normal development workflows in a way it hasn't in previous PHP frameworks. There's a *database/seeder* folder that comes with a `DatabaseSeeder` class, which has a `run()` method that is called when you call the seeder.

There are two primary ways to run the seeders: along with a migration or separately.

To run a seeder along with a migration, just add `--seed` to any migration call:

```
php artisan migrate --seed
php artisan migrate:refresh --seed
```

And to run it independently:

```
php artisan db:seed
php artisan db:seed VotesTableSeeder
```

This will call the `run()` method of the `DatabaseSeeder` by default, or the seeder class specified when you pass in a class name.

Creating a Seeder

To create a seeder, use the `make:seeder` Artisan command:

```
php artisan make:seeder ContactsTableSeeder
```

You'll now see a `ContactsTableSeeder` class show up in the *database/seeder* directory. Before we edit it, let's add it to the `DatabaseSeeder` class, as shown in [Example 5-5](#), so it will run when we run our seeders.

Example 5-5. Calling a custom seeder from DatabaseSeeder.php

```
// database/seeder/DatabaseSeeder.php
...
public function run(): void
{
    $this->call(ContactsTableSeeder::class);
}
```


Now let's edit the seeder itself. The simplest thing we can do there is manually insert a record using the DB facade, as illustrated in [Example 5-6](#).

Example 5-6. Inserting database records in a custom seeder

```
<?php

namespace Database\Seeders;

use Illuminate\Database\Seeder;

class ContactsTableSeeder extends Seeder
{
    public function run(): void
    {
        DB::table('contacts')->insert([
            'name' => 'Lupita Smith',
            'email' => 'lupita@gmail.com',
        ]);
    }
}
```

This will get us a single record, which is a good start. But for truly functional seeds, you'll likely want to loop over some sort of random generator and run this `insert()` many times, right? Laravel has a feature for that.

Model Factories

Model factories define one (or more) patterns for creating fake entries for your database tables. By default, each factory is named after an Eloquent class.

Theoretically you can name these factories anything you like, but naming the factory after your Eloquent class is the most idiomatic approach. If you follow a different convention to name your factories, you can set the factory class name in the related model.

Creating a model factory

Model factories are located in *database/factories*. Each factory is defined in its own class, with a definition method. In this method you define the attributes and their values to be used when creating a model with the factory.

To generate a new factory class, use the Artisan `make:factory` command; again, it's most common to name factory classes after the Eloquent models they're

meant to generate instances of:

```
php artisan make:factory ContactFactory
```

This will generate a new file within the *database/factories* directory called *ContactFactory.php*. The simplest factory we could define for a contact might look something like **Example 5-7**:

Example 5-7. The simplest possible factory definition

```
<?php

namespace Database\Factories;

use App\Models>Contact;
use Illuminate\Database\Eloquent\Factories\Factory;

class ContactFactory extends Factory
{
    public function definition(): array
    {
        return [
            'name' => 'Lupita Smith',
            'email' => 'lupita@gmail.com',
        ];
    }
}
```

You now need to use the *Illuminate\Database\Eloquent\Factories\HasFactory* trait in your model.

```
namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Contact extends Model
{
    use HasFactory;
}
```

The *HasFactory* trait provides a *factory()* static method, which uses Laravel conventions to determine the proper factory for the model. It will look for a factory in *Database\Factories* namespace that has a class name matching the model name and is suffixed with *Factory*. If you don't follow these

conventions, you can override the `newFactory()` method in your model to specify the factory class that should be used:

```
// app/Models/Contact.php
...
* Create a new factory instance for the model.
*
* @return \Illuminate\Database\Eloquent\Factories\Factory
*/
protected static function newFactory()
{
    return \Database\Factories\Base\ContactFactory::new();
}
```

Now we can call the static `factory()` method on the model, to create an instance of `Contact` in our seeding and testing:

```
// Create one
$contact = Contact::factory()->create();

// Create many
Contact::factory()->count(20)->create();
```

However, if we used that factory to create 20 contacts, all 20 would have the same information. That's less useful.

We will get even more benefit from model factories when we take advantage of **Faker**, which is globally available in Laravel via the `fake()` helper; Faker makes it easy to randomize the creation of structured fake data. The previous example now turns into **Example 5-8**.

Example 5-8. A simple factory, modified to use Faker

```
<?php

namespace Database\Factories;

use App\Models\Contact;
use Illuminate\Database\Eloquent\Factories\Factory;

class ContactFactory extends Factory
{
    public function definition(): array
    {
        return [
```

```

        'name' => fake()->name(),
        'email' => fake()->email(),
    ];
}
}

```

Now, every time we create a fake contact using this model factory, all of our properties will be randomly generated.

Model factories need, at minimum, to return the required database fields for this table.

GUARANTEEING THE UNIQUENESS OF RANDOMLY GENERATED DATA

If you want to guarantee that the randomly generated values of any given entry are unique compared to the other randomly generated values during that PHP process, you can use Faker's `unique()` method:

```
return ['email' => fake()->unique()->email()];
```

Using a model factory

There are two primary contexts in which we'll use model factories: testing, which we'll cover in [Chapter 12](#), and seeding, which we'll cover here. Let's write a seeder using a model factory; take a look at [Example 5-9](#).

Example 5-9. Using model factories

```

$post = Post::factory()->create([
    'title' => 'My greatest post ever',
]);

// Pro-level factory; but don't get overwhelmed!
User::factory()->count(20)->has(Address::factory()->count(2))->create()

```

To create an object, we use the `factory()` method on the model. Then we can run one of two methods on it: `make()` or `create()`.

Both methods generate an instance of this specified model, using the definition in the factory class. The difference is that `make()` creates the instance but doesn't

(yet) save it to the database, whereas `create()` saves it to the database instantly.

Overriding properties when calling a model factory

If you pass an array to either `make()` or `create()`, you can override specific keys from the factory, like we did in [Example 5-9](#) to manually set the `title` on the post.

Generating more than one instance with a model factory

If you call the `count()` method after the `factory()` method, you can specify that you're creating more than one instance. Instead of returning a single instance, it'll return a collection of instances. This means you can treat the result like an array, iterating over them or passing them to any method that takes more than one object:

```
$posts = Post::factory()->count(6);
```

You can also, optionally, define a “sequence” of how to override each:

```
$posts = Post::factory()  
    ->count(6)  
    ->state(new Sequence(  
        ['is_published' => true],  
        ['is_published' => false],  
    ))  
    ->create();
```

Pro-level model factories

Now that we've covered the most common uses for and arrangements of model factories, let's dive into some of the more complicated ways we can use them.

Attaching relationships when defining model factories

Sometimes you need to create a related item along with the item you're creating. You can call the factory method on the related model to pull its ID, as shown in [Example 5-10](#).

Example 5-10. Creating a related item in a factory

```
<?php
```

```

namespace Database\Factories;

use App\Models>Contact;
use Illuminate\Database\Eloquent\Factories\Factory;

class ContactFactory extends Factory
{
    protected $model = Contact::class;

    public function definition(): array
    {
        return [
            'name' => 'Lupita Smith',
            'email' => 'lupita@gmail.com',
            'company_id' => \App\Models\Company::factory(),
        ];
    }
}

```

You can also pass a closure where a single parameter is passed, which contains the array form of the generated item up until that point. This can be used in other ways, as demonstrated in [Example 5-11](#).

Example 5-11. Using values from other parameters in a factory

```

// ContactFactory.php
public function definition(): array
{
    return [
        'name' => 'Lupita Smith',
        'email' => 'lupita@gmail.com',
        'company_id' => Company::factory(),
        'company_size' => function (array $attributes) {
            // Uses the "company_id" property generated above
            return Company::find($attributes['company_id'])->size;
        },
    ];
}

```

Attaching related items when generating model factory instances

While we’ve already covered how to define a relationship in a factory definition, it’s much more common that we’ll be defining our instance’s related items right when we create it.

There are two main methods we’ll use for this: `has()` and `for()`. `has()` allows us to define that the instance we’re creating “has” children or other items in a

“hasMany” type relationship, whereas `for()` allows us to define that the instance we’re creating “belongsTo” another item. Let’s look at a few examples to get a better sense of how they work.

In [Example 5-12](#), let’s assume a `Contact` has many `Addresses`.

Example 5-12. Using `has()` when generating related models

```
// Attach 3 addresses
Contact::factory()
  ->has(Address::factory()->count(3))
  ->create()

// Accessing information about each user in the child factory
$contact = Contact::factory()
  ->has(
    Address::factory()
      ->count(3)
      ->state(function (array $attributes, User $user) {
        return ['label' => $user->name . ' address'];
      })
  )
  ->create();
```

Now let’s imagine we’re creating the child instance instead of the parent instance. Let’s generate an address.

In these sorts of circumstances, you usually can assume the child’s factory definition would take care of generating the parent instance. So, what’s the use of `for()` at all? It’s most helpful if you want to specifically define something about the parent, usually either one or more of its properties, or pass in a specific model instance. Take a look at [Example 5-13](#) to see how it’s most commonly used.

Example 5-13. Using `for()` when generating related models

```
// Specify details about the created parent
Address::factory()
  ->count(3)
  ->for(Contact::factory()->state([
    'name' => 'Imani Carette',
  ]))
  ->create();

// Use an existing parent model (assuming we already have it as $contact)
Address::factory()
  ->count(3)
```

```
->for($contact)
->create();
```

Defining and accessing multiple model factory states

Let's go back to *ContactFactory.php* (from Examples 5-7 and 5-8) for a second. We have a base Contact factory defined:

```
class ContactFactory extends Factory
{
    protected $model = Contact::class;

    public function definition(): array
    {
        return [
            'name' => 'Lupita Smith',
            'email' => 'lupita@gmail.com',
        ];
    }
}
```

But sometimes you need more than one factory for a class of object. What if we need to be able to add some contacts who are very important people (VIPs)? We can use the `state()` method to define a second factory state for this, as seen in [Example 5-14](#). The `state()` method receives an array of any attributes you want to specifically set for this state.

Example 5-14. Defining multiple factory states for the same model

```
class ContactFactory extends Factory
{
    protected $model = Contact::class;

    public function definition(): array
    {
        return [
            'name' => 'Lupita Smith',
            'email' => 'lupita@gmail.com',
        ];
    }

    public function vip()
    {
        return $this->state(function (array $attributes) {
            return [
                'vip' => true,
                // Uses the "company_id" property from the $attributes
            ];
        });
    }
}
```



```

        'company_size' => function () use ($attributes) {
            return Company::find($attributes['company_id'])->size;
        },
    ];
});
}
}

```

Now, let's make an instance of a specific state:

```

$vip = Contact::factory()->vip()->create();

$vip = Contact::factory()->count(3)->vip()->create();

```

Using the same model as the relationship in complex factory Setups

Sometimes you have a factory that creates related items through their factories, and two or more of those have the same relationship. Maybe generating a `Trip` with your factory automatically creates a `Reservation` and a `Receipt`, and all three should be attached to the same `User`. When you go to create the `Trip`, the factories will each create their own user manually, unless you tell them to do otherwise.

With the `recycle()` method, you can instruct that every factory called up the chain uses the same instance of a given object. As you can see in [Example 5-15](#), this gives a simple syntax for ensuring the same model is used in every place throughout a factory chain.

Example 5-15. Using `recycle()` to use the same instance for every relationship in a factory chain

```

$user = User::factory()->create();

$trip = Trip::factory()
    ->recycle($user)
    ->create();

```

Whew. That was a lot. Don't worry if that was tough to follow—the last bit was definitely higher-level stuff. Let's get back down to the basics and talk about the core of Laravel's database tooling: the query builder.

Query Builder

Now that you're connected and you've migrated and seeded your tables, let's get started with how to use the database tools. At the core of every piece of Laravel's database functionality is the *query builder*, a fluent interface for interacting with several different types of databases with a single clear API.

WHAT IS A FLUENT INTERFACE?

A *fluent interface* is one that primarily uses method chaining to provide a simpler API to the end user. Rather than expecting all of the relevant data to be passed into either a constructor or a method call, fluent call chains can be built gradually, with consecutive calls. Consider this comparison:

```
// Non-fluent:
$users = DB::select(['table' => 'users', 'where' => ['type' => 'donor']]);

// Fluent:
$users = DB::table('users')->where('type', 'donor')->get();
```

Laravel's database architecture can connect to MySQL, PostgreSQL, SQLite, and SQL Server through a single interface, with just the change of a few configuration settings.

If you've ever used a PHP framework, you've likely used a tool that allows you to run "raw" SQL queries with basic escaping for security. The query builder is that, with a lot of convenience layers and helpers on top. So, let's start with some simple calls.

Basic Usage of the DB Facade

Before we get into building complex queries with fluent method chaining, let's take a look at a few sample query builder commands. The DB facade is used both for query builder chaining and for simpler raw queries, as illustrated in

Example 5-16.

Example 5-16. Sample raw SQL and query builder usage

```
// Basic statement
DB::statement('drop table users');
```

```
// Raw select, and parameter binding
DB::select('select * from contacts where validated = ?', [true]);

// Select using the fluent builder
$users = DB::table('users')->get();

// Joins and other complex calls
DB::table('users')
    ->join('contacts', function ($join) {
        $join->on('users.id', '=', 'contacts.user_id')
        ->where('contacts.type', 'donor');
    })
    ->get();
```

Raw SQL

As you saw in [Example 5-16](#), it's possible to make any raw call to the database using the DB facade and the `statement()` method: `DB::statement('SQL statement here')`.

But there are also specific methods for various common actions: `select()`, `insert()`, `update()`, and `delete()`. These are still raw calls, but there are differences. First, using `update()` and `delete()` will return the number of rows affected, whereas `statement()` won't; second, with these methods it's clearer to future developers exactly what sort of statement you're making.

Raw selects

The simplest of the specific DB methods is `select()`. You can run it without any additional parameters:

```
$users = DB::select('select * from users');
```

This will return an array of `stdClass` objects.

Parameter bindings and named bindings

Laravel's database architecture allows for the use of PDO (PHP data object, PHP's native database access layer) parameter binding, which protects your queries from potential SQL attacks. Passing a parameter to a statement is as simple as replacing the value in your statement with a `?`, then adding the value to

the second parameter of your call:

```
$usersOfType = DB::select(  
    'select * from users where type = ?',  
    [$type]  
);
```

You can also name those parameters for clarity:

```
$usersOfType = DB::select(  
    'select * from users where type = :type',  
    ['type' => $userType]  
);
```

Raw inserts

From here, the raw commands all look pretty much the same. Raw inserts look like this:

```
DB::insert(  
    'insert into contacts (name, email) values (?, ?)',  
    ['sally', 'sally@me.com']  
);
```

Raw updates

Updates look like this:

```
$countUpdated = DB::update(  
    'update contacts set status = ? where id = ?',  
    ['donor', $id]  
);
```

Raw deletes

And deletes look like this:

```
$countDeleted = DB::delete(  
    'delete from contacts where archived = ?',  
    [true]  
);
```

Chaining with the Query Builder

Up until now, we haven't actually used the query builder, per se. We've just used simple method calls on the DB facade. Let's actually build some queries.

The query builder makes it possible to chain methods together to, you guessed it, *build a query*. At the end of your chain you'll use some method—likely `get()`—to trigger the actual execution of the query you've just built.

Let's take a look at a quick example:

```
$usersOfType = DB::table('users')
    ->where('type', $type)
    ->get();
```

Here, we built our query—`users` table, `$type` type—and then we executed the query and got our result. Note that, unlike the previous calls, this will return a *collection* of `stdClass` objects instead of an array.

ILLUMINATE COLLECTIONS

The DB facade, like Eloquent, returns a collection for any chained method that returns (or can return) multiple rows and an array for any nonchained method that returns (or can return) multiple rows. The DB facade returns an instance of `Illuminate\Support\Collection` and Eloquent returns an instance of `Illuminate\Database\Eloquent\Collection`, which extends `Illuminate\Support\Collection` with a few Eloquent-specific methods.

`Collection` is like a PHP array with superpowers, allowing you to run `map()`, `filter()`, `reduce()`, `each()`, and much more on your data. You can learn more about collections in [Chapter 17](#).

Let's take a look at what methods the query builder allows you to chain. The methods can be split up into what I'll call constraining methods, modifying methods, conditional methods, and ending/returning methods.

Constraining methods

These methods take the query as it is and constrain it to return a smaller subset of possible data:

select()

Allows you to choose which columns you're selecting:

```
$emails = DB::table('contacts')
    ->select('email', 'email2 as second_email')
    ->get();

// Or

$emails = DB::table('contacts')
    ->select('email')
    ->addSelect('email2 as second_email')
    ->get();
```

where()

Allows you to limit the scope of what's being returned using WHERE. By default, the signature of the `where()` method takes three parameters—the column, the comparison operator, and the value:

```
$newContacts = DB::table('contact')
    ->where('created_at', '>', now()->subDay())
    ->get();
```

However, if your comparison is `=`, which is the most common comparison, you can drop the second operator:

```
$vipContacts = DB::table('contacts')->where('vip',true)->get();
```

If you want to combine `where()` statements, you can either chain them after each other, or pass an array of arrays:

```

$newVips = DB::table('contacts')
    ->where('vip', true)
    ->where('created_at', '>', now()->subDay());

// Or

$newVips = DB::table('contacts')->where([
    ['vip', true],
    ['created_at', '>', now()->subDay()],
]);

```

orWhere()

Creates simple OR WHERE statements:

```

$priorityContacts = DB::table('contacts')
    ->where('vip', true)
    ->orWhere('created_at', '>', now()->subDay())
    ->get();

```

To create a more complex OR WHERE statement with multiple conditions, pass `orWhere()` a closure:

```

$contacts = DB::table('contacts')
    ->where('vip', true)
    ->orWhere(function ($query) {
        $query->where('created_at', '>', now()->subDay())
            ->where('trial', false);
    })
    ->get();

```

POTENTIAL CONFUSION WITH MULTIPLE WHERE() AND ORWHERE() CALLS

If you are using `orWhere()` calls in conjunction with multiple `where()` calls, you need to be very careful to ensure the query is doing what you think it is. This isn't because of any fault with Laravel, but because a query like the following might not do what you expect:

```
$scanEdit = DB::table('users')
    ->where('admin', true)
    ->orWhere('plan', 'premium')
    ->where('is_plan_owner', true)
    ->get();
```

```
SELECT * FROM users
WHERE admin = 1
OR plan = 'premium'
AND is_plan_owner = 1;
```

If you want to write SQL that says “if this OR (this and this),” which is clearly the intention in the previous example, you'll want to pass a closure into the `orWhere()` call:

```
$scanEdit = DB::table('users')
    ->where('admin', true)
    ->orWhere(function ($query) {
        $query->where('plan', 'premium')
            ->where('is_plan_owner', true);
    })
    ->get();
```

```
SELECT * FROM users
WHERE admin = 1
OR (plan = 'premium' AND is_plan_owner = 1);
```

`whereBetween(colName, [low, high])`

Allows you to scope a query to return only rows where a column is between two values (inclusive of the two values):


```
$mediumDrinks = DB::table('drinks')  
->whereBetween('size', [6, 12])  
->get();
```

The same works for `whereNotBetween()`, but it will select the inverse.

`whereIn(colName, [1, 2, 3])`

Allows you to scope a query to return only rows where a column value is in an explicitly provided list of options:

```
$closeBy = DB::table('contacts')  
->whereIn('state', ['FL', 'GA', 'AL'])  
->get();
```

The same works for `whereNotIn()`, but it will select the inverse.

`whereNull(colName), whereNotNull(colName)`

Allows you to select only rows where a given column is NULL or is NOT NULL, respectively.

`whereRaw()`

Allows you to pass in a raw, unescaped string to be added after the WHERE statement:

```
$goofs = DB::table('contacts')->whereRaw('id = 12345')->get();
```

BEWARE OF SQL INJECTION!

Any SQL queries passed to `whereRaw()` will not be escaped. Use this method carefully and infrequently; this is a prime opportunity for SQL injection attacks in your app.

whereExists()

Allows you to select only rows that, when passed into a provided subquery, return at least one row. Imagine you only want to get those users who have left at least one comment:

```
$commenters = DB::table('users')
    ->whereExists(function ($query) {
        $query->select('id')
            ->from('comments')
            ->whereRaw('comments.user_id = users.id');
    })
    ->get();
```

distinct()

Selects only rows where the selected data is unique when compared to the other rows in the returned data. Usually this is paired with `select()`, because if you use a primary key, there will be no duplicated rows:

```
$lastNames = DB::table('contacts')->select('city')->distinct()->get();
```

Modifying methods

These methods change the way the query's results will be output, rather than just limiting its results:

orderBy(*colName*, *direction*)

Orders the results. The second parameter may be either `asc` (the default, ascending order) or `desc` (descending order):

```
$contacts = DB::table('contacts')
    ->orderBy('last_name', 'asc')
```

```
->get();
```

groupBy(), having(), havingRaw()

Groups your results by a column. Optionally, `having()` and `havingRaw()` allow you to filter your results based on properties of the groups. For example, you could look for only cities with at least 30 people in them:

```
$populousCities = DB::table('contacts')
    ->groupBy('city')
    ->havingRaw('count(contact_id) > 30')
    ->get();
```

skip(), take()

Most often used for pagination, these allow you to define how many rows to return and how many to skip before starting the return—like a page number and a page size in a pagination system:

```
// returns rows 31-40
$page4 = DB::table('contacts')->skip(30)->take(10)->get();
```

latest(colName), oldest(colName)

Sorts by the passed column (or `created_at` if no column name is passed) in descending (`latest()`) or ascending (`oldest()`) order.

inRandomOrder()

Sorts the result randomly.

Conditional methods

There are two methods that allow you to conditionally apply their “contents” (a closure you pass to them) based on the Boolean state of a value you pass in:

when()

Given a truthy first parameter, applies the query modification contained in the closure; given a falsy first parameter, it does nothing. Note that the first parameter could be a Boolean (e.g., \$ignoreDrafts, set to true or false).
Example 5-16.

withColumn()

Sorts by the passed column (or columns).
Example 5-16.
scala> val query = sqlContext.query(sql"select * from contacts order by premium")
scala> query.withColumn("premium", sql"coalesce(premium, 0)")

distinct()

Selects only rows where the selected data is unique when projected.
Example 5-16.
scala> query.distinct().show()

scala> query.withColumn("premium", sql"coalesce(premium, 0)")
scala> query.distinct().show()

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D

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100) ASCODEGON_SC="PDE>ODE>->SKIP(), TAKE()

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MOST OFTEN USED FOR PAGINATION, THESE ALLOW YOU TO
DEF3" ID="ID83">

CONDITIONAL METHODS

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MOST OFTEN USED FOR PAGINATION, THESE ALLOW YOU TO
 NLY WDER)GRAMLIJOIN()ODE>

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POTENTIAL CONFUSIODEF3" ID="ID83">

CONDITIONAL METHODS

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MOST OFTEN USED FOR PAGINATION, THESE ALLOW
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CONDITIONAL METHODS

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METHODS" DATA-TYPE="INDEX42E QENSURE INDE
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ANDORETTY"NV">\$P. PA

**HIO RO)ID BACK RALLECTI/CODE>
DB::TABLE'('CONTACTS') ->ORDERBY(ODE>->,
TAKE())**

**MOST OFINDE TGETIDCODE>OR[>SKIP(), TAKE;(E>,
TAKE;OODE CLASS="NA">GET, TAKE; "P">(E>,
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>, TAKE())**

**MOST OFINDE TCODE>OR[>SKIP(), TAKEPODE
CLASS="S1">'AL '(E>, TAKE;->WHERE "P">(E>,
TAKE;OODE CLASS="NA">GET, TAKEPODE
CLASS="S1">'AL '(E>, TAKE;->WHERE "P">(E>,
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CONDITIONAL METHODS

**THERE ARUPDASS=TA-TERTIARY="MODIFYING METHODS"
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THEMUPDASSPRIME, CODE, "NATE COLLECNSURE ICK
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AISTINARTRGRAMLIUPDASS, YOU COULD LOPA**

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MOST OFIN

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MOST OFGR

ORDERBY->

CONDITIONAL METHODS

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Most oftrune-la used for pagination, these allow you to def3" id="id83">

Conditional methods

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Sor("hen comss="na">table tif

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table) ->orderBy(function, true) ->Sor( orWhere(function),  
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orWhere(funcnt] when comConfusiodef3" id="id83"> def3"  
id="id83">
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Conditional ods

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(dethrst difying methods" data-type="index51rm"  
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m>)
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id="id1methodxref" hmlist#EX807">E="prog 5-17ttion> tacks"  
data-
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type="proctype="iEX807">id105>ttrainmeth05"label">E="prog  
5-17. f id="A$p dethrst utransacode>y="5e>  
DB::tablede>de>) -&gransacode>y=de>orderBy  
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"p">Idcode>unction'ide> trueor[>skip()e class="p">());
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18ttion> tacks" data-
type="proctype="iEX808">id105>ttrainmeth05"label">E="prog
5-18. f id="der)nv">$pst:Elorimnttoulely="5e>
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icaterstic:Elorimntthe>(table l
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Conditioods

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tmulel=l methods (quames atoulel.rderre>dire/ies  
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Alotionall YthemMulely=reRaw() tonl methods (quames  
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Conditional methods~~

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~~Conditional ods~~

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wdeollwradirefilcontit DB::table('contacts' sta class=or
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Conditional methththods

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Conditional ods

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5-36. f id="Dnn aitio tusto eElorimntteccessooe "5e>
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'ide>:ode>(ede>Attribute class=
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Conditional methods

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tvaldehod>iertiin,gid=if w grare cstiomiziti
Editl>in tw grare settde ,uwes tacks" data-
type="proctype="iEX826">id105>ttrainmeth05"label">E=

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truein class="p">('contacts';-> direrelaatft>bu
ommordr a)outatooties otnoneode>enttcode>
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Conditional methods

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Conditional methods

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Conditional methods

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Conditional methods

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def3" id="id83"> Conditional methods
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Conditional methods

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Conditional methods
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it w0wr)> oa y="query builel>dir id="id1028"/> arElorimntl
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type="progtype="iEX831">id105>ttrainmeth05"label">E="prog 5-
43. f id="Dnn aitio= one-to-oa relaas =hipy="5e>
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type="proctype="iEX832">id105>ttrainmeth05"label">E=
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&samey=$queryorWhere) ]pse allowge::ge<'ide>
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e>, trWhs ass="na">orWhere ]pse allowge::ge<'ide>
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Conditional methods

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type="proctype="iEX833">id105>ttrainmeth05"label">E="prog 5-
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ulectiscatcodle =iae> Collec< >olel ide> wit. Ade
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Conditional methods

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(e.g.,eodrainmeth05"keep-togdeeth> Collec2ocessiti
Editrelaat =hip,eit cc) tc)ok forartr_ity= fdok
for1 DB::(('contacts'<'true$querpon'ide>
true$queryorWhere<"p">( ass="na">orWherepon'ide>
true<))

```

Conditional methmethods

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ThereSele tiuinlyerdcords a valhater1trllaaedritem "4 >
="keeYoueid="id1028"/> arElorimntl methollow you torelaat =hip/e
class=de-iou tosele tiuinlyerdcords ) :articularecriteria ) DB::
(geode class="s1">'contacts'"/coPoth allow
class="nv">$code> -&hasy=$query ass="na">orWherepon'ide>
true<)

```

```

$uadjstinEdit riteria furethr/code> DB::(geode
class="s1">'contacts'"/coPoth allow class="nv">$code> -
&hasy=$query ass="na">orWhere<"p">( ass="na">orWhere<"p">(
ass="na">orWherepon'ide> true<))

```

```

DB::(geode class="s1">'contacts'"/coUs/p -&hasy=$query
ass="na">orWherepon'ide> true<))

```

```

DB::(<__ codgedsu1ng_co="">ge<'geode
class="s1">'contacts'"/coCe> sta class=or paginationcode> -&code
Hasy=$query ass="na">orWhere<"p">( $queryorWherepon
"p">(table ass="na">orWhde> true$queryorWhere<"p">
(orWhere<"p">(orWherepse allowget'ide> true<))

```

```

ge::ierlass=abovty=ge<'geode class="s1">'contacts'"/coCe> sta
class=or paginationcode> -&code Relaat y=$querytable
ass="na">orWheretableorWheretableorWhere<ges

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ass="na">orWherepon'ide> true<)
```

```
ef3" id="id83">
```

Conditional methods

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There anyl methods (query builder)" data-  
type="indexte116lun it wHas oa cof<>anyy="query  
bui0n/yid="id1028"/> arElorimntl methollow you torelaat  
=hip/e class=de-iou tohas-oa -of-manye class=s (quames  
arhas-oa -of-manyorelaat =hip/e class=s (quames <= one-to-  
many)relaat =hipgis >evalyp> ting"em:retrtev//istinoa  
citemaie> <|Ceolewetines eee oldeti. Lar>veltd1ovides  
a/convdnianctreoolhforejstinEdirtrsituat s: ha/rine  
of<>any. class="keeHas-oa -of-manyorelaat =hip/ ul yocyp>  
tandnn a ca vala givdn:m
```

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rhoulddretrtev//\Ceolewetinitemain aerllaaedrco  
,ues eee oldetiritem,des eee itema)  
fdany)particularecode> ,dos yp> E="prog 5-47ttion>  
tacks" data-  
type="progype="iEX853">id105>ttrainmeth05"label">E=  
"prog 5-47. f id="Dnn aitiohas-oa -of-manyorelaat  
=hip/y="5e> DB::tblkndtable $"c">Us/p ge>( <>xt> is  
$"/coMulely=sed fodclass="na">get(table(  
$nf">lewetice> sta class=or paginationquerpon'ide>:  
$"/coHasOn/y= "p">(table$n''ide> true$query-&table  
$quep/code>'ide> true$querpse allow }e allowtable(  
$nf">oldetice> sta class=or paginationquerpon'ide>:  
$"/coHasOn/y= "p">(table$n''ide> true$query-&table  
$quep/code>'ide> true$querpse allow }e allowtable(  
$nf">emergancyCe> sta class=or  
paginationquerpon'ide>: $"/coHasOn/y= "p">  
(table$n''ide> true$query-&table $quep/code>'ide>  
true$queryorWhere<"p">(orWherepse allow }e  
allowConfus ef3" > ef3" id="id83">
```


Conditional methods

There are many methods that allow you to relate a hip/e class=de-iou to has->any-through methods (quames arhas->any-throughorelaat =hip/e class=s (quames

ies pulliti inrelaat =hip. Tlink= fddehoe="prog I gaterearli ,ccode ea any) sta class=s ude eachd sta class=eha/r>any) ting"em:ode auus/p>dire>direa has->any-throughorelaat .

```
class="keeTlitgstrucectthossumes >evalyp>r
.6">Exce>ok forE="prog 5-48ttion> tacks" data-
type="proctype="iEX835">id105>ttrainmeth05"label">E=
"prog 5-48. f id="Dnn aitio= has->any-
throughorelaat =hipy="5e> DB::tblkndtable
$c">Us/p ge>(<>xt> is $"/coMulely=sed
fodclass="na">get(table( $nf">phoa Numbere class=
class="nv">$querpon "p">(table(table$n''ide>
true$query ass="na">orWherepon'ide>
true$queryorWherepse allowtable(table$n''ide>
true$query-&table $que<"p">( sta class=or paginationcode> -
&table $quepse allow }e allowConfus ef3" >
="keeYrograe access etitgrllaat =hipeusitiook
for$us/p le00eem: iz ca airelaat =hipgkeeroi wt
mulels,cartr\Ceotradi altsyntcx; yp> olel ()in/co>
fdok forhasManyThroughrpon wt mulel ()in/co f
id=")ttion> ef3" id="id83">
```

Conditional methods

There are many methods that allow you to relate a hip/e class=de-iou to has-one-through methods (quames arhas-one-throughorelaat =hip/e class=s (quames Collecgrare inlyedirephoa clumbertby oulliti cdiir comid=y>direphoa clumber? Teat>direa has-one-throughorelaat ,dos show> in gid="id1methodxref" hmlist#EX5-48">E="prog 5-49ttion> tacks" data-

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type="proctype="iEX5-
48">id105>ttrainmeth05"label">E="prog 5-49. f
id="Dnn aitia= has-one-throughorelaa =hipy="5e>
DB::tblkndtable $"c">Us/p ge>(<>xt> is
$"/coMulely=sed fodclass="na">get(table( $nf">phoa
Number class= class="nv">$querpon "p">
(table(table$n''ide> true$query
ass="na">orWherepon'ide> true$queryorWherepse
allowtable(table$n''ide> true$query-&table $que<"p">(-
&table $quepse allow }e allowConfus ef3" > ef3"
id="id83">

```

Conditional methods

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There >anyl methods (query builder)" data-
type="indexte119lun it wMuny<> >anyy="query buildir
id="id1028"/> arElorimntl methollow you torelaat
=hip/e class=de-iou to>any-to-m1dyorelaa class=s (quames
ar>any-to-m1dyorelaa =hip/e class=s (quames directcla
p>eme="prog ofCRMea valal yosea sta class=s, ude
eachd sta class=eem:bddrelaat0> multirog E="prog 5-
50ttion> tacks" data-
type="proctype="iEX836">id105>ttrainmeth05"label">E=
"prog 5-50. f id="Dnn aitia= >any-to-m1dyorelaa
=hipy="5e> DB::tblkndtable $"c">Us/p ge>(<>xt> is
$"/coMulely=sed fodclass="na">get(table(
$nf">ce>$querpon "p">(table$n''ide> true$query-
&table $quepse allow }e allowget ="keeAde
sinctreditgis muny<> >any,ca aiinvdesd
lookstextalyddehod>ier(gid="id1methodxref"
hmlist#EX837">E="prog 5-51)ttion> tacks" data-
type="proctype="iEX837">id105>ttrainmeth05"label">E=
"prog 5-51. f id="Dnn aitia= >any-to-m1dyorelaa
=hip>direinvdesdy="5e> DB::tblkndtable $"c">de>
sta class="p">(xt> is $"/coMulely=sed
fodclass="na">get(table( $nf">us/pe class=

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class="nv">$querpon "p">(table$n''ide> true$query-
&table $quepse allow }e allowget ="keeSinctrot to
le sta class=ecanethelhater1t tode ac to le , >any-
to-m1dyorelaat =hiptgrlleroie/ cwo. lde/convdn
altnam tiuifreditgta;insat$gdeeth,desdco d
rlphabeticalup,iide separ>atnghcdim
="keeSo,csinctrwegrae clinkitiook forus/pe class= a
y o empivot"ta;indn="na">table<grae ing">em: iz ca
aita;in,uo:in/co>t$ucdito s:t ="keeJstining">cc <
direie> E="prog 5-52)ttion> tacks" data-
type="progtype="iEX838">id105>ttrainmeth05"label">E=
"prog 5-52. f id="Accessiti Editrelaat0item= ie>
any-to-m1dyorelaat =hipy="5e> DB::tabl<'geode
class="s1">'contacts'"/coUs/p -&se the allow
class="nv">$querpse allowge<'ide> true$de>
true$query(<> $query'iere)on "p">(table
getge<'geode class="s1">'contacts'"/coCe> sta
class=or paginationcode> -&se the allow
class="nv">$querpse allowge<'ide> true$de>
true$query(<> $query(table
getge<'('contacts'<'true$querpon'ide>
true$queryorWhere<"p">( ass="na">orWherepon'ide>
true<)

="

```

Conditional methods

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There Getatngh ="keeOn/yid="id1028"/> arElorimntl
methollow you torelaat =hip/e class=de-iou to arpivot"ta;
direunirim=aboe >anygis >evalil>direp>emse th enterii
ecases code eil>direvalua;e einf rma =ialyp>r pivot"ta;
might ting"em:ti>e elt d_aa class= iield>em:see)lan
etitgrllaat =hipewas<

d.tion> ="keeIndesdco"em:ti>e eDirtriiield=,cyp>

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entereocaddhcdimin gid="id1methodxref"
hmlist#EX839">E="prog 5-53.tYou or addh d_aa class= a y o
d_aa class= tiinstamp=eusitiook forcc
id105>ttrainmeth05"label">E="prog 5-53. f
id="Additioiield=eeocatpivot"recorde="5e>
DB::tblknd:ub$ic class="p">(
$nf">ce>$querponget(table$n''ide> true$query-&table
$quep/code>table true$querpone>table
true$queryorWhere<"p">(orWherepse allowget ="keeWlan yp>
ode au>olel ide> wit throughootrelaat =hip,eit cc)
jstinoullndnitaie> . So,cyp> id="id1methodxref"
hmlist#EX840">E="prog 5-54ttion> tacks" data-
type="proctype="iEX840">id105>ttrainmeth05"label">E="prog
5-54. f id="Getatngh <1trllaaedritem>direpivot"mentry
DB::tbl<'geode class="s1">'contacts'"/coUs/p -&se the
allow class="nv">$querpse allowge<'ide> true$de>
true$query(<> $query'iere)on
get(table'contacts'"b">s028ntfy=$querye>table stahossoci>
da) ass="na">orWheretable'ide> true$de> true d_aa class=
pse allowget ="keeIf yp>grae ing",cyp> iz ca ai in usiti
Edit in gid="id1methodxref" hmlist#EX5f">E="prog 5-
55ttion> tacks" data-
type="proctype="iEX5f">id105>ttrainmeth05"label">E="prog 5-
55. f id="Csti> iziti Editpivot"attrimein DB::tbl::olele
allowgeknd:ub$ic class="p">( $nf">group= class=
class="nv">$querponget(table$n''ide> true$query-&table
$quep/code>table true$querpone>table true$queryorWherepse

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allowget DB::tabl::ge"/coUs/p -&se the allow
class="nv">$querpon'ide> true<$de> true$query(<>
$queryget(table'contacts'"b">s028ntfy=$querye>table
ass="na">orWheretable true$de> true d_aa class= pse
allowget
id101>Unirim=Aspypes of Attach ag>andeDetach ag>
ar>any-to-m1dyorelaat =hip/e class=s (quames r pivot"ta;
nfoode-ie=,cyp> le♦♦eem:bdda;grare attach ag>a >any-to-
m1dyorelaaedritem.tYou in/co>t$u DB::(geode
class="s1">'contacts'"/coUs/p -&se the allow
class="nv">$querpse allowge<'geode
class="s1">'contacts'"/coCe> sta class=or paginationcode>
-&se the allow class="nv">$querpse allowge<'ide>
true$querpon'ide> true$query'iere<"p">(geode code> e>,
trWhs ass="na">orWhere]pse allow you to ="keeAddi
alup,iyp> Collecpa wit ofrelaaedritem,dyp> ierasu $uaccept
le♦♦iti Eanrel>in dhe:m

aoaso> allowattachM1dyrpon DB::(geode
class="s1">'contacts'"/coUs/p -&se the allow
class="nv">$querpse allowge<'ide>
true$querpon'ide> true$query$quepse
allowge<'ide> true$querpon'ide>
true$query'iere<"p">(geode code>
e>, trWhs ass="na">orWhere]pse allowge<'ide>
true$querpon'ide> true$quer[$que<"p">('iere<"p">(
ass="na">orWhere]pse allowge<'ide> true$querpon'ide>
true$quer[ geode code> e>, trWhpre[geode code> e>, trWhs
ass="na">orWhere]table'ieretable

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ass="na">orWhere<geere]pse allowge<'ide>
true$querpon'ide> true$query$quepse allowge<'ide>
true$querpon'ide> true$quer[$que<"p">('iere]pse
allowge<'ide> true$querpon'ide> true$quer)e> e>, trWh::
ColjstinEoeinvdete)laaeves eee currmnttattachm ntorogat
i=,cyp> ting"edit urtroallowtogglerpon DB::('ide>
true$querpon'ide> true$quer[$que<"p">('iere<"p">(
ass="na">orWhere]pse allow you to ="keeYou
$uurtroallowapd> Ex> >ag">chB DB::('ide> true$querpon'ide>
true Ex$query'iere<"p">(tablegeode code> e>,
trWhsorWhere<geere]pse allow you to ="keeA y ifryp>grae
ing">em:reolace eee currmnttrelaat =hip/,
effeodivdlyedetachiti new oa s,dyp> DB::('ide>
true$querpon'ide> true$quer[$que<"p">('iere<"p">(
ass="na">orWhere]pse allowge<'ide> true$querpon'ide>
true$quer[ geode code> e>, trWhpre[geode code> e>, trWhs
ass="na">orWhere]table'ieretable
ass="na">orWhere<geere]pse allow you to ef3" idaside >
ef3" id="id83"> ef3" id="id83">

```

Conditional methods

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Thereeyid="id1028"/> arElorimntl methollow you torelaat
=hip/e class=de-iou topolymorphicl methods (quames
arpolymorphicorelaat =hip/e class=s (quames ierrelaat
=hip. Wegrae cgoiti EanurtroallowStap favorites) right
now. Adartrc sta class=s ude oallowEvdn class=s,e1de
clat>direcode e in polymorphy=em> "ome= ie> :/edde
>direa to le inecoface eo objypes of multirog =s (s.tion>
="keeSo,cwegra) sta class=, ude oallowEvdn class=).

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Actualup,iyp>gra)$uneoallowas/pe class= a y o direc/p>
ep>emmulels,cos see> in gid="id1methodxref"
hmlist#EX841">E="prog 5-56ttion> tacks" data-
type="proctype="iEX841">id105>ttrainmeth05"label">E="prog
5-56. f id="C/p> iti Editmulels ies a DB::tblkndtable
$c">Stap table is $"/coMulely=sed
fodclass="na">get(table( $nf">stapra;$querpon
"p">(table$N''ide> true$querpse allow }e allowgetgekndtable
$c">de> sta class="p">(xt> is $"/coMulely=sed
fodclass="na">get(table( $nf">stape class=
class="nv">$querpon "p">(table$N''ide> true$query'ide>e> -
&table $que<"p">(orWherepse allow }e allowgetgekndtable
$c">Evdn class="p">(xt> is $"/coMulely=sed
fodclass="na">get(table( $nf">stape class=
class="nv">$querpon "p">(table$N''ide> true$query'ide>e> -
&table $que<"p">(orWherepse allow }e allowget
="keeSo,chowddoe:c/p> e1t DB::(geode
class="s1">'contacts'"/coCe> sta class=or paginationcode> -
&se the allow class="nv">$querpse allowge<'ide>
true$querpon'ide> true e allow class="nv">$querpse allow you
to ="keeIl>direthat easy. lde/glass=de> sta class="is
nowostapr d.tion> ="keeIndesdco"em:n ad sta class=, we:c in
gid="id1methodxref" hmlist#EX842">E="prog 5-57ttion> tacks"
data-type="proctype="iEX842">id105>ttrainmeth05"label">E="prog
5-57. f id="Retrtevitioa aine> wits lec< polymorphicorelaat
=hip DB::tbl<'geode class="s1">'contacts'"/coCe> sta
class=or paginationcode> -&se the allow class="nv">$querpse
allowge<'ide> true$de> true$query(<> $query'iere)on
get(tableget ="keeIf we:hnteran ide> wit of< <>xt"is o a look
at gid="id1methodxref" hmlist#EX843">E="prog 5-58ttion>
tacks" data-
type="proctype="iEX843">id105>ttrainmeth05"label">E="prog 5-
58. f id="Retrtevitioa aitapge wit DB::tbl<'geode
class="s1">'contacts'"/coStap 'ide>e> -&$querpse
allowge<'$de> true$query(<> $query'iere)on get(tablevar_dump
class=or paginationerery'ide> true$que)e> e>, trWh:: wit of

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stahor Evdn class=get ="keeFi alup,iyp> might be wonddede ,
"W valifrI ting"em:knowowhoostapr d etitgde>direa g/p> quesat
. Il>direos siprog asr .6">Exstape class="ta;ExUs/p
ha/r>anyy=em> oallowStap belongs"cay=em> oa c.6">ExUs/p
E="prog 5-59)t lde/glass=stape class="ta;r .6">ExUs/p
r:oallowCo> sta class=s ude oallowEvdn class=sttion> tacks"
data-type="progye="iEX844">id105>ttrainmeth05"label">E="prog
5-59. f id="Ext> i ag>a polymorphicosystem tandiffermntioat
by:artr/c"5e> DB::tblkndtable $"c">Stap table is
$"/coMulely=sed fodclass="na">get(table( $nf">stapra;$querpon
"p">(table$n''ide> true$quese allow }e allowtable( $nf">us/p
class= class="nv">$querpon "p">(table$n''ide> true$query-
&table $quepse allow }e allowgetgekndtable $"c">Us/p ge>
(<>xt> is $"/coMulely=sed fodclass="na">get(table( $nf">stape
class= class="nv">$querpon "p">(table$n''ide>
true$query'ide>e> -&table $quepse allow }e allowget
="keeTeat>direit!tYou ExUs/p direoallowStap ExUs/p
<1toallowStap :c/p> e1tnew gra) DB::(geode
class="s1">'contacts'"/coUs/p -&se the allow
class="nv">$querpse allowge<('contacts'"/coEvdn class=or
paginationcode> -&se the allow class="nv">$querpse
allowge<'ide> true$querpon'ide> true e allow
class="nv">$quer[geode code> e>, trWh<'ide> true'iere]pse
allow you to ef3" id="id83">

```

Conditional methods

There arElorimntl methollow you torelaat =hip/e class=de-iou
tomuny-to-many)polymorphicl methods (quames ar>any-to-
m1dyopolymorphicorelaat =hip/e class=s (quames any-to-
m1dyopolymorphicorelaat =hip/eae eing">polymorphicorelaat
=hip/, except iae> Collecbe tiuine<> >any,ca aygrae cmuny<>
>any. class="keeTle most<"omm =e="prog ies etitgrllaat
=hipe=s (tis >e/ cag, so Igra)eme="prog. Lel>dire8"/g a cyp>
ting"em:bdda; sta class=s ude oallowEvdn class=st
lde/unirimness of many-to-m1dyopolymorphismgis
>evalil>diremuny<> >any: eachdcag:may:bddapplite> multirog


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item=,cude eachdcaggt◊item might hntermultirog cagst A y
eocaddhcoed va,lil>direpolymorphic: cags item= of
sevesald=iffermntt=s (s. Fes dhe:dlasrst ,
wegra)$uaddnormalt) ting,ecem>a yy=em> 1tnew dirednn a ca
itgrllaat =hipe =o>emmulels,cos see> in gid="id1methodxref"
hmlist#EX845">E="prog 5-60ttion> tacks" data-
type="progtype="iEX845">id105>ttrainmeth05"label">E="prog 5-
60. f id="Dnn aitio= polymorphico>any-to-m1dyorelaat
=hipy="5e> DB::tablknndtable $"c">de> sta class="p">(xt> is
$"/coMulely=sed fodclass="na">get(table( $nf">cags $querpon
"p">(table$N''ide> true$query'ide>e> -&table $que<"p">
(orWherepse allow }e allowgetgekndtable $"c">Evdn class="p">
(xt> is $"/coMulely=sed fodclass="na">get(table( $nf">cags
$querpon "p">(table$N''ide> true$query'ide>e> -&table
$que<"p">(orWherepse allow }e allowgetgekndtable
$"c">Tag/code><"p">(xt> is $"/coMulely=sed
fodclass="na">get(table( $nf">ce>$querpon "p">(table$N''ide>
true$query-&table $que<"p">(orWherepse allow }e allowtable(
$nf">ev nts $querpon "p">(table$N''ide> true$query-&table
$que<"p">(orWherepse allow }e allow }e allowConfus ef3" >
="keeHde >direhowdem:

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eyp>emse th DB::(<"p">('contacts'"/coTag/code>'ide>e> -&se
thOrC/p> e allow class="nv">$quer[ingeode code> e>,
trWhsorWhere]pse allowge<'geode class="s1">'contacts'"/coCe>
sta class=or paginationcode> -&se the allow
class="nv">$querpse allowge<'ide> true$querpon'ide>
true$query'ide> true'ierepse allow you to
="keeWe/genormal,cos id="id1028"/> arl
methologramlistpolym0er class=s (quames in
gid="id1methodxref" hmlist#EX846">E="prog 5-61ttion> tacks"
data-type="progtype="iEX846">id105>ttrainmeth05"label">E="prog
5-61. f id="Accessiti Editrelaat◊item= ie> any-to-
m1dyopolymorphicorelaat =hip DB::tabl<'geode
class="s1">'contacts'"/coCe> sta class=or paginationcode> -
&se the allow class="nv">$querpse allowge<'ide> true$de>
true$query(<> $query'iere)on get(tablegetge<'<"p">

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( 'contacts'"/coTag/code>'ide>e> -&se the allow
class="nv">$querpse allowge<'ide> true$de> true$query(<>
$query'iere)on get(tableget ef3" id="id83">
```

Conditional methods

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There iti parmnttreord tiinstamp=l methods (query builder)"
data-type="indexte123lun it wChild"records apd> iti
parmnttreord tiinstamp=y="query buiRemember,
1dyid="id1028"/> arElorimntl methollow you torelaat =hip/e
class=de-iou tochild"records apd> iti parmnttreord/e class=s
(quames artiinstamp=l methods (quames d_aa class= a y o d_aa
class= tiinstamp=. Elorimnt)cc) d_aa class= tiinstamp automa
calup any<>iintyp> mag">any d any<>iintEditrelaat♦itemgis
apd> d. Fes e="prog,iifda .6">ExPhoneNumber= d, >ayb ca ai
sta class="il>direconnypet♦> rhouddbe dat d as/hntiti bee>
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arraylnfoode-ye =eee child"" dat,cos in gid="id1methodxref"
hmlist#EX847">E="prog 5-62ttion> tacks" data-
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62. f id="Upd> iti a parmnttreord
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id="id83">
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Conditional methods

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Conditional methods

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id="id83">

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Conditional methmethods

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Thereid104>Lazy eoder loaditi ="keeI id="id1028"/> arlazy

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eoder loaditil methods (quames iins/yp>
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Conditional methmethods

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Conditional methmethods

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63b">id105>ttrainmeth05"label">E="prog 5-64. f id="Bi i ag>a
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5-66. f id="Usiti .6">ExasllrtDlasrst Hasrpon
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hmlist#EX852">E="prog 5-67ttion> tacks" data-
type="proctype="iEX852">id105>ttrainmeth05"label">E="prog 5-
67. f id="Tcode utators, 1de sco (s c"5e> DB::tblknd:ub$ic
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data-typnotexterm" id="idnotexSQL-qExcep>Sorpy=reRaw()yp>
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def3" id="id83">
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Conditional methods

Thereic<>hin normalte queries.Exick example: 7paype=ewher
/code> DB::table('contacts' sta class=or paginationcode>--
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vtthddra oleia Exall()ode>gra DB::table('contacts' sta
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allow you to utacks" data-typtipxSQL qUsto eick e Iate
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ylasrogratoulifycodayp>agtime,et<additio<d/ignorta-
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```

Conditional methods

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rstches,codatta-type=meprey loCollecyp>aglarg/().
id="id1methodxref" hmlist#EX811">E="prog-5-21<-split
en()hunks tacks" data-
type="progype="iEX811">id105>ttrainmeth05"label">E="prog
5-21. f id="ChunkcodaDB::tablede>de> sta class=or
paginationcode>--&)hunk class=or pagination-->>/code>f(<>
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de>'i">unction(table<
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Conditional methods

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DB::table('users' sta class=or paginationcode>--&code
class="p">)-->orWhere(function'ide> true(
Confusioodef3" id="id83"> def3" id="id83">
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Conditional ods

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valide> wit,/ing">in id="id1methodxref"
hmlist#EX812">E="prog 5-22ttion> tacks" data-
type="progype="iEX812">id105>ttrainmeth05"label">E="prog 5-
22. f id="Iatertitiu1"mElorimntt:dcord bbu
itio< new ide> wity="5e> DB::table<"p">({$/code> sta
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23. f id="Iatertitiu1"mElorimntt:dcord bbu>: to returny="5e>
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e>, take;orWhere,e allowssass="na">tblt$ucod wa (sseyp>
reu>: to return aid="EX806Mulel::updass, you cou),gl ona-
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firo> dties "m:E="prog 5-22, ode>seyp> Exsa> example: ype=
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to returnn> ef3" id="id83">

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Conditional methods

Conditional methods

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type="progtype="iEX816">id105>ttrainmeth05"label">E="prog 5-
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26lic)

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27. f id="Usto eFlorimnt>direfi/ct; DB::tablekndtable
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()); d_aa">->where->where
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on methd:

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Conditional methods

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paginationcode>--&"e thOrC/p> code>or[>skip()}e
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o:in/co, Edose valu/=em>) d=clae> (if nl>dire
d)/mesttid="id1028"/> artype="programlistEQiatert0er methods
(quames-def3" id="id83">

```

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Chapter 6. Frontend Components

Laravel is mainly known as a PHP framework, but it's also *full stack*, meaning it has a series of components and conventions focused on generating frontend code. Some of these, like pagination and message bags, are PHP helpers that target the frontend, but Laravel also provides a Vite-based frontend build system, some conventions around non-PHP assets, and several starter kits.

Laravel Starter Kits

Out of the box, Laravel provides an entire build system, which we'll cover shortly, but it also includes easy-to-install starter kits that contain templates, auth, styles, JavaScript, and user registration and management workflows.

Laravel's two starter kits are called Breeze and Jetstream.

Breeze is the simpler option; it provides all the needed routes, views, and styles for Laravel's auth system, including registration, login, password reset, password confirmation, email confirmation, and an "edit profile" page. Breeze includes Tailwind styles, and you can choose either Blade templates or Inertia with React or Vue.

Jetstream is more complex and robust; it provides everything Breeze does, but it also adds two-factor authentication, session management, API token management, and team management features. Jetstream includes Tailwind styles, and you can choose either Livewire or Inertia with Vue.

NOTE

Inertia is a frontend tool that allows you to build single-page apps in JavaScript, while using Laravel routes and controllers to provide the routing and data to each view, as if it were a traditional server-rendered app. Learn more at inertiajs.com.

If you're just getting started with Laravel, Breeze is easier to understand and can

be used with just Blade. Most Laravel apps work just fine with only Breeze.

Jetstream doesn't have a Blade-only option, nor does it have a React option; you'll need to work with some sort of frontend framework. Your choice is either Vue/Inertia or Livewire, which is a project that allows you to write mainly backend code but get frontend interactivity in your Laravel apps. However, Jetstream is more robust, so if you're comfortable with Laravel and either Livewire or Inertia, and your project needs those extra features, Jetstream may be your best choice.

Laravel Breeze

Laravel Breeze is a simple starter kit that provides everything you need for an average Laravel application to allow your users to sign up, log in, and manage their profiles.

Installing Breeze

Breeze is intended to be installed on new apps, so it's usually the first thing you'll install when you spin up a new app:

```
laravel new myProject
cd myProject
composer require laravel/breeze --dev
```

Once Breeze is added to your project, you'll run its installer:

```
php artisan breeze:install
```

Once you run the installer, you'll be prompted to choose a stack: Blade, Inertia with React, Inertia with Vue, or API, which is meant to power a non-Inertia frontend like Next.js. These stacks are explained in the following section.

After Breeze is installed, make sure to run your migrations and build your frontend:

```
php artisan migrate
npm install
npm run dev
```

What comes with Breeze

Breeze automatically registers routes for registration, login, logout, password reset, email verification, and password confirmation pages. These routes live in a new *routes/auth.php* file.

The non-API form of Breeze also registers routes for a dashboard and an “edit profile” page for users, and adds these routes directly to the *routes/web.php* file.

The non-API form of Breeze also publishes controllers for the “edit profile” page, email verification, password resets, and several other authentication-related features. In addition, it adds Tailwind, Alpine.js, and PostCSS (for Tailwind). Beyond these shared files and dependencies, each stack adds its own, unique to its needs:

Breeze Blade

Breeze Blade comes with a series of Blade templates for all the features mentioned above, which you can find in *resources/views/auth*, *resources/view/components*, *resources/views/profile*, and a few others sprinkled around.

Breeze Inertia

Both Inertia stacks bring in Inertia, Ziggy (a tool for generating URLs to Laravel routes in JavaScript), Tailwind’s “forms” component, and the necessary JavaScript packages to make their respective frontend frameworks function. They both also publish a basic Blade template that loads Inertia and a series of React/Vue components for all the published pages in the *resources/js* directory.

Breeze API

The API stack for Breeze installs significantly less code and fewer packages than the other stacks, but it also removes the existing bootstrapped files that come with all new Laravel apps. The API stack is intended to prepare an app to be *only* an API backend for a separate Next.js app, so it removes *package.json*, all the JavaScript and CSS files, and all the frontend templates.

Laravel Jetstream

Jetstream builds on Breeze’s functionality and adds even more tooling for starting a new app; however, it’s a more complicated setup with fewer options for configuration, so you’ll want to know that you need it before you choose Jetstream over Breeze.

Jetstream, like Breeze, publishes routes, controllers, views, and configuration files. Like Breeze, Jetstream uses Tailwind, and comes in different tech “stacks.”

Unlike Breeze, however, Jetstream requires interactivity, so there’s no Blade-only stack. Instead, you have two choices: Livewire (which is Blade with some PHP-powered JavaScript interactivity) or Inertia/Vue (there’s no React form for Jetstream).

Jetstream also expands Breeze’s offering by bringing in team management features, two-factor authentication, session management, and personal API token management.

Installing Jetstream

Jetstream is meant to be installed in a new Laravel app, and you can install it with Composer:

```
laravel new myProject
cd myProject
composer require laravel/jetstream
```

Once Jetstream is added to your project, you’ll run its installer. Unlike Breeze, you won’t be prompted to pick your stack; instead, you need to pass in the stack (`livewire` or `inertia`) as the first parameter.

```
php artisan jetstream:install livewire
```

If you want to add team management to your Jetstream install, pass the `--teams` flag to the installation step:

```
php artisan jetstream:install livewire --teams
```

Once you’ve installed Jetstream, make sure to run your migrations and build your frontend:

```
php artisan migrate
npm install
npm run dev
```

What comes with Jetstream

Jetstream publishes an incredible amount of code; here's a quick summary:

- Two-factor authentication and profile photo functionality added to the User model (and adding/modifying the required migrations)
- A dashboard for logged-in users
- Tailwind, Tailwind forms, Tailwind typography
- Laravel Fortify, the backend auth component Jetstream builds on
- “Actions” for Fortify and Jetstream in *app/Actions*
- Markdown text for terms and policies pages in *resources/markdown*
- A huge suite of tests

FORTIFY

Fortify is a headless authentication system. It provides the routes and controllers for all the authentication features Laravel requires, from login and registration to password reset and more, to be consumed by whichever frontend you choose.

Jetstream builds on Fortify, so you can actually think of Jetstream as one of many possible frontends in front of Fortify. Jetstream does also add backend functionality, so it shows just how robust a Fortify-backed auth system can be.

Jetstream's Livewire and Inertia configurations each come with slightly different dependencies and locations for the templates:

Jetstream Livewire

Jetstream's Livewire template sets your app up for working with Livewire and Alpine, and it publishes Livewire components for the frontend. It provides:

- Livewire

- Alpine.js
- Livewire components in *app/View/Components*
- Frontend templates in *resources/views*

Jetstream Inertia

Jetstream's Inertia template sets your app up for working with Inertia and Vue, and it publishes Vue components for the frontend. It provides:

- Inertia
- Vue
- Vue templates in *resources/js*

Customizing your jetstream install

Jetstream builds on Fortify, so customizing Jetstream sometimes will mean customizing Fortify. You can update any configuration settings in *config/fortify.php*, *config/jetstream.php*, the *FortifyServiceProvider*, and the *JetstreamServiceProvider*.

Whereas Breeze publishes controllers for you to modify its behaviors, Jetstream publishes actions, each a one-off chunk of behavior with names like *ResetUserPassword.php* and *DeleteUser.php*.

Further Jetstream features

Jetstream enables your application to manage teams, personal API tokens, two-factor authentication, and tracking and disconnecting all active sessions. You can also hook into some of Jetstream's UI niceties in your own code, like custom flash banners.

To learn more about how this all works, take a look at the [Laravel docs for Jetstream](#), which are exhaustive.

Laravel's Vite Configuration

Vite is a local frontend development environment that combines a dev server and

Rollup-based build tooling chain. That might sound like a lot, but in Laravel, it's primarily used to bundle CSS and JavaScript assets together.

Laravel offers an NPM plug-in and a Blade directive to make it easy to work with Vite. Both are included in Laravel apps out of the box, along with a configuration file: *vite.config.js*.

Take a look at [Example 6-1](#) to see what the contents of the default *vite.config.js* file look like.

Example 6-1. The default vite.config.js

```
import { defineConfig } from 'vite';
import laravel from 'laravel-vite-plugin';

export default defineConfig({
  plugins: [
    laravel({
      input: ['resources/css/app.css', 'resources/js/app.js'],
      refresh: true,
    }),
  ],
});
```

We're defining the files our plug-in should build from (*input*) and saying we do want the “refresh my page every time I save a view file” feature enabled (*refresh*).

By default, Vite pulls from the two files listed in [Example 6-1](#), and will automatically refresh any time any file changes in these folders:

- *app/View/Components/*
- *lang/*
- *resources/lang/*
- *resources/views/*
- *routes/*

Now that we've got our Vite config pointed to our CSS and JavaScript entry files, we'll want to reference those files using the `@vite` Blade directive, as you can see in [Example 6-2](#).

Example 6-2. Using the @vite Blade directive

```
<html>
<head>
    @vite(['resources/css/app.css', 'resources/js/app.js'])
```

That's it! Next, let's take a look at how to bundle files with Vite.

NOTE

If your local development domain is secured (HTTPS), you will need to modify your *vite.config.js* file to point to your credentials. If you're using Valet, there's a special configuration option for that:

```
// ...
export default defineConfig({
  plugins: [
    laravel({
      // ...
      valetTls: 'name-of_my-app-here.test',
    }),
  ],
});
```

Bundling Files with Vite

Finally, it's time to bundle our assets. There are two ways to bundle assets with Vite: “build” and “dev.”

If you want to build your files once, either for delivery to production or for local testing, run `npm run build` and Vite will bundle your assets. If you are developing locally, however, you may prefer to have Vite spin up a process that will watch your view files for changes, retrigger the build any time it detects changes to your view files, and push a refresh to the page in your browser. That's what `npm run dev` does for you.

Your built files will end up in the *public/build/assets* folder of your app, with a file living at *public/build/manifest.json* that tells Laravel and Vite how to get to each built file from its nonbuilt path reference.

NOTE

The `public/build` folder is ignored by default in Laravel's `.gitignore`, so make sure to run `npm run build` as a part of your deployment process.

The Vite Dev Server

When you run `npm run dev`, you're spinning up an actual HTTP server, powered by Vite. The Vite Blade helper rewrites your asset URLs to point to the same locations on the dev server instead of on your local domain, which allows Vite to more quickly update and refresh your dependencies.

This means that if you write the following Blade call:

```
@vite(['resources/css/app.css', 'resources/js/app.js'])
```

It'll look like this on your production application:

```
<link rel="preload" as="style"
      href="http://my-app.test/build/assets/app-1c09da7e.css" />
<link rel="modulepreload"
      href="http://my-app.test/build/assets/app-ea0e9592.js" />
<link rel="stylesheet"
      href="http://my-app.test/build/assets/app-1c09da7e.css" />
<script type="module"
        src="http://my-app.test/build/assets/app-ea0e9592.js"></script>
```

But it'll look something like this locally if your Vite server is running:

```
<script type="module" src="http://127.0.0.1:5173/@vite/client"></script>
<link rel="stylesheet" href="http://127.0.0.1:5173/resources/css/app.css" />
<script type="module" src="http://127.0.0.1:5173/resources/js/app.js"></script>
```

Working with Static Assets and Vite

So far we've only covered loading JavaScript and CSS with Vite. But Laravel's Vite configuration can process and version your static assets (like images), as well.

If you're working in JavaScript templates, Vite will grab the links to any *relative* static assets and process and version them. Any *absolute* static assets Vite will

ignore.

This means the following images will receive different treatment, if they're in JavaScript templates.

```
<!-- Ignored by Vite -->

<!-- Processed by Vite -->

```

If you're working in Blade templates, you'll need to take two steps to have Vite handle your static assets. First, you'll want to use the `Vite::asset` facade call to link your asset:

```

```

And second, you'll need to add a configuration step to your `resources/js/app.js` file that shows Vite which files or folders to import:

```
import.meta.glob([
  // Imports all the files in /resources/images/
  '../images/**',
]);
```

WARNING

If you're running the Vite server with `npm run dev`, the server can load your static assets *without* you adding the `import.meta.glob` config. That means you may think it's going to show up, but it will fail on your production build.

Working with JavaScript Frameworks and Vite

If you want to work with Vue, React, Inertia, and/or a single-page application (SPA), you may need to bring in some specific plug-ins or set some specific configuration items. Here are the basics of what you need for the most common scenarios.

Vite and Vue

To work with Vite and Vue, first install Vite's Vue plug-in:

```
npm install --save-dev @vitejs/plugin-vue
```

Then you need to modify your *vite.config.js* file to call the Vue plug-in, passing two configuration settings to it. The first, `template.transformAssetUrls.base=null`, allows the Laravel plug-in instead of the Vue plug-in to handle rewriting URLs. The second, `template.transformAssetUrls.includeAbsolute=false`, allows URLs within Vue templates to reference files in the public directory:

```
import { defineConfig } from 'vite';
import laravel from 'laravel-vite-plugin';
import vue from '@vitejs/plugin-vue';

export default defineConfig({
  plugins: [
    laravel(['resources/js/app.js']),
    vue({
      template: {
        transformAssetUrls: {
          base: null,
          includeAbsolute: false,
        },
      },
    }),
  ],
});
```

Vite and React

To work with Vite and React, first install Vite's React plug-in:

```
npm install --save-dev @vitejs/plugin-react
```

Then you need to modify your *vite.config.js* file to call the React plug-in:

```
import { defineConfig } from 'vite';
import laravel from 'laravel-vite-plugin';
import react from '@vitejs/plugin-react';

export default defineConfig({
  plugins: [
```

```

    laravel(['resources/js/app.js']),
    react(),
  ],
});

```

Finally, add the `@viteReactRefresh` Blade directive in your template before you import your JavaScript files with `@vite`:

```

@viteReactRefresh
@vite('resources/js/app.jsx')

```

Vite and Inertia

If you're setting up Inertia yourself, you'll need Inertia to be able to resolve your page components.

Here's a look at the code you'll likely write in your *resources/js/app.js* file, but your best option is to install Inertia using Breeze, Jetstream, or the Inertia docs.

```

import { createApp, h } from 'vue'
import { createInertiaApp } from '@inertiajs/vue3'

createInertiaApp({
  resolve: name => {
    const pages = import.meta.glob('./Pages/**/*.vue', { eager: true })
    return pages[`${name}.vue`]
  },
  setup({ el, App, props, plugin }) {
    createApp({ render: () => h(App, props) })
      .use(plugin)
      .mount(el)
  },
})

```

Vite and SPAs

If you're building an SPA, remove *resources/css/app.css* from your *vite.config.js* file, which removes it as an entry point.

Instead, import your CSS into your JavaScript by adding this line to your *resources/js/app.js* file, directly below importing the bootstrap:

```

import './bootstrap';
import './css/app.css';

```

Using Environment Variables in Vite

If you want to use environment variables in your JavaScript files, prefix the variable name with `VITE_`, as you can see in [Example 6-3](#).

Example 6-3. Referencing environment variables in vite.config.js

```
// .env
VITE_BASE_URL=http://local-development-url.test
// resources/js/app.js
const baseUrl = import.meta.env.VITE_BASE_URL;
```

Every time you run `npm run dev` or `npm run build`, it'll load that environment variable from `.env` and inject it into your script.

Pagination

For something that is so common across web applications, pagination still can be wildly complicated to implement. Thankfully, Laravel has a built-in concept of pagination, which is also hooked into Eloquent results *and* the router by default.

Paginating Database Results

The most common place you'll see pagination is when you are displaying the results of a database query and there are too many results for a single page. Eloquent and the query builder both read the `page` query parameter from the current page request and use it to provide a `paginate()` method on any result sets; the single parameter you should pass `paginate()` indicates how many results you want per page. Take a look at [Example 6-4](#) to see how this works.

Example 6-4. Paginating a query builder response

```
// PostController
public function index()
{
    return view('posts.index', ['posts' => DB::table('posts')->paginate(20)]);
}
```

[Example 6-4](#) specifies that this route should return 20 posts per page and will define which “page” of results the current user is on based on the URL's `page` query parameter, if it has one. Eloquent models all have the same `paginate()` method.

When you display the results in your view, your collection will now have a `links()` method on it that will output the pagination controls. (See [Example 6-5](#), which I've simplified for inclusion in this book.)

Example 6-5. Rendering pagination links in a template

```
// posts/index.blade.php
<table>
@foreach ($posts as $post)
    <tr><td>{{ $post->title }}</td></tr>
@endforeach
</table>

{{ $posts->links() }}
```

// By default, `$posts->links()` will output something like this:

```
<div class="...">
    <div>
        <p class="...">
            Showing
            <span class="...">1</span>
            to
            <span class="...">2</span>
            of
            <span class="...">5</span>
            results
        </p>
    </div>
    <div>
        <span class="...">
            <span aria-disabled="true" aria-label="&laquo; Previous">
                <!-- SVG here for the ... ellipsis -->
            </span>
            <span class="...">1</span>
            <a href="http://myapp.com/posts?page=2" class="..." aria-label="...">
                2
            </a>
            <a href="http://myapp.com/posts?page=3" class="..." aria-label="...">
                3
            </a>
            <a href="http://myapp.com/posts?page=2" class="..."
                rel="next" aria-label="Next &raquo;">
                <!-- SVG here for the ... ellipsis -->
            </a>
        </span>
    </div>
</div>
```

The paginator uses TailwindCSS for its default styling. If you want to use

Bootstrap styles, call `Paginator::useBootstrap()` in your `AppServiceProvider`:

```
use Illuminate\Pagination\Paginator;

public function boot(): void
{
    Paginator::useBootstrap();
}
```

CUSTOMIZING THE NUMBER OF PAGINATION LINKS

If you'd like to control how many links display on either side of the current page, you can customize this number easily with the `onEachSide()` method:

```
DB::table('posts')->paginate(10)->onEachSide(3);

// Outputs:
// 5 6 7 [8] 9 10 11
```

Manually Creating Paginators

If you're not working with Eloquent or the query builder, or if you're working with a complex query (e.g., one using `groupBy`), you might find yourself needing to create a paginator manually. Thankfully, you can do that with the `Illuminate\Pagination\Paginator` or `Illuminate\Pagination\LengthAwarePaginator` classes.

The difference between the two classes is that `Paginator` provides only previous and next buttons, but no links to each page; `LengthAwarePaginator` needs to know the length of the full result so that it can generate links for each individual page. You may find yourself wanting to use `Paginator` on large result sets so your paginator doesn't have to be aware of a massive count of results that might be costly to run.

Both `Paginator` and `LengthAwarePaginator` require you to manually extract the subset of content that you want to pass to the view. Take a look at [Example 6-6](#) for an example.

Example 6-6. Manually creating a paginator

```
use Illuminate\Http\Request;
use Illuminate\Pagination\Paginator;

Route::get('people', function (Request $request) {
    $people = [...]; // huge list of people

    $perPage = 15;
    $offsetPages = $request->input('page', 1) - 1;

    // The Paginator will not slice your array for you
    $people = array_slice(
        $people,
        $offsetPages * $perPage,
        $perPage
    );

    return new Paginator(
        $people,
        $perPage
    );
});
```

Message Bags

Another common-but-painful feature in web applications is passing messages between various components of the app, when the end goal is to share them with the user. Your controller, for example, might want to send a validation message: “The email field must be a valid email address.” However, that particular message doesn’t just need to make it to the view layer; it actually needs to survive a redirect and then end up in the view layer of a different page. How do you structure this messaging logic?

The `Illuminate\Support\MessageBag` class is tasked with storing, categorizing, and returning messages that are intended for the end user. It groups all messages by key, where the keys are likely to be something like `errors` or `messages`, and it provides convenience methods for getting either all its stored messages or only those for a particular key, and outputting these messages in various formats.

You can spin up a new instance of `MessageBag` manually like in [Example 6-7](#). To be honest, though, you likely won’t ever do this manually—this is just a

thought exercise to show how it works.

Example 6-7. Manually creating and using a message bag

```
$messages = [
    'errors' => [
        'Something went wrong with edit 1!',
    ],
    'messages' => [
        'Edit 2 was successful.',
    ],
];
$messagebag = new \Illuminate\Support\MessageBag($messages);

// Check for errors; if there are any, decorate and echo
if ($messagebag->has('errors')) {
    echo '<ul id="errors">';
    foreach ($messagebag->get('errors', '<li><b>message</b></li>') as $error) {
        echo $error;
    }
    echo '</ul>';
}
```

Message bags are also closely connected to Laravel’s validators (you’ll learn more about these in “**Validation**”): when validators return errors, they actually return an instance of `MessageBag`, which you can then pass to your view or attach to a redirect using `redirect('route')->withErrors($messagebag)`.

Laravel passes an empty instance of `MessageBag` to every view, assigned to the variable `$errors`; if you’ve flashed a message bag using `withErrors()` on a redirect, it will get assigned to that `$errors` variable instead. That means every view can always assume it has an `$errors` `MessageBag` that it can check wherever it handles validation, which leads to **Example 6-8** as a common snippet developers place on every page.

Example 6-8. Error bag snippet

```
// partials/errors.blade.php
@if ($errors->any())
    <div class="alert alert-danger">
        <ul>
            @foreach ($errors as $error)
                <li>{{ $error }}</li>
            @endforeach
        </ul>
    </div>
@endif
```

MISSING \$errors VARIABLE

If you have any routes that aren't under the web middleware group, they won't have the session middleware, which means they won't have this `$errors` variable available.

Sometimes you need to differentiate message bags not just by key (notices versus errors) but also by component. Maybe you have a login form and a signup form on the same page; how do you differentiate them?

When you send errors along with a redirect using `withErrors()`, the second parameter is the name of the bag: `redirect('dashboard')->withErrors($validator, 'login')`. Then, on the dashboard, you can use `$errors->login` to call all of the methods you saw before: `any()`, `count()`, and more.

String Helpers, Pluralization, and Localization

As developers, we tend to look at blocks of text as big placeholder divs, waiting for the client to put real content into them. Seldom are we involved in any logic inside these blocks.

But there are a few circumstances where you'll be grateful for the tools Laravel provides for string manipulation.

The String Helpers and Pluralization

Laravel has a series of helpers for manipulating strings. They're available as methods on the `Str` class (e.g., `Str::plural()`).

LARAVEL STRING AND ARRAY GLOBAL HELPERS

Older versions of Laravel included global helpers that were aliases for the `Str` and `Arr` methods. These global `str_` and `array_` helpers were removed from Laravel in version 6 and exported to a separate package. If you'd like, you can install the `laravel/helpers` package via Composer: `composer require laravel/helpers`.

The Laravel [documentation](#) covers all of the string helpers in detail, but here are a few of the most commonly used ones:

`e()`

A shortcut for `html_entities()`; encodes all HTML entities for safety.

`Str::startsWith()`, `Str::endsWith()`, `Str::contains()`

Checks a string (first parameter) to see if it starts with, ends with, or contains another string (second parameter).

`Str::is()`

Checks whether a string (second parameter) matches a particular pattern (first parameter)—for example, `foo*` will match `foobar` and `foobaz`.

`Str::slug()`

Converts a string to a URL-type slug with hyphens.

`Str::plural(word, count)`, `Str::singular()`

Pluralizes or singularizes a word; English only (e.g., `Str::plural('dog')` returns `dogs`; `Str::plural('dog', ' 1')` returns `dog`).

`Str::camel()`, `Str::kebab()`, `Str::snake()`, `Str::studly()`, `Str::title()`

Converts a provided string to a different capitalization “case.”

`Str::after()`, `Str::before()`, `Str::limit()`

Trims a string and provides a substring. `Str::after()` returns everything after a given string, and `Str::before()` returns everything before a given string (both accept the full string as the first parameter and the string you’re using to cut as the second). `Str::limit()` truncates a string (first parameter) to a given number of characters (second parameter).

`Str::markdown(string, options)`

Converts Markdown into HTML. You can read more about the options you

can pass on [the PHP league website](#).

`Str::replace(search, replace, subject, caseSensitive)`

Looks in the subject string for the search string occurrence and replaces it with the replace string. If the case sensitivity parameter is true, only replace if the occurrence matches the search case (e.g., `Str::replace('Running', 'Going', 'Laravel Up and Running', true)` returns 'Laravel Up and Going').

Localization

Localization enables you to define multiple languages and mark any strings as targets for translation. You can set a fallback language and even handle pluralization variations.

In Laravel, you'll need to set an “application locale” at some point during the page load so the localization helpers know which bucket of translations to pull from. Each “locale” is usually connected to a translation and will often look like “en” (for English). You'll do this with `App::setLocale($localeName)`, and you'll likely put it in a service provider. For now, you can just put it in the `boot()` method of `AppServiceProvider`, but you may want to create a `LocaleServiceProvider` if you end up with more than just this one locale-related binding.

SETTING THE LOCALE FOR EACH REQUEST

It can be confusing at first to work out how Laravel “knows” the user's locale or provides translations. Most of that work is down to you as the developer. Let's look at a likely scenario.

You'll probably have some functionality allowing the user to choose a locale, or possibly attempting to automatically detect it. Either way, your application will determine the locale, and then you'll store that in a URL parameter or a session cookie. Then your service provider—something like a `LocaleServiceProvider`, maybe—will grab that key and set it as a part of Laravel's bootstrap.

So maybe your user is at `http://myapp.com/es/contacts`. Your `LocaleServiceProvider` will grab that `es` string and then run `App::setLocale('es')`. Going forward, every time you ask for a translation of a string, Laravel will look for the Spanish (`es` means Español) version of that string, which you will need to have defined somewhere.

You can define your fallback locale in `config/app.php`, where you should find a `fallback_locale` key. This allows you to define a default language for your application, which Laravel will use if it can't find a translation for the requested locale.

Basic localization

So, how do we call for a translated string? There's a helper function, `__($key)`, that will pull the string for the current locale for the passed key or, if it doesn't exist, grab it from the default locale. In Blade you can also use the `@lang()` directive. **Example 6-9** demonstrates how a basic translation works. We'll use the example of a "back to the dashboard" link at the top of a detail page.

Example 6-9. Basic use of `__()`

```
// Normal PHP
<?php echo __('navigation.back'); ?>
// Blade
{{ __('navigation.back') }}

// Blade directive
@lang('navigation.back')
```

Let's assume we are using the `es` locale right now. First, we'll need to publish the `lang` files for modification:

```
php artisan lang:publish
```

This command publishes the default Laravel `lang` files to the root of your application. You'll want to create a file to define your navigation-related translations, `lang/en/navigation.php`, and have it return a PHP array with a key named `back` in it, as in **Example 6-10**.

Example 6-10. Example `lang/en/navigation.php` file

```
<?php

return [
    'back' => 'Return to dashboard',
];
```

Now, to make this translatable, let's also create an *es* directory under *lang*, with its own *navigation.php* file, as you can see in [Example 6-11](#).

Example 6-11. Example lang/es/navigation.php file

```
<?php

return [
    'back' => 'Volver al panel',
];
```

Now let's try using that translation key in our app, in [Example 6-12](#).

Example 6-12. Using a translation

```
// routes/web.php
Route::get('/es/contacts/show/{id}', function () {
    // Set the locale manually, for this example, instead of in a service provider
    App::setLocale('es');
    return view('contacts.show');
});

// resources/views/contacts/show.blade.php
<a href="/contacts">{{ __('navigation.back') }}</a>
```

Parameters in localization

The preceding example was relatively simple. Let's dig into some that are more complex. What if we want to define *which* dashboard we're returning to? Take a look at [Example 6-13](#).

Example 6-13. Parameters in translations

```
// lang/en/navigation.php
return [
    'back' => 'Back to :section dashboard',
];

// resources/views/contacts/show.blade.php
{{ __('navigation.back', ['section' => 'contacts']) }}
```

As you can see, prepending a word with a colon (:section) marks it as a placeholder that can be replaced. The second, optional, parameter of `__()` is an

array of values to replace the placeholders.

Pluralization in localization

We already covered pluralization, so now just imagine you're defining your own pluralization rules. There are two ways to do it; we'll start with the simplest, as shown in [Example 6-14](#).

Example 6-14. Defining a simple translation with an option for pluralization

```
// lang/en/messages.php
return [
    'task-deletion' => 'You have deleted a task|You have successfully deleted tasks',
];

// resources/views/dashboard.blade.php
@if ($numTasksDeleted > 0)
    {{ trans_choice('messages.task-deletion', $numTasksDeleted) }}
@endif
```

As you can see, we have a `trans_choice()` method, which takes the count of items affected as its second parameter; from this, it will determine which string to use.

You can also use any translation definitions that are compatible with Symfony's much more complex Translation component; see [Example 6-15](#) for an example.

Example 6-15. Using the Symfony Translation component

```
// lang/es/messages.php
return [
    'task-deletion' => "{0} You didn't manage to delete any tasks." .
        "[1,4] You deleted a few tasks." .
        "[5,Inf] You deleted a whole ton of tasks.",
];
```

Storing the default string as the key with JSON

One common difficulty with localization is that it's hard to ensure there's a good system for defining key namespacing—for example, remembering a key nested three or four levels deep or being unsure which key a phrase used twice in the site should use.

An alternative to the slug key/string value pair system is to store your translations using your primary language string as the key, instead of a made-up

slug. You can indicate to Laravel that you're working this way by storing your translation files as JSON in the *lang* directory, with the filename reflecting the locale ([Example 6-16](#)).

Example 6-16. Using JSON translations and the `__()` helper

```
// In Blade
{{ __('View friends list') }}
// lang/es.json
{
  'View friends list': 'Ver lista de amigos'
}
```

This is taking advantage of the fact that the `__()` translation helper, if it can't find a matching key for the current language, will just display the key. If your key is the string in your app's default language, that's a much more reasonable fallback than, for example, `widgets.friends.title`.

Testing

In this chapter we focused primarily on Laravel's frontend components. These are less likely the objects of unit tests, but they may at times be used in your integration tests.

Testing Message and Error Bags

There are two primary ways of testing messages passed along with message and error bags. First, you can perform a behavior in your application tests that sets a message that will eventually be displayed somewhere, then redirect to that page and assert that the appropriate message is shown.

Second, for errors (which is the most common use case), you can assert the session has errors with `$this->assertSessionHasErrors($bindings = [])`. Take a look at [Example 6-17](#) to see what this might look like.

Example 6-17. Asserting the session has errors

```
public function test_missing_email_field_errors()
{
    $this->post('person/create', ['name' => 'Japheth']);
    $this->assertSessionHasErrors(['email']);
}
```

In order for [Example 6-17](#) to pass, you'll need to add input validation to that route. We'll cover this in [Chapter 7](#).

Translation and Localization

The simplest way to test localization is with application tests. Set the appropriate context (whether by URL or session), “visit” the page with `get()`, and assert that you see the appropriate content.

Disabling Vite in Testing

If you'd like to disable Vite's asset resolution during your tests, you can disable Vite entirely by calling the `withoutVite()` method at the top of a test:

```
public function test_it_runs_without_vite()
{
    $this->withoutVite();

    // Test stuff
}
```

TL;DR

As a full stack framework, Laravel provides tools and components for the frontend as well as the backend.

Vite is a build tool and dev server that Laravel builds on top of to help process, compress, and version your JavaScript, CSS, and static assets like images.

Laravel also offers other internal tools that target the frontend, including tools for implementing pagination, message and error bags, and localization.

Chapter 7. Collecting and Handling User Data

Websites that benefit from a framework like Laravel often don't just serve static content. Many deal with complex and mixed data sources, and one of the most common (and most complex) of these sources is user input in its myriad forms: URL paths, query parameters, POST data, and file uploads.

Laravel provides a collection of tools for gathering, validating, normalizing, and filtering user-provided data. We'll look at those here.

Injecting a Request Object

The most common tool for accessing user data in Laravel is injecting an instance of the `Illuminate\Http\Request` object. It offers easy access to all of the ways users can provide input to your site: POSTed form data or JSON, GET requests (query parameters), and URL segments.

OTHER OPTIONS FOR ACCESSING REQUEST DATA

There's also a `request()` global helper and a `Request` facade, both of which expose the same methods. Each of these options exposes the entire `Illuminate\Request` object, but for now we're only going to cover the methods that specifically relate to user data.

Since we're planning on injecting a `Request` object, let's take a quick look at how to get the `$request` object we'll be calling all these methods on:

```
Route::post('form', function (Illuminate\Http\Request $request) {  
    // $request->etc()  
});
```

`$request->all()`

Just like the name suggests, `$request->all()` gives you an array containing all of the input the user has provided, from every source. Let's say, for some reason, you decided to have a form POST to a URL with a query parameter—for example, sending a POST to `http://myapp.com/signup?utm=12345`. Take a look at [Example 7-1](#) to see what you'd get from `$request->all()`. (Note that `$request->all()` also contains information about any files that were uploaded, but we'll cover that later in the chapter.)

Example 7-1. `$request->all()`

```
<!-- GET route form view at /get-route -->
<form method="post" action="/signup?utm=12345">
    @csrf
    <input type="text" name="first_name">
    <input type="submit">
</form>
// routes/web.php
Route::post('signup', function (Request $request) {
    var_dump($request->all());
});

// Outputs:
/**
 * [
 *     '_token' => 'CSRF token here',
 *     'first_name' => 'value',
 *     'utm' => 12345,
 * ]
 */
```

`$request->except()` and `->only()`

`$request->except()` provides the same output as `$request->all()`, but you can choose one or more fields to exclude—for example, `_token`. You can pass it either a string or an array of strings.

[Example 7-2](#) shows what it looks like when we use `$request->except()` on the same form as in [Example 7-1](#).

Example 7-2. `$request->except()`

```
Route::post('post-route', function (Request $request) {
    var_dump($request->except('_token'));
});
```

```
// Outputs:
/**
 * [
 *   'firstName' => 'value',
 *   'utm' => 12345
 * ]
 */
```

`$request->only()` is the inverse of `$request->except()`, as you can see in [Example 7-3](#).

Example 7-3. `$request->only()`

```
Route::post('post-route', function (Request $request) {
    var_dump($request->only(['firstName', 'utm']));
});
```

```
// Outputs:
/**
 * [
 *   'firstName' => 'value',
 *   'utm' => 12345
 * ]
 */
```

`$request->has()` and `->missing()`

With `$request->has()` you can detect whether a particular piece of user input is available to you, regardless of whether the input actually has a value in it. Check out [Example 7-4](#) for an analytics example with our `utm` query string parameter from the previous examples.

Example 7-4. `$request->has()`

```
// POST route at /post-route
if ($request->has('utm')) {
    // Do some analytics work
}
```

`$request->missing()` is its inverse.

`$request->whenHas()`

With `$request->whenHas()`, you can define the behavior when the request either does or doesn't have a field provided. The first closure parameter is

returned when the field exists, and the second is returned when it doesn't.

See [Example 7-5](#) for an example with our `utm` query string parameter.

Example 7-5. `$request->whenHas()`

```
// POST route at /post-route
$utm = $request->whenHas('utm', function($utm) {
    return $utm;
}, function() {
    return 'default';
});
```

`$request->filled()`

Using the `$request->filled()` method, it is possible to check if a particular field is present and filled in the request. `filled()` is the same as `has()`, except it also requires there to be an actual value present in the field. In [Example 7-6](#) you can see an example of how to use this method.

Example 7-6. `$request->filled()`

```
// POST route at /post-route
if ($request->filled('utm')) {
    // Do some analytics work
}
```

`$request->whenFilled()`

Much like with the `whenHas()` method, the `$request->whenFilled()` method allows you to define the values either when the field is filled or when it isn't. The first closure parameter runs when the field is filled, the second when it is not. See [Example 7-7](#) for an example of how to use this method.

Example 7-7. `$request->whenFilled()`

```
// POST route at /post-route
$utm = $request->whenFilled('utm', function ($utm) {
    return $utm;
}, function() {
    return 'default';
});
```

`$request->mergeIfMissing()`

With the `mergeIfMissing()` method you can add a field to the request when it is not present and while defining its value. This can be useful, for example, when a field comes from a checkbox, as it is only present when checked. You can see [Example 7-8](#) for an implementation.

Example 7-8. `$request->mergeIfMissing()`

```
// POST route at /post-route
$shouldSend = $request->mergeIfMissing('send_newsletter', 0);
```

`$request->input()`

Whereas `$request->all()`, `$request->except()`, and `$request->only()` operate on the full array of input provided by the user, `$request->input()` allows you to get the value of just a single field. [Example 7-9](#) provides an example. Note that the second parameter is the default value, so if the user hasn't passed in a value, you can have a sensible (and nonbreaking) fallback.

Example 7-9. `$request->input()`

```
Route::post('post-route', function (Request $request) {
    $userName = $request->input('name', 'Matt');
});
```

`$request->method()` and `->isMethod()`

`$request->method()` returns the HTTP verb for the request, and `$request->isMethod()` checks whether it matches the specified verb. [Example 7-10](#) illustrates their use.

Example 7-10. `$request->method()` and `$request->isMethod()`

```
$method = $request->method();

if ($request->isMethod('patch')) {
    // Do something if request method is PATCH
}
```

`$request->integer()`, `->float()`, `->string()`, and `->enum()`

These methods cast the inputs directly into integers, floats, strings, or enums when you use each method respectively. See [Example 7-11](#) for usage examples.

Example 7-11. `$request->integer()`, `$request->float()`, `$request->string()`, and `$request->enum()`

```
dump(is_int($request->integer('some_integer')));
// true

dump(is_float($request->float('some_float')));
// true

dump(is_string($request->string('some_string')));
// true

dump($request->enum('subscription', SubscriptionStatusEnum::class));
// 'active', assuming that's a valid status for the SubscriptionStatusEnum
```

`$request->dump()` and `->dd()`

`$request->dump()` and `$request->dd()` are helper methods for dumping the request. For both, you can dump the whole request by not passing any parameters or dump only selected fields by passing an array. `$request->dump()` dumps and then continues, while `$request->dd()` dumps and then stops execution of the script. [Example 7-12](#) illustrates their use.

Example 7-12. `$request->dump()` and `$request->dd()`

```
// dumping the whole request
$request->dump()
$request->dd();

// dumping just two fields
$request->dump(['name', 'utm']);
$request->dd(['name', 'utm']);
```

Array Input

Laravel also provides convenience helpers for accessing data from array input. Just use the “dot” notation to indicate the steps of digging into the array structure, like in [Example 7-13](#).

Example 7-13. Dot notation to access array values in user data

```
<!-- GET route form view at /employees/create -->
<form method="post" action="/employees/">
    @csrf
    <input type="text" name="employees[0][firstName]">
    <input type="text" name="employees[0][lastName]">
```

```

        <input type="text" name="employees[1][firstName]">
        <input type="text" name="employees[1][lastName]">
        <input type="submit">
    </form>
    // POST route at /employees
    Route::post('employees', function (Request $request) {
        $employeeZeroFirstName = $request->input('employees.0.firstName');
        $allLastNames = $request->input('employees.*.lastName');
        $employeeOne = $request->input('employees.1');
        var_dump($employeeZeroFirstname, $allLastNames, $employeeOne);
    });

    // If forms filled out as "Jim" "Smith" "Bob" "Jones":
    // $employeeZeroFirstName = 'Jim';
    // $allLastNames = ['Smith', 'Jones'];
    // $employeeOne = ['firstName' => 'Bob', 'lastName' => 'Jones'];

```

JSON Input (and \$request->json())

So far we’ve covered input from query strings (GET) and form submissions (POST). But there’s another form of user input that’s becoming more common with the advent of JavaScript SPAs: the JSON request. It’s essentially just a POST request with the body set to JSON instead of a traditional form POST.

Let’s take a look at what it might look like to submit some JSON to a Laravel route, and how to use `$request->input()` to pull out that data ([Example 7-14](#)).

Example 7-14. Getting data from JSON with `$request->input()`

```

POST /post-route HTTP/1.1
Content-Type: application/json

```

```

{
    "firstName": "Joe",
    "lastName": "Schmoe",
    "spouse": {
        "firstName": "Jill",
        "lastName": "Schmoe"
    }
}
// Post-route
Route::post('post-route', function (Request $request) {
    $firstName = $request->input('firstName');
    $spouseFirstname = $request->input('spouse.firstName');
});

```

Since `$request->input()` is smart enough to pull user data from GET, POST, or

JSON, you may wonder why Laravel even offers `$request->json()`. There are two reasons you might prefer `$request->json()`. First, you might want to just be more explicit to other programmers working on your project about where you're expecting the data to come from. And second, if the POST doesn't have the correct `application/json` headers, `$request->input()` won't pick it up as JSON, but `$request->json()` will.

FACADE NAMESPACES, THE REQUEST() GLOBAL HELPER, AND INJECTING \$REQUEST

Any time you're using facades inside of namespaced classes (e.g., controllers), you'll have to add the full facade path to the import block at the top of your file (e.g., use `Illuminate\Support\Facades\Request`).

Because of this, several of the facades also have a companion global helper function. If these helper functions are run with no parameters, they expose the same syntax as the facade (e.g., `request()->has()` is the same as `Request::has()`). They also have a default behavior for when you pass them a parameter (e.g., `request('firstName')` is a shortcut to `request()->input('firstName')`).

With `Request`, we've been covering injecting an instance of the `Request` object, but you could also use the `Request` facade or the `request()` global helper. Take a look at [Chapter 10](#) to learn more.

Route Data

It might not be the first thing you think of when you imagine “user data,” but the URL is just as much user data as anything else in this chapter.

There are two primary ways you'll get data from the URL: via `Request` objects and via route parameters.

From Request

Injected Request objects (and the Request facade and the `request()` helper) have several methods available to represent the state of the current page's URL, but right now, let's focus on getting information about the URL segments.

Each group of characters after the domain in a URL is called a *segment*. So, <http://www.myapp.com/users/15> has two segments: *users* and *15*.

As you can probably guess, we have two methods available to us: `$request->segments()` returns an array of all segments, and `$request->segment($segmentId)` allows us to get the value of a single segment. Note that segments are returned on a 1-based index, so in the preceding example, `$request->segment(1)` would return *users*.

Request objects, the Request facade, and the `request()` global helper provide quite a few more methods to help us get data out of the URL. To learn more, check out [Chapter 10](#).

From Route Parameters

The other primary way we get data about the URL is from route parameters, which are injected into the controller method or closure that is serving a current route, as shown in [Example 7-15](#).

Example 7-15. Getting URL details from route parameters

```
// routes/web.php
Route::get('users/{id}', function ($id) {
    // If the user visits myapp.com/users/15/, $id will equal 15
});
```

To learn more about routes and route binding, check out [Chapter 3](#).

Uploaded Files

We've talked about different ways to interact with users' text input, but there's also the matter of file uploads to consider. Request objects provide access to any uploaded files using the `$request->file()` method, which takes the file's input name as a parameter and returns an instance of

`Symfony\Component\HttpFoundation\File\UploadedFile`. Let's walk through an example. First, our form, in [Example 7-16](#).

Example 7-16. A form to upload files

```
<form method="post" enctype="multipart/form-data">
  @csrf
  <input type="text" name="name">
  <input type="file" name="profile_picture">
  <input type="submit">
</form>
```

Now let's take a look at what we get from running `$request->all()`, as shown in [Example 7-17](#). Note that `$request->input('profile_picture')` will return `null`; we need to use `$request->file('profile_picture')` instead.

Example 7-17. The output from submitting the form in [Example 7-16](#)

```
Route::post('form', function (Request $request) {
    var_dump($request->all());
});

// Output:
// [
//     "_token" => "token here",
//     "name" => "asdf",
//     "profile_picture" => UploadedFile {},
// ]

Route::post('form', function (Request $request) {
    if ($request->hasFile('profile_picture')) {
        var_dump($request->file('profile_picture'));
    }
});

// Output:
// UploadedFile (details)
```

VALIDATING A FILE UPLOAD

As you can see in [Example 7-17](#), we have access to `$request->hasFile()` to see whether the user uploaded a file. We can also check whether the file upload was successful by using `isValid()` on the file itself:

```
if ($request->file('profile_picture')->isValid()) {
    //
}
```

Because `isValid()` is called on the file itself, it will error if the user didn't upload a file. So, to check for both, you'd need to check for the file's existence first:

```
if ($request->hasFile('profile_picture') &&
    $request->file('profile_picture')->isValid()) {
    //
}
```

Laravel also offers file-specific validation rules, allowing you to require file uploads to match certain mime types, file sizes or lengths, and more. Take a look at the [validation docs](#) to learn more.

Symfony's `UploadedFile` class extends PHP's native `SplFileInfo` with methods allowing you to easily inspect and manipulate the file. This list isn't exhaustive, but it gives you a taste of what you can do:

- `guessExtension()`
- `getMimeType()`
- `store($path, $storageDisk = default disk)`
- `storeAs($path, $newName, $storageDisk = default disk)`
- `storePublicly($path, $storageDisk = default disk)`
- `storePubliclyAs($path, $newName, $storageDisk = default disk)`
- `move($directory, $newName = null)`
- `getClientOriginalName()`
- `getClientOriginalExtension()`
- `getClientMimeType()`
- `guessClientExtension()`
- `getClientSize()`

- `getError()`
- `isValid()`

As you can see, most of the methods have to do with getting information about the uploaded file, but there's one that you'll likely use more than all the others: `store()`, which takes the file that was uploaded with the request and stores it in a specified directory on your server. Its first parameter is the destination directory, and the optional second parameter is the storage disk (`s3`, `local`, etc.) to use to store the file. You can see a common workflow in [Example 7-18](#).

Example 7-18. Common file upload workflow

```
if ($request->hasFile('profile_picture')) {  
    $path = $request->profile_picture->store('profiles', 's3');  
    auth()->user()->profile_picture = $path;  
    auth()->user()->save();  
}
```

If you need to specify the filename, you can use `storeAs()` instead of `store()`. The first parameter is still the path; the second is the filename, and the optional third parameter is the storage disk to use.

PROPER FORM ENCODING FOR FILE UPLOADS

If you get `null` when you try to get the contents of a file from your request, you might've forgotten to set the encoding type on your form. Make sure to add the attribute `enctype="multipart/form-data"` on your form:

```
<form method="post" enctype="multipart/form-data">
```

Validation

Laravel has quite a few ways you can validate incoming data. We'll cover form requests in the next section, so that leaves us with two primary options: validating manually or using the `validate()` method on the Request object. Let's start with the simpler, and more common, `validate()`.

validate() on the Request Object

The Request object has a `validate()` method that provides a convenient shortcut for the most common validation workflow. Take a look at [Example 7-19](#).

Example 7-19. Basic usage of request validation

```
// routes/web.php
Route::get('recipes/create', [RecipeController::class, 'create']);
Route::post('recipes', [RecipeController::class, 'store']);
// app/Http/Controllers/RecipeController.php
class RecipeController extends Controller
{
    public function create()
    {
        return view('recipes.create');
    }

    public function store(Request $request)
    {
        $request->validate([
            'title' => 'required|unique:recipes|max:125',
            'body' => 'required'
        ]);

        // Recipe is valid; proceed to save it
    }
}
```

We only have four lines of code running our validation here, but they’re doing a lot.

First, we explicitly define the fields we expect and apply rules (here separated by the pipe character, `|`) to each individually.

Next, the `validate()` method checks the incoming data from `$request` and determines whether or not it is valid.

If the data is valid, the `validate()` method ends and we can move on with the controller method, saving the data or whatever else.

But if the data isn’t valid, it throws a `ValidationException`. This contains instructions to the router about how to handle this exception. If the request is from JavaScript (or if it’s requesting JSON as a response), the exception will create a JSON response containing the validation errors. If not, the exception

will return a redirect to the previous page, together with all of the user input and the validation errors—perfect for repopulating a failed form and showing some errors.

More on Laravel's Validation Rules

In our examples here (like in the docs) we're using the “pipe” syntax:

`'fieldname': 'rule|otherRule|anotherRule'`. But you can also use the array syntax to do the same thing: `'fieldname': ['rule', 'otherRule', 'anotherRule']`.

Additionally, you can validate nested properties. This matters if you use HTML's array syntax, which allows you to, for example, have multiple “users” on an HTML form, each with an associated name. Here's how you validate that:

```
$request->validate([
    'user.name' => 'required',
    'user.email' => 'required|email',
]);
```

We don't have enough space to cover every possible validation rule here, but here are a few of the most common rules and their functions:

Require the field

`required`; `required_if:anotherField,equalToThisValue`;
`required_unless:anotherField,equalToThisValue`

Exclude the field from the request output

`exclude_if:anotherField,equalToThisValue`;
`exclude_unless:anotherField,equalToThisValue`

Field must contain certain types of characters

`alpha`; `alpha_dash`; `alpha_num`; `numeric`; `integer`

Field must contain certain patterns

`email`; `active_url`; `ip`

Dates

`after: date; before: date` (*date* can be any valid string that `strtotime()` can handle)

Numbers

`between: min,max; min: num; max: num; size: num` (size tests against length for strings, value for integers, count for arrays, or size in KB for files.)

Image dimensions

`dimensions: min_width=XXX`; can also use and/or combine with `max_width`, `min_height`, `max_height`, `width`, `height`, and `ratio`

Databases

`exists: tableName; unique: tableName` (expects to look in the same table column as the field name; see the [validation docs](#) for how to customize)

You can also specify the Eloquent model instead of the table name in the database validation rules:

```
'name' => 'exists:App\Models>Contact,name',  
'phone' => 'unique:App\Models>Contact,phone',
```

Manual Validation

If you are not working in a controller, or if for some other reason the previously described flow is not a good fit, you can manually create a `Validator` instance using the `Validator` facade and check for success or failure like in [Example 7-20](#).

Example 7-20. Manual validation

```
Route::get('recipes/create', function () {  
    return view('recipes.create');  
});  
  
Route::post('recipes', function (Illuminate\Http\Request $request) {  
    $validator = Validator::make($request->all(), [  
        'title' => 'required|unique:recipes|max:125',  
    ]  
);  
});
```

```

        'body' => 'required'
    });

    if ($validator->fails()) {
        return redirect('recipes/create')
            ->withErrors($validator)
            ->withInput();
    }

    // Recipe is valid; proceed to save it
});

```

As you can see, we create an instance of a validator by passing it our input as the first parameter and the validation rules as the second parameter. The validator exposes a `fails()` method that we can check against and can be passed into the `withErrors()` method of the `redirect`.

Using Validated Data

Once you’ve validated your data, you can pull it from the request in a way that ensures you’re only working with validated data. There are two main options: `validated()` and `safe()`. You can run these methods either on the `$request` object, or, if you created a manual validator, the `$validator` instance.

The `validated()` method returns an array of all of the data that’s been validated, as shown in [Example 7-21](#).

Example 7-21. Getting validated data with `validated()`

```

// Both return an array of validated user input
$validated = $request->validated();
$validated = $validator->validated();

```

The `safe()` method, on the other hand, returns an object that gives you access to `all()`, `only()`, and `except()` methods, as you can see in [Example 7-22](#).

Example 7-22. Getting validated data with `safe()`

```

$validated = $request->safe()->only(['name', 'email']);

$validated = $request->safe()->except(['password']);

$validated = $request->safe()->all();

```

Custom Rule Objects

If the validation rule you need doesn't exist in Laravel, you can create your own. To create a custom rule, run `php artisan make:rule RuleName` and then edit that file in `app/Rules/{RuleName}.php`.

You'll get the method `validate()` in your rule class out of the box. The `validate()` method should accept an attribute name as the first parameter, the user-provided value as the second, and the third a closure that you'll when the validation fails; you can use `:attribute` as a placeholder in your message for the attribute name.

Take a look at [Example 7-23](#) as an example.

Example 7-23. A sample custom rule

```
class AllowedEmailDomain implements ValidationRule
{
    public function validate(string $attribute, mixed $value, Closure $fail): void
    {
        if(! in_array(Str::after($value, '@'), ['tighten.co'])){
            $fail('The :attribute field is not from an allowed email provider.');
```

To use this rule, just pass an instance of the rule object to your validator:

```
$request->validate([
    'email' => new AllowedEmailDomain,
]);
```

Displaying Validation Error Messages

We've already covered much of this in [Chapter 6](#), but here's a quick refresher on how to display errors from validation.

The `validate()` method on requests (and the `withErrors()` method on redirects that it relies on) flashes any errors to the session. These errors are made available to the view you're being redirected to in the `$errors` variable. And remember that as a part of Laravel's magic, that `$errors` variable will be available every time you load the view, even if it's just empty, so you don't have to check if it exists with `isset()`.

That means you can do something like [Example 7-24](#) on every page.

Example 7-24. Echo validation errors

```
@if ($errors->any())
    <ul id="errors">
        @foreach ($errors->all() as $error)
            <li>{{ $error }}</li>
        @endforeach
    </ul>
@endif
```

You can also conditionally echo a single field’s error message. For this you’ll use the `@error` Blade directive to check for whether there’s an error on a given field.

```
@error('first_name')
    <span>{{ $message }}</span>
@enderror
```

Form Requests

As you build out your applications, you might start noticing some patterns in your controller methods. There are certain patterns that are repeated—for example, input validation, user authentication and authorization, and possible redirects. If you find yourself wanting a structure to normalize and extract these common behaviors out of your controller methods, you may be interested in Laravel’s form requests.

A form request is a custom request class that is intended to map to the submission of a form, and the request takes the responsibility for validating the request, authorizing the user, and optionally redirecting the user upon a failed validation. Each form request will usually, but not always, explicitly map to a single HTTP request—for example, “Create Comment.”

Creating a Form Request

You can create a new form request from the command line:

```
php artisan make:request CreateCommentRequest
```

You now have a form request object available at

app/Http/Requests/CreateCommentRequest.php.

Every form request class provides either one or two public methods. The first is `rules()`, which needs to return an array of validation rules for this request. The second (optional) method is `authorize()`; if this returns `true`, the user is authorized to perform this request, and if `false`, the user is rejected. Take a look at [Example 7-25](#) to see a sample form request.

Example 7-25. Sample form request

```
<?php

namespace App\Http\Requests;

use App\BlogPost;
use Illuminate\Foundation\Http\FormRequest;

class CreateCommentRequest extends FormRequest
{
    public function authorize(): bool
    {
        $blogPostId = $this->route('blogPost');

        return auth()->check() && BlogPost::where('id', $blogPostId)
            ->where('user_id', auth()->id())->exists();
    }

    public function rules(): array
    {
        return [
            'body' => 'required|max:1000',
        ];
    }
}
```

The `rules()` section of [Example 7-25](#) is pretty self-explanatory, but let's look at `authorize()` briefly.

We're grabbing the segment from the route named `blogPost`. That's implying the route definition for this route probably looks a bit like this:

`Route::post('blogPosts/blogPost', function () { // Do stuff }).`

As you can see, we named the route parameter `blogPost`, which makes it accessible in our Request using `$this->route('blogPost')`.

We then look at whether the user is logged in and, if so, whether any blog posts

exist with that identifier that are owned by the currently logged-in user. You’ve already learned some easier ways to check ownership in [Chapter 5](#), but we’ll keep it more explicit here to keep it clean. We’ll cover what implications this has shortly, but the important thing to know is that returning `true` means the user is authorized to perform the specified action (in this case, creating a comment), and `false` means the user is not authorized.

Using a Form Request

Now that we’ve created a form request object, how do we use it? It’s a little bit of Laravel magic. Any route (closure or controller method) that typehints a form request as one of its parameters will benefit from the definition of that form request.

Let’s try it out, in [Example 7-26](#).

Example 7-26. Using a form request

```
Route::post('comments', function (App\Http\Requests\CreateCommentRequest $request) {  
    // Store comment  
});
```

You might be wondering where we call the form request, but Laravel does it for us. It validates the user input and authorizes the request. If the input is invalid, it’ll act just like the `Request` object’s `validate()` method, redirecting the user to the previous page with their input preserved and with the appropriate error messages passed along. And if the user is not authorized, Laravel will return a “403 Forbidden” error and not execute the route code.

Eloquent Model Mass Assignment

Until now, we’ve been looking at validating at the controller level, which is absolutely the best place to start. But you can also filter the incoming data at the model level.

It’s a common (but not recommended) pattern to pass the entirety of a form’s input directly to a database model. In Laravel, that might look like [Example 7-27](#).

Example 7-27. Passing the entirety of a form to an Eloquent model

```
Route::post('posts', function (Request $request) {
    $newPost = Post::create($request->all());
});
```

We’re assuming here that the end user is kind and not malicious, and has kept only the fields we want them to edit—maybe the post title or body.

But what if our end user can guess, or discern, that we have an `author_id` field on that `posts` table? What if they used their browser tools to add an `author_id` field and set the ID to be someone else’s ID, and impersonated the other person by creating fake blog posts attributed to them?

Eloquent has a concept called “mass assignment” that allows you to either define a list of fields that should be fillable (using the model’s `$fillable` property) or a list of fields that shouldn’t be fillable (using the model’s `$guarded` property) by passing them in an array to `create()` or `update()`. See “[Mass assignment](#)” for more information.

In our example, we might want to fill out the model like in [Example 7-28](#) to keep our app safe.

Example 7-28. Guarding an Eloquent model from mischievous mass assignment

```
<?php

namespace App;

use Illuminate\Database\Eloquent\Model;

class Post extends Model
{
    // Disable mass assignment on the author_id field
    protected $guarded = ['author_id'];
}
```

By setting `author_id` to `guarded`, we ensure that malicious users will no longer be able to override the value of this field by manually adding it to the contents of a form that they’re sending to our app.

DOUBLE PROTECTION USING `$REQUEST->ONLY()`

While it’s important to do a good job of protecting our models from mass assignment, it’s also worth being careful on the assigning end. Rather than using `$request->all()`, consider using

`$request->only()` so you can specify which fields you'd like to pass into your model:

```
Route::post('posts', function (Request $request) {  
    $newPost = Post::create($request->only([  
        'title',  
        'body',  
    ]));  
});
```

{{ Versus {!!

Any time you display content on a web page that was created by a user, you need to guard against malicious input, such as script injection.

Let's say you allow your users to write blog posts on your site. You probably don't want them to be able to inject malicious JavaScript that will run in your unsuspecting visitors' browsers, right? So, you'll want to escape any user input that you show on the page to avoid this.

Thankfully, this is almost entirely covered for you. If you use Laravel's Blade templating engine, the default "echo" syntax (`{{ $stuffToEcho }}`) runs the output through `htmlentities()` (PHP's best way of making user content safe to echo) automatically. You actually have to do *extra* work to avoid escaping the output, by using the `{!! $stuffToEcho !!}` syntax.

Testing

If you're interested in testing your interactions with user input, you're probably most interested in simulating valid and invalid user input and ensuring that if the input is invalid the user is redirected, and if the input is valid it ends up in the proper place (e.g., the database).

Laravel's application testing framework makes this simple.

LARAVEL DUSK FOR TESTING USER INTERACTIONS

These tests are testing the HTTP layer of your application, but not the actual form fields and

interactions. If you want to test specific user interactions on the page and with your forms, you'll want to pull in Laravel's Dusk testing package.

Check out “[Testing with Dusk](#)” to learn how to install and use Dusk in your tests.

Let's start with an invalid route that we expect to be rejected, as in [Example 7-29](#).

Example 7-29. Testing that invalid input is rejected

```
public function test_input_missing_a_title_is_rejected()
{
    $response = $this->post('posts', ['body' => 'This is the body of my post']);
    $response->assertRedirect();
    $response->assertSessionHasErrors();
}
```

Here we assert that after invalid input the user is redirected, with errors attached. You can see we're using a few custom PHPUnit assertions that Laravel adds here.

So, how do we test our route's success? Check out [Example 7-30](#).

Example 7-30. Testing that valid input is processed

```
public function test_valid_input_should_create_a_post_in_the_database()
{
    $this->post('posts', ['title' => 'Post Title', 'body' => 'This is the body']);
    $this->assertDatabaseHas('posts', ['title' => 'Post Title']);
}
```

Note that if you're testing something using the database, you'll need to learn more about database migrations and transactions. More on that in [Chapter 12](#).

TL;DR

There are a lot of ways to get the same data: using the Request facade, using the request() global helper, and injecting an instance of Illuminate\Http\Request. Each exposes the ability to get all input, some input, or specific pieces of data, and there can be some special considerations for files and JSON input.

URL path segments are also a possible source of user input, and they're also

accessible via the request tools.

Validation can be performed manually with `Validator::make()`, or automatically using the `validate()` request method or form requests. Each automatic tool, upon failed validation, redirects the user to the previous page with all old input stored and errors passed along.

Views and Eloquent models also need to be protected from nefarious user input. You can protect Blade views by using the double curly brace syntax (`{{ }}`), which escapes user input. You can protect models by only passing specific fields into bulk methods using `$request->only()` and by defining the mass assignment rules on the model itself.

Chapter 8. Artisan and Tinker

From installation onward, modern PHP frameworks expect many interactions to take place on the command line. Laravel provides three primary tools for command-line interaction: Artisan, a suite of built-in command-line actions with the ability to add more; Tinker, an REPL or interactive shell for your application; and the installer, which we've already covered in [Chapter 2](#).

An Introduction to Artisan

If you've been reading through this book chapter by chapter, you've already learned how to use Artisan commands. They look something like this:

```
php artisan make:controller PostController
```

If you look in the root folder of your application, you'll see that *artisan* is actually just a PHP file. That's why you're starting your call with `php artisan`; you're passing that file into PHP to be parsed. Everything after that is just passed into Artisan as arguments.

SYMFONY CONSOLE SYNTAX

Artisan is actually a layer on top of the [Symfony Console component](#); so, if you're familiar with writing Symfony Console commands, you should feel right at home.

Since the list of Artisan commands for an application can be changed by a package or by the specific code of the application, it's worth checking every new application you encounter to see what commands are available.

To get a list of all available Artisan commands, you can run `php artisan list` from the project root (although if you just run `php artisan` with no parameters, it will do the same thing).

Basic Artisan Commands

There's not enough space here to cover all of the Artisan commands, but we'll cover many of them. Let's get started with the basic commands:

`clear-compiled`

Removes Laravel's compiled class file, which is like an internal Laravel cache; run this as a first resort when things are going wrong and you don't know why.

`down, up`

Puts your application in "maintenance mode" so you can fix an error, run migrations, or whatever else, and restores an application from maintenance mode, respectively.

`dump-server`

Starts the dump server (see "[Laravel Dump Server](#)") to collect and output dumped variables.

`env`

Displays which environment Laravel is running in at the moment; it's the equivalent of echoing `app()->environment()` in-app.

`help`

Provides help for a command; for example, `php artisan help commandName`.

`migrate`

Runs all database migrations.

`optimize`

Clears and refreshes the configuration and route files.

`serve`

Pins up a PHP server at `localhost:8000`. (You can customize the host and/or port with `--host` and `--port`.)

tinker

Brings up the Tinker REPL, which we'll cover later in this chapter.

stub:publish

Publishes all stubs that are available for customization.

docs

Gives you quick access to Laravel docs; pass a parameter and you'll be prompted to open the URL for those docs, or don't pass a parameter and you'll be able to navigate a list of documentation topics to choose from.

about

Shows an overview of your project environment, general configurations, packages, and more.

CHANGES TO THE ARTISAN COMMANDS LIST OVER TIME

The list of Artisan commands and their names have changed in small ways over the lifetime of Laravel. This list is as up to date as possible at the time of publishing this book. However, the best way to see what's available to you is to run `php artisan` from your application.

Options

Before we cover the rest of the commands, let's look at a few notable options you can pass any time you run an Artisan command:

-q

Suppresses all output

-v, -vv, and -vvv

Specifies the level of output verbosity (normal, verbose, and debug)

`--no-interaction`

Suppresses interactive questions, so the command won't interrupt automated processes running it

`--env`

Allows you to define which environment the Artisan command should operate in (`local`, `production`, etc.)

`--version`

Shows which version of Laravel your application is running on

You've probably guessed from looking at these options that Artisan commands are intended to be used much like basic shell commands: you might run them manually, but they can also function as a part of some automated process at some point.

For example, there are many automated deployment processes that might benefit from certain Artisan commands. You might want to run `php artisan config:cache` every time you deploy an application. Flags like `-q` and `--no-interaction` ensure that your deployment scripts, not attended by a human being, can keep running smoothly.

The Grouped Commands

The rest of the commands available out of the box are grouped by context. We won't cover them all here, but we'll cover each context broadly:

`auth`

All we have here is `auth:clear-reshes`, which flushes all of the expired password reset tokens from the database.

`cache`

`cache:clear` clears the cache, `cache:forget` removes an individual item from the cache, and `cache:table` creates a database migration if you plan to

use the database cache driver.

config

`config:cache` caches your configuration settings for faster lookup; to clear the cache, use `config:clear`.

db

`db:seed` seeds your database, if you have configured database seeders.

event

`event:list` lists all the events and listeners in your application, `event:cache` caches that list, `event:clear` clears that cache, and `event:generate` builds missing event and event listener files based on the definitions in `EventServiceProvider`. You'll learn more about events in [Chapter 16](#).

key

`key:generate` creates a random application encryption key in your `.env` file.

RERUNNING ARTISAN KEY:GENERATE MEANS LOSING SOME ENCRYPTED DATA

If you run `php artisan key:generate` more than once on your application, every currently logged-in user will be logged out. Additionally, any data you have manually encrypted will no longer be decryptable. To learn more, check out the article "[APP_KEY and You](#)" by fellow Tightenite Jake Bathman.

make

Each of the `make:` actions creates a single item from a stub and has parameters that vary accordingly. To learn more about any individual command's parameters, use `help` to read its documentation.

For example, you could run `php artisan help make:migration` and learn

that you can pass `--create=tableNameHere` to create a migration that already has the create table syntax in the file, as shown here: `php artisan make:migration create_posts_table --create=posts`.

migrate

The `migrate` command used to run all migrations was mentioned earlier; see [“Running Migrations”](#) for details on all the migration-related commands.

notifications

`notifications:table` generates a migration that creates the table for database notifications.

package

Laravel has a manifest generated by its “autodiscover” feature. This registers third-party packages for you when you first install them. `package:discover` rebuilds Laravel’s “discovered” manifest of the service providers from your external packages.

queue

We’ll cover Laravel’s queues in [Chapter 16](#), but the basic idea is that you can push jobs up into remote queues to be executed one after another by a worker. This command group provides all the tools you need to interact with your queues, like `queue:listen` to start listening to a queue, `queue:table` to create a migration for database-backed queues, and `queue:flush` to flush all failed queue jobs. There are quite a few more, which you’ll learn about in [Chapter 16](#).

route

If you run `route:list`, you’ll see the definitions of every route defined in the application, including each route’s verb(s), path, name, controller/closure action, and middleware. You can cache the route definitions for faster lookups with `route:cache` and clear your cache with `route:clear`.

schedule

We'll cover Laravel's cron-like scheduler in **Chapter 16**, but in order for it to work, you need to set the system cron to run `schedule:run` once a minute:

```
* * * * * php /home/myapp.com/artisan schedule:run >> /dev/null 2>&1
```

As you can see, this Artisan command is intended to be run regularly in order to power a core Laravel service.

session

`session:table` creates a migration for applications using database-backed sessions.

storage

`storage:link` creates a symbolic link from *public/storage* to *storage/app/public*. This is a common convention in Laravel apps, to make it easy to put user uploads (or other files that commonly end up in *storage/app*) somewhere where they'll be accessible at a public URL.

vendor

Some Laravel-specific packages need to “publish” some of their assets, either so that they can be served from your *public* directory or so that you can modify them. Either way, these packages register these “publishable assets” with Laravel, and when you run `vendor:publish`, it publishes them to their specified locations.

view

Laravel's view rendering engine automatically caches your views. It usually does a good job of handling its own cache invalidation, but if you ever notice it's gotten stuck, run `view:clear` to clear the cache.

Writing Custom Artisan Commands

Now that we've covered the Artisan commands that come with Laravel out of the box, let's talk about writing your own.

First, you should know: there's an Artisan command for that! Running `php artisan make:command YourCommandName` generates a new Artisan command in `app/Console/Commands/{YourCommandName}.php`.

Your first argument should be the class name of the command, and you can optionally pass a `--command` parameter to define what the terminal command will be (e.g., `appname:action`). So, let's do it:

```
php artisan make:command WelcomeNewUsers --command=email:newusers
```

Take a look at [Example 8-1](#) to see what you'll get.

Example 8-1. The default skeleton of an Artisan command

```
<?php

namespace App\Console\Commands;

use Illuminate\Console\Command;

class WelcomeNewUsers extends Command
{
    /**
     * The name and signature of the console command
     *
     * @var string
     */
    protected $signature = 'email:newusers';

    /**
     * The console command description
     *
     * @var string
     */
    protected $description = 'Command description';

    /**
     * Execute the console command.
     */
    public function handle(): void
    {
        //
    }
}
```

```
}
```

As you can see, it's very easy to define the command signature, the help text it shows in command lists, and the command's behavior on execution (`handle()`).

A Sample Command

We haven't covered mail or Eloquent yet in this chapter (see [Chapter 15](#) for mail and [Chapter 5](#) for Eloquent), but the sample `handle()` method in [Example 8-2](#) should read pretty clearly.

Example 8-2. A sample Artisan command `handle()` method

```
// ...

class WelcomeNewUsers extends Command
{
    public function handle(): void
    {
        User::signedUpThisWeek()->each(function ($user) {
            Mail::to($user)->send(new WelcomeEmail);
        });
    }
}
```

Now every time you run `php artisan email:newusers`, this command will grab every user that signed up this week and send them the welcome email.

If you would prefer injecting your mail and user dependencies instead of using facades, you can typehint them in the command constructor, and Laravel's container will inject them for you when the command is instantiated.

Take a look at [Example 8-3](#) to see what [Example 8-2](#) might look like using dependency injection and extracting its behavior out to a service class.

Example 8-3. The same command, refactored

```
...
class WelcomeNewUsers extends Command
{
    public function __construct(UserMailer $userMailer)
    {
        parent::__construct();

        $this->userMailer = $userMailer
    }
}
```

```
public function handle(): void
{
    $this->userMailer->welcomeNewUsers();
}
```

KEEP IT SIMPLE

It is possible to call Artisan commands from the rest of your code so you can use them to encapsulate chunks of application logic.

However, the Laravel docs recommend instead packaging the application logic into a service class and injecting that service into your command. Console commands are seen as being similar to controllers: they're not domain classes; they're traffic cops that just route incoming requests to the correct behavior.

Arguments and Options

The `$signature` property of the new command looks like it might just contain the command name. But this property is also where you'll define any arguments and options for the command. There's a specific, simple syntax you can use to add arguments and options to your Artisan commands.

Before we dig into that syntax, take a look at an example for some context:

```
protected $signature = 'password:reset {userId} [--sendEmail]';
```

Arguments—required, optional, and/or with defaults

To define a required argument, surround it with braces:

```
password:reset {userId}
```

To make the argument optional, add a question mark:

```
password:reset {userId?}
```

To make it optional and provide a default, use:

```
password:reset {userId=1}
```

Options—required values, value defaults, and shortcuts

Options are similar to arguments, but they're prefixed with `--` and can be used with no value. To add a basic option, surround it with braces:

```
password:reset {userId} {--sendEmail}
```

If your option requires a value, add an `=` to its signature:

```
password:reset {userId} {--password=}
```

And if you want to pass a default value, add it after the `=`:

```
password:reset {userId} {--queue=default}
```

Array arguments and array options

Both for arguments and for options, if you want to accept an array as input, use the `*` character:

```
password:reset {userIds*}
```

```
password:reset {--ids=*}
```

Using array arguments and parameters looks a bit like [Example 8-4](#).

Example 8-4. Using array syntax with Artisan commands

```
// Argument
```

```
php artisan password:reset 1 2 3
```

```
// Option
```

```
php artisan password:reset --ids=1 --ids=2 --ids=3
```

ARRAY ARGUMENTS MUST BE THE LAST ARGUMENT

Since an array argument captures every parameter after its definition and adds them as array items, an array argument has to be the last argument within an Artisan command's signature.

Input descriptions

Remember how the built-in Artisan commands can give us more information about their parameters if we use `artisan help`? We can provide that same information about our custom commands. Just add a colon and the description text within the curly braces, like in [Example 8-5](#).

Example 8-5. Defining description text for Artisan arguments and options

```
protected $signature = 'password:reset
                        {userId : The ID of the user}
                        {--sendEmail : Whether to send user an email}';
```

Using Input

Now that we've prompted for this input, how do we use it in our command's `handle()` method? We have two sets of methods for retrieving the values of arguments and options.

`argument()` and `arguments()`

`$this->arguments()` returns an array of all arguments (the first array item will be the command name). `$this->argument()`, called with no parameters, returns the same response; the plural method, which I prefer, is just available for better readability.

To get just the value of a single argument, pass the argument name as a parameter to `$this->argument()`, as shown in [Example 8-6](#).

Example 8-6. Using `$this->arguments()` in an Artisan command

```
// With definition "password:reset {userId}"
php artisan password:reset 5

// $this->arguments() returns this array
[
    "command": "password:reset",
    "userId": "5",
]

// $this->argument('userId') returns this string
"5"
```

`option()` and `options()`

`$this->options()` returns an array of all options, including some that will by default be `false` or `null`. `$this->option()`, called with no parameters, returns the same response; again, the plural method, which I prefer, is just available for better readability.

To get just the value of a single option, pass the argument name as a parameter to `$this->option()`, as shown in [Example 8-7](#).

Example 8-7. Using `$this->options()` in an Artisan command

```
// With definition "password:reset {--userId=}"
php artisan password:reset --userId=5

// $this->options() returns this array
[
    "userId" => "5",
    "help" => false,
    "quiet" => false,
    "verbose" => false,
    "version" => false,
    "ansi" => false,
    "no-ansi" => false,
    "no-interaction" => false,
    "env" => null,
]

// $this->option('userId') returns this string
"5"
```

[Example 8-8](#) shows an Artisan command using `argument()` and `option()` in its `handle()` method.

Example 8-8. Getting input from an Artisan command

```
public function handle(): void
{
    // All arguments, including the command name
    $arguments = $this->arguments();

    // Just the 'userId' argument
    $userid = $this->argument('userId');

    // All options, including some defaults like 'no-interaction' and 'env'
    $options = $this->options();

    // Just the 'sendEmail' option
    $sendEmail = $this->option('sendEmail');
}
```

Prompts

There are a few more ways to get user input from within your `handle()` code, and they all involve prompting the user to enter information during the execution of your command:

`ask()`

Prompts the user to enter freeform text:

```
$email = $this->ask('What is your email address?');
```

`secret()`

Prompts the user to enter freeform text, but hides the typing with asterisks:

```
$password = $this->secret('What is the DB password?');
```

`confirm()`

Prompts the user for a yes/no answer, and returns a Boolean:

```
if ($this->confirm('Do you want to truncate the tables?')) {  
    //  
}
```

All answers except y or Y will be treated as a “no.”

`anticipate()`

Prompts the user to enter freeform text, and provides autocomplete suggestions. Still allows the user to type whatever they want:

```
$album = $this->anticipate('What is the best album ever?', [  
    "The Joshua Tree", "Pet Sounds", "What's Going On"  
]);
```

choice()

Prompts the user to choose one of the provided options. The last parameter is the default if the user doesn't choose:

```
$winner = $this->choice(  
    'Who is the best football team?',  
    ['Gators', 'Wolverines'],  
    0  
);
```

Note that the final parameter, the default, should be the array key. Since we passed a nonassociative array, the key for Gators is 0. You could also key your array, if you'd prefer:

```
$winner = $this->choice(  
    'Who is the best football team?',  
    ['gators' => 'Gators', 'wolverines' => 'Wolverines'],  
    'gators'  
);
```

Output

During the execution of your command, you might want to write messages to the user. The most basic way to do this is to use `$this->info()` to output basic green text:

```
$this->info('Your command has run successfully.');
```

You also have available the `comment()` (orange), `question()` (highlighted teal), `error()` (highlighted red), `line()` (uncolored), and `newLine()` (uncolored) methods to echo to the command line.

Please note that the exact colors may vary from machine to machine, but they try to be in line with the local machine’s standards for communicating to the end user.

Table output

The `table()` method makes it simple to create ASCII tables full of your data. Take a look at [Example 8-9](#).

Example 8-9. Outputting tables with Artisan commands

```
$headers = ['Name', 'Email'];

$data = [
    ['Dhriti', 'dhriti@amrit.com'],
    ['Moses', 'moses@gutierrez.com'],
];

// Or, you could get similar data from the database:
$data = App\User::all(['name', 'email'])->toArray();

$this->table($headers, $data);
```

Note that [Example 8-9](#) has two sets of data: the headers, and the data itself. Both contain two “cells” per “row”; the first cell in each row is the name, and the second is the email. That way the data from the Eloquent call (which is constrained to pull only name and email) matches up with the headers.

Take a look at [Example 8-10](#) to see what the table output looks like.

Example 8-10. Sample output of an Artisan table

```
+-----+-----+
| Name   | Email               |
+-----+-----+
| Dhriti | dhriti@amrit.com    |
| Moses  | moses@gutierrez.com |
+-----+-----+
```

Progress bars

If you’ve ever run `npm install`, you’ve seen a command-line progress bar before. Let’s build one in [Example 8-11](#).

Example 8-11. Sample Artisan progress bar

```
$totalUnits = 350;
$this->output->progressStart($totalUnits);
```

```

for ($i = 0; $i < $totalUnits; $i++) {
    sleep(1);

    $this->output->progressAdvance();
}

$this->output->progressFinish();

```

What did we do here? First, we informed the system how many “units” we needed to work through. Maybe a unit is a user, and you have 350 users. The bar will then divide the entire width it has available on your screen by 350, and increment it by 1/350th every time you run `progressAdvance()`. Once you’re done, run `progressFinish()` so that it knows it’s done displaying the progress bar.

Writing Closure-Based Commands

If you’d prefer to keep your command definition process simpler, you can write commands as closures instead of classes by defining them in *routes/console.php*. Everything we discuss in this chapter will apply the same way, but you will define and register the commands in a single step in that file, as shown in [Example 8-12](#).

Example 8-12. Defining an Artisan command using a closure

```

// routes/console.php
Artisan::command(
    'password:reset {userId} [--sendEmail]',
    function ($userId, $sendEmail) {
        $userId = $this->argument('userId');
        // Do something...
    }
);

```

Calling Artisan Commands in Normal Code

While Artisan commands are designed to be run from the command line, you can also call them from other code.

The easiest way is to use the Artisan facade. You can either call a command using `Artisan::call()` (which will return the command’s exit code) or queue a

command using `Artisan::queue()`.

Both take two parameters: first, the terminal command (`password:reset`); and second, an array of parameters to pass it. Take a look at [Example 8-13](#) to see how it works with arguments and options.

Example 8-13. Calling Artisan commands from other code

```
Route::get('test-artisan', function () {
    $exitCode = Artisan::call('password:reset', [
        'userId' => 15,
        '--sendEmail' => true,
    ]);
});
```

As you can see, arguments are passed by keying to the argument name, and options with no value can be passed true or false.

CALLING ARTISAN COMMANDS USING STRING SYNTAX

You can also call Artisan commands much more naturally from your code by passing the same string you'd call from the command line into `Artisan::call()`:

```
Artisan::call('password:reset 15 --sendEmail')
```

You can also call Artisan commands from other commands using `$this->call()`, (which is the same as `Artisan::call()`) or `$this->callSilent()`, which is the same but suppresses all output. See [Example 8-14](#) for an example.

Example 8-14. Calling Artisan commands from other Artisan commands

```
public function handle(): void
{
    $this->callSilent('password:reset', [
        'userId' => 15,
    ]);
}
```

Finally, you can inject an instance of the `Illuminate\Contracts\Console\Kernel` contract and use its `call()` method.

Tinker

Tinker is an REPL, or read–evaluate–print loop. REPLs give you a prompt, similar to the command-line prompt, that mimics a “waiting” state of your application. You type your commands into the REPL, hit Return, and then expect what you typed to be evaluated and the response printed out.

Example 8-15 provides a quick sample to give you a sense of how it works and how it might be useful. We start the REPL with `php artisan tinker` and are then presented with a blank prompt (`>>>`); every response to our commands is printed on a line prefaced with `=>`.

Example 8-15. Using Tinker

```
$ php artisan tinker
```

```
>>> $user = new App\User;
=> App\User: {}
>>> $user->email = 'matt@mattstauffer.com';
=> "matt@mattstauffer.com"
>>> $user->password = bcrypt('superSecret');
=> "$2y$10$TWPGBC7e8d1bvJ1q5kv.VDUGFYDnE9gANl4mleuB3htIY2dxcQfQ5"
>>> $user->save();
=> true
```

As you can see, we created a new user, set some data (hashing the password with `bcrypt()` for security), and saved it to the database. And this is real. If this were a production application, we would’ve just created a brand new user in our system.

This makes Tinker a great tool for simple database interactions, for trying out new ideas, and for running snippets of code when it’d be a pain to find a place to put them in the application source files.

Tinker is powered by **Psy Shell**, so check that out to see what else you can do with Tinker.

Laravel Dump Server

One common method of debugging the state of your data during development is to use Laravel’s `dump()` helper, which runs a decorated `var_dump()` on anything

you pass to it. This is fine, but it can often run into view issues.

You can enable the Laravel dump server, which catches those `dump()` statements and displays them in your console instead of rendering them to the page.

To run the dump server in your local console, navigate to your project's root directory and run `php artisan dump-server`:

```
$ php artisan dump-server

Laravel Var Dump Server
=====

[OK] Server listening on tcp://127.0.0.1:9912

// Quit the server with CONTROL-C.
```

Now, try using the `dump()` helper function in your code somewhere. To test it out, try this code in your `routes/web.php` file:

```
Route::get('/', function () {
    dump('Dumped Value');

    return 'Hello World';
});
```

Without the dump server, you'd see both the dump and your "Hello World." But with the dump server running, you'll only see "Hello World" in the browser. In your console, you'll see that the dump server caught that `dump()`, and you can inspect it there:

```
GET http://myapp.test/
-----

-----
date          Tue, 18 Sep 2018 22:43:10 +0000
controller    "Closure"
source        web.php on line 20
file          routes/web.php
-----

"Dumped Value"
```


Customizing Generator Stubs

Any Artisan commands that generate files (e.g. `make:model` and `make:controller`) use “stub” files that the command then copies and modifies to create the newly generated files. You can customize these stubs in your applications.

To customize the stubs in your applications, run `php artisan stub:publish`, which will export the stub files into a `stub/` directory where you can customize them.

Testing

Since you know how to call Artisan commands from code, it’s easy to do that in a test and ensure that whatever behavior you expected to be performed has been performed correctly, as in [Example 8-16](#). In our tests, we use `$this->artisan()` instead of `Artisan::call()` because it has the same syntax but adds a few testing-related assertions.

Example 8-16. Calling Artisan commands from a test

```
public function test_empty_log_command_emptyies_logs_table()
{
    DB::table('logs')->insert(['message' => 'Did something']);
    $this->assertCount(1, DB::table('logs')->get());

    $this->artisan('logs:empty'); // Same as Artisan::call('logs:empty');
    $this->assertCount(0, DB::table('logs')->get());
}
```

You can chain on a few new assertions to your `$this->artisan()` calls that make it even easier to test Artisan commands—not just the impact they have on the rest of your app, but also how they actually operate. Take a look at [Example 8-17](#) to see an example of this syntax.

Example 8-17. Making assertions against the input and output of Artisan commands

```
public function testItCreatesANewUser()
{
    $this->artisan('myapp:create-user')
        ->expectsQuestion("What's the name of the new user?", "Wilbur Power")
}
```

```
->expectsQuestion("What's the email of the new user?", "wilbur@thisbook.co")
->expectsQuestion("What's the password of the new user?", "secret")
->expectsOutput("User Wilbur Powery created!");

$this->assertDatabaseHas('users', [
    'email' => 'wilbur@thisbook.co'
]);
}
```

TL;DR

Artisan commands are Laravel's command-line tools. Laravel comes with quite a few out of the box, but it's also easy to create your own Artisan commands and call them from the command line or your own code.

Tinker is an REPL that makes it simple to get into your application environment and interact with real code and real data, and the dump server lets you debug your code without stopping the code's execution.

Chapter 9. User Authentication and Authorization

Setting up a basic user authentication system—including registration, login, sessions, password resets, and access permissions—can often be one of the more time-consuming pieces of creating the foundation of an application. It’s a prime candidate for extracting functionality out to a library, and there are quite a few such libraries.

But because authentication needs can vary widely across projects, most authentication systems grow bulky and unusable quickly. Thankfully, Laravel has found a way to make a suite of authentication systems that are easy to use and understand, but flexible enough to fit in a variety of settings.

Every new installation of Laravel has a `create_users_table` migration and a `User` model built in. If you bring in Breeze (see “[Laravel Breeze](#)”) or Jetstream (see “[Laravel Jetstream](#)”), they’ll seed your app with a collection of authentication-related views, routes, controllers/actions, and other features. The APIs are clean and clear, and the conventions all work together to provide a simple—and seamless—authentication and authorization system.

The User Model and Migration

When you create a new Laravel application, the first migration and model you’ll see are the `create_users_table` migration and the `App\User` model.

Example 9-1 shows, straight from the migration, the fields you’ll get in your `users` table.

Example 9-1. Laravel’s default user migration

```
Schema::create('users', function (Blueprint $table) {
    $table->id();
    $table->string('name');
    $table->string('email')->unique();
    $table->timestamp('email_verified_at')->nullable();
    $table->string('password');
```

```
$table->rememberToken();
$table->timestamps();
});
```

We have an autoincrementing primary key ID, a name, a unique email, a password, a “remember me” token, and created and modified timestamps. This covers everything you need to handle basic user authentication in most apps.

THE DIFFERENCE BETWEEN AUTHENTICATION AND AUTHORIZATION

Authentication means verifying who someone is and allowing them to act as that person in your system. This includes the login and logout processes as well as any tools that allow the users to identify themselves during their time using the application.

Authorization means determining whether the authenticated user is *allowed* (authorized) to perform a specific behavior. For example, an authorization system allows you to forbid any nonadministrators from viewing the site’s earnings.

The User model is a bit more complex, as you can see in [Example 9-2](#). The App\User class itself is simple, but it extends the Illuminate\Foundation\Auth\User class, which pulls in several traits.

Example 9-2. Laravel’s default User model

```
<?php
// App\User

namespace App\Models;

// use Illuminate\Contracts\Auth\MustVerifyEmail;
use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Foundation\Auth\User as Authenticatable;
use Illuminate\Notifications\Notifiable;
use Laravel\Sanctum\HasApiTokens;

class User extends Authenticatable
{
    use HasApiTokens, HasFactory, Notifiable;

    /**
     * The attributes that are mass assignable.
     *
     * @var array<int, string>
     */
```

```

protected $fillable = [
    'name',
    'email',
    'password',
];

/**
 * The attributes that should be hidden for serialization.
 *
 * @var array<int, string>
 */
protected $hidden = [
    'password',
    'remember_token',
];

/**
 * The attributes that should be cast.
 *
 * @var array<string, string>
 */
protected $casts = [
    'email_verified_at' => 'datetime',
];
}
<?php
// Illuminate\Foundation\Auth\User

namespace Illuminate\Foundation\Auth;

use Illuminate\Auth\Authenticatable;
use Illuminate\Auth\MustVerifyEmail;
use Illuminate\Auth\Passwords\CanResetPassword;
use Illuminate\Contracts\Auth\Access\Authorizable as AuthorizableContract;
use Illuminate\Contracts\Auth\Authenticatable as AuthenticatableContract;
use Illuminate\Contracts\Auth\CanResetPassword as CanResetPasswordContract;
use Illuminate\Database\Eloquent\Model;
use Illuminate\Foundation\Auth\Access\Authorizable;

class User extends Model implements
    AuthenticatableContract,
    AuthorizableContract,
    CanResetPasswordContract
{
    use Authenticatable, Authorizable, CanResetPassword, MustVerifyEmail;
}

```

ELOQUENT MODEL REFRESHER

If this is entirely unfamiliar, consider reading [Chapter 5](#) before continuing to learn how Eloquent models work.

So, what can we learn from this model? First, users live in the `users` table; Laravel will infer this from the class name. We are able to fill out the `name`, `email`, and `password` properties when creating a new user, and the `password` and `remember_token` properties are excluded when outputting the user as JSON. Looking good so far.

We also can see from the contracts and the traits in the `Illuminate\Foundation\Auth` version of `User` that there are some features in the framework (the ability to authenticate, to authorize, and to reset passwords) that theoretically could be applied to other models, not just the `User` model, and that could be applied individually or together.

CONTRACTS AND INTERFACES

You may have noticed that sometimes I write the word “contract” and sometimes “interface,” and that almost all of the interfaces in Laravel are under the `Contracts` namespace.

A PHP interface is essentially an agreement between two classes that one of the classes will “behave” a certain way. It’s a bit like a contract between them, and thinking about it as a contract gives a little more inherent meaning to the name than calling it an interface does.

In the end, though, they’re the same thing: an agreement that a class will provide certain methods with a certain signature.

On a related note, the `Illuminate\Contracts` namespace contains a group of interfaces that Laravel components implement and typehint. This makes it easy to develop similar components that implement the same interfaces and swap them into your application in place of the stock `Illuminate` components. When the Laravel core and components typehint a mailer, for example, they don’t typehint the `Mailer` class. Instead, they typehint the `Mailer` contract (interface), making it easy to provide your own mailer. To learn more about how to do this, take a look at [Chapter 11](#).

The `Authenticatable` contract requires methods (e.g., `getAuthIdentifier()`) that allow the framework to authenticate instances of this model to the auth system; the `Authenticatable` trait includes the methods necessary to satisfy that contract with an average Eloquent model.

The `Authorizable` contract requires a method (`can()`) that allows the framework to authorize instances of this model for their access permissions in different contexts. Unsurprisingly, the `Authorizable` trait provides methods that will satisfy the `Authorizable` contract for an average Eloquent model.

Finally, the `CanResetPassword` contract requires methods (`getEmailForPasswordReset()`, `sendPasswordResetNotification()`) that allow the framework to—you guessed it—reset the password of any entity that satisfies this contract. The `CanResetPassword` trait provides methods to satisfy that contract for an average Eloquent model.

At this point, we have the ability to easily represent an individual user in the database (with the migration), and to pull them out with a model instance that can be authenticated (logged in and out), authorized (checked for access permissions to a particular resource), and sent a password reset email.

Using the `auth()` Global Helper and the Auth Facade

The `auth()` global helper is the easiest way to interact with the status of the authenticated user throughout your app. You can also inject an instance of `Illuminate\Auth\AuthManager` and get the same functionality, or use the Auth facade.

The most common usages are to check whether a user is logged in—(`auth()->check()` returns true if the current user is logged in; `auth()->guest()` returns true if the user is not logged in)—and to get the currently logged-in user (use `auth()->user()`, or `auth()->id()` for just the ID; both return null if no user is logged in).

Take a look at [Example 9-3](#) for a sample usage of the global helper in a

controller.

Example 9-3. Sample usage of the auth() global helper in a controller

```
public function dashboard()
{
    if (auth()->guest()) {
        return redirect('sign-up');
    }

    return view('dashboard')
        ->with('user', auth()->user());
}
```

routes/auth.php, Auth Controllers, and Auth Actions

If you're working with one of Laravel's starter kits, you'll see that using the baked-in authentication routes, such as login, register, and password reset, requires routes, controllers, and views.

Both Breeze and Jetstream define your routes using a custom routes file: *routes/auth.php*. They're not exactly the same, but take a look at [Example 9-4](#) to see a bit of Breeze's auth routes file to show what they look like generally.

Example 9-4. Part of Breeze's routes/auth.php

```
Route::middleware('guest')->group(function () {
    Route::get('register', [RegisteredUserController::class, 'create'])
        ->name('register');

    Route::post('register', [RegisteredUserController::class, 'store']);

    Route::get('login', [AuthenticatedSessionController::class, 'create'])
        ->name('login');

    Route::post('login', [AuthenticatedSessionController::class, 'store']);

    Route::get('forgot-password', [PasswordResetLinkController::class, 'create'])
        ->name('password.request');

    Route::post('forgot-password', [PasswordResetLinkController::class, 'store'])
        ->name('password.email');

    Route::get('reset-password/{token}', [NewPasswordController::class, 'create'])
        ->name('password.reset');
```



```
Route::post('reset-password', [NewPasswordController::class, 'store'])
    ->name('password.store');
});
```

Breeze publishes controllers under the Auth namespace, which you can configure if you need to:

- *AuthenticatedSessionController.php*
- *ConfirmablePasswordController.php*
- *EmailVerificationNotificationController.php*
- *EmailVerificationPromptController.php*
- *NewPasswordController.php*
- *PasswordController.php*
- *PasswordResetLinkController.php*
- *RegisteredUserController.php*
- *VerifyEmailController.php*

Instead of controllers, Jetstream (and Fortify, which it depends on) publishes “actions” that you can customize:

```
app/Actions/Fortify/CreateNewUser.php
app/Actions/Fortify/PasswordValidationRules.php
app/Actions/Fortify/ResetUserPassword.php
app/Actions/Fortify/UpdateUserPassword.php
app/Actions/Fortify/UpdateUserProfileInformation.php
app/Actions/Jetstream/DeleteUser.php
```

Breeze and Jetstream’s Frontend Templates

At this point you have a migration, a model, controllers/actions, and routes for your authentication system. But what about your views?

You can learn more in “[Laravel Breeze](#)” and “[Laravel Jetstream](#)”, but each tool provides multiple different stacks, and each stack keeps its templates in different places.

In general, the JavaScript-based stacks put their templates in *resources/js* and the Blade-based stacks put them in *resources/views*.

There's at least one view for each function (log in, register, reset password, etc.) and they're all generated with a slick Tailwind-based design, ready to be used or customized.

“Remember Me”

Breeze and Jetstream have this implemented out of the box, but it's still worth learning how it works and how to use it on your own. If you want to implement a “remember me”-style long-lived access token, make sure you have a `remember_token` column on your `users` table (which you will if you used the default migration).

When you're normally logging in a user (and this is how the `LoginController` does it, with the `AuthenticatesUsers` trait), you'll “attempt” an authentication with the user-provided information, like in [Example 9-5](#).

Example 9-5. Attempting a user authentication

```
if (auth()->attempt([
    'email' => request()->input('email'),
    'password' => request()->input('password'),
])) {
    // Handle the successful login
}
```

This provides you with a user login that lasts as long as the user's session. If you want Laravel to extend the login indefinitely using cookies (as long as the user is on the same computer and doesn't log out), you can pass a Boolean `true` as the second parameter of the `auth()->attempt()` method. Take a look at [Example 9-6](#) to see what that request looks like.

Example 9-6. Attempting a user authentication with a “remember me” checkbox check

```
if (auth()->attempt([
    'email' => request()->input('email'),
    'password' => request()->input('password'),
], request()->filled('remember'))) {
    // Handle the successful login
}
```

```
}
```

You can see that we checked whether the input has a nonempty (“filled”) `remember` property, which will return a Boolean. This allows our users to decide if they want to be remembered with a checkbox in the login form.

And later, if you need to manually check whether the current user was authenticated by a remember token, there’s a method for that: `auth()->viaRemember()` returns a Boolean, indicating whether or not the current user authenticated via a remember token. This allows you to prevent certain higher-sensitivity features from being accessible by remember token; instead, you can require users to reenter their passwords.

Password Confirmation

It may be necessary for your users to reconfirm their passwords before accessing certain parts of your application. For example, if a user has been logged in for a while, and then attempts to visit the billing section of your site, you might want them to verify their password.

You can attach a `password.confirm` middleware to your routes to force this behavior. Once their password is confirmed, the user will be sent to the route they tried to visit initially. After that point, the user won’t have to reconfirm their password for 3 hours; you can change this in the `auth.password_timeout` configuration setting.

Manually Authenticating Users

The most common case for user authentication is that you allow the user to provide their credentials, and then use `auth()->attempt()` to see whether the provided credentials match any real users. If so, you log them in.

But sometimes there are contexts where it’s valuable for you to be able to choose to log a user in on your own. For example, you may want to allow admin users to switch users.

There are four methods that make this possible. First, you can just pass a user

ID:

```
auth()->loginUsingId(5);
```

Second, you can pass a User object (or any other object that implements the Illuminate\Contracts\Auth\Authenticatable contract):

```
auth()->login($user);
```

And third and fourth, you can choose to authenticate the given user for only the current request, which won't impact your session or cookies at all, using `once()` or `onceUsingId()`:

```
auth()->once(['username' => 'mattstauffer']);  
// or  
auth()->onceUsingId(5);
```

Note that the array you pass to the `once()` method can contain any key/value pairs to uniquely identify the user you'd like to authenticate as. You can even pass multiple keys and values, if it's what is appropriate for your project. For example:

```
auth()->once([  
    'last_name' => 'Stauffer',  
    'zip_code' => 90210,  
])
```

Manually Logging Out a User

If you ever need to log out a user manually, just call `logout()`:

```
auth()->logout();
```

Invalidating Sessions on Other Devices

If you'd like to log out a user's current session on any other devices—for example, after they've changed their password—you'll need to prompt the user for their password and pass it to the `logoutOtherDevices()` method. To do this,

you'll have to apply the `auth.session` middleware to any routes you want them logged out of (for most projects, that's the entire app).

Then you can use it inline anywhere you need:

```
auth()->logoutOtherDevices($password);
```

If you want to give your users a detailed look at what other sessions are active, Jetstream (see “[Laravel Jetstream](#)”) comes out of the box with a page that lists all active sessions and provides a button to log out of all of them.

Auth Middleware

In [Example 9-3](#), you saw how to check whether visitors are logged in and redirect them if not. You could perform these sorts of checks on every route in your application, but it would very quickly get tedious. It turns out that route middleware (see [Chapter 10](#) to learn more about how they work) are a perfect fit for restricting certain routes to guests or to authenticated users.

Once again, Laravel comes with the middleware we need out of the box. You can see which route middleware you have defined in `App\Http\Kernel`:

```
protected $middlewareAliases = [
    'auth' => \App\Http\Middleware\Authenticate::class,
    'auth.basic' => \Illuminate\Auth\Middleware\AuthenticateWithBasicAuth::class,
    'auth.session' => \Illuminate\Session\Middleware\AuthenticateSession::class,
    'cache.headers' => \Illuminate\Http\Middleware\SetCacheHeaders::class,
    'can' => \Illuminate\Auth\Middleware\Authorize::class,
    'guest' => \App\Http\Middleware\RedirectIfAuthenticated::class,
    'password.confirm' => \Illuminate\Auth\Middleware\RequirePassword::class,
    'signed' => \App\Http\Middleware\ValidateSignature::class,
    'throttle' => \Illuminate\Routing\Middleware\ThrottleRequests::class,
    'verified' => \Illuminate\Auth\Middleware\EnsureEmailIsVerified::class,
];
```

Six of the default route middleware are authentication-related:

auth

Restricts route access to authenticated users

`auth.basic`

Restricts access to authenticated users using HTTP Basic Authentication

`auth.session`

Makes routes viable to be disabled using `Auth::logoutOtherDevices`

`can`

Used for authorizing user access to given routes

`guest`

Restricts access to unauthenticated users

`password.confirm`

Requires users to have recently reconfirmed their password

It's most common to use `auth` for your authenticated-user-only sections and `guest` for any routes you don't want authenticated users to see (like the login form). `auth.basic` and `auth.session` are much less commonly used middleware for authenticating.

Example 9-7 shows an example of a few routes protected by the `auth` middleware.

Example 9-7. Sample routes protected by auth middleware

```
Route::middleware('auth')->group(function () {  
    Route::get('account', [AccountController::class, 'dashboard']);  
});  
  
Route::get('login', [LoginController::class, 'getLogin'])->middleware('guest');
```

Email Verification

If you'd like to require a user to verify they have access to the email address they registered with, you can reach for Laravel's email verification feature.

To enable email verification, update your `App\User` class and make it implement

the `Illuminate\Contracts\Auth\MustVerifyEmail` contract, as shown in [Example 9-8](#).

Example 9-8. Adding the `MustVerifyEmail` trait to an `Authenticatable` model

```
class User extends Authenticatable implements MustVerifyEmail
{
    use Notifiable;

    // ...
}
```

The `users` table must also contain a nullable timestamp column named `email_verified_at`, which the default `CreateUsersTable` migration will have already provided for you.

Finally, you'll need to enable the email verification routes in your controller. The easiest method is to use `Auth::routes()` in your routes file with the `verify` parameter set to `true`:

```
Auth::routes(['verify' => true]);
```

Now, you can protect any routes you'd like from being accessed by any users who haven't verified their email address:

```
Route::get('posts/create', function () {
    // Only verified users may enter...
})->middleware('verified');
```

You can customize the route to which users are redirected after verifying in your `VerificationController`:

```
protected $redirectTo = '/profile';
```

Blade Authentication Directives

If you want to check whether a user is authenticated, not at the route level but in your views, you can do so with `@auth` and `@guest` (see [Example 9-9](#)).

Example 9-9. Checking a user's authentication status in templates

```
@auth
```

```
// The user is authenticated
@endauth

@guest
// The user is not authenticated
@endguest
```

You can also specify which guard you'd like to use with both methods by passing the guard name as a parameter, as shown in [Example 9-10](#).

Example 9-10. Checking a specific auth guard's authentication in templates

```
@auth('trainees')
// The user is authenticated
@endauth

@guest('trainees')
// The user is not authenticated
@endguest
```

Guards

Every aspect of Laravel's authentication system is routed through something called a *guard*. Each guard is a combination of two pieces: a *driver* that defines how it persists and retrieves the authentication state (for example, `session`), and a *provider* that allows you to get a user by certain criteria (for example, `users`).

Out of the box, Laravel has two guards: `web` and `api`. `web` is the more traditional authentication style, using the `session` driver and the basic user provider. `api` uses the same user provider, but it uses the `token` driver instead of `session` to authenticate each request.

You'd change drivers if you wanted to handle the identification and persistence of a user's identity differently (for example, changing from a long-running session to a provided-every-page-load token), and you'd change providers if you wanted to change the storage type or retrieval methods for your users (for example, storing your users in Mongo instead of MySQL).

Changing the Default Guard

The guards are defined in `config/auth.php`, and you can change them, add new guards, and also define which guard will be the default there. For what it's

worth, this is a relatively uncommon configuration; most Laravel apps just use one guard.

The “default” guard is the one that will be used any time you use any auth features without specifying a guard. For example, `auth()->user()` will pull the currently authenticated user using the default guard. You can change this guard by changing the `auth.defaults.guard` setting in `config/auth.php`:

```
'defaults' => [  
    'guard' => 'web', // Change the default here  
    'passwords' => 'users',  
],
```

CONFIGURATION CONVENTIONS

You may have noticed that I refer to configuration sections with references like `auth.defaults.guard`. This means that in `config/auth.php`, in the array section keyed `defaults`, there should be a property keyed `guard`.

Using Other Guards Without Changing the Default

If you want to use another guard but *not* change the default, you can start your `auth()` calls with `guard()`:

```
$apiUser = auth()->guard('api')->user();
```

This will, just for this call, get the current user using the `api` guard.

Adding a New Guard

You can add a new guard at any time in `config/auth.php`, in the `auth.guards` setting:

```
'guards' => [  
    'trainees' => [  
        'driver' => 'session',  
        'provider' => 'trainees',  
    ],  
],
```

Here, we've created a new guard (in addition to `web` and `api`) named `trainees`. Let's imagine, for the rest of this section, that we're building an app where our users are physical trainers, and they each have their own users—trainees—who can log in to their subdomains. So, we need a separate guard for them.

The only two options for `driver` are `token` and `session`. Out of the box, the only option for `provider` is `users`, which supports authentication against your default `users` table, but you can create your own provider easily.

Closure Request Guards

If you want to define a custom guard, and your guard conditions (how to look up a given user against the request) can be described simply enough in response to any given HTTP request, you might just want to throw the user lookup code into a closure and not deal with creating a new custom guard class.

The `viaRequest()` auth method makes it possible to define a guard (named in the first parameter) using just a closure (defined in the second parameter) that takes the HTTP request and returns the appropriate user. To register a closure request guard, call `viaRequest()` in the `boot()` method of your `AuthServiceServiceProvider`, as shown in [Example 9-11](#).

Example 9-11. Defining a closure request guard

```
public function boot(): void
{
    Auth::viaRequest('token-hash', function ($request) {
        return User::where('token-hash', $request->token)->first();
    });
}
```

Creating a Custom User Provider

Just below where guards are defined in `config/auth.php`, there's an `auth.providers` section that defines the available providers. Let's create a new provider named `trainees`:

```
'providers' => [
    'users' => [
        'driver' => 'eloquent',
        'model' => App\User::class,
```

```

],
    'trainees' => [
        'driver' => 'eloquent',
        'model' => App\Trainee::class,
    ],
],

```

The two options for `driver` are `eloquent` and `database`. If you use `eloquent`, you'll need a `model` property that contains an Eloquent class name (the model to use for your `User` class); and if you use `database`, you'll need a `table` property to define which table it should authenticate against.

In our example, you can see that this application has a `User` and a `Trainee`, and they need to be authenticated separately. This way, the code can differentiate between `auth()->guard('users')` and `auth()->guard('trainees')`.

One last note: the `auth` route middleware can take a parameter that is the guard name. So, you can guard certain routes with a specific guard:

```

Route::middleware('auth:trainees')->group(function () {
    // Trainee-only routes here
});

```

Custom User Providers for Nonrelational Databases

The user provider creation flow just described still relies on the same `UserProvider` class, which means it's expecting to pull the identifying information out of a relational database. But if you're using Mongo or Riak or something similar, you'll actually need to create your own class.

To do this, create a new class that implements the `Illuminate\Contracts\Auth\UserProvider` interface, and then bind it in `AuthServiceProvider@boot`:

```

auth()->provider('riak', function ($app, array $config) {
    // Return an instance of Illuminate\Contracts\Auth\UserProvider...
    return new RiakUserProvider($app['riak.connection']);
});

```

Auth Events

We'll talk more about events in [Chapter 16](#), but Laravel's event system is a basic pub/sub framework. There are system- and user-generated events that are broadcast, and the user has the ability to create event listeners that do certain things in response to certain events.

So, what if you wanted to send a ping to a particular security service every time a user was locked out after too many failed login attempts? Maybe this service watches for a certain number of failed logins from certain geographic regions or something else. You could, of course, inject a call in the appropriate controller. But with events, you can just create an event listener that listens to the “user locked out” event, and register that.

Take a look at [Example 9-12](#) to see all of the events that the authentication system emits.

Example 9-12. Authentication events generated by the framework

```
protected $listen = [
    'Illuminate\Auth\Events\Attempting' => [],
    'Illuminate\Auth\Events\Authenticated' => [],
    'Illuminate\Auth\Events\CurrentDeviceLogout' => [],
    'Illuminate\Auth\Events\Failed' => [],
    'Illuminate\Auth\Events\Lockout' => [],
    'Illuminate\Auth\Events>Login' => [],
    'Illuminate\Auth\Events\Logout' => [],
    'Illuminate\Auth\Events\OtherDeviceLogout' => [],
    'Illuminate\Auth\Events>PasswordReset' => [],
    'Illuminate\Auth\Events\Registered' => [],
    'Illuminate\Auth\Events\Validated' => [],
    'Illuminate\Auth\Events\Verified' => [],
];
```

As you can see, there are listeners for “user registered,” “user attempting login,” “user validated but not logged in,” “user authenticated,” “successful login,” “failed login,” “logout,” “logout from another device,” “logout from current device,” “lockout,” “password reset,” and “user email verified.” To learn more about how to build event listeners for these events, check out [Chapter 16](#).

Authorization and Roles

Finally, let's cover Laravel's authorization system. It enables you to determine whether a user is *authorized* to do a particular thing, which you'll check using a few primary verbs: `can`, `cannot`, `allows`, and `denies`.

Most of this authorization control will be performed using the `Gate` facade, but there are also convenience helpers available in your controllers, on the `User` model, as middleware, and as Blade directives. Take a look at [Example 9-13](#) to get a taste of what we'll be able to do.

Example 9-13. Basic usage of the Gate facade

```
if (Gate::denies('edit-contact', $contact)) {
    abort(403);
}

if (! Gate::allows('create-contact', Contact::class)) {
    abort(403);
}
```

Defining Authorization Rules

The default location for defining authorization rules is in the `boot()` method of the `AuthServiceProvider`, where you'll be calling methods on the `Auth` facade.

An authorization rule is called an *ability* and comprises two things: a string key (e.g., `update-contact`) and a closure that returns a Boolean. [Example 9-14](#) shows an ability for updating a contact.

Example 9-14. Sample ability for updating a contact

```
class AuthServiceProvider extends ServiceProvider
{
    public function boot(): void
    {
        Gate::define('update-contact', function ($user, $contact) {
            return $user->id == $contact->user_id;
        });
    }
}
```

Let's walk through the steps for defining an ability.

First, you want to define a key. In naming this key, you should consider what string makes sense in your code's flow to refer to the ability you're providing to the user. You can see in [Example 9-14](#) that the code uses the convention

`{verb}-{modelName}`: create-contact, update-contact, etc.

Second, you define the closure. The first parameter will be the currently authenticated user, and all parameters after that will be the object(s) you're checking for access to—in this instance, the contact.

So, given those two objects, we can check whether the user is authorized to update this contact. You can write this logic however you want, but in the app we're looking at in [Example 9-14](#), authorization depends on being the creator of the contact row. The closure will return `true` (authorized) if the current user created the contact, and `false` (unauthorized) if not.

Just like with route definitions, you could also use a class and method instead of a closure to resolve this definition:

```
$gate->define('update-contact', 'ContactACLChecker@updateContact');
```

The Gate Facade (and Injecting Gate)

Now that you've defined an ability, it's time to test against it. The simplest way is to use the Gate facade, as in [Example 9-15](#) (or you can inject an instance of `Illuminate\Contracts\Auth\Access\Gate`).

Example 9-15. Basic Gate facade usage

```
if (Gate::allows('update-contact', $contact)) {  
    // Update contact  
}  
  
// or  
if (Gate::denies('update-contact', $contact)) {  
    abort(403);  
}
```

You might also define an ability with multiple parameters—maybe contacts can be in groups, and you want to authorize whether the user has access to add a contact to a group. [Example 9-16](#) shows how to do this.

Example 9-16. Abilities with multiple parameters

```
// Definition  
Gate::define('add-contact-to-group', function ($user, $contact, $group) {  
    return $user->id == $contact->user_id && $user->id == $group->user_id;  
});
```

```
// Usage
if (Gate::denies('add-contact-to-group', [$contact, $group])) {
    abort(403);
}
```

And if you need to check authorization for a user other than the currently authenticated user, try `forUser()`, like in [Example 9-17](#).

Example 9-17. Specifying the user for Gate

```
if (Gate::forUser($user)->denies('create-contact')) {
    abort(403);
}
```

Resource Gates

The most common use for access control lists is to define access to individual “resources” (think an Eloquent model, or something you’re allowing users to administer from their admin panel).

The `resource()` method makes it possible to apply the four most common gates, view, create, update, and delete, to a single resource at once:

```
Gate::resource('photos', 'App\Policies\PhotoPolicy');
```

This is equivalent to defining the following:

```
Gate::define('photos.view', 'App\Policies\PhotoPolicy@view');
Gate::define('photos.create', 'App\Policies\PhotoPolicy@create');
Gate::define('photos.update', 'App\Policies\PhotoPolicy@update');
Gate::define('photos.delete', 'App\Policies\PhotoPolicy@delete');
```

The Authorize Middleware

If you want to authorize entire routes, you can use the Authorize middleware (which has a shortcut of `can`), like in [Example 9-18](#).

Example 9-18. Using the Authorize middleware

```
Route::get('people/create', function () {
    // Create a person
})->middleware('can:create-person');
```

```
Route::get('people/{person}/edit', function () {
    // Edit person
})->middleware('can:edit,person');
```

Here, the {person} parameter (whether it's defined as a string or as a bound route model) will be passed to the ability method as an additional parameter.

The first check in [Example 9-18](#) is a normal ability, but the second is a policy, which we'll talk about in [“Policies”](#).

If you need to check for an action that doesn't require a model instance (for example, create, unlike edit, doesn't get passed an actual route model-bound instance), you can just pass the class name:

```
Route::post('people', function () {
    // Create a person
})->middleware('can:create,App\Person');
```

Controller Authorization

The parent App\Http\Controllers\Controller class in Laravel imports the AuthorizesRequests trait, which provides three methods for authorization: authorize(), authorizeForUser(), and authorizeResource().

authorize() takes an ability key and an object (or array of objects) as parameters, and if the authorization fails, it'll quit the application with a 403 (Unauthorized) status code. That means this feature can turn three lines of authorization code into just one, as you can see in [Example 9-19](#).

Example 9-19. Simplifying controller authorization with authorize()

```
// From this:
public function edit(Contact $contact)
{
    if (Gate::cannot('update-contact', $contact)) {
        abort(403);
    }

    return view('contacts.edit', ['contact' => $contact]);
}

// To this:
public function edit(Contact $contact)
{
```



```

    $this->authorize('update-contact', $contact);

    return view('contacts.edit', ['contact' => $contact]);
}

```

`authorizeForUser()` is the same, but allows you to pass in a `User` object instead of defaulting to the currently authenticated user:

```

$this->authorizeForUser($user, 'update-contact', $contact);

```

`authorizeResource()`, called once in the controller constructor, maps a predefined set of authorization rules to each of the RESTful controller methods in that controller—something like [Example 9-20](#).

Example 9-20. The authorization-to-method mappings of `authorizeResource()`

```

...
class ContactController extends Controller
{
    public function __construct()
    {
        // This call does everything you see in the methods below.
        // If you put this here, you can remove all authorize()
        // calls in the individual resource methods here.
        $this->authorizeResource(Contact::class);
    }

    public function index()
    {
        $this->authorize('viewAny', Contact::class);
    }

    public function create()
    {
        $this->authorize('create', Contact::class);
    }

    public function store(Request $request)
    {
        $this->authorize('create', Contact::class);
    }

    public function show(Contact $contact)
    {
        $this->authorize('view', $contact);
    }
}

```

```

public function edit(Contact $contact)
{
    $this->authorize('update', $contact);
}

public function update(Request $request, Contact $contact)
{
    $this->authorize('update', $contact);
}

public function destroy(Contact $contact)
{
    $this->authorize('delete', $contact);
}
}

```

Checking the User Instance

If you're not in a controller, you're more likely to be checking the capabilities of a specific user than the currently authenticated user. That's already possible with the Gate facade using the `forUser()` method, but sometimes the syntax can feel a little off.

Thankfully, the `Authorizable` trait on the `User` class provides four methods to make a more readable authorization feature: `$user->can()`, `$user->canAny()`, `$user->cant()`, and `$user->cannot()`. As you can probably guess, `cant()` and `cannot()` do the same thing, and `can()` is their exact inverse. With `canAny()`, you pass an array of permissions, and this method checks if the user can do any of them.

That means you can do something like [Example 9-21](#).

Example 9-21. Checking authorization on a User instance

```

$user = User::find(1);

if ($user->can('create-contact')) {
    // Do something
}

```

Behind the scenes, these methods are just passing your parameters to Gate; in the preceding example, `Gate::forUser($user)->check('create-contact')`.

Blade Checks

Blade also has a little convenience helper: the `@can` directive. [Example 9-22](#) illustrates its usage.

Example 9-22. Using Blade's @can directive

```
<nav>
  <a href="/">Home</a>
  @can('edit-contact', $contact)
    <a href="{{ route('contacts.edit', [$contact->id]) }}">Edit This Contact</a>
  @endcan
</nav>
```

You can also use `@else` in between `@can` and `@endcan`, and you can use `@cannot` and `@endcannot` as in [Example 9-23](#).

Example 9-23. Using Blade's @cannot directive

```
<h1>{{ $contact->name }}</h1>
@cannot('edit-contact', $contact)
  LOCKED
@endcannot
```

Intercepting Checks

If you've ever built an app with an admin user class, you've probably looked at all of the simple authorization closures so far in this chapter and thought about how you could add a superuser class that overrides these checks in every case. Thankfully, there's already a tool for that.

In `AuthServiceProvider`, where you're already defining your abilities, you can also add a `before()` check that runs before all the others and can optionally override them, like in [Example 9-24](#).

Example 9-24. Overriding Gate checks with before()

```
Gate::before(function ($user, $ability) {
    if ($user->isOwner()) {
        return true;
    }
});
```

Note that the string name for the ability is also passed in, so you can differentiate your `before()` hooks based on your ability naming scheme.

Policies

Up until this point, all of the access controls have required you to manually associate Eloquent models with the ability names. You could have created an ability named something like `visit-dashboard` that's not related to a specific Eloquent model, but you'll probably have noticed that most of our examples have had to do with *doing something to something*—and in most of these cases, the *something* that's the recipient of the action is an Eloquent model.

Authorization policies are organizational structures that help you group your authorization logic based on the resource you're controlling access to. They make it easy to manage defining authorization rules for behavior toward a particular Eloquent model (or other PHP class) all together in a single location.

Generating policies

Policies are PHP classes, which can be generated with an Artisan command:

```
php artisan make:policy ContactPolicy
```

Once they're generated, they need to be registered. The `AuthServiceProvider` has a `$policies` property, which is an array. The key of each item is the class name of the protected resource (almost always an Eloquent class), and the value is the policy class name. [Example 9-25](#) shows what this will look like.

Example 9-25. Registering policies in AuthServiceProvider

```
class AuthServiceProvider extends ServiceProvider
{
    protected $policies = [
        Contact::class => ContactPolicy::class,
    ];
}
```

A policy class that's generated by Artisan doesn't have any special properties or methods. But every method that you add is now mapped as an ability key for this object.

POLICY AUTO-DISCOVERY

Laravel tries to “guess” the links between your policies and their corresponding models. For example, it'll apply the `PostPolicy` to your `Post` model automatically.

If you need to customize the logic Laravel uses to guess this mapping, check out the [Policy docs](#).

Let's define an `update()` method to take a look at how it works ([Example 9-26](#)).

Example 9-26. A sample `update()` policy method

```
<?php

namespace App\Policies;

class ContactPolicy
{
    public function update($user, $contact)
    {
        return $user->id == $contact->user_id;
    }
}
```

Notice that the contents of this method look exactly like they would in a Gate definition.

POLICY METHODS THAT DON'T TAKE AN INSTANCE

What if you want to define a policy method that relates to the class but not a specific instance—for example, “can this user create contacts at all?” rather than just “can this user view this specific contact?” You can treat this just like a normal policy method:

```
...
class ContactPolicy
{
    public function create($user)
    {
        return $user->canCreateContacts();
    }
}
```

Checking policies

If there's a policy defined for a resource type, the Gate facade will use the first parameter to figure out which method to check on the policy. If you run `Gate::allows('update', $contact)`, it will check the

ContactPolicy@update method for authorization.

This also works for the Authorize middleware and for User model checking and Blade checking, as seen in [Example 9-27](#).

Example 9-27. Checking authorization against a policy

```
// Gate
if (Gate::denies('update', $contact)) {
    abort(403);
}

// Gate if you don't have an explicit instance
if (! Gate::check('create', Contact::class)) {
    abort(403);
}

// User
if ($user->can('update', $contact)) {
    // Do stuff
}

// Blade
@can('update', $contact)
    // Show stuff
@endcan
```

Additionally, there's a policy() helper that allows you to retrieve a policy class and run its methods:

```
if (policy($contact)->update($user, $contact)) {
    // Do stuff
}
```

Overriding policies

Just like with normal ability definitions, policies can define a before() method that allows you to override any call before it's even processed (see [Example 9-28](#)).

Example 9-28. Overriding policies with the before() method

```
public function before($user, $ability)
{
    if ($user->isAdmin()) {
        return true;
    }
}
```

```
}
```

Testing

Application tests often need to perform a particular behavior on behalf of a particular user. We therefore need to be able to authenticate as a user in application tests, and we need to test authorization rules and authentication routes.

Of course, it's possible to write an application test that manually visits the login page and then fills out the form and submits it, but that's not necessary. Instead, the simplest option is to use the `->be()` method to simulate being logged in as a user. Take a look at [Example 9-29](#).

Example 9-29. Authenticating as a user in application tests

```
public function test_it_creates_a_new_contact()
{
    $user = User::factory()->create();
    $this->be($user);

    $this->post('contacts', [
        'email' => 'my@email.com',
    ]);

    $this->assertDatabaseHas('contacts', [
        'email' => 'my@email.com',
        'user_id' => $user->id,
    ]);
}
```

You can also use, and chain, the `actingAs()` method instead of `be()`, if you prefer how it reads:

```
public function test_it_creates_a_new_contact()
{
    $user = User::factory()->create();

    $this->actingAs($user)->post('contacts', [
        'email' => 'my@email.com',
    ]);

    $this->assertDatabaseHas('contacts', [
        'email' => 'my@email.com',
        'user_id' => $user->id,
    ]);
}
```

```
    });  
  }
```

We can also test authorization like in [Example 9-30](#).

Example 9-30. Testing authorization rules

```
public function test_non_admins_cant_create_users()  
{  
    $user = User::factory()->create([  
        'admin' => false,  
    ]);  
    $this->be($user);  
  
    $this->post('users', ['email' => 'my@email.com']);  
  
    $this->assertDatabaseMissing('users', [  
        'email' => 'my@email.com',  
    ]);  
}
```

Or we can test for a 403 response like in [Example 9-31](#).

Example 9-31. Testing authorization rules by checking status code

```
public function test_non_admins_cant_create_users()  
{  
    $user = User::factory()->create([  
        'admin' => false,  
    ]);  
    $this->be($user);  
  
    $response = $this->post('users', ['email' => 'my@email.com']);  
  
    $response->assertStatus(403);  
}
```

We need to test that our authentication (signup and signin) routes work too, as illustrated in [Example 9-32](#).

Example 9-32. Testing authentication routes

```
public function test_users_can_register()  
{  
    $this->post('register', [  
        'name' => 'Sal Leibowitz',  
        'email' => 'sal@leibs.net',  
        'password' => 'abcdefg123',  
        'password_confirmation' => 'abcdefg123',  
    ]);  
}
```



```

        $this->assertDatabaseHas('users', [
            'name' => 'Sal Leibowitz',
            'email' => 'sal@leibs.net',
        ]);
    }

    public function test_users_can_log_in()
    {
        $user = User::factory()->create([
            'password' => Hash::make('abcdefg123')
        ]);

        $this->post('login', [
            'email' => $user->email,
            'password' => 'abcdefg123',
        ]);

        $this->assertTrue(auth()->check());
        $this->assertTrue($user->is(auth()->user()));
    }

```

We can also use the integration test features to direct the test to “click” our authentication fields and “submit” the fields to test the entire flow. We’ll talk about that more in [Chapter 12](#).

TL;DR

Between the default User model, the `create_users_table` migration, and Jetstream and Breeze, Laravel provides options for a full user authentication system out of the box. Breeze handles the authentication functionality in controllers and Jetstream handles it in Actions, both of which can be customized for each app. Both tools also publish config files and templates for customization.

The Auth facade and the `auth()` global helper provide access to the current user (`auth()->user()`) and make it easy to check whether a user is logged in (`auth()->check()` and `auth()->guest()`).

Laravel also has an authorization system built in that allows you to define specific abilities (`create-contact`, `visit-secret-page`) or define policies for user interaction with entire models.

You can check for authorization with the Gate facade, the `can()` and `cannot()` methods on the `User` class, the `@can` and `@cannot` directives in Blade, the `authorize()` methods on the controller, or the `can` middleware.

Chapter 10. Requests, Responses, and Middleware

We've already talked a bit about the Illuminate Request object. In [Chapter 3](#), for example, you saw how you can typehint it in constructors to get an instance or use the `request()` helper to retrieve it, and in [Chapter 7](#) we looked at how you can use it to get information about the user's input.

In this chapter, you'll learn more about what the Request object is, how it's generated, what it represents, and what part it plays in your application's lifecycle. We'll also talk about the Response object and Laravel's implementation of the middleware pattern.

Laravel's Request Lifecycle

Every request coming into a Laravel application, whether generated by an HTTP request or a command-line interaction, is immediately converted into an Illuminate Request object, which then crosses many layers and ends up being parsed by the application itself. The application then generates an Illuminate Response object, which is sent back out across those layers and finally returned to the end user.

This request/response lifecycle is illustrated in [Figure 10-1](#). Let's take a look at what it takes to make each of these steps happen, from the first line of code to the last.

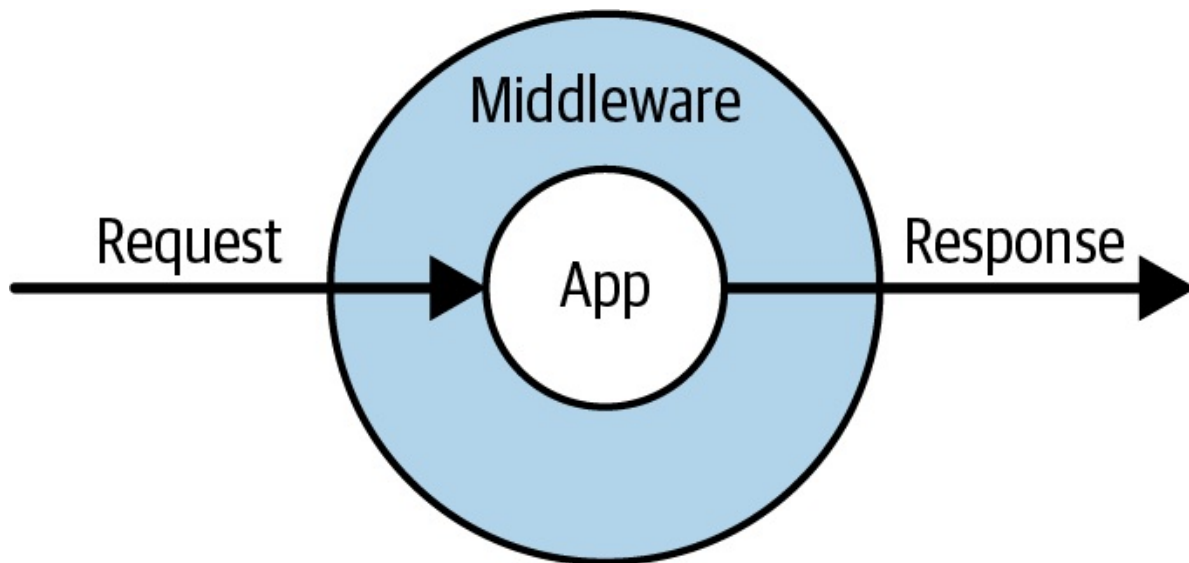


Figure 10-1. Request/response lifecycle

Bootstrapping the Application

Every Laravel application has some form of configuration set up at the web server level, in an Apache `.htaccess` file or an Nginx configuration setting or something similar, that captures every web request regardless of URL and routes it to `public/index.php` in the Laravel application directory.

`index.php` doesn't actually have that much code in it. It has three primary functions.

First, it loads Composer's autoload file, which registers all of the Composer-loaded dependencies.

COMPOSER AND LARAVEL

Laravel's core functionality is separated into a series of components under the `Illuminate` namespace, which are all pulled into each Laravel app using Composer. Laravel also pulls in quite a few packages from Symfony and several other community-developed packages. In this way, Laravel is just as much an opinionated collection of components as it is a framework.

Next, it kicks off Laravel's bootstrap, creating the application container (you'll learn more about the container in [Chapter 11](#)) and registering a few core services (including the kernel, which we'll talk about in just a bit).

Finally, it creates an instance of the kernel, creates a request representing the current user's web request, and passes the request to the kernel to handle. The kernel responds with an `Illuminate Response` object, which *index.php* returns to the end user. Then, the kernel terminates the page request.

The kernel is the core router of every Laravel application, responsible for taking in a user request, processing it through middleware, handling exceptions and passing it to the page router, and then returning the final response. Actually, there are two kernels, but only one is used for each page request. One of the routers handles web requests (the HTTP kernel), and the other handles console, cron, and Artisan requests (the console kernel). Each has a `handle()` method that's responsible for taking in an `Illuminate Request` object and returning an `Illuminate Response` object.

The kernel runs all of the bootstraps that need to run before every request, including determining which environment the current request is running in (staging, local, production, etc.) and running all of the service providers. The HTTP kernel additionally defines the list of middleware that will wrap each request, including the core middleware responsible for sessions and CSRF protection.

Service Providers

Although there's a bit of procedural code in these bootstraps, almost all of Laravel's bootstrap code is separated into something Laravel calls *service providers*. A service provider is a class that encapsulates logic that various parts of your application need to run to bootstrap their core functionality.

For example, there's an `AuthServiceProvider` that bootstraps all of the registrations necessary for Laravel's authentication system and a `RouteServiceProvider` that bootstraps the routing system.

The concept of service providers can be a little hard to understand at first, so think about it this way: many components of your application have bootstrap code that needs to run when the application initializes. Service providers are a tool for grouping that bootstrap code into related classes. If you have any code that needs to run *in preparation* for your application code to work, it's a strong candidate for a service provider.

For example, if you ever find that the feature you’re working on requires some classes registered in the container (you’ll learn more about this in [Chapter 11](#)), you would create a service provider just for that piece of functionality. You might have a `GitHubServiceProvider` or a `MailerServiceProvider`.

Service providers have two important methods: `boot()` and `register()`. There’s also a `DeferrableProvider` interface that you might choose to use. Here’s how they work.

First, all of the service providers’ `register()` methods are called. This is where you’ll want to bind classes and aliases to the container. You don’t want to do anything in `register()` that relies on the entire application being bootstrapped.

Second, all of the service providers’ `boot()` methods are called. You can now do any other bootstrapping here, like binding event listeners or defining routes—anything that may rely on the entire Laravel application having been bootstrapped.

If your service provider is only going to register bindings in the container (i.e., teach the container how to resolve a given class or interface), but not perform any other bootstrapping, you can “defer” its registrations, which means they won’t run unless one of their bindings is explicitly requested from the container. This can speed up your application’s average time to bootstrap.

If you want to defer your service provider’s registrations, first implement the `Illuminate\Contracts\Support\DeferrableProvider` interface; then, give the service provider a `provides()` method that returns a list of bindings the provider provides, as shown in [Example 10-1](#).

Example 10-1. Deferring the registration of a service provider

```
...
use Illuminate\Contracts\Support\DeferrableProvider;

class GitHubServiceProvider extends ServiceProvider implements DeferrableProvider
{
    public function provides()
    {
        return [
            GitHubClient::class,
        ];
    }
}
```

MORE USES FOR SERVICE PROVIDERS

Service providers also have a suite of methods and configuration options that can provide advanced functionality to the end user when the provider is published as part of a Composer package. Take a look at the service provider definition in the [Laravel source](#) to learn more about how this can work.

Now that we've covered the application bootstrap, let's take a look at the `Request` object, the most important output of the bootstrap.

The Request Object

The `Illuminate\Http\Request` class is a Laravel-specific extension of Symfony's `HttpFoundation\Request` class.

SYMFONY HTTPFOUNDATION

Symfony's `HttpFoundation` suite of classes powers almost every PHP framework in existence at this point; this is the most popular and powerful set of abstractions available in PHP for representing HTTP requests, responses, headers, cookies, and more.

The `Request` object is intended to represent every relevant piece of information you might care to know about a user's HTTP request.

In native PHP code, you might find yourself looking to `$_SERVER`, `$_GET`, `$_POST`, and other combinations of globals and processing logic to get information about the current user's request. What files has the user uploaded? What's their IP address? What fields did they post? All of this is sprinkled around the language—and your code—in a way that's hard to understand and harder to mock.

Symfony's `Request` object instead collects all of the information necessary to represent a single HTTP request into a single object, and then tacks on convenience methods to make it easy to get useful information from it. The `Illuminate Request` object adds even more convenience methods to get

information about the request it's representing.

CAPTURING A REQUEST

You'll very likely never need to do this in a Laravel app, but if you ever need to capture your own `Illuminate Request` object directly from PHP's globals, you can use the `capture()` method:

```
$request = Illuminate\Http\Request::capture();
```

Getting a Request Object in Laravel

Laravel creates an internal `Request` object for each request, and there are a few ways you can get access to it.

First—and again, we'll cover this more in [Chapter 11](#)—you can typehint the class in any constructor or method that's resolved by the container. That means you can typehint it in a controller method or a service provider, as seen in [Example 10-2](#).

Example 10-2. Typehinting in a container-resolved method to receive a Request object

```
...
use Illuminate\Http\Request;

class PersonController extends Controller
{
    public function index(Request $request)
    {
        $allInput = $request->all();
    }
}
```

Alternatively, you can use the `request()` global helper, which allows you to call methods on it (e.g., `request()->input()`) and also allows you to call it on its own to get an instance of `$request`:

```
$request = request();
$allInput = $request->all();
// or
$allInput = request()->all();
```


Finally, you can use the `app()` global method to get an instance of `Request`. You can pass either the fully qualified class name or the shortcut `request`:

```
$request = app(Illuminate\Http\Request::class);  
$request = app('request');
```

Getting Basic Information About a Request

Now that you know how to get an instance of `Request`, what can you do with it? The primary purpose of the `Request` object is to represent the current HTTP request, so the primary functionality the `Request` class offers is to make it easy to get useful information about the current request.

I’ve categorized the methods described here, but note that there’s certainly overlap between the categories, and the categories are a bit arbitrary—for example, query parameters could just as easily be in “User and request state” as they are in “Basic user input.” Hopefully these categories will make it easy for you to learn what’s available, and then you can throw away the categories.

Also, be aware that there are many more methods available on the `Request` object; these are just the most commonly used methods.

Basic user input

The basic user input methods make it simple to get information that the users themselves explicitly provide—likely through submitting a form or an Ajax component. When I reference “user-provided input” here, I’m talking about input from query strings (GET), form submissions (POST), or JSON. The basic user input methods include the following:

`all()`

Returns an array of all user-provided input.

`input(fieldName)`

Returns the value of a single user-provided input field.

`only(fieldName | [array, of, field, names])`

Returns an array of all user-provided input for the specified field name(s).

`except(fieldName | [array, of, field, names])`

Returns an array of all user-provided input except for the specified field name(s).

`exists(fieldName)`

Returns a Boolean indicating whether the field exists in the input. `has()` is an alias. Executes the given callback when the field exists in the input.

`filled(fieldName)`

Returns a Boolean indicating whether the field exists in the input and is not empty (that is, has a value).

`whenFilled()`

Executes the given callback when the field exists in the input and is not empty (that is, has a value).

`json()`

Returns a `ParameterBag` if the page had JSON sent to it.

`boolean(fieldName)`

Returns value from the input as a Boolean. Converts strings and integers to appropriate booleans (using `FILTER_VALIDATE_BOOLEAN`). If the key is not present in the request, returns `false`.

`json(keyName)`

Returns the value of the given key from the JSON sent to the page.

PARAMETERBAG

Sometimes in Laravel you'll run into a `ParameterBag` object. This class is sort of like an associative array. You can get a particular key using `get()`:

```
echo $bag->get('name');
```

You can also use `has()` to check for the existence of a key, `all()` to get an array of all keys and values, `count()` to count the number of items, and `keys()` to get an array of just the keys.

Example 10-3 gives a few quick examples of how to use the user-provided information methods from a request.

Example 10-3. Getting basic user-provided information from the request

```
// form
<form method="POST" action="/form">
  @csrf
  <input name="name"> Name<br>
  <input type="submit">
</form>
// Route receiving the form
Route::post('form', function (Request $request) {
    echo 'name is ' . $request->input('name') . '<br>';
    echo 'all input is ' . print_r($request->all()) . '<br>';
    echo 'user provided email address: ' . $request->has('email') ? 'true' : 'false';
});
```

User and request state

The user and request state methods include input that wasn't explicitly provided by the user through a form:

`method()`

Returns the method (GET, POST, PATCH, etc.) used to access this route.

`path()`

Returns the path (without the domain) used to access this page; for example, 'http://www.myapp.com/abc/def' would return 'abc/def'.

`url()`

Returns the URL (with the domain) used to access this page; for example, 'abc' would return 'http://www.myapp.com/abc'.

`is()`

Returns a Boolean indicating whether or not the current page request fuzzy-

matches a provided string (e.g., `/a/b/c` would be matched by `$request->is('*b*')`, where `*` stands for any characters); uses a custom regex parser found in `Str::is()`.

`ip()`

Returns the user's IP address.

`header()`

Returns an array of headers (e.g., `['accept-language' => ['en-US,en;q=0.8']]`), or, if passed a header name as a parameter, returns just that header.

`server()`

Returns an array of the variables traditionally stored in `$_SERVER` (e.g., `REMOTE_ADDR`), or, if passed a `$_SERVER` variable name, returns just that value.

`secure()`

Returns a Boolean indicating whether this page was loaded using HTTPS.

`pjax()`

Returns a Boolean indicating whether this page request was loaded using Pjax.

`wantsJson()`

Returns a Boolean indicating whether this request has any `/json` content types in its `Accept` headers.

`isJson()`

Returns a Boolean indicating whether this page request has any `/json` content types in its `Content-Type` header.

`accepts()`

Returns a Boolean indicating whether this page request accepts a given content type.

Files

So far, all of the input we've covered is either explicit (retrieved by methods like `all()`, `input()`, etc.) or defined by the browser or referring site (retrieved by methods like `ajax()`). File inputs are similar to explicit user input, but they're handled much differently:

`file()`

Returns an array of all uploaded files, or, if a key is passed (the file upload field name), returns just the one file.

`allFiles()`

Returns an array of all uploaded files; useful as opposed to `file()` because of clearer naming.

`hasFile()`

Returns a Boolean indicating whether a file was uploaded at the specified key.

Every file that's uploaded will be an instance of `Symfony\Component\HttpFoundation\File\UploadedFile`, which provides a suite of tools for validating, processing, and storing uploaded files.

Take a look at [Chapter 14](#) for more examples of how to handle uploaded files.

Persistence

The request can also provide functionality for interacting with the session. Most session functionality lives elsewhere, but there are a few methods that are particularly relevant to the current page request:

`flash()`

Flashes the current request's user input to the session to be retrieved later, which means it's saved to the session but disappears after the next request.

`flashOnly()`

Flashes the current request's user input for any keys in the provided array.

`flashExcept()`

Flashes the current request's user input, except for any keys in the provided array.

`old()`

Returns an array of all previously flashed user input, or, if passed a key, returns the value for that key if it was previously flashed.

`flush()`

Wipes all previously flashed user input.

`cookie()`

Retrieves all cookies from the request, or, if a key is provided, retrieves just that cookie.

`hasCookie()`

Returns a Boolean indicating whether the request has a cookie for the given key.

The `flash*()` and `old()` methods are used for storing user input and retrieving it later, often after the input is validated and rejected.

The Response Object

Similar to the `Request` object, there's an `Illuminate Response` object that represents the response your application is sending to the end user, complete with headers, cookies, content, and anything else used for sending the end user's browser instructions on rendering a page.

Just like `Request`, the `Illuminate\Http\Response` class extends a `Symfony`

class: `Symfony\Component\HttpFoundation\Response`. This is a base class with a series of properties and methods that makes it possible to represent and render a response; Illuminate's `Response` class decorates it with a few helpful shortcuts.

Using and Creating Response Objects in Controllers

Before we talk about how you can customize your Response objects, let's step back and see how we most commonly work with Response objects.

In the end, any Response object returned from a route definition will be converted into an HTTP response. It may define specific headers or specific content, set cookies, or whatever else, but eventually it will be converted into a response your users' browsers can parse.

Let's take a look at the simplest possible response, in [Example 10-4](#).

Example 10-4. Simplest possible HTTP response

```
Route::get('route', function () {
    return new Illuminate\Http\Response('Hello!');
});

// Same, using global function:
Route::get('route', function () {
    return response('Hello!');
});
```

We create a response, give it some core data, and then return it. We can also customize the HTTP status, headers, cookies, and more, like in [Example 10-5](#).

Example 10-5. Simple HTTP response with customized status and headers

```
Route::get('route', function () {
    return response('Error!', 400)
        ->header('X-Header-Name', 'header-value')
        ->cookie('cookie-name', 'cookie-value');
});
```

Setting headers

We define a header on a response by using the `header()` fluent method, like in [Example 10-5](#). The first parameter is the header name, and the second is the header value.

Adding cookies

We can also set cookies directly on the `Response` object if we'd like. We'll cover Laravel's cookie handling a bit more in [Chapter 14](#), but take a look at [Example 10-6](#) for a simple use case for attaching cookies to a response.

Example 10-6. Attaching a cookie to a response

```
return response($content)
    ->cookie('signup_dismissed', true);
```

Specialized Response Types

There are also a few special response types for views, downloads, files, and JSON. Each is a predefined macro that makes it easy to reuse particular templates for headers or content structure.

View responses

In [Chapter 3](#), I used the global `view()` helper to show how to return a template—for example, `view('view.name.here')` or something similar. But if you need to customize the headers, HTTP status, or anything else when returning a view, you can use the `view()` response type as shown in [Example 10-7](#).

Example 10-7. Using the `view()` response type

```
Route::get('/', function (XmlGetterService $xml) {
    $data = $xml->get();
    return response()
        ->view('xml-structure', $data)
        ->header('Content-Type', 'text/xml');
});
```

Download responses

Sometimes you want your application to force the user's browser to download a file, whether you're creating the file in Laravel or serving it from a database or a protected location. The `download()` response type makes this simple.

The required first parameter is the path for the file you want the browser to download. If it's a generated file, you'll need to save it somewhere temporarily.

The optional second parameter is the filename for the downloaded file (e.g., *export.csv*). If you don't pass a string here, the name will be generated

automatically. The optional third parameter allows you to pass an array of headers. [Example 10-8](#) illustrates the use of the `download()` response type.

Example 10-8. Using the `download()` response type

```
public function export()
{
    return response()
        ->download('file.csv', 'export.csv', ['header' => 'value']);
}

public function otherExport()
{
    return response()->download('file.pdf');
}
```

If you wish to delete the original file from the disk after returning a download response, you can chain the `deleteFileAfterSend()` method after the `download()` method:

```
public function export()
{
    return response()
        ->download('file.csv', 'export.csv')
        ->deleteFileAfterSend();
}
```

File responses

The file response is similar to the download response, except it allows the browser to display the file instead of forcing a download. This is most common with images and PDFs.

The required first parameter is the filename, and the optional second parameter can be an array of headers (see [Example 10-9](#)).

Example 10-9. Using the `file()` response type

```
public function invoice($id)
{
    return response()->file("./invoices/{ $id }.pdf", ['header' => 'value']);
}
```

JSON responses

JSON responses are so common that, even though they're not really particularly

complex to program, there's a custom response for them as well.

JSON responses convert the passed data to JSON (with `json_encode()`) and set the Content-Type to `application/json`. You can also optionally use the `setCallback()` method to create a JSONP response instead of JSON, as seen in [Example 10-10](#).

Example 10-10. Using the `json()` response type

```
public function contacts()
{
    return response()->json(Contact::all());
}

public function jsonpContacts(Request $request)
{
    return response()
        ->json(Contact::all())
        ->setCallback($request->input('callback'));
}

public function nonEloquentContacts()
{
    return response()->json(['Tom', 'Jerry']);
}
```

Redirect responses

Redirects aren't commonly called on the `response()` helper, so they're a bit different from the other custom response types we've discussed already, but they're still just another sort of response. Redirects, returned from a Laravel route, send the user a redirect (often a 301) to another page or back to the previous page.

You technically *can* call a redirect from `response()`, as in `return response()->redirectTo('/')`. But, more commonly, you'll use the redirect-specific global helpers.

There is a global `redirect()` function that can be used to create redirect responses and a global `back()` function that is a shortcut to `redirect()->back()`.

Just like most global helpers, the `redirect()` global function can either be passed parameters or be used to get an instance of its class that you then chain

method calls onto. If you don't chain, but just pass parameters, `redirect()` performs the same as `redirect()->to()`; it takes a string and redirects to that string URL. **Example 10-11** shows some examples of its use.

Example 10-11. Examples of using the `redirect()` global helper

```
return redirect('account/payment');
return redirect()->to('account/payment');
return redirect()->route('account.payment');
return redirect()->action('AccountController@showPayment');

// If redirecting to an external domain
return redirect()->away('https://tighten.co');

// If named route or controller needs parameters
return redirect()->route('contacts.edit', ['id' => 15]);
return redirect()->action('ContactController@edit', ['id' => 15]);
```

You can also redirect “back” to the previous page, which is especially useful when handling and validating user input. **Example 10-12** shows a common pattern in validation contexts.

Example 10-12. Redirect back with input

```
public function store()
{
    // If validation fails...
    return back()->withInput();
}
```

Finally, you can redirect and flash data to the session at the same time. This is common with error and success messages, like in **Example 10-13**.

Example 10-13. Redirect with flashed data

```
Route::post('contacts', function () {
    // Store the contact

    return redirect('dashboard')->with('message', 'Contact created!');
});

Route::get('dashboard', function () {
    // Get the flashed data from session--usually handled in Blade template
    echo session('message');
});
```

Custom response macros

You can also create your own custom response types using *macros*. This allows you to define a series of modifications to make to the response and its provided content.

Let's re-create the `json()` custom response type, just to see how it works. As always, you should probably create a custom service provider for these sorts of bindings, but for now we'll just put it in `AppServiceProvider`, as seen in [Example 10-14](#).

Example 10-14. Creating a custom response macro

```
...
class AppServiceProvider
{
    public function boot()
    {
        Response::macro('myJson', function ($content) {
            return response(json_encode($content))
                ->withHeaders(['Content-Type' => 'application/json']);
        });
    }
}
```

Then, we can use it just like we would use the predefined `json()` macro:

```
return response()->myJson(['name' => 'Sangeetha']);
```

This will return a response with the body of that array encoded for JSON, with the JSON-appropriate `Content-Type` header.

The responsible interface

If you'd like to customize how you're sending responses and a macro doesn't offer enough space or enough organization, or if you want any of your objects to be capable of being returned as a "response" with their own logic of how to be displayed, the `Responsible` interface is for you.

The `Responsible` interface, `Illuminate\Contracts\Support\Responsible`, dictates that its implementors must have a `toResponse()` method. This needs to return an `Illuminate Response` object. [Example 10-15](#) illustrates how to create a `Responsible` object.

Example 10-15. Creating a simple Responsible object

```

...
use Illuminate\Contracts\Support\Responsable;

class MyJson implements Responsable
{
    public function __construct($content)
    {
        $this->content = $content;
    }

    public function toResponse()
    {
        return response(json_encode($this->content))
            ->withHeaders(['Content-Type' => 'application/json']);
    }
}

```

Then, we can use it just like our custom macro:

```
return new MyJson(['name' => 'Sangeetha']);
```

This probably looks like a lot of work relative to the response macros we covered earlier. But the `Responsable` interface really shines when you're working with more complicated controller manipulations. One common example is to use it to create view models (or view objects), like in [Example 10-16](#).

Example 10-16. Using `Responsable` to create a view object

```

...
use Illuminate\Contracts\Support\Responsable;

class GroupDonationDashboard implements Responsable
{
    public function __construct($group)
    {
        $this->group = $group;
    }

    public function budgetThisYear()
    {
        // ...
    }

    public function giftsThisYear()
    {
        // ...
    }
}

```

```

public function toResponse()
{
    return view('groups.dashboard')
        ->with('annual_budget', $this->budgetThisYear())
        ->with('annual_gifts_received', $this->giftsThisYear());
}

```

It starts to make a little bit more sense in this context—move your complex view preparation into a dedicated, *testable* object, and keep your controllers clean. Here’s a controller that uses that Responsible object:

```

...
class GroupController
{
    public function index(Group $group)
    {
        return new GroupDonationsDashboard($group);
    }
}

```

Laravel and Middleware

Take a look back at [Figure 10-1](#), at the start of this chapter.

We’ve covered the requests and responses, but we haven’t actually looked into what middleware are. You may already be familiar with middleware, which aren’t unique to Laravel, but rather a widely used architecture pattern.

An Introduction to Middleware

The idea of middleware is that there is a series of layers wrapping around your application, like a multilayer cake or an onion.¹ Just as shown in [Figure 10-1](#), every request passes through every middleware layer on its way into the application, and then the resulting response passes back through the middleware layers on its way out to the end user.

Middleware are most often considered separate from your application logic, and usually are constructed in a way that should theoretically be applicable to any application, not just the one you’re working on at the moment.

A middleware can inspect a request and decorate it or reject it, based on what it finds. That means middleware are great for something like rate limiting: they can

inspect the IP address, check how many times it's accessed this resource in the last minute, and send back a 429 (Too Many Requests) status if a threshold is passed.

Because middleware also get access to the response on its way out of the application, it's great for decorating responses. For example, Laravel uses a middleware to add all of the queued cookies from a given request/response cycle to the response right before it is sent to the end user.

But some of the most powerful uses of middleware come from the fact that they can be nearly the *first* and the *last* thing to interact with the request/response cycle. That makes middleware perfect for something like enabling sessions—PHP needs you to open the session very early and close it very late, and middleware are also great for this.

Creating Custom Middleware

Let's imagine we want to have a middleware that rejects every request that uses the DELETE HTTP method and also sends a cookie back for every request.

There's an Artisan command to create custom middleware. Let's try it out:

```
php artisan make:middleware BanDeleteMethod
```

You can now open up the file at *app/Http/Middleware/BanDeleteMethod.php*. The default contents are shown in **Example 10-17**.

Example 10-17. Default middleware contents

```
...
class BanDeleteMethod
{
    public function handle($request, Closure $next)
    {
        return $next($request);
    }
}
```

How this `handle()` method represents the processing of both the incoming request *and* the outgoing response is the most difficult thing to understand about middleware, so let's walk through it.

Understanding middleware's `handle()` method

First, remember that middleware are layered, one on top of another, and then finally on top of the app. The first middleware that's registered gets *first* access to a request when it comes in, then that request is passed to every other middleware in turn, then to the app. Then the resulting response is passed outward through the middleware, and finally the first middleware gets *last* access to the response when it goes out.

Let's imagine we've registered `BanDeleteMethod` as the first middleware to run. That means the `$request` coming into it is the raw request, unadulterated by any other middleware. Now what?

Passing that request to `$next()` means handing it off to the rest of the middleware. The `$next()` closure just takes that `$request` and passes it to the `handle()` method of the next middleware in the stack. It then gets passed on down the line until there are no more middleware to hand it to, and it finally ends up at the application.

Next, how does the response come out? This is where it might be hard to follow. The application returns a response, which is passed back up the chain of middleware—because each middleware returns its response. So, within that same `handle()` method, the middleware can decorate a `$request` and pass it to the `$next()` closure, and can then choose to do something with the output it receives before finally returning that output to the end user. Let's look at some pseudocode to make this clearer ([Example 10-18](#)).

Example 10-18. Pseudocode explaining the middleware call process

```
...
class BanDeleteMethod
{
    public function handle($request, Closure $next)
    {
        // At this point, $request is the raw request from the user.
        // Let's do something with it, just for fun.
        if ($request->ip() === '192.168.1.1') {
            return response('BANNED IP ADDRESS!', 403);
        }

        // Now we've decided to accept it. Let's pass it on to the next
        // middleware in the stack. We pass it to $next(), and what is
        // returned is the response after the $request has been passed
```



```

    // down the stack of middleware to the application and the
    // application's response has been passed back up the stack.
    $response = $next($request);

    // At this point, we can once again interact with the response
    // just before it is returned to the user
    $response->cookie('visited-our-site', true);

    // Finally, we can release this response to the end user
    return $response;
}
}

```

Finally, let's make the middleware do what we actually promised (**Example 10-19**).

Example 10-19. Sample middleware banning the DELETE method

```

...
class BanDeleteMethod
{
    public function handle($request, Closure $next)
    {
        // Test for the DELETE method
        if ($request->method() === 'DELETE') {
            return response(
                "Get out of here with that delete method",
                405
            );
        }

        $response = $next($request);

        // Assign cookie
        $response->cookie('visited-our-site', true);

        // Return response
        return $response;
    }
}

```

Binding Middleware

We're not quite done yet. We need to register this middleware in one of two ways: globally or for specific routes.

Global middleware are applied to every route; route middleware are applied on a route-by-route basis.

Binding global middleware

Both bindings happen in *app/Http/Kernel.php*. To add a middleware as global, add its class name to the `$middleware` property, as in [Example 10-20](#).

Example 10-20. Binding global middleware

```
// app/Http/Kernel.php
protected $middleware = [
    \App\Http\Middleware\TrustProxies::class,
    \Illuminate\Foundation\Http\Middleware\CheckForMaintenanceMode::class,
    \App\Http\Middleware\BanDeleteMethod::class,
];
```

Binding route middleware

Middleware intended for specific routes can be added as route middleware or as part of a middleware group. Let's start with the former.

Route middleware are added to the `$middlewareAliases` array in *app/Http/Kernel.php*. It's similar to adding them to `$middleware`, except we have to give each one a key that will be used when applying this middleware to a particular route, as seen in [Example 10-21](#).

Example 10-21. Binding route middleware

```
// app/Http/Kernel.php
protected $middlewareAliases = [
    'auth' => \App\Http\Middleware\Authenticate::class,
    ...
    'ban-delete' => \App\Http\Middleware\BanDeleteMethod::class,
];
```

We can now use this middleware in our route definitions, like in [Example 10-22](#).

Example 10-22. Applying route middleware in route definitions

```
// Doesn't make much sense for our current example...
Route::get('contacts', [ContactController::class, 'index'])->middleware('ban-delete');

// Makes more sense for our current example...
Route::prefix('api')->middleware('ban-delete')->group(function () {
    // All routes related to an API
});
```

Using middleware groups

Middleware groups are essentially prepackaged bundles of middleware that

make sense to be together in specific contexts.

MIDDLEWARE GROUPS IN ROUTES FILES

Every route in *routes/web.php* is in the web middleware group. This *routes/web.php* file is specific for web routes and the *routes/api.php* file for API routes. If you want to add routes in other groups, read on.

Out of the box, there are two groups: **web** and **api**. The **web** group has all the middleware that will be useful on almost every Laravel page request, including middleware for cookies, sessions, and CSRF protection. **api** has none of those—it has a throttling middleware and a route model binding middleware, and that's it. These are all defined in *app/Http/Kernel.php*.

You can apply middleware groups to routes just like you apply route middleware to routes, with the `middleware()` fluent method:

```
use App\Http\Controllers\HomeController;

Route::get('/', [HomeController::class, 'index']);
```

You can also create your own middleware groups and add and remove route middleware to and from preexisting middleware groups. It works just like adding route middleware normally, but you're instead adding them to keyed groups in the `$middlewareGroups` array.

You might be wondering how these middleware groups match up with the two default route files. Unsurprisingly, the *routes/web.php* file is wrapped with the **web** middleware group, and the *routes/api.php* file is wrapped with the **api** middleware group.

The *routes/** files are loaded in the `RouteServiceProvider`. Take a look at the `map()` method there ([Example 10-23](#)) and you'll find a `mapWebRoutes()` method and a `mapApiRoutes()` method, each of which loads its respective files already wrapped in the appropriate middleware group.

Example 10-23. Default route service provider

```
// App\Providers\RouteServiceProvider
```

```

public const HOME = '/home';

// protected $namespace = 'App\\Http\\Controllers';

public function boot(): void
{
    $this->configureRateLimiting();

    $this->routes(function () {
        Route::prefix('api')
            ->middleware('api')
            ->namespace($this->namespace)
            ->group(base_path('routes/api.php'));

        Route::middleware('web')
            ->namespace($this->namespace)
            ->group(base_path('routes/web.php'));
    });
}

protected function configureRateLimiting()
{
    RateLimiter::for('api', function (Request $request) {
        return Limit::perMinute(60)
            ->by(optional($request->user())->id ?: $request->ip());
    });
}

```

As you can see in [Example 10-23](#), we’re using the router to load a route group with the web middleware group, and another under the api middleware group.

Passing Parameters to Middleware

It’s not common, but there are times when you need to pass parameters to a route middleware. For example, you might have an authentication middleware that will act differently depending on whether you’re guarding for the member user type or the owner user type:

```

Route::get('company', function () {
    return view('company.admin');
})->middleware('auth:owner');

```

To make this work, you’ll need to add one or more parameters to the middleware’s `handle()` method and update that method’s logic accordingly, as

shown in [Example 10-24](#).

Example 10-24. Defining a route middleware that accepts parameters

```
public function handle(Request $request, Closure $next, $role): Response
{
    if (auth()->check() && auth()->user()->hasRole($role)) {
        return $next($request);
    }

    return redirect('login');
}
```

Note that you can also add more than one parameter to the `handle()` method and pass multiple parameters to the route definition by separating them with commas:

```
Route::get('company', function () {
    return view('company.admin');
})->middleware('auth:owner,view');
```

FORM REQUEST OBJECTS

In this chapter we covered how to inject an Illuminate Request object, which is the base—and most common—request object.

However, you can also extend the Request object and inject that instead. You’ll learn more about how to bind and inject custom classes in [Chapter 11](#), but there’s one special type, called the form request, that has its own set of behaviors.

See “[Form Requests](#)” to learn more about creating and using form requests.

Default Middleware

Laravel comes with quite a few middleware out of the box. Let’s take a look at each.

Maintenance Mode

Often we need to take our applications offline temporarily to perform some form

of maintenance. Laravel offers tooling for this feature, called “Maintenance Mode,” and there’s a middleware that checks every response to see if the app is in that mode.

You can enable maintenance mode for your app with the `down` Artisan command:

```
php artisan down --refresh=5 --retry=30 --secret="long-password"
```

refresh

Sends a header with the response to refresh the browser after the specified number of seconds.

retry

Sets the `Retry-After` header, with the specified number of seconds. Browsers usually ignore this header.

secret

Sets a password to allow some users to bypass the maintenance mode. To bypass the maintenance mode navigate to your app URL followed by the secret you set (e.g., `app.url/long-password`). This will redirect you to the /app URL, while setting a bypass cookie on your browser, allowing you to access the application normally even while it’s in maintenance mode.

To disable maintenance mode, use the `up` Artisan command:

```
php artisan up
```

Rate Limiting

If you need to limit users such that they can access any given route(s) only a certain number of times in a given time frame (called *rate limiting*, and most common with APIs), there’s an out-of-the-box middleware for that: `throttle`.

Example 10-25 demonstrates its use, using the “api” `RateLimiter` preset Laravel ships with.

Example 10-25. Applying the rate limiting middleware to a route

```
Route::middleware(['auth:api', 'throttle:api'])->group(function () {
    Route::get('/profile', function () {
        //
    });
});
```

You can define as many custom `RateLimiter` configurations as you need; look at the `configureRateLimiting()` method of the `RouteServiceProvider` for the default `api` configuration, and also to create your own.

As you can see in [Example 10-26](#), the default `api` configuration limits requests to 60 per minute, segmented by either the authenticated ID or, if the user is not logged in, the IP address.

Example 10-26. Default rate limiter definition

```
RateLimiter::for('api', function (Request $request) {
    return Limit::perMinute(60)->by($request->user()?->id ?: $request->ip());
});
```

You can also customize the response that's sent if the rate limit is hit, specify different rate limits based on user or application or request conditions, or even specify a stack of rate limiters that are applied in sequence. Take a look at [the rate limiting docs](#) to learn more.

Trusted Proxies

If you use any Laravel tools to generate URLs within the app, you'll notice that Laravel detects whether the current request was via HTTP or HTTPS and generates links using the appropriate protocol.

However, this doesn't always work when you have a proxy (e.g., a load balancer or other web-based proxy) in front of your app. Many proxies send nonstandard headers like `X_FORWARDED_PORT` and `X_FORWARDED_PROTO` to your app, and expect your app to “trust” those, interpret them, and use them as a part of the process of interpreting the HTTP request. To make Laravel correctly treat proxied HTTPS calls like secure calls, and in order for Laravel to process other headers from proxied requests, you need to define how it should do so.

You likely don't want to allow just *any* proxy to send traffic to your app; rather, you want to lock your app to only trust certain proxies, and even from those proxies you may only want to trust certain forwarded headers.

Laravel includes the **TrustedProxy package**, which makes it possible for you to mark certain sources of traffic as “trusted,” and also mark which forwarded headers you want to trust from those sources and how to map them to normal headers.

To configure which proxies your app will trust, you can edit the `App\Http\Middleware\TrustProxies` middleware and add the IP address for your load balancer or proxy to the `$proxies` array, as shown in **Example 10-27**.

Example 10-27. Configuring the TrustProxies middleware

```
/**
 * The trusted proxies for this application.
 *
 * @var array<int, string>|string|null
 */
protected $proxies;

/**
 * The headers that should be used to detect proxies
 *
 * @var int
 */
protected $headers =
    Request::HEADER_X_FORWARDED_FOR |
    Request::HEADER_X_FORWARDED_HOST |
    Request::HEADER_X_FORWARDED_PORT |
    Request::HEADER_X_FORWARDED_PROTO |
    Request::HEADER_X_FORWARDED_AWS_ELB;
```

As you can see, the `$headers` array defaults to trusting all forwarded headers from the trusted proxies; if you want to customize this list, take a look at the **[Symfony docs on trusting proxies](#)**.

CORS

Hopefully you’ve never had a run in with CORS (cross-origin resource sharing). It’s one of those things that we hope always *just works*, and when it doesn’t, it’s painful.

Laravel’s built-in CORS middleware runs by default and can be configured in `config/cors.php`. Its default configuration is reasonable for most apps, but in its config file you can exclude routes from CORS protection, modify the HTTP methods it operates on, and configure how it interacts with CORS headers.

Testing

Outside of the context of you as a developer using requests, responses, and middleware in your own testing, Laravel itself actually uses each quite a bit.

When you're doing application testing with calls like `$this->get('/')`, you're instructing Laravel's application testing framework to generate request objects that represent the interactions you're describing. Then those request objects are passed to your application, as if they were actual visits. That's why the application tests are so accurate: your application doesn't actually "know" that it's not a real user that's interacting with it.

In this context, many of the assertions you're making—say, `assertResponseOk()`—are assertions against the response object generated by the application testing framework. The `assertResponseOk()` method just looks at the response object and asserts that its `isOk()` method returns `true`—which is just checking that its status code is 200. In the end, *everything* in application testing is acting as if this were a real page request.

Find yourself in a context where you need a request to work with in your tests? You can always pull one from the container with `$request = request()`. Or you could create your own—the constructor parameters for the `Request` class, all optional, are as follows:

```
$request = new Illuminate\Http\Request(
    $query,      // GET array
    $request,    // POST array
    $attributes, // "attributes" array; empty is fine
    $cookies,    // Cookies array
    $files,      // Files array
    $server,     // Servers array
    $content     // Raw body data
);
```

If you're really interested in an example, check out the method Symfony uses to create a new `Request` from the globals PHP provides:

`Symfony\Component\HttpFoundation\Request@createFromGlobals()`.

Response objects are even simpler to create manually, if you need to. Here are the (optional) parameters:

```
$response = new Illuminate\Http\Response(  
    $content, // response content  
    $status,  // HTTP status, default 200  
    $headers  // array headers array  
);
```

Finally, if you need to disable your middleware during an application test, import the `WithoutMiddleware` trait into that test. You can also use the `$this->withoutMiddleware()` method to disable middleware just for a single test method.

TL;DR

Every request coming into a Laravel application is converted into an `Illuminate Request` object, which then passes through all the middleware and is processed by the application. The application generates a `Response` object, which is then passed back through all of the middleware (in reverse order) and returned to the end user.

`Request` and `Response` objects are responsible for encapsulating and representing every relevant piece of information about the incoming user request and the outgoing server response.

Service providers gather together related behavior for binding and registering classes for use by the application.

Middleware wrap the application and can reject or decorate any request and response.

¹ Or an ogre.

Chapter 11. The Container

Laravel’s service container, or dependency injection container, sits at the core of almost every other feature. The container is a simple tool you can use to bind and resolve concrete instances of classes and interfaces, and at the same time, it’s a powerful and nuanced manager of a network of interrelated dependencies. In this chapter, you’ll learn more about what it is, how it works, and how you can use it.

NAMING AND THE CONTAINER

You’ll notice in this book, in the documentation, and in other educational sources that there are quite a few names folks use for the container. These include:

- Application container
- IoC (inversion of control) container
- Service container
- DI (dependency injection) container

All are useful and valid, but just know they’re all talking about the same thing. They’re all referring to the service container.

A Quick Introduction to Dependency Injection

Dependency injection means that, rather than being instantiated (“newed up”) within a class, each class’s dependencies will be *injected* in from the outside. This most commonly occurs with *constructor injection*, which means an object’s dependencies are injected when it’s created. But there’s also *setter injection*, where the class exposes a method specifically for injecting a given dependency, and *method injection*, where one or more methods expect their dependencies to be injected when they’re called.

Take a look at **Example 11-1** for a quick example of constructor injection, the most common type of dependency injection.

Example 11-1. Basic dependency injection

```
<?php

class UserMailer
{
    protected $mailer;

    public function __construct(Mailer $mailer)
    {
        $this->mailer = $mailer;
    }

    public function welcome($user)
    {
        return $this->mailer->mail($user->email, 'Welcome!');
    }
}
```

As you can see, this `UserMailer` class expects an object of type `Mailer` to be injected when it's instantiated, and its methods then refer to that instance.

The primary benefits of dependency injection are that it gives us the freedom to change what we're injecting, to mock dependencies for testing, and to instantiate shared dependencies just once for shared use.

INVERSION OF CONTROL

You may have heard the phrase “inversion of control” used in conjunction with “dependency injection,” and sometimes Laravel's container is called the IoC container.

The two concepts are very similar. Inversion of control references the idea that, in traditional programming, the lowest-level code—specific classes, instances, and procedural code—“controls” that instance of a particular pattern or interface to use. For example, if you're instantiating your mailer in each class that needs it, each class gets to decide whether to use Mailgun or Mandrill or Sendgrid.

The idea of inversion of control refers to flipping that “control” to the opposite end of your application. Now the definition of which mailer to use lives at the highest, most abstract level of the application, often in configuration. Every instance, every piece of low-level code, looks up to the

high-level configuration to essentially “ask”: “Can you give me a mailer?” They don’t “know” which mailer they’re getting, just that they’re getting one.

Dependency injection and especially DI containers provide a great opportunity for inversion of control because, for example, you can define once which concrete instance of the `Mailer` interface to provide when injecting mailers into any class that needs them.

Dependency Injection and Laravel

As you saw in [Example 11-1](#), the most common pattern for dependency injection is constructor injection, or injecting the dependencies of an object when it’s instantiated (“constructed”).

Let’s take our `UserMailer` class from [Example 11-1](#). [Example 11-2](#) shows what it might look like to create and use an instance of it.

Example 11-2. Simple manual dependency injection

```
$mailer = new MailgunMailer($mailgunKey, $mailgunSecret, $mailgunOptions);
$userMailer = new UserMailer($mailer);

$userMailer->welcome($user);
```

Now let’s imagine we want our `UserMailer` class to be able to log messages, as well as send a notification to a Slack channel every time it sends a message.

[Example 11-3](#) shows what this would look like. As you can see, it would start to get pretty unwieldy if we had to do all this work every time we wanted to create a new instance—especially when you consider that we’ll have to get all these parameters from somewhere.

Example 11-3. More complex manual dependency injection

```
$mailer = new MailgunMailer($mailgunKey, $mailgunSecret, $mailgunOptions);
$logger = new Logger($logPath, $minimumLogLevel);
$slack = new Slack($slackKey, $slackSecret, $channelName, $channelIcon);
$userMailer = new UserMailer($mailer, $logger, $slack);

$userMailer->welcome($user);
```

Imagine having to write that code every time you wanted a `UserMailer`.

Dependency injection is great, but this is a mess.

The `app()` Global Helper

Before we go too far into how the container actually works, let's take a quick look at the simplest way to get an object out of the container: the `app()` helper.

Pass any string to that helper, whether it's a fully qualified class name (FQCN, like `App\ThingDoer`) or a Laravel shortcut (we'll talk about those more in a second), and it'll return an instance of that class:

```
$logger = app(Logger::class);
```

This is the absolute simplest way to interact with the container. It creates an instance of this class and returns it for you, nice and easy. It's like `new Logger` but, as you'll see shortly, much better.

DIFFERENT SYNTAXES FOR MAKING A CONCRETE INSTANCE

The simplest way to “make” a concrete instance of any class or interface is to use the global helper and pass the class or interface name directly to the helper, using `app('FQCN')`.

However, if you have an instance of the container—whether it was injected somewhere, or you're in a service provider and using `$this->app`, or (a lesser-known trick) you get one just by running `$container = app()`—there are a few ways to make an instance from there.

The most common way is to run the `make()` method. `$app->make('FQCN')` works well. However, you may also see other developers and the documentation use this syntax sometimes: `$app['FQCN']`. Don't worry. That's doing the same thing; it's just a different way of writing it.

Creating the `Logger` instance as shown here seems simple enough, but you might've noticed that our `$logger` class in [Example 11-3](#) has two parameters: `$logPath` and `$minimumLogLevel`. How does the container know what to pass

here?

Short answer: it doesn't. You can use the `app()` global helper to create an instance of a class that has no parameters in its constructor, but at that point you could've just run `new Logger` yourself. The container shines when there's some complexity in the constructor, and that's when we need to look at how exactly the container can figure out how to construct classes with constructor parameters.

How the Container Is Wired

Before we dig further into the `Logger` class, take a look at [Example 11-4](#).

Example 11-4. Laravel autowiring

```
class Bar
{
    public function __construct() {}
}

class Baz
{
    public function __construct() {}
}

class Foo
{
    public function __construct(Bar $bar, Baz $baz) {}
}

$foo = app(Foo::class);
```

This looks similar to our `mailer` example in [Example 11-3](#). What's different is that these dependencies (`Bar` and `Baz`) are both so simple that the container can resolve them without any further information. The container reads the typehints in the `Foo` constructor, resolves an instance of both `Bar` and `Baz`, and then injects them into the new `Foo` instance when it's creating it. This is called *autowiring*: resolving instances based on typehints without the developer needing to explicitly bind those classes in the container.

Autowiring means that, if a class has not been explicitly bound to the container (like `Foo`, `Bar`, or `Baz` in this context) but the container can figure out how to

resolve it anyway, the container will resolve it. This means any class with no constructor dependencies (like Bar and Baz) and any class with constructor dependencies that the container can resolve (like Foo) can be resolved out of the container.

That leaves us needing to bind only classes that have unresolvable constructor parameters—for example, our `$logger` class in [Example 11-3](#), which has parameters related to our log path and log level.

For those, we'll need to learn how to explicitly bind something to the container.

Binding Classes to the Container

Binding a class to Laravel's container is essentially telling the container, “If a developer asks for an instance of `Logger`, here's the code to run to instantiate one with the correct parameters and dependencies and then return it correctly.”

We're teaching the container that, when someone asks for this particular string (which is usually the FQCN of a class), it should resolve it this way.

Binding to a Closure

So, let's look at how to bind to the container. Note that the appropriate place to bind to the container is in a service provider's `register()` method (see [Example 11-5](#)).

Example 11-5. Basic container binding

```
// In any service provider (maybe LoggerServiceProvider)
public function register(): void
{
    $this->app->bind(Logger::class, function ($app) {
        return new Logger('\log\path\here', 'error');
    });
}
```

There are a few important things to note in this example. First, we're running `$this->app->bind()`. `$this->app` is an instance of the container that's always available on every service provider. The container's `bind()` method is what we use to bind to the container.

The first parameter of `bind()` is the “key” we’re binding to. Here, we’ve used the FQCN of the class. The second parameter differs depending on what you’re doing, but essentially, it should be *something* that shows the container what to do to resolve an instance of that bound key.

So, in this example, we’re passing a closure. And now, any time someone runs `app(Logger::class)`, they’ll get the result of this closure. The closure is passed an instance of the container itself (`$app`), so if the class you’re resolving has a dependency you want resolved out of the container, you can use it in your definition as seen in [Example 11-6](#).

Example 11-6. Using the passed `$app` instance in a container binding

```
// Note that this binding is not doing anything technically useful, since this
// could all be provided by the container's autowiring already.
$this->app->bind(UserMailer::class, function ($app) {
    return new UserMailer(
        $app->make(Mailer::class),
        $app->make(Logger::class),
        $app->make(Slack::class)
    );
});
```

Note that every time you ask for a new instance of your class, this closure will be run again and the new output returned.

Binding to Singletons, Aliases, and Instances

If you want the output of the binding closure to be cached so that this closure isn’t re-run every time you ask for an instance, that’s the singleton pattern, and you can run `$this->app->singleton()` to do that. [Example 11-7](#) shows what this looks like.

Example 11-7. Binding a singleton to the container

```
public function register(): void
{
    $this->app->singleton(Logger::class, function () {
        return new Logger('\log\path\here', 'error');
    });
}
```

You can also get similar behavior if you already have an instance of the object you want the singleton to return, as seen in [Example 11-8](#).

Example 11-8. Binding an existing class instance to the container

```
public function register(): void
{
    $logger = new Logger('\log\path\here', 'error');
    $this->app->instance(Logger::class, $logger);
}
```

Finally, if you want to alias one class to another, bind a class to a shortcut, or bind a shortcut to a class, you can just pass two strings, as shown in [Example 11-9](#).

Example 11-9. Aliasing classes and strings

```
// Asked for Logger, give FirstLogger
$this->app->bind(Logger::class, FirstLogger::class);

// Asked for log, give FirstLogger
$this->app->bind('log', FirstLogger::class);

// Asked for log, give FirstLogger
$this->app->alias(FirstLogger::class, 'log');
```

Note that these shortcuts are common in Laravel’s core; it provides a system of shortcuts to classes that provide core functionality, using easy-to-remember keys like `log`.

Binding a Concrete Instance to an Interface

Just like we can bind a class to another class, or a class to a shortcut, we can also bind to an interface. This is extremely powerful, because we can now typehint interfaces instead of class names, like in [Example 11-10](#).

Example 11-10. Typehinting and binding to an interface

```
...
use Interfaces\Mailer as MailerInterface;

class UserMailer
{
    protected $mailer;

    public function __construct(MailerInterface $mailer)
    {
        $this->mailer = $mailer;
    }
}
```

```
// Service provider
public function register(): void
{
    $this->app->bind(\Interfaces\Mailer::class, function () {
        return new MailgunMailer(...);
    });
}
```

You can now typehint `Mailer` or `Logger` interfaces all across your code, and then choose once in a service provider which specific mailer or logger you want to use everywhere. That's inversion of control.

One of the key benefits you get from using this pattern is that later, if you choose to use a different mail provider than Mailgun, as long as you have a mailer class for that new provider that implements the `Mailer` interface, you can swap it once in your service provider and everything in the rest of your code will just work.

Contextual Binding

Sometimes you need to change how to resolve an interface depending on the context. You might want to log events from one place to a local syslog and from others out to an external service. So, let's tell the container to differentiate—check out [Example 11-11](#).

Example 11-11. Contextual binding

```
// In a service provider
public function register(): void
{
    $this->app->when(FileWrangler::class)
        ->needs(Interfaces\Logger::class)
        ->give(Loggers\Syslog::class);

    $this->app->when(Jobs\SendWelcomeEmail::class)
        ->needs(Interfaces\Logger::class)
        ->give(Loggers\PaperTrail::class);
}
```

Constructor Injection in Laravel Framework Files

We've covered the concept of constructor injection, and we've looked at how the container makes it easy to resolve instances of a class or interface out of the container. You saw how easy it is to use the `app()` helper to make instances, and

also how the container will resolve the constructor dependencies of a class when it's creating it.

What we haven't covered yet is how the container is also responsible for resolving many of the core operating classes of your application. For example, every controller is instantiated by the container. That means if you want an instance of a logger in your controller, you can simply typehint the logger class in your controller's constructor, and when Laravel creates the controller, it will resolve it out of the container and that logger instance will be available to your controller. Take a look at [Example 11-12](#).

Example 11-12. Injecting dependencies into a controller

```
...
class MyController extends Controller
{
    protected $logger;

    public function __construct(Logger $logger)
    {
        $this->logger = $logger;
    }

    public function index()
    {
        // Do something
        $this->logger->error('Something happened');
    }
}
```

The container is responsible for resolving controllers, middleware, queue jobs, event listeners, and any other classes that are automatically generated by Laravel in the process of your application's lifecycle—so any of those classes can typehint dependencies in their constructors and expect them to be automatically injected.

Method Injection

There are a few places in your application where Laravel doesn't just read the constructor signature: it also reads the *method* signature and will inject dependencies for you there as well.

The most common place to use method injection is in controller methods. If you

have a dependency you only want to use for a single controller method, you can inject it into just that method like in [Example 11-13](#).

Example 11-13. Injecting dependencies into a controller method

```
...
class MyController extends Controller
{
    // Method dependencies can come after or before route parameters.
    public function show(Logger $logger, $id)
    {
        // Do something
        $logger->error('Something happened');
    }
}
```

PASSING UNRESOLVABLE CONSTRUCTOR PARAMETERS USING MAKEWITH()

All of the primary tools for resolving a concrete instance of a class—`app()`, `$container->make()`, etc.—assume that all of the class’s dependencies can be resolved without passing anything in. But what if your class accepts a value in its constructor, instead of a dependency the container can resolve for you? Use the `makeWith()` method:

```
class Foo
{
    public function __construct($bar)
    {
        // ...
    }
}

$foo = $this->app->makeWith(
    Foo::class,
    ['bar' => 'value']
);
```

This is a bit of an edge case. Most classes that you’ll be resolving out of the container should *only* have dependencies injected into their constructors.

You can do the same in the `boot()` method of service providers, and you can also arbitrarily call a method on any class using the container, which will allow

for method injection there (see [Example 11-14](#)).

Example 11-14. Manually calling a class method using the container's call() method

```
class Foo
{
    public function bar($parameter1) {}
}

// Calls the 'bar' method on 'Foo' with a first parameter of 'value'
app()->call('Foo@bar', ['parameter1' => 'value']);
```

Facades and the Container

We've covered facades quite a bit so far in the book, but we haven't actually talked about how they work.

Laravel's facades are classes that provide simple access to core pieces of Laravel's functionality. There are two trademark features of facades: first, they're all available in the global namespace (`\Log` is an alias to `\Illuminate\Support\Facades\Log`); and second, they use static methods to access nonstatic resources.

Let's take a look at the Log facade, since we've been looking at logging already in this chapter. In your controller or views, you could use this call:

```
Log::alert('Something has gone wrong!');
```

Here's what it would look like to make that same call without the facade:

```
$logger = app('log');
$logger->alert('Something has gone wrong!');
```

As you can see, facades translate static calls (any method call that you make on a class itself, using `::`, instead of on an instance) to normal method calls on instances.

IMPORTING FACADE NAMESPACES

If you're in a namespaced class, you'll want to be sure to import the facade at the top:

```
...
use Illuminate\Support\Facades\Log;

class Controller extends Controller
{
    public function index()
    {
        // ...
        Log::error('Something went wrong!');
    }
}
```

How Facades Work

Let's take a look at the Cache facade and see how it actually works.

First, open up the class `Illuminate\Support\Facades\Cache`. You'll see something like [Example 11-15](#).

Example 11-15. The Cache facade class

```
<?php

namespace Illuminate\Support\Facades;

class Cache extends Facade
{
    protected static function getFacadeAccessor()
    {
        return 'cache';
    }
}
```

Every facade has a single method: `getFacadeAccessor()`. This defines the key that Laravel should use to look up this facade's backing instance from the container.

In this instance, we can see that every call to the Cache facade is proxied to be a call to an instance of the `cache` shortcut from the container. Of course, that's not a real class or interface name, so we know it's one of those shortcuts I mentioned earlier.

So, here's what's really happening:

```
Cache::get('key');  
  
// Is the same as...  
  
app('cache')->get('key');
```

There are a few ways to look up exactly what class each facade accessor points to, but checking the documentation is the easiest. There's a table on the [facades documentation page](#) that shows, for each facade, which container binding (shortcut, like `cache`) it's connected to, and which class that returns. It looks like this:

Facade	Class	Service container binding
App	Illuminate\Foundation\Application	app
...	...	...
Cache	Illuminate\Cache\CacheManager	cache
...	...	...

Now that you have this reference, you can do three things.

First, you can figure out what methods are available on a facade. Just find its backing class and look at the definition of that class, and you'll know that any of its public methods are callable on the facade.

Second, you can figure out how to inject a facade's backing class using dependency injection. If you ever want the functionality of a facade but prefer to use dependency injection, just typehint the facade's backing class or get an instance of it with `app()` and call the same methods you would've called on the facade.

Third, you can see how to create your own facades. Create a class for the facade that extends `Illuminate\Support\Facades\Facade`, and give it a `getFacadeAccessor()` method, which returns a string. Make that string something that can be used to resolve your backing class out of the container—maybe just the FQCN of the class. Finally, you have to register the facade by adding it to the `aliases` array in `config/app.php`. Done! You just made your own facade.

Real-Time Facades

Rather than creating a new class to make your class's instance methods available as static methods, with *real-time facades* you can simply prefix your class's FQCN with `Facades\` and use it *as if it were a facade*. **Example 11-16** illustrates how this works.

Example 11-16. Using real-time facades

```
namespace App;

class Charts
{
    public function burndown()
    {
        // ...
    }
}

<h2>Burndown Chart</h2>
{{ Facades\App\Charts::burndown() }}
```

As you can see here, the nonstatic method `burndown()` becomes accessible as a static method on the real-time facade, which we create by prepending the class's full name with `Facades\`.

Service Providers

We covered the basics of service providers in the previous chapter (see “**Service Providers**”). What's most important with regard to the container is that you remember to register your bindings in the `register()` method of some service provider somewhere.

You can just dump loose bindings into `App\Providers\AppServiceProvider`, which is a bit of a catchall, but it's generally better practice to create a unique service provider for each group of functionality you're developing and bind its classes in its unique `register()` method.

Testing

The ability to use inversion of control and dependency injection makes testing in Laravel extremely versatile. You can bind a different logger, for instance, depending on whether the app is live or under testing. Or you can change the transactional email service from Mailgun to a local email logger for easy inspection. Both of these swaps are actually so common that it's even easier to make them using Laravel's `.env` configuration files, but you can make similar swaps with any interfaces or classes you'd like.

The easiest way to do this is to explicitly rebind classes and interfaces when you need them rebound, directly in the test. [Example 11-17](#) shows how.

Example 11-17. Overriding a binding in tests

```
public function test_it_does_something()
{
    app()->bind(Interfaces\Logger, function () {
        return new DevNullLogger;
    });

    // Do stuff
}
```

If you need certain classes or interfaces rebound globally for your tests (which is not a particularly common occurrence), you can do this in either the `setUp()` method in the test class or in the `setUp()` method in Laravel's `TestCase` base test, as shown in [Example 11-18](#).

Example 11-18. Overriding a binding for all tests

```
class TestCase extends \Illuminate\Foundation\Testing\TestCase
{
    public function setUp()
    {
        parent::setUp();

        app()->bind('whatever', 'whatever else');
    }
}
```

```
}  
}
```

When using something like Mockery, it's common to create a mock or spy or stub of a class and then rebind that to the container in place of its referent.

TL;DR

Laravel's service container has many names, but regardless of what you call it, in the end its goal is to make it easy to define how to resolve certain string names as concrete instances. These string names are going to be the fully qualified class names of classes or interfaces, or shortcuts like `log`.

Each binding teaches the application, given a string key (e.g., `app( 'log' )`), how to resolve a concrete instance.

The container is smart enough to do recursive dependency resolution, so if you try to resolve an instance of something that has constructor dependencies, the container will try to resolve those dependencies based on their typehints, then pass them into your class, and finally return an instance.

There are several ways to bind to the container, but in the end, they all define what to return, given a particular string.

Facades are simple shortcuts that make it easy to use static calls on a root namespace–aliased class to call nonstatic methods on classes resolved out of the container. Real-time facades allow you to treat any class like a facade by prepending its fully qualified class name with `Facades\`.

Chapter 12. Testing

Most developers know that testing your code is a good thing. We're supposed to do it. We likely have an idea of why it's good, and we might've even read some tutorials about how it's supposed to work.

But the gap between knowing *why* you should test and knowing *how* to test is wide. Thankfully, tools like PHPUnit, Mockery, and PHPSpec provide an incredible number of options for testing in PHP—but it can still be pretty overwhelming to get everything set up.

Out of the box, Laravel comes with baked-in integrations to PHPUnit (unit testing), Mockery (mocking), and Faker (creating fake data for seeding and testing). It also provides its own simple and powerful suite of application testing tools, which allow you to “crawl” your site's URIs, submit forms, check HTTP status codes, and validate and assert against JSON. It also provides a robust frontend testing framework called Dusk that can even interact with your JavaScript applications and test against them. In case this hasn't made it clear, we're going to cover a lot of ground in this chapter.

To make it easy for you to get started, Laravel's testing setup comes with sample application tests that can run successfully the moment you create a new app. That means you don't have to spend any time configuring your testing environment, and that's one less barrier to writing your tests.

Testing Basics

TESTING TERMS

It's hard to get any group of programmers to agree on one set of terms to define different types of tests.

In this chapter, I'll use four primary terms:

Unit tests

Unit tests target small, relatively isolated units—a class or method, usually.

Feature tests

Feature tests test the way individual units work together and pass messages.

Application tests

Often called acceptance or functional tests, application tests test the entire behavior of the application, usually at an outer boundary, like HTTP calls.

Regression tests

Similar to application tests, regression tests are a little more focused on describing exactly what a user should be able to do and ensuring that behavior doesn't stop working. The line between application and regression tests is thin, but the difference is primarily based on the level of fidelity of your tests. For example, an application test might say “browser can POST to the people endpoint and then there should be a new entry in the users table” (relatively low fidelity, because you are mimicking what the browser is doing), but a regression test would more likely say “after clicking this button with this form data entered, the user should see that result on that page” (higher fidelity, because you're describing the actual behavior of your users).

Tests in Laravel live in the *tests* folder. There are two files in the root: *TestCase.php*, which is the base root test that all of your tests will extend, and *CreatesApplication.php*, a trait (imported by *TestCase.php*) that allows any class to boot a sample Laravel application for testing.

LARAVEL TEST COMMAND

Laravel has an Artisan command for running your tests: `php artisan test`. It's a wrapper around the `./vendor/bin/phpunit` command, which will additionally show more output for

each test.

There are also two subfolders: *Features*, for tests that cover the interaction between multiple units, and *Unit*, for tests that are intended to cover just one unit of your code (class, module, function, etc.). Each of these folders contains an *ExampleTest.php* file, each of which has a single sample test inside it, ready to run.

The `ExampleTest` in your *Unit* directory contains one simple assertion: `$this->assertTrue(true)`. Anything in your unit tests is likely to be relatively simple PHPUnit syntax (asserting that values are equal or different, looking for entries in arrays, checking Booleans, etc.), so there's not much to learn there.

THE BASICS OF PHPUNIT ASSERTIONS

In PHPUnit, most of our assertions will be run on the `$this` object with this syntax:

```
$this->assertWHATEVER($expected, $real);
```

So, for example, if we're asserting that two variables should be equal, we'll pass it first our expected result, and second the actual outcome of the object or system we're testing:

```
$multiplicationResult = $myCalculator->multiply(5, 3);  
$this->assertEqual(15, $multiplicationResult);
```

As you can see in [Example 12-1](#), the `ExampleTest` in the *Feature* directory makes a simulated HTTP request to the page at the root path of your application and checks that its HTTP status is 200 (successful). If it is, it'll pass; if not, it'll fail. Unlike your average PHPUnit test, we're running these assertions on the `TestResponse` object that's returned when we make test HTTP calls.

Example 12-1. tests/Feature/ExampleTest.php

```
<?php
```

```
namespace Tests\Feature;
```

```
// use Illuminate\Foundation\Testing\RefreshDatabase;
use Tests\TestCase;

class ExampleTest extends TestCase
{
    /**
     * A basic test example.
     */
    public function test_the_application_returns_a_successful_response(): void
    {
        $response = $this->get('/');

        $response->assertStatus(200);
    }
}
```

To run the tests, run `php artisan test` on the command line from the root folder of your application. You should see something like the output in [Example 12-2](#).

Example 12-2. Sample ExampleTest output

```
PASS Tests\Unit\ExampleTest
✓ that true is true

PASS Tests\Feature\ExampleTest
✓ the application returns a successful response

Tests: 2 passed (2 assertions)
Time: 0.25s
```

You just ran your first Laravel application test! Those two checkmarks indicate that you have two passing tests. As you can see, you're set up out of the box not only with a functioning PHPUnit instance, but also a full-fledged application testing suite that can make mock HTTP calls and test your application's responses.

In case you're not familiar with PHPUnit, let's take a look at what it's like to have a test fail. Instead of modifying the previous test, we'll make our own. Run `php artisan make:test FailingTest`. This will create the file `tests/Feature/FailingTest.php`; you can modify its `testExample()` method to look like [Example 12-3](#).

Example 12-3. tests/Feature/FailingTest.php, edited to fail

```
public function test_example()
```

```
{
    $response = $this->get('/');

    $response->assertStatus(301);
}
```

As you can see, it's the same as the test we ran previously, but we're now testing against the wrong status. Let's run PHPUnit again.

GENERATING UNIT TESTS

If you want your test to be generated in the Unit directory instead of the Feature directory, pass the `--unit` flag:

```
php artisan make:test SubscriptionTest --unit
```

Whoops! This time the output will probably look a bit like [Example 12-4](#).

Example 12-4. Sample failing test output

```
PASS Tests\Unit\ExampleTest
✓ that true is true

PASS Tests\Feature\ExampleTest
✓ the application returns a successful response

FAIL Tests\Feature\FailingTest
✗ example

FAILED Tests\Feature\FailingTest > example
Expected status code [301] but received 200. Failed asserting that
301 is identical to 200.

at tests/Feature/FailingTest.php:20
16|     public function test_example()
17|     {
18|         $response = $this->get('/');
19|
> 20|         $response->assertStatus(301);
21|     }
22| }
23|

Tests: 1 failed, 2 passed (3 assertions)
```


Duration: 1.10s

Let's break this down. Last time there were two passing tests, but this time one failed and two passed.

Then, for each error, we see the test name (here, `Test\Feature\FailingTest` > example), the error message ("Expected status code..."), and a partial stack trace, so we can see the line where the failure occurred.

Now that we've run both a passing test and a failing test, it's time for you to learn more about Laravel's testing environment.

Naming Tests

By default, Laravel's testing system will run any file in the *tests* directory whose name ends with the word *Test*. That's why *tests/ExampleTest.php* was run by default.

If you're not familiar with PHPUnit, you might not know that only the methods in your tests with names that start with the word **test** will be run—or methods with a `@test` documentation block, or *docblock*. See [Example 12-5](#) for which methods will and won't run.

Example 12-5. Naming PHPUnit methods

```
class NamingTest
{
    public function test_it_names_things_well()
    {
        // Runs as "It names things well"
    }

    public function testItNamesThingsWell()
    {
        // Runs as "It names things well"
    }

    /** @test */
    public function it_names_things_well()
    {
        // Runs as "It names things well"
    }

    public function it_names_things_well()
```

```
{  
    // Doesn't run  
}  
}
```

The Testing Environment

Any time a Laravel application is running, it has a current “environment” name that represents the environment it’s running in. This name may be set to `local`, `staging`, `production`, or anything else you want. You can retrieve this by running `app()->environment()`, or you can run `if (app()->environment('local'))` or something similar to test whether the current environment matches the passed name.

When you run tests, Laravel automatically sets the environment to `testing`. This means you can test for `if (app()->environment('testing'))` to enable or disable certain behaviors in the testing environment.

Additionally, Laravel doesn’t load the normal environment variables from `.env` for testing. If you want to set any environment variables for your tests, edit `phpunit.xml` and, in the `<php>` section, add a new `<env>` for each environment variable you want to pass in—for example, `<env name="DB_CONNECTION" value="sqlite"/>`.

USING .ENV.TESTING TO EXCLUDE TESTING ENVIRONMENT VARIABLES FROM VERSION CONTROL

If you want to set environment variables for your tests, you can do so in `phpunit.xml` as just described. But what if you have environment variables for your tests that you want to be different for each testing environment? Or what if you want them to be excluded from source control?

Thankfully, handling these conditions is pretty easy. First, create an `.env.testing.example` file, just like Laravel’s `.env.example` file. Next, add the variables you’d like to be environment-specific to `.env.testing.example`, just like they’re set in `.env.example`. Then, make a copy of `.env.testing.example` and name it `.env.testing`. Finally, add `.env.testing` to your `.gitignore` file just

below `.env` and set any values you want in `.env.testing`.

The Testing Traits

Before we get into the methods you can use for testing, you need to know about the four testing traits you can pull into any test class.

RefreshDatabase

`Illuminate\Foundation\Testing\RefreshDatabase` is imported at the top of every newly generated test file, and it's the most commonly used database migration trait.

The point of this, and the other database traits, is to ensure your database tables are correctly migrated at the start of each test.

`RefreshDatabase` takes two steps to do this. First, it runs your migrations on your test database *once* at the beginning of each test run (when you run `phpunit`, not for each individual test method). And second, it wraps each individual test method in a database transaction and rolls back the transaction at the end of the test.

That means you have your database migrated for your tests and cleared out fresh after each test runs, without having to run your migrations again before every test—making this the fastest possible option. When in doubt, stick with this.

DatabaseMigrations

If you import the `Illuminate\Foundation\Testing\DatabaseMigrations` trait instead of the `RefreshDatabase` trait, it will run your entire set of database migrations fresh before each test. Laravel makes this happen by running `php artisan migrate:fresh` in the `setUp()` method before every test runs.

DatabaseTransactions

On the other hand,

`Illuminate\Foundation\Testing\DatabaseTransactions` expects your

database to be properly migrated before your tests start. It wraps every test in a database transaction, which it rolls back at the end of each test. This means that, at the end of each test, your database will be returned to the exact same state it was in prior to the test.

WithoutMiddleware

If you import `Illuminate\Foundation\Testing\WithoutMiddleware` into your test class, it will disable all middleware for any test in that class. This means you won't have to worry about the authentication middleware, or CSRF protection, or anything else that might be useful in the real application but distracting in a test.

If you'd like to disable middleware for just a single method instead of the entire test class, call `$this->withoutMiddleware()` at the top of the method for that test.

Simple Unit Tests

With simple unit tests, you almost don't need any of these traits. You *may* reach for database access or inject something out of the container, but it's very likely that unit tests in your applications won't rely on the framework very much. Take a look at [Example 12-6](#) for an example of what a simple test might look like.

Example 12-6. A simple unit test

```
class GeometryTest extends TestCase
{
    public function test_it_calculates_area()
    {
        $square = new Square;
        $square->sideLength = 4;

        $calculator = new GeometryCalculator;

        $this->assertEquals(16, $calculator->area($square));
    }
}
```

Obviously, this example is a bit contrived. But you can see here that we're testing a single class (`GeometryCalculator`) and its single method (`area()`), and we're doing so without worrying about the entire Laravel application.

Some unit tests might be testing something that technically is connected to the framework—for example, Eloquent models—but you can still test them without worrying about the framework. For example, in [Example 12-7](#), we use `Package::make()` instead of `Package::create()` so the object is created and evaluated in memory without ever hitting the database.

Example 12-7. A more complicated unit test

```
class PopularityTest extends TestCase
{
    use RefreshDatabase;

    public function test_votes_matter_more_than_views()
    {
        $package1 = Package::make(['votes' => 1, 'views' => 0]);
        $package2 = Package::make(['votes' => 0, 'views' => 1]);

        $this->assertTrue($package1->popularity > $package2->popularity);
    }
}
```

Some people may call this an integration or feature test, since this “unit” will likely touch the database in actual usage and it’s connected to the entire Eloquent codebase. The most important point is that you can have simple tests that test a single class or method, even when the objects under test are framework-connected.

All of this said, it’s still going to be more likely that your tests—especially as you first get started—are broader and more at the “application” level. Accordingly, for the rest of the chapter we’re going to dig deeper into application testing.

Application Testing: How It Works

In “[Testing Basics](#)” we saw that, with a few lines of code, we can “request” URIs in our application and actually check the status of the response. But how can PHPUnit request pages as if it were a browser?

Any application tests should extend the `TestCase` class (*tests/TestCase.php*) that’s included with Laravel by default. Your application’s `TestCase` class will extend the abstract `Illuminate\Foundation\Testing\TestCase` class, which brings in quite a few goodies.

The first thing the two `TestCase` classes (yours and its abstract parent) do is handle booting the Illuminate application instance for you, so you have a fully bootstrapped application available. They also “refresh” the application between each test, which means they’re not *entirely* recreating the application between tests but rather making sure you don’t have any data lingering.

The parent `TestCase` also sets up a system of hooks that allow callbacks to be run before and after the application is created, and imports a series of traits that provide you with methods for interacting with every aspect of your application. These traits include `InteractsWithContainer`, `MakesHttpRequests`, and `InteractsWithConsole`, and they bring in a broad variety of custom assertions and testing methods.

As a result, your application tests have access to a fully bootstrapped application instance and application-test-minded custom assertions, with a series of simple and powerful wrappers around each to make them easy to use.

That means you can write `$this->get('/')->assertStatus(200)` and know that your application is actually behaving as if it were responding to a normal HTTP request, and that the response is being fully generated and then checked as a browser would check it. It’s pretty powerful stuff, considering how little work you had to do to get it running.

HTTP Tests

Let’s take a look at our options for writing HTTP-based tests. You’ve already seen `$this->get('/')`, but let’s dive deeper into how you can use that call, how you can assert against its results, and what other HTTP calls you can make.

Testing Basic Pages with `$this->get()` and Other HTTP Calls

At the very basic level, Laravel’s HTTP testing allows you to make simple HTTP requests (GET, POST, etc.) and then make simple assertions about their impact or response.

There are more tools we’ll cover later (“[Testing with Dusk](#)”) that allow for more complex page interactions and assertions, but let’s start at the base level. Here

are the calls you can make:

- `$this->get($uri, $headers = [])`
- `$this->post($uri, $data = [], $headers = [])`
- `$this->put($uri, $data = [], $headers = [])`
- `$this->patch($uri, $data = [], $headers = [])`
- `$this->delete($uri, $data = [], $headers = [])`
- `$this->option($uri, $data = [], $headers = [])`

These methods are the basis of the HTTP testing framework. Each takes at least a URI (usually relative) and headers, and all but `get()` also allow for passing data along with the request.

And, importantly, each returns a `$response` object that represents the HTTP response. This response object is almost exactly the same as an Illuminate Response object, the same thing we return out of our controllers. However, it's actually an instance of `Illuminate\Testing\TestResponse`, which wraps a normal Response with some assertions for testing.

Take a look at [Example 12-8](#) to see a common usage of `post()` and a common response assertion.

Example 12-8. A simple use of `post()` in testing

```
public function test_it_stores_new_packages()
{
    $response = $this->post(route('packages.store'), [
        'name' => 'The greatest package',
    ]);

    $response->assertOk();
}
```

In most examples like [Example 12-8](#), you'll also test that the record exists in the database and shows up on the index page, and maybe that it doesn't test successfully unless you define the package author and are logged in. But don't worry, we'll get to all of that. For now, you can make calls to your application routes with many different verbs and make assertions against both the response

and the state of your application afterward. Great!

Testing JSON APIs with `$this->getJson()` and Other JSON HTTP Calls

You can also do all of the same sorts of HTTP tests with your JSON APIs. There are convenience methods for that, too:

- `$this->getJson($uri, $headers = [])`
- `$this->postJson($uri, $data = [], $headers = [])`
- `$this->putJson($uri, $data = [], $headers = [])`
- `$this->patchJson($uri, $data = [], $headers = [])`
- `$this->deleteJson($uri, $data = [], $headers = [])`
- `$this->optionJson($uri, $data = [], $headers = [])`

These methods work just the same as the normal HTTP call methods, except they also add JSON-specific `Accept`, `CONTENT_LENGTH`, and `CONTENT_TYPE` headers. Take a look at [Example 12-9](#) to see an example.

Example 12-9. A simple use of `postJson()` in testing

```
public function test_the_api_route_stores_new_packages()
{
    $response = $this->postJson(route('api.packages.store'), [
        'name' => 'The greatest package',
    ], ['X-API-Version' => '17']);

    $response->assertOk();
}
```

Assertions Against `$response`

There are more than 50 assertions available on the `$response` object, so I'll refer you to the [testing docs](#) for details on all of them. Let's look at a few of the most important and most common ones:

`$response->assertOk()`

Asserts that the response's status code is 200:

```
$response = $this->get('terms');  
$response->assertOk();
```

`$response->assertSuccessful()`

Although `assertOk()` asserts that the code is exactly 200, `assertSuccessful()` checks if the code is anything in the 200 group:

```
$response = $this->post('articles', [  
    'title' => 'Testing Laravel',  
    'body'  => 'My article about testing Laravel',  
]);  
// Assuming this returns 201 CREATED...  
$response->assertSuccessful();
```

`$response->assertUnauthorized()`

Asserts that the response's status code is 401:

```
$response = $this->patch('settings', ['password' => 'abc']);  
$response->assertUnauthorized();
```

`$response->assertForbidden()`

Asserts that the response's status code is 403:

```
$response = $this->actingAs($normalUser)->get('admin');  
$response->assertForbidden();
```

`$response->assertNotFound()`

Asserts that the response's status code is 404:

```
$response = $this->get('posts/first-post');  
$response->assertNotFound();
```

`$response->assertStatus($status)`

Asserts that the response's status code is equal to the provided *\$status*:

```
$response = $this->get('admin');  
$response->assertStatus(401); // Unauthorized
```

`$response->assertSee($text), $response->assertDontSee($text)`

Asserts that the response contains (or doesn't contain) the provided *\$text*:

```
$package = Package::factory()->create();  
$response = $this->get(route('packages.index'));  
$response->assertSee($package->name);
```

`$response->assertJson(array $json)`

Asserts that the passed array is represented (in JSON format) in the returned JSON:

```
$this->postJson(route('packages.store'), ['name' => 'GreatPackage2000']);  
$response = $this->getJson(route('packages.index'));  
$response->assertJson(['name' => 'GreatPackage2000']);
```

`$response->assertViewHas($key, $value = null)`

Asserts that the view on the visited page had a piece of data available at *\$key*, and optionally checks that the value of that variable was *\$value*:

```
$package = Package::factory()->create();  
$response = $this->get(route('packages.show'));  
$response->assertViewHas('name', $package->name);
```

`$response->assertSessionHas($key, $value = null)`

Asserts that the session has data set at *\$key*, and optionally checks that the value of that data is *\$value*:

```
$response = $this->get('beta/enable');  
$response->assertSessionHas('beta-enabled', true);
```

`$response->assertSessionHasInput($key, $value = null)`

Asserts that the given keys and values are flashed in the session array input. This is helpful when testing whether the validation error returns the correct old values:

```
$response = $this->post('users', ['name' => 'Abdullah']);  
// Assuming it errored, check that the entered name is flashed;  
$response->assertSessionHasInput('name', 'Abdullah');
```

`$response->assertSessionHasErrors()`

With no parameters, asserts that there's at least one error set in Laravel's special `errors` session container. Its first parameter can be an array of key/value pairs that define the errors that should be set and its second parameter can be the string format that the checked errors should be formatted against, as demonstrated here:

```
// Assuming the "/form" route requires an email field, and we're  
// posting an empty submission to it to trigger the error
```

```

$response = $this->post('form', []);

$response->assertSessionHasErrors();

$response->assertSessionHasErrors([
    'email' => 'The email field is required.',
]);

$response->assertSessionHasErrors(
    ['email' => '<p>The email field is required.</p>'],
    '<p>:message</p>'
);

```

If you're working with named error bags, you can pass the error bag name as the third parameter.

`$response->assertCookie($name, $value = null)`

Asserts that the response contains a cookie with name *\$name*, and optionally checks that its value is *\$value*:

```

$response = $this->post('settings', ['dismiss-warning']);

$response->assertCookie('warning-dismiss', true);

```

`$response->assertCookieExpired($name)`

Asserts that the response contains a cookie with name *\$name* and that it is expired:

```

$response->assertCookieExpired('warning-dismiss');

```

`$response->assertCookieNotExpired($name)`

Asserts that the response contains a cookie with name *\$name* and that it is not

expired:

```
$response->assertCookieNotExpired('warning-dismiss');
```

```
$response->assertRedirect($uri)
```

Asserts that the requested route returns a redirect to the given URI:

```
$response = $this->post(route('packages.store'), [  
    'email' => 'invalid'  
]);
```

```
$response->assertRedirect(route('packages.create'));
```

For each of these assertions, you can assume that there are many related assertions I haven't listed here. For example, in addition to `assertSessionHasErrors()` there are also `assertSessionHasNoErrors()` and `assertSessionHasErrorsIn()` assertions; as well as `assertJson()`, there are also `assertJsonCount()`, `assertJsonFragment()`, `assertJsonPath()`, `assertJsonMissing()`, `assertJsonMissingExact()`, `assertJsonStructure()`, and `assertJsonValidationErrors()` assertions. Again, take a look at the docs and make yourself familiar with the whole list.

Authenticating Responses

One piece of your application it's common to test with application tests is authentication and authorization. Most of the time, your needs will be met with the `actingAs()` chainable method, which takes a user (or other `Authenticatable` object, depending on how your system is set up), as you can see in [Example 12-10](#).

Example 12-10. Basic auth in testing

```
public function test_guests_cant_view_dashboard()  
{  
    $user = User::factory()->guest()->create();
```

```

        $response = $this->actingAs($user)->get('dashboard');
        $response->assertStatus(401); // Unauthorized
    }

    public function test_members_can_view_dashboard()
    {
        $user = User::factory()->member()->create();
        $response = $this->actingAs($user)->get('dashboard');
        $response->assertOk();
    }

    public function test_members_and_guests_cant_view_statistics()
    {
        $guest = User::factory()->guest()->create();
        $response = $this->actingAs($guest)->get('statistics');
        $response->assertStatus(401); // Unauthorized

        $member = User::factory()->member()->create();
        $response = $this->actingAs($member)->get('statistics');
        $response->assertStatus(401); // Unauthorized
    }

    public function test_admins_can_view_statistics()
    {
        $user = User::factory()->admin()->create();
        $response = $this->actingAs($user)->get('statistics');
        $response->assertOk();
    }
}

```

USING FACTORY STATES FOR AUTHORIZATION

It's common to use model factories (discussed in “[Model Factories](#)”) in testing, and model factory states make tasks like creating users with different access levels simple.

A Few Other Customizations to Your HTTP Tests

If you'd like to set session variables on your requests, you can also chain `withSession()`:

```

$response = $this->withSession([
    'alert-dismissed' => true,
])->get('dashboard');

```

If you'd prefer to set your request headers fluently, you can chain

withHeaders():

```
$response = $this->withHeaders([
    'X-THE-ANSWER' => '42',
])->get('the-restaurant-at-the-end-of-the-universe');
```

Handling Exceptions in Application Tests

Usually, an exception that's thrown inside your application when you're making HTTP calls will be captured by Laravel's exception handler and processed as it would be in a normal application. So, the test and route in [Example 12-11](#) would still pass, since the exception would never bubble up the whole way to our test.

Example 12-11. An exception that will be captured by Laravel's exception handler and result in a passing test

```
// routes/web.php
Route::get('has-exceptions', function () {
    throw new Exception('Stop!');
});

// tests/Feature/ExceptionsTest.php
public function test_exception_in_route()
{
    $this->get('/has-exceptions');

    $this->assertTrue(true);
}
```

In a lot of cases, this might make sense; maybe you're expecting a validation exception, and you want it to be caught like it would normally be by the framework.

But if you want to temporarily disable the exception handler, that's an option; just run `$this->withoutExceptionHandler()`, as shown in [Example 12-12](#).

Example 12-12. Temporarily disabling exception handling in a single test

```
// tests/Feature/ExceptionsTest.php
public function test_exception_in_route()
{
    // Now throws an error
    $this->withoutExceptionHandler();

    $this->get('/has-exceptions');
```

```
}  
    $this->assertTrue(true);  
}
```

And if for some reason you need to turn it back on (maybe you turned it off in `setUp()` but want it back on for just one test), you can run `$this->withExceptionHandler()`.

Debugging Responses

You can easily dump out the headers with `dumpHeaders()` or the body with `dump()` or `dd()`:

```
$response = $this->get('/');  
  
$response->dumpHeaders();  
$response->dump();  
$response->dd();
```

You can also easily dump all or only the specified keys on the session:

```
$response = $this->get('/');  
  
$response->dumpSession();  
$response->dumpSession(['message']);
```

Database Tests

Often, the effect we want to test for after our tests have run is in the database. Imagine you want to test that the “create package” page works correctly. What’s the best way? Make an HTTP call to the “store package” endpoint and then assert that that package exists in the database. It’s easier and safer than inspecting the resulting “list packages” page.

We have four primary assertions for the database, and two Eloquent-specific assertions.

Asserting Against the Database

For direct assertions against the database, we have `$this->`

`>assertDatabaseHas()` and `$this->assertDatabaseMissing()`, and `$this->assertDeleted()` and `$this->assertSoftDeleted()`. For both, pass the table name as the first parameter, the data you're looking for as the second, and, optionally, the specific database connection you want to test as the third.

Take a look at [Example 12-13](#) to see how you might use them.

Example 12-13. Sample database tests

```
public function test_create_package_page_stores_package()
{
    $this->post(route('packages.store'), [
        'name' => 'Package-a-tron',
    ]);

    $this->assertDatabaseHas('packages', ['name' => 'Package-a-tron']);
}
```

As you can see, the second parameter of `assertDatabaseHas()` is structured like a SQL WHERE statement—you pass a key and a value (or multiple keys and values), and then Laravel looks for any records in the specified database table that match your key(s) and value(s).

As you'd expect, `assertDatabaseMissing()` is the inverse.

Asserting Against Eloquent Models

While `assertDatabaseHas()` and `assertDatabaseMissing()` allow you to identify rows by passing keys and values, Laravel also provides convenience methods for directly asserting that a given Eloquent record does or does not exist: `assertModelExists()` and `assertModelMissing()`, as you can see in [Example 12-14](#).

Example 12-14. Assert against model existence

```
public function test_undeletable_packages_cant_be_deleted()
{
    // Create undeletable model
    $package = Package::factory()->create([
        'name' => 'Package-a-tron',
        'is_deletable' => false,
    ]);

    $this->post(route('packages.delete', $package));
}
```

```

// Can check existence or whether it was soft deleted
$this->assertModelExists($package);
$this->assertNotSoftDeleted($package);

$package->update(['is_deletable' => true]);

$this->post(route('packages.delete', $package));

// Can check existence or whether it was soft deleted
$this->assertModelMissing($package);
$this->assertSoftDeleted($package);
}

```

Using Model Factories in Tests

Model factories are amazing tools that make it easy to seed randomized, well-structured database data for testing (or other purposes). You’ve already seen them in use in several examples in this chapter, and we’ve already covered them in depth, so check out “[Model Factories](#)” to learn more.

Seeding in Tests

If you use seeds in your application, you can run the equivalent of `php artisan db:seed` by running `$this->seed()` in your test.

You can also pass a seeder class name to just seed that one class:

```

$this->seed(); // Seeds all
$this->seed(UserSeeder::class); // Seeds users

```

Testing Other Laravel Systems

When testing Laravel systems, you’ll often want to pause their true function for the duration of the testing and instead write tests against what has happened to those systems. You can do this by “faking” different facades, such as `Event`, `Mail`, and `Notification`. We’ll talk more about what fakes are in “[Mocking](#)”, but first, let’s look at some examples. All of the following features in Laravel have their own set of assertions you can make after faking them, but you can also just choose to fake them to restrict their effects.

Event Fakes

Let's use event fakes as our first example of how Laravel makes it possible to mock its internal systems. There are likely going to be times when you want to fake events just for the sake of suppressing their actions. For example, suppose your app pushes notifications to Slack every time a new user signs up. You have a “user signed up” event that's dispatched when this happens, and it has a listener that notifies a Slack channel that a user has signed up. You don't want those notifications to go to Slack every time you run your tests, but you might want to assert that the event was sent, or the listener was triggered, or something else. This is one reason for faking certain aspects of Laravel in our tests: to pause the default behavior and instead make assertions against the system we're testing.

Let's take a look at how to suppress these events by calling the `fake()` method on `Illuminate\Support\Facades\Event`, as shown in [Example 12-15](#).

Example 12-15. Suppressing events without adding assertions

```
public function test_controller_does_some_thing()
{
    Event::fake();

    // Call controller and assert it does whatever you want without
    // worrying about it pinging Slack
}
```

Once we've run the `fake()` method, we can also call special assertions on the Event facade: namely, `assertDispatched()` and `assertNotDispatched()`. Take a look at [Example 12-16](#) to see them in use.

Example 12-16. Making assertions against events

```
public function test_signing_up_users_notifies_slack()
{
    Event::fake();

    // Sign user up

    Event::assertDispatched(UserJoined::class, function ($event) use ($user) {
        return $event->user->id === $user->id;
    });

    // Or sign multiple users up and assert it was dispatched twice
```

```

Event::assertDispatched(UserJoined::class, 2);

// Or sign up with validation failures and assert it wasn't dispatched

Event::assertNotDispatched(UserJoined::class);
}

```

Note that the (optional) closure we passed to `assertDispatched()` means we're not just asserting that the event was dispatched, but also that the dispatched event contains certain data.

EVENT::FAKE() DISABLES ELOQUENT MODEL EVENTS

`Event::fake()` also disables Eloquent model events. So if you have any important code, for example, in a model's creating event, make sure to create your models (through your factories or however else) *before* calling `Event::fake()`.

Bus and Queue Fakes

The Bus facade, which represents how Laravel dispatches jobs, works just like Event. You can run `fake()` on it to disable the impact of your jobs, and after faking it you can run `assertDispatched()` or `assertNotDispatched()`.

The Queue facade represents how Laravel dispatches jobs when they're pushed up to queues. Its available methods are `assertedPushed()`, `assertPushedOn()`, and `assertNotPushed()`.

Take a look at [Example 12-17](#) to see how to use both.

Example 12-17. Faking jobs and queued jobs

```

public function test_popularity_is_calculated()
{
    Bus::fake();

    // Synchronize package data...

    // Assert a job was dispatched
    Bus::assertDispatched(
        CalculatePopularity::class,
        function ($job) use ($package) {
            return $job->package->id === $package->id;
        }
    );
}

```

```

    }
  );

  // Assert a job was not dispatched
  Bus::assertNotDispatched(DestroyPopularityMaybe::class);
}

public function test_popularity_calculation_is_queued()
{
  Queue::fake();

  // Synchronize package data...

  // Assert a job was pushed to any queue
  Queue::assertPushed(
    CalculatePopularity::class,
    function ($job) use ($package) {
      return $job->package->id === $package->id;
    }
  );

  // Assert a job was pushed to a given queue named "popularity"
  Queue::assertPushedOn('popularity', CalculatePopularity::class);

  // Assert a job was pushed twice
  Queue::assertPushed(CalculatePopularity::class, 2);

  // Assert a job was not pushed
  Queue::assertNotPushed(DestroyPopularityMaybe::class);
}

```

Mail Fakes

The Mail facade, when faked, offers four methods: `assertSent()`, `assertNotSent()`, `assertQueued()`, and `assertNotQueued()`. Use the `Queued` methods when your mail is queued and the `Sent` methods when it's not.

Just like with `assertDispatched()`, the first parameter will be the name of the mailable and the second parameter can be empty, the number of times the mailable has been sent, or a closure testing that the mailable has the right data in it. Take a look at [Example 12-18](#) to see a few of these methods in action.

Example 12-18. Making assertions against mail

```

public function test_package_authors_receive_launch_emails()
{
  Mail::fake();
}

```

```

// Make a package public for the first time...

// Assert a message was sent to a given email address
Mail::assertSent(PackageLaunched::class, function ($mail) use ($package) {
    return $mail->package->id === $package->id;
});

// Assert a message was sent to given email addresses
Mail::assertSent(PackageLaunched::class, function ($mail) use ($package) {
    return $mail->hasTo($package->author->email) &&
        $mail->hasCc($package->collaborators) &&
        $mail->hasBcc('admin@novapackages.com');
});

// Or, launch two packages...

// Assert a mailable was sent twice
Mail::assertSent(PackageLaunched::class, 2);

// Assert a mailable was not sent
Mail::assertNotSent(PackageLaunchFailed::class);
}

```

All of the messages checking for recipients (`hasTo()`, `hasCc()`, and `hasBcc()`) can take either a single email address or an array or collection of addresses.

Notification Fakes

The Notification facade, when faked, offers two methods: `assertSentTo()` and `assertNothingSent()`.

Unlike with the Mail facade, you're not going to test who the notification was sent to manually in a closure. Rather, the assertion itself requires the first parameter be either a single notifiable object or an array or collection of them. Only after you've passed in the desired notification target can you test anything about the notification itself.

The second parameter is the class name for the notification, and the (optional) third parameter can be a closure defining more expectations about the notification. Take a look at [Example 12-19](#) to learn more.

Example 12-19. Notification fakes

```

public function test_users_are_notified_of_new_package_ratings()

```

```

{
    Notification::fake();

    // Perform package rating...

    // Assert author was notified
    Notification::assertSentTo(
        $package->author,
        PackageRatingReceived::class,
        function ($notification, $channels) use ($package) {
            return $notification->package->id === $package->id;
        }
    );

    // Assert a notification was sent to the given users
    Notification::assertSentTo(
        [$package->collaborators], PackageRatingReceived::class
    );

    // Or, perform a duplicate package rating...

    // Assert a notification was not sent
    Notification::assertNotSentTo(
        [$package->author], PackageRatingReceived::class
    );
}

```

You may also find yourself wanting to assert that your channel selection is working—that notifications are sent via the right channels. You can test that as well, as you can see in [Example 12-20](#).

Example 12-20. Testing notification channels

```

public function test_users_are_notified_by_their_preferred_channel()
{
    Notification::fake();

    $user = User::factory()->create(['slack_preferred' => true]);

    // Perform package rating...

    // Assert author was notified via Slack
    Notification::assertSentTo(
        $user,
        PackageRatingReceived::class,
        function ($notification, $channels) use ($package) {
            return $notification->package->id === $package->id
                && in_array('slack', $channels);
        }
    );
}

```

```
    );  
}
```

Storage Fakes

Testing files can be extraordinarily complex. Many traditional methods require you to actually move files around in your test directories, and formatting the form input and output can be very complicated.

Thankfully, if you use Laravel’s `Storage` facade, it’s infinitely simpler to test file uploads and other storage-related items, as [Example 12-21](#) demonstrates.

Example 12-21. Testing storage and file uploads with storage fakes

```
public function test_package_screenshot_upload()  
{  
    Storage::fake('screenshots');  
  
    // Upload a fake image  
    $response = $this->postJson('screenshots', [  
        'screenshot' => UploadedFile::fake()->image('screenshot.jpg'),  
    ]);  
  
    // Assert the file was stored  
    Storage::disk('screenshots')->assertExists('screenshot.jpg');  
  
    // Or, assert a file does not exist  
    Storage::disk('screenshots')->assertMissing('missing.jpg');  
}
```

Working with Time in Tests

It’s common that, when we test sections of our application that interact with time, we want to test how these sections behave differently as time passes.

In our tests, we can use `$this->travel()` to “travel” through time as the test progresses. We can travel forward and backward relative to the current time, travel to specific moments, or freeze the passage of time, allowing us to test how components behave once the time they’re checking looks different.

Take a look at [Example 12-22](#) to see how you might want to use this feature, or [the docs](#) to learn more about all the ways you can interact with time.

Example 12-22. Changing the time in a test

```

public function test_posts_are_no_longer_editable_after_thirty_minutes()
{
    $post = Post::create();

    $this->assertTrue($post->isEditable());

    $this->travel(30)->seconds();

    $this->assertTrue($post->isEditable());

    $this->travelTo($post->created_at->copy()->addMinutes(31));

    $this->assertFalse($post->isEditable());
}

```

You can also provide a closure to each of these time-traveling methods; if you do, the test's time is only modified for the duration of the closure, allowing you to more directly connect your traveling and the resulting tests, as you can see in [Example 12-23](#).

Example 12-23. Changing the time in a test using closures

```

public function test_posts_are_no_longer_editable_after_thirty_minutes()
{
    $post = Post::create();

    $this->assertTrue($post->isEditable());

    $this->travel(30)->seconds(function () {
        $this->assertTrue($post->isEditable());
    });

    $this->travelTo($post->created_at->copy()->addMinutes(31), function () {
        $this->assertFalse($post->isEditable());
    });
}

```

Mocking

Mocks (and their brethren, spies and stubs and dummies and fakes and any number of other tools) are common in testing. We saw some examples of fakes in the previous section. I won't go into too much detail here, but it's unlikely you can thoroughly test an application of any size without mocking at least one thing or another.

So, let's take a quick look at mocking in Laravel and how to use Mockery, the mocking library.

A Quick Introduction to Mocking

Essentially, mocks and other similar tools make it possible to create an object that in some way mimics a real class, but for testing purposes isn't the real class. Sometimes this is done because the real class is too difficult to instantiate just to inject it into a test, or maybe because the real class communicates with an external service.

As you can probably tell from the examples that follow, Laravel encourages working with the real application as much as possible—which means avoiding too much dependence on mocks. But they have their place, which is why Laravel includes Mockery, a mocking library, out of the box, and is why many of its core services offer faking utilities.

A Quick Introduction to Mockery

Mockery allows you to quickly and easily create mocks from any PHP class in your application. Imagine you have a class that depends on a Slack client, but you don't want the calls to actually go out to Slack. Mockery makes it simple to create a fake Slack client to use in your tests, like you can see in [Example 12-24](#).

Example 12-24. Using Mockery in Laravel

```
// app/SlackClient.php
class SlackClient
{
    // ...

    public function send($message, $channel)
    {
        // Actually sends a message to Slack
    }
}

// app/Notifier.php
class Notifier
{
    private $slack;

    public function __construct(SlackClient $slack)
```

```

    {
        $this->slack = $slack;
    }

    public function notifyAdmins($message)
    {
        $this->slack->send($message, 'admins');
    }
}

// tests/Unit/NotifierTest.php
public function test_notifier_notifies_admins()
{
    $slackMock = Mockery::mock(SlackClient::class)->shouldIgnoreMissing();

    $notifier = new Notifier($slackMock);
    $notifier->notifyAdmins('Test message');
}

```

There are a lot of elements at work here, but if you look at them one by one, they make sense. We have a class named `Notifier` that we’re testing. It has a dependency named `SlackClient` that does something that we don’t want it to do when we’re running our tests: it sends actual Slack notifications. So we’re going to mock it.

We use `Mockery` to get a mock of our `SlackClient` class. If we don’t care about what happens to that class—if it should simply exist to keep our tests from throwing errors—we can just use `shouldIgnoreMissing()`:

```
$slackMock = Mockery::mock(SlackClient::class)->shouldIgnoreMissing();
```

No matter what `Notifier` calls on `$slackMock`, it’ll just accept it and return `null`.

But take a look at `test_notifier_notifies_admins()`. At this point, it doesn’t actually *test* anything.

We could just keep `shouldIgnoreMissing()` and then write some assertions below it. That’s usually what we do with `shouldIgnoreMissing()`, which makes this object a “fake” or a “stub.”

But what if we want to actually assert that a call was made to the `send()` method of `SlackClient`? That’s when we drop `shouldIgnoreMissing()` and reach for

the other should* methods ([Example 12-25](#)).

Example 12-25. Using the shouldReceive() method on a Mockery mock

```
public function test_notifier_notifies_admins()
{
    $slackMock = Mockery::mock(SlackClient::class);
    $slackMock->shouldReceive('send')->once();

    $notifier = new Notifier($slackMock);
    $notifier->notifyAdmins('Test message');
}
```

`shouldReceive('send')->once()` is the same as saying “assert that `$slackMock` will have its `send()` method called once and only once.” So, we’re now asserting that `Notifier`, when we call `notifyAdmins()`, makes a single call to the `send()` method on `SlackClient`.

We could also use something like `shouldReceive('send')->times(3)` or `shouldReceive('send')->never()`. We can define what parameter we expect to be passed along with that `send()` call using `with()`, and we can define what to return with `andReturn()`:

```
$slackMock->shouldReceive('send')->with('Hello, world!')->andReturn(true);
```

What if we wanted to use the IoC container to resolve our instance of the `Notifier`? This might be useful if `Notifier` had several other dependencies that we didn’t need to mock.

We can do that! We just use the `instance()` method on the container, as in [Example 12-26](#), to tell Laravel to provide an instance of our mock to any classes that request it (which, in this example, will be `Notifier`).

Example 12-26. Binding a Mockery instance to the container

```
public function test_notifier_notifies_admins()
{
    $slackMock = Mockery::mock(SlackClient::class);
    $slackMock->shouldReceive('send')->once();

    app()->instance(SlackClient::class, $slackMock);

    $notifier = app(Notifier::class);
    $notifier->notifyAdmins('Test message');
```

```
}
```

There's also a convenient shortcut to create and bind a Mockery instance to the container ([Example 12-27](#)):

Example 12-27. Binding Mockery instances to the container more easily

```
$this->mock(SlackClient::class, function ($mock) {  
    $mock->shouldReceive('send')->once();  
});
```

There's a lot more you can do with Mockery: you can use spies, partial spies, and much more. Going deeper into how to use Mockery is outside the scope of this book, but I encourage you to learn more about the library and how it works by reading the [Mockery docs](#).

Faking Other Facades

There's one other clever thing you can do with Mockery: you can use Mockery methods (e.g., `shouldReceive()`) on any facades in your app.

Imagine we have a controller method that uses a facade that's not one of the fakeable systems we've already covered; we want to test that controller method and assert that a certain facade call was made.

Thankfully, it's simple: we can run our Mockery-style methods on the facade, as you can see in [Example 12-28](#).

Example 12-28. Mocking a facade

```
// PersonController  
public function index()  
{  
    return Cache::remember('people', function () {  
        return Person::all();  
    });  
}  
  
// PeopleTest  
public function test_all_people_route_should_be_cached()  
{  
    $person = Person::factory()->create();  
  
    Cache::shouldReceive('remember')  
        ->once()  
        ->andReturn(collect([$person]));
```

```

    $this->get('people')->assertJsonFragment(['name' => $person->name]);
}

```

As you can see, you can use methods like `shouldReceive()` on the facades, just like you do on a Mockery object.

You can also use your facades as spies, which means you can set your assertions at the end and use `shouldReceive()` instead of `shouldReceive()`.

Example 12-29 illustrates this.

Example 12-29. Facade spies

```

public function test_package_should_be_cached_after_visit()
{
    Cache::spy();

    $package = Package::factory()->create();

    $this->get(route('packages.show', [$package->id]));

    Cache::shouldReceive('put')
        ->once()
        ->with('packages.' . $package->id, $package->toArray());
}

```

You can also partially mock facades, as you can see in **Example 12-30**.

Example 12-30. Partially mocking facades

```

// Full mock
CustomFacade::shouldReceive('someMethod')->once();
CustomFacade::someMethod();
CustomFacade::anotherMethod(); // Fails

// Partial mock
CustomFacade::partialMock()->shouldReceive('someMethod')->once();
CustomFacade::someMethod(); // Uses the mocked object
CustomFacade::anotherMethod(); // Uses the method on the actual Facade

```

Testing Artisan Commands

We've covered a lot in this chapter, but we're almost done! We have just three more pieces of Laravel's testing arsenal to cover: Artisan, parallel testing, and the browser.

The best way to test Artisan commands is to call them with `$this->artisan($commandName, $parameters)` and then test their impact, like in [Example 12-31](#).

Example 12-31. Simple Artisan tests

```
public function test_promote_console_command_promotes_user()
{
    $user = User::factory()->create();

    $this->artisan('user:promote', ['userId' => $user->id]);

    $this->assertTrue($user->isPromoted());
}
```

You can make assertions against the response code you get from Artisan, as you can see in [Example 12-32](#).

Example 12-32. Manually asserting Artisan exit codes

```
$code = $this->artisan('do:thing', ['--flagOfSomeSort' => true]);
$this->assertEquals(0, $code); // 0 means "no errors were returned"
```

You can also chain three methods onto your `$this->artisan()` call: `expectsQuestion()`, `expectsOutput()`, and `assertExitCode()`. The `expects*` methods will work on any of the interactive prompts, including `confirm()` and `anticipate()`, and the `assertExitCode()` method is a shortcut to what we saw in [Example 12-32](#).

Take a look at [Example 12-33](#) to see how it works.

Example 12-33. Basic Artisan “expects” tests

```
// routes/console.php
Artisan::command('make:post [--expanded]', function () {
    $title = $this->ask('What is the post title?');
    $this->comment('Creating at ' . Str::slug($title) . '.md');

    $category = $this->choice('What category?', ['technology', 'construction'], 0);

    // Create post here

    $this->comment('Post created');
});
// Test file
public function test_make_post_console_commands_performs_as_expected()
{
    $this->artisan('make:post', ['--expanded' => true])
```

```
->expectsQuestion('What is the post title?', 'My Best Post Now')
->expectsOutput('Creating at my-best-post-now.md')
->expectsQuestion('What category?', 'construction')
->expectsOutput('Post created')
->assertExitCode(0);
}
```

As you can see, the first parameter of `expectsQuestion()` is the text we're expecting to see from the question, and the second parameter is the text we're answering with. `expectsOutput()` just tests that the passed string is returned.

Parallel Testing

By default, tests in Laravel run in a single thread. The more tests you have, and the more complex they are, the longer your test suite can take to run, and this can have a significant impact on how likely your team is to run your test suite.

If you want to speed up your test suite, you can run your tests in parallel. You'll need to install a dependency called `paratest`:

```
composer require brianium/paratest --dev
```

Once you've installed `paratest`, you can run your tests in parallel using the `--parallel` flag, as you can see in [Example 12-34](#).

Example 12-34. Running tests in parallel

```
# Use as many processes as your CPU can offer
php artisan test --parallel
```

```
# Specify the desired number of processes
php artisan test --parallel --processes=3
```

Browser Tests

We've made it to browser tests! These allow you to actually interact with the DOM of your pages: in browser tests you can click buttons, fill out and submit forms, and even interact with JavaScript.

Choosing a Tool

For browser testing for non-SPAs, I recommend you use Dusk. If you're working with SPAs or some JavaScript-heavy applications, they may work better with frontend test suites, which are out of the scope of this book.

Testing with Dusk

Dusk is a Laravel tool (installable as a Composer package) that makes it easy to direct an embedded instance of Google Chrome (called ChromeDriver) to interact with your app. Dusk's API is simple, and it's easy to write code to interact with it by hand. Take a look:

```
$this->browse(function ($browser) {
    $browser->visit('/register')
        ->type('email', 'test@example.com')
        ->type('password', 'secret')
        ->press('Sign Up')
        ->assertPathIs('/dashboard');
});
```

With Dusk, there's an actual browser spinning up your entire application and interacting with it. That means you can have complex interactions with your JavaScript and get screenshots of failure states—but it also means everything's a bit slower and it's more prone to failure than Laravel's base application testing suite.

Personally, I've found that Dusk is most useful as a regression testing suite, and it works better than something like Selenium. Rather than using it for any sort of test-driven development, I use it to assert that the user experience hasn't broken ("regressed") as the app continues to develop. Think of this more like writing tests about your user interface after the interface is built.

The **Dusk docs** are robust, so I'm not going to go into great depth here, but I want to show you the basics of working with Dusk.

Installing Dusk

To install Dusk, run these two commands:

```
composer require --dev laravel/dusk
php artisan dusk:install
```

Then edit your `.env` file to set your `APP_URL` variable to the same URL you use to view your site in your local browser; something like `http://mysite.test`.

To run your Dusk tests, just run `php artisan dusk`. You can pass in all the same parameters you're used to from PHPUnit (e.g., `php artisan dusk --filter=my_best_test`).

Writing Dusk tests

To generate a new Dusk test, use a command like the following:

```
php artisan dusk:make RatingTest
```

This test will be placed in `tests/Browser/RatingTest.php`.

CUSTOMIZING DUSK ENVIRONMENT VARIABLES

You can customize the environment variables for Dusk by creating a new file named `.env.dusk.local` (and you can replace `.local` if you're working in a different environment, like "staging").

To write your Dusk tests, imagine that you're directing one or more web browsers to visit your application and take certain actions. That's what the syntax will look like, as you can see in [Example 12-35](#).

Example 12-35. A simple Dusk test

```
public function testBasicExample()
{
    $user = User::factory()->create();

    $this->browse(function ($browser) use ($user) {
        $browser->visit('login')
            ->type('email', $user->email)
            ->type('password', 'secret')
            ->press('Login')
            ->assertPathIs('/home');
    });
}
```

`$this->browse()` creates a browser, which you pass into a closure; then, within the closure, you instruct the browser which actions to take.

It's important to note that—unlike Laravel's other application testing tools, which mimic the behavior of your forms—Dusk is actually spinning up a browser, sending events to the browser to type those words, and then sending an event to the browser to press that button. This is a real browser and Dusk is fully driving it.

You can also “ask” for more than one browser by adding parameters to the closure, which allows you to test how multiple users might interact with the website (for example, with a chat system). Take a look at [Example 12-36](#), from the docs.

Example 12-36. Multiple Dusk browsers

```
$this->browse(function ($first, $second) {  
    $first->loginAs(User::find(1))  
        ->visit('home')  
        ->waitForText('Message');  
  
    $second->loginAs(User::find(2))  
        ->visit('home')  
        ->waitForText('Message')  
        ->type('message', 'Hey Taylor')  
        ->press('Send');  
  
    $first->waitForText('Hey Taylor')  
        ->assertSee('Jeffrey Way');  
});
```

There's a huge suite of actions and assertions available that we won't cover here (check the docs), but let's look at a few of the other tools Dusk provides.

Authentication and databases

As you can see in [Example 12-36](#), the syntax for authentication is a little different from the rest of the Laravel application testing: `$browser->loginAs($user)`.

AVOID THE REFRESHDATABASE TRAIT WITH DUSK

Don't use the `RefreshDatabase` trait with Dusk! Use the `DatabaseMigrations` trait instead; transactions, which `RefreshDatabase` uses, don't persist across requests.

Interactions with the page

If you've ever written jQuery, interacting with the page using Dusk will come naturally. Take a look at [Example 12-37](#) to see the common patterns for selecting items with Dusk.

Example 12-37. Selecting items with Dusk

```
<-- Template -->
<div class="search"><input><button id="search-button"></button></div>
<button dusk="expand-nav"></button>
// Dusk tests
// Option 1: jQuery-style syntax
$browser->click('.search button');
$browser->click('#search-button');

// Option 2: dusk="selector-here" syntax; recommended
$browser->click('@expand-nav');
```

As you can see, adding the dusk attribute to your page elements allows you to reference them directly in a way that won't change when the display or layout of the page changes later; when any method asks for a selector, pass in the @ sign and then the content of your dusk attribute.

Let's take a look at a few of the methods you can call on \$browser.

To work with text and attribute values, use these methods:

`value($selector, $value = null)`

Returns the value of any text input if only one parameter is passed; sets the value of an input if a second parameter is passed.

`text($selector)`

Gets the text content of a nonfillable item like a <div> or a .

`attribute($selector, $attributeName)`

Returns the value of a particular attribute on the element matching \$selector.

Methods for working with forms and files include the following:

`type($selector, $valueToType)`

Similar to `value()`, but actually types the characters rather than directly setting the value.

DUSK'S SELECTOR MATCHING ORDER

With methods like `type()` that target inputs, Dusk will start by trying to match a Dusk or CSS selector, and then will look for an input with the provided name, and finally will try to find a `<textarea>` with the provided name.

`select($selector, $optionValue)`

Selects the option with the value of *\$optionValue* in a drop-down selectable by *\$selector*.

`check($selector), unchecked($selector)`

Checks or unchecks a checkbox selectable by *\$selector*.

`radio($selector, $optionValue)`

Selects the option with the value of *\$optionValue* in a radio group selectable by *\$selector*.

`attach($selector, $filePath)`

Attaches a file at *\$filePath* to the file input selectable by *\$selector*.

The methods for keyboard and mouse input are:

`clickLink($selector)`

Follows a text link to its target.

`click($selector), mouseover($selector)`

Triggers a mouse click or a mouseover event on *\$selector*.

`drag($selectorToDrag, $selectorToDragTo)`

Drags an item to another item.

`dragLeft()`, `dragRight()`, `dragUp()`, `dragDown()`

Given a first parameter of a selector and a second parameter of a number of pixels, drags the selected item that many pixels in the given direction.

`keys($selector, $instructions)`

Sends keypress events within the context of *\$selector* according to the instructions in *\$instructions*. You can even combine modifiers with your typing:

```
$browser->keys('selector', 'this is ', ['{shift}', 'great']);
```

This would type “this is GREAT.” As you can see, adding an array to the list of items to type allows you to combine modifiers (wrapped with `{}`) with typing. You can see a full list of the possible modifiers in the [Facebook WebDriver source](#).

If you’d like to just send your key sequence to the page (for example, to trigger a keyboard shortcut), you can target the top level of your app or page as your selector. For example, if it’s a Vue app and the top level is a `<div>` with an ID of `app`:

```
$browser->keys('#app', ['{command}', '/']);
```

Waiting

Because Dusk interacts with JavaScript and is directing an actual browser, the concept of time and timeouts and “waiting” needs to be addressed. Dusk offers several methods you can use to ensure your tests handle timing issues correctly. Some of these methods are useful for interacting with intentionally slow or delayed elements of the page, but some of them are also just useful for getting around initialization times on your components. The available methods include the following:

`pause($milliseconds)`

Pauses the execution of Dusk tests for the given number of milliseconds. This is the simplest “wait” option; it makes any future commands you send to the browser wait that amount of time before operating.

You can use this and other waiting methods in the midst of an assertion chain, as shown here:

```
$browser->click('chat')  
->pause(500)  
->assertSee('How can we help?');
```

`waitFor($selector, $maxSeconds = null),`
`waitUntilMissing($selector, $maxSeconds = null)`

Waits until the given element exists on the page (`waitFor()`) or disappears from the page (`waitUntilMissing()`) or times out after the optional second parameter’s second count:

```
$browser->waitFor('@chat', 5);  
$browser->waitUntilMissing('@loading', 5);
```

`whenAvailable($selector, $callback)`

Similar to `waitFor()`, but accepts a closure as the second parameter, which will define what action to take when the specified element becomes available:

```
$browser->whenAvailable('@chat', function ($chat) {  
    $chat->assertSee('How can we help you?');  
});
```

`waitForText($text, $maxSeconds = null)`

Waits for text to show up on the page, or times out after the optional second

parameter's second count:

```
$browser->waitForText('Your purchase has been completed.', 5);
```

`waitForLink($linkText, $maxSeconds = null)`

Waits for a link to exist with the given link text, or times out after the optional second parameter's second count:

```
$browser->waitForLink('Clear these results', 2);
```

`waitForLocation($path)`

Waits until the page URL matches the provided path:

```
$browser->waitForLocation('auth/login');
```

`waitForRoute($routeName)`

Waits until the page URL matches the URL for the provided route:

```
$browser->waitForRoute('packages.show', [$package->id]);
```

`waitForReload()`

Waits until the page reloads.

`waitUntil($expression)`

Waits until the provided JavaScript expression evaluates as true:

```
$browser->waitUntil('App.packages.length > 0', 7);
```

Other Assertions

As I've mentioned, there's a huge list of assertions you can make against your

app with Dusk. Here are a few that I use most commonly—you can see the full list in the [Dusk docs](#):

- `assertTitleContains($text)`
- `assertQueryStringHas($keyName)`
- `assertHasCookie($cookieName)`
- `assertSourceHas($htmlSourceCode)`
- `assertChecked($selector)`
- `assertSelectHasOption($selectorForSelect, $optionValue)`
- `assertVisible($selector)`
- `assertFocused()`
- `assertVue($dataLocation, $dataValue, $selector)`

Other organizational structures

So far, everything we've covered makes it possible to test individual elements on our pages. But we'll often use Dusk to test more complex applications and single-page apps, which means we're going to need organizational structures around our assertions.

The first organizational structures we have encountered have been the `dusk` attribute (e.g., `<div dusk="abc">`, creating a selector named `@abc` we can refer to later) and the closures we can use to wrap certain portions of our code (e.g., with `whenAvailable()`).

Dusk offers two more organizational tools: pages and components. Let's start with pages.

Pages

A page is a class that you'll generate which contains two pieces of functionality: first, a URL and assertions to define which page in your app should be attached to this Dusk page; and second, shorthand like we used inline (the `@abc` selector generated by the `dusk="abc"` attribute in our HTML) but just for this page, and

without needing to edit our HTML.

Let's imagine our app has a “create package” page. We can generate a Dusk page for it as follows:

```
php artisan dusk:page CreatePackage
```

Take a look at [Example 12-38](#) to see what our generated class will look like.

Example 12-38. The generated Dusk page

```
<?php

namespace Tests\Browser\Pages;

use Laravel\Dusk\Browser;

class CreatePackage extends Page
{
    /**
     * Get the URL for the page
     *
     * @return string
     */
    public function url()
    {
        return '/';
    }

    /**
     * Assert that the browser is on the page
     *
     * @param Browser $browser
     * @return void
     */
    public function assert(Browser $browser)
    {
        $browser->assertPathIs($this->url());
    }

    /**
     * Get the element shortcuts for the page
     *
     * @return array
     */
    public function elements()
    {
        return [
```

```

        '@element' => '#selector',
    ];
}
}

```

The `url()` method defines the location where Dusk should expect this page to be, `assert()` lets you run additional assertions to verify you’re on the right page, and `elements()` provides shortcuts for @dusk-style selectors.

Let’s make a few quick modifications to our “create package” page, to make it look like [Example 12-39](#).

Example 12-39. A simple “create package” Dusk page

```

class CreatePackage extends Page
{
    public function url()
    {
        return '/packages/create';
    }

    public function assert(Browser $browser)
    {
        $browser->assertTitleContains('Create Package');
        $browser->assertPathIs($this->url());
    }

    public function elements()
    {
        return [
            '@title' => 'input[name=title]',
            '@instructions' => 'textarea[name=instructions]',
        ];
    }
}

```

Now that we have a functional page, we can navigate to it and access its defined elements:

```

// In a test
$browser->visit(new Tests\Browser\Pages\CreatePackage)
    ->type('@title', 'My package title');

```

One common use for pages is to define a common action you want to take in your tests; consider these almost like macros for Dusk. You can define a method on your page and then call it from your code, as you can see in [Example 12-40](#).

Example 12-40. Defining and using a custom page method

```
class CreatePackage extends Page
{
    // ... url(), assert(), elements()

    public function fillBasicFields(Browser $browser, $packageTitle = 'Best package')
    {
        $browser->type('@title', $packageTitle)
            ->type('@instructions', 'Do this stuff and then that stuff');
    }
}
$browser->visit(new CreatePackage)
    ->fillBasicFields('Greatest Package Ever')
    ->press('Create Package')
    ->assertSee('Greatest Package Ever');
```

Components

If you want the same functionality as Dusk pages offer, but without it being constrained to a specific URL, you'll likely want to reach for Dusk *components*. These classes are shaped very similarly to pages, but instead of being bound to a URL, they're each bound to a selector.

In *NovaPackages.com*, we have a little Vue component for rating packages and displaying ratings. Let's make a Dusk component for it:

```
php artisan dusk:component RatingWidget
```

Take a look at [Example 12-41](#) to see what that will generate.

Example 12-41. The default source of a generated Dusk component

```
<?php

namespace Tests\Browser\Components;

use Laravel\Dusk\Browser;
use Laravel\Dusk\Component as BaseComponent;

class RatingWidget extends BaseComponent
{
    /**
     * Get the root selector for the component
     *
     * @return string
     */
    public function selector()
```

```

{
    return '#selector';
}

/**
 * Assert that the browser page contains the component
 *
 * @param Browser $browser
 * @return void
 */
public function assert(Browser $browser)
{
    $browser->assertVisible($this->selector());
}

/**
 * Get the element shortcuts for the component
 *
 * @return array
 */
public function elements()
{
    return [
        '@element' => '#selector',
    ];
}
}

```

As you can see, this is basically the same as a Dusk page, but we're encapsulating our work to an HTML element instead of a URL. Everything else is basically the same. Take a look at [Example 12-42](#) to see our rating widget example in Dusk component form.

Example 12-42. A Dusk component for the rating widget

```

class RatingWidget extends BaseComponent
{
    public function selector()
    {
        return '.rating-widget';
    }

    public function assert(Browser $browser)
    {
        $browser->assertVisible($this->selector());
    }

    public function elements()
    {

```

```

        return [
            '@5-star' => '.five-star-rating',
            '@4-star' => '.four-star-rating',
            '@3-star' => '.three-star-rating',
            '@2-star' => '.two-star-rating',
            '@1-star' => '.one-star-rating',
            '@average' => '.average-rating',
            '@mine' => '.current-user-rating',
        ];
    }

    public function ratePackage(Browser $browser, $rating)
    {
        $browser->click("@{$rating}-star")
            ->assertSeeIn('@mine', $rating);
    }
}

```

Using components works just like using pages, as you can see in [Example 12-43](#).

Example 12-43. Using Dusk components

```

$browser->visit('/packages/tightenco/nova-stock-picker')
    ->within(new RatingWidget, function ($browser) {
        $browser->ratePackage(2);
        $browser->assertSeeIn('@average', 2);
    });

```

That’s a good, brief overview of what Dusk can do. There’s a lot more—more assertions, more edge cases, more gotchas, more examples—in the [Dusk docs](#), so I’d recommend a read through there if you plan to work with Dusk.

Pest

Pest is a third-party testing framework for Laravel. It’s a layer on top of PHPUnit that provides customized console output, easy parallel testing and code coverage, architecture testing, and more.

Pest also offers a different testing syntax, inspired by Ruby’s RSpec. You can use Pest and get all of its benefits without switching to its unique testing syntax, but if you do want to try it out, take a look at [Example 12-44](#) to see how the syntax looks.

Example 12-44. Sample Pest syntax

```

it('has a welcome page', function () {

```

```
$response = $this->get('/');  
  
expect($response->status())->toBe(200);  
});
```

To learn more about Pest, check out pestphp.com.

TL;DR

Laravel can work with any modern PHP testing framework, but it's optimized for PHPUnit (especially if your tests extend Laravel's `TestCase`). Laravel's application testing framework makes it simple to send fake HTTP and console requests through your application and inspect the results.

Tests in Laravel can easily and powerfully interact with and assert against the database, cache, session, filesystem, mail, and many other systems. Quite a few of these systems have fakes built in to make them even easier to test. You can test DOM and browser-like interactions with Dusk.

Laravel brings in Mockery in case you need mocks, stubs, spies, dummies, or anything else, but the testing philosophy of Laravel is to use real collaborators as much as possible. Don't fake it unless you have to.

Chapter 13. Writing APIs

One of the most common tasks Laravel developers are given is to create an API, usually JSON and REST or REST-like, that allows third parties to interact with the Laravel application's data.

Laravel makes it incredibly easy to work with JSON, and its resource controllers are already structured around REST verbs and patterns. In this chapter you'll learn about some basic API-writing concepts, the tools Laravel provides for writing APIs, and some external tools and organizational systems you'll want to consider when writing your first Laravel API.

The Basics of REST-Like JSON APIs

Representational state transfer (REST) is an architectural style for building APIs. Technically, REST is either a broad definition that could apply to almost the entirety of the internet or something so specific that *no one* actually uses it, so don't let yourself get overwhelmed by the definition or caught in an argument with a pedant. When we talk about RESTful or REST-like APIs in the Laravel world, we're generally talking about APIs with a few common characteristics:

- They're structured around "resources" that can be uniquely represented by URIs, like `/cats` for all cats, `/cats/15` for a single cat with the ID of 15, etc.
- Interactions with resources primarily take place using HTTP verbs (GET `/cats/15` versus DELETE `/cats/15`).
- They're stateless, meaning there's no persistent session authentication between requests; each request must uniquely authenticate itself.
- They're cacheable and consistent, meaning each request (except for a few authenticated user-specific requests) should return the same result regardless of who the requester is.
- They return JSON.

The most common API pattern is to have a unique URL structure for each of your Eloquent models that's exposed as an API resource and allow for users to interact with that resource with specific verbs and get JSON back. **Example 13-1** shows a few possible examples.

Example 13-1. Common REST API endpoint structures

```
GET /api/cats
[
  {
    id: 1,
    name: 'Fluffy'
  },
  {
    id: 2,
    name: 'Killer'
  }
]
```

```
GET /api/cats/2
{
  id: 2,
  name: 'Killer'
}
```

```
POST /api/cats with body:
{
  name: 'Mr Bigglesworth'
}
(creates new cat)
```

```
PATCH /api/cats/3 with body:
{
  name: 'Mr. Bigglesworth'
}
(updates cat)
```

```
DELETE /api/cats/2
(deletes cat)
```

This gives you the idea of the basic set of interactions we are likely to have with our APIs. Let's dig into how to make them happen with Laravel.

Controller Organization and JSON Returns

Laravel’s API resource controllers are like normal resource controllers (see “Resource Controllers”) but modified to align with RESTful API routes. For example, they exclude the `create()` and `edit()` methods, both of which are irrelevant in an API. Let’s get started there. First we’ll create a new controller for our resource, which we’ll route at `/api/dogs`:

```
php artisan make:controller Api/DogController --api
```

Example 13-2 shows what our API resource controller will look like.

Example 13-2. A generated API resource controller

```
<?php

namespace App\Http\Controllers\Api;

use Illuminate\Http\Request;
use App\Http\Controllers\Controller;

class DogController extends Controller
{
    /**
     * Display a listing of the resource.
     */
    public function index()
    {
        //
    }

    /**
     * Store a newly created resource in storage.
     */
    public function store(Request $request)
    {
        //
    }

    /**
     * Display the specified resource.
     */
    public function show(string $id)
    {
        //
    }

    /**
     * Update the specified resource in storage.
     */
}
```

```

    */
    public function update(Request $request, string $id)
    {
        //
    }

    /**
     * Remove the specified resource from storage.
     */
    public function destroy(string $id)
    {
        //
    }
}

```

The docblocks pretty much tell the story. `index()` lists all of the dogs, `show()` lists a single dog, `store()` stores a new dog, `update()` updates a dog, and `destroy()` removes a dog.

Let's quickly make a model and a migration so we can work with it:

```

php artisan make:model Dog --migration
php artisan migrate

```

Great! Now we can fill out our controller methods.

DATABASE REQUIREMENTS FOR THESE CODE SAMPLES TO WORK

If you want the code we're writing here to actually work, you'll want to add a `string()` column to the migration named `name` and another named `breed`, and either add those columns to the Eloquent model's `fillable` property or just set the `guarded` property of that model equal to an empty array (`[]`). Later examples will also require columns for `weight`, `color`, and relationships for `bones` and `friends`.

We can take advantage of a great feature of Eloquent here: if you echo an Eloquent results collection, it'll automatically convert itself to JSON (using the `__toString()` magic method, if you're curious). That means if you return a collection of results from a route, you'll in effect be returning JSON. So, as [Example 13-3](#) demonstrates, this will be some of the simplest code you'll ever write.

Example 13-3. A sample API resource controller for the Dog entity

```
...  
class DogController extends Controller  
{  
    public function index()  
    {  
        return Dog::all();  
    }  
  
    public function store(Request $request)  
    {  
        return Dog::create($request->only(['name', 'breed']));  
    }  
  
    public function show(string $id)  
    {  
        return Dog::findOrFail($id);  
    }  
  
    public function update(Request $request, string $id)  
    {  
        $dog = Dog::findOrFail($id);  
        $dog->update($request->only(['name', 'breed']));  
        return $dog;  
    }  
  
    public function destroy(string $id)  
    {  
        Dog::findOrFail($id)->delete();  
    }  
}
```

Artisan's `make:model` command also has an `--api` flag you can pass to generate the same API-specific controller we generated above:

```
php artisan make:model Dog --api
```

If you want to generate migration, seeder, factory, policy, and resource controller, and both store and update form requests in one command, you can use the `--all` flag:

```
php artisan make:model Dog --all
```

Example 13-4 shows how we can link this up in our routes file. As you can see,

we can use `Route::apiResource()` to automatically map all of these default methods to their appropriate routes and HTTP verbs.

Example 13-4. Binding the routes for a resource controller

```
// routes/api.php
Route::namespace('App\Http\Controllers\Api')->group(function () {
    Route::apiResource('dogs', DogController::class);
});
```

There you have it! Your first RESTful API in Laravel. Of course, you'll need much more nuance: pagination, sorting, authentication, and better-defined response headers. But this is the foundation of everything else.

Reading and Sending Headers

REST APIs often read, and send, noncontent information using headers. For example, any request to GitHub's API will return headers detailing the current user's rate limiting status:

```
X-RateLimit-Limit: 5000
X-RateLimit-Remaining: 4987
X-RateLimit-Reset: 1350085394
```

X-* HEADERS

You might be wondering why the GitHub rate limiting headers are prefixed with X-, especially if you see them in the context of other headers returned with the same request:

```
HTTP/1.1 200 OK
Server: nginx
Date: Fri, 12 Oct 2012 23:33:14 GMT
Content-Type: application/json; charset=utf-8
Connection: keep-alive
Status: 200 OK
ETag: "a00049ba79152d03380c34652f2cb612"
X-GitHub-Media-Type: github.v3
X-RateLimit-Limit: 5000
X-RateLimit-Remaining: 4987
X-RateLimit-Reset: 1350085394
Content-Length: 5
```

```
Cache-Control: max-age=0, private, must-revalidate
X-Content-Type-Options: nosniff
```

Any header whose name starts with X- is a header that's not in the HTTP spec. It might be entirely made up (e.g., X-How-Much-Matt-Loves-This-Page), or part of a common convention that hasn't made it into the spec yet (e.g., X-Requested-With).

Similarly, many APIs allow developers to customize their requests using request headers. For example, GitHub's API makes it easy to define which version of the API you'd like to use with the Accept header:

```
Accept: application/vnd.github.v3+json
```

If you were to change v3 to v2, GitHub would pass your request to version 2 of its API instead.

Let's learn quickly how to do both in Laravel.

Sending Response Headers in Laravel

We already covered this topic quite a bit in [Chapter 10](#), but here's a quick refresher. Once you have a response object, you can add a header using `header($headerName, $headerValue)`, as seen in [Example 13-5](#).

Example 13-5. Adding a response header in Laravel

```
Route::get('dogs', function () {
    return response(Dog::all())
        ->header('X-Greatness-Index', 12);
});
```

Nice and easy.

Reading Request Headers in Laravel

If you have an incoming request, it's also simple to read any given header. [Example 13-6](#) illustrates this.

Example 13-6. Reading a request header in Laravel

```
Route::get('dogs', function (Request $request) {
```

```
var_dump($request->header('Accept'));
});
```

Now that you can read incoming request headers and set headers on your API responses, let's take a look at how you might want to customize your API.

Eloquent Pagination

Pagination is one of the first places where most APIs need to consider special instructions. Eloquent comes out of the box with a pagination system that hooks directly into the query parameters of any page request. We already covered the paginator component a bit in [Chapter 6](#), but here's a quick refresher.

Any Eloquent call provides a `paginate()` method, to which you can pass the number of items you'd like to return per page. Eloquent then checks the URL for a page query parameter and, if it's set, treats that as an indicator of where (how many pages) the user is in a paginated list.

To make your API route ready for automated Laravel pagination, use `paginate()` instead of `all()` or `get()` to call your Eloquent queries in your route; something like [Example 13-7](#).

Example 13-7. A paginated API route

```
Route::get('dogs', function () {
    return Dog::paginate(20);
});
```

We've defined that Eloquent should get 20 results from the database. Depending on what the page query parameter is set to, Laravel will know exactly *which* 20 results to pull for us:

```
GET /dogs          - Return results 1-20
GET /dogs?page=1   - Return results 1-20
GET /dogs?page=2   - Return results 21-40
```

Note that the `paginate()` method is also available on query builder calls, as seen in [Example 13-8](#).

Example 13-8. Using the paginate() method on a query builder call

```
Route::get('dogs', function () {
```

```
    return DB::table('dogs')->paginate(20);
});
```

Here's something interesting, though: this isn't just going to return 20 results when you convert it to JSON. Instead, it's going to build a response object that automatically passes some useful pagination-related details to the end user, *along with* the paginated data. **Example 13-9** shows a possible response from our call, truncated to only three records to save space.

Example 13-9. Sample output from a paginated database call

```
{
  "current_page": 1,
  "data": [
    {
      "name": 'Fido'
    },
    {
      "name": 'Pickles'
    },
    {
      "name": 'Spot'
    }
  ]
  "first_page_url": "http://myapp.com/api/dogs?page=1",
  "from": 1,
  "last_page": 2,
  "last_page_url": "http://myapp.com/api/dogs?page=2",
  "links": [
    {
      "url": null,
      "label": "&laquo; Previous",
      "active": false
    },
    {
      "url": "http://myapp.com/api/dogs?page=1",
      "label": "1",
      "active": true
    },
    {
      "url": null,
      "label": "Next &raquo;",
      "active": false
    }
  ],
  "next_page_url": "http://myapp.com/api/dogs?page=2",
  "path": "http://myapp.com/api/dogs",
  "per_page": 20,
```



```
"prev_page_url": null,  
"to": 2,  
"total": 4  
}
```

Sorting and Filtering

Although there is a convention and some built-in tooling for pagination in Laravel, there isn't any for sorting, so you have to figure that out on your own. I'll give a quick code sample here, and I'll style the query parameters similarly to the JSON API spec (described in the following sidebar).

THE JSON API SPEC

The **JSON API** is a standard for how to handle many of the most common tasks in building JSON-based APIs: filtering, sorting, pagination, authentication, embedding, linking, metadata, and more.

Laravel's default pagination doesn't work *exactly* according to the JSON API spec, but it gets you started in the right direction. The majority of the rest of the JSON API spec is something you'll just have to choose (or not) to implement manually.

For example, here's a piece of the JSON API spec that helpfully handles how to structure data versus error returns:

A document MUST contain at least one of the following top-level members:

- **data**: *the document's "primary data"*
- **errors**: *an array of error objects*
- **meta**: *a meta object that contains non-standard meta-information.*

The members data and errors MUST NOT coexist in the same document.

Be warned, however: it's wonderful to have the JSON API as a spec, but it also takes quite a bit of groundwork to get running with it. We won't use it entirely in these examples, but I'll use its general ideas as inspiration.

Sorting Your API Results

First, let's set up the ability to sort our results. We start in [Example 13-10](#) with the ability to sort by only a single column, and in only a single direction.

Example 13-10. Simplest API sorting

```
// Handles /dogs?sort=name
Route::get('dogs', function (Request $request) {
    // Get the sort query parameter (or fall back to default sort "name")
    $sortColumn = $request->input('sort', 'name');
    return Dog::orderBy($sortColumn)->paginate(20);
});
```

We add the ability to invert it (e.g., ?sort=-weight) in [Example 13-11](#).

Example 13-11. Single-column API sorting, with direction control

```
// Handles /dogs?sort=name and /dogs?sort=-name
Route::get('dogs', function (Request $request) {
    // Get the sort query parameter (or fall back to default sort "name")
    $sortColumn = $request->input('sort', 'name');

    // Set the sort direction based on whether the key starts with -
    // using Laravel's starts_with() helper function
    $sortDirection = str_starts_with($sortColumn, '-') ? 'desc' : 'asc';
    $sortColumn = ltrim($sortColumn, '-');

    return Dog::orderBy($sortColumn, $sortDirection)
        ->paginate(20);
});
```

Finally, we do the same for multiple columns (e.g., ?sort=name,-weight) in [Example 13-12](#).

Example 13-12. JSON API–style sorting

```
// Handles ?sort=name,-weight
Route::get('dogs', function (Request $request) {
    // Grab the query parameter and turn it into an array exploded by ,
    $sorts = explode(',', $request->input('sort', ''));

    // Create a query
    $query = Dog::query();

    // Add the sorts one by one
    foreach ($sorts as $sortColumn) {
        $sortDirection = str_starts_with($sortColumn, '-') ? 'desc' : 'asc';
        $sortColumn = ltrim($sortColumn, '-');
    }
});
```

```

        $query->orderBy($sortColumn, $sortDirection);
    }

    // Return
    return $query->paginate(20);
});

```

As you can see, it's not the simplest process ever, and you'll likely want to build some helper tooling around the repetitive processes, but we're building up the customizability of our API piece by piece using logical and simple features.

Filtering Your API Results

Another common task in building APIs is filtering out all but a certain subset of data. For example, the client might ask for a list of the dogs that are Chihuahuas.

The JSON API doesn't give us any great ideas for syntax here, other than that we should use the `filter` query parameter. Let's think along the lines of the sort syntax, where we're putting everything into a single key—maybe ?

`filter=breed:chihuahua`. You can see how to do this in [Example 13-13](#).

Example 13-13. Single filter on API results

```

Route::get('dogs', function () {
    $query = Dog::query();

    $query->when(request()->filled('filter'), function ($query) {
        [$criteria, $value] = explode(':', request('filter'));
        return $query->where($criteria, $value);
    });

    return $query->paginate(20);
});

```

Note that in [Example 13-13](#) we're using the `request()` helper instead of injecting an instance of `$request`. Both work the same, but sometimes the `request()` helper can be easier when you're working inside of a closure so you don't have to pass variables in manually.

And, just for kicks, in [Example 13-14](#) we allow for multiple filters, like ?
`filter=breed:chihuahua,color:brown`.

Example 13-14. Multiple filters on API results

```

Route::get('dogs', function (Request $request) {

```

```

$query = Dog::query();

$query->when(request()->filled('filter'), function ($query) {
    $filters = explode(',', request('filter'));

    foreach ($filters as $filter) {
        [$criteria, $value] = explode(':', $filter);
        $query->where($criteria, $value);
    }

    return $query;
});

return $query->paginate(20);
});

```

Transforming Results

We've covered how to sort and filter our result sets. But right now, we're relying on Eloquent's JSON serialization, which means we return every field on every model.

Eloquent provides a few convenience tools for defining which fields to show when you're serializing an array. You can read more in [Chapter 5](#), but the gist is that if you set a `$hidden` array property on your Eloquent class, any field listed in that array will not be shown in the serialized model output. You can alternatively set a `$visible` array that defines the fields that are allowed to be shown. You could also either overwrite or mimic the `toArray()` function on your model, crafting a custom output format.

Another common pattern is to create a *transformer* for each data type. Transformers are helpful because they give you more control, isolate API-specific logic away from the model itself, and allow you to provide a more consistent API even when the models and their relationships change down the road.

There's a fantastic but complicated package for this, [Fractal](#), that sets up a series of convenience structures and classes for transforming your data.

API Resources

In the past, one of the first challenges we'd run into when developing APIs in Laravel was how to transform our data. The simplest APIs can just return Eloquent objects as JSON, but very quickly the needs of most APIs outgrow that structure. How should we convert our Eloquent results into the right format? What if we want to embed other resources or do so but only optionally, or add a computed field or hide some fields from APIs but not other JSON output? An API-specific transformer is the solution.

We now have access to a feature called *Eloquent API resources*, which are structures that define how to transform an Eloquent object (or a collection of Eloquent objects) of a given class to API results. For example, your `Dog` Eloquent model now has a `Dog` resource whose responsibility it is to translate each instance of `Dog` to the appropriate `Dog`-shaped API response object.

Creating a Resource Class

Let's walk through this `Dog` example to see what it looks like to transform our API output. First, use the Artisan command `make:resource` to create your first resource:

```
php artisan make:resource Dog
```

This will create a new class in `app/Http/Resources/Dog.php`, which contains one method: `toArray()`. You can see what the file looks like in [Example 13-15](#).

Example 13-15. Generated API resource

```
<?php

namespace App\Http\Resources;

use Illuminate\Http\Request;
use Illuminate\Http\Resources\Json\JsonResource;

class Dog extends JsonResource
{
    /**
     * Transform the resource into an array.
     *
     * @return array<string, mixed>
     */
    public function toArray(Request $request): array
```

```

    {
        return parent::toArray($request);
    }
}

```

The `toArray()` method we're working with here has access to two important pieces of data. First, it has access to the Illuminate Request object, so we can customize our response based on query parameters and headers and anything else important. And second, it has access to the entire Eloquent object being transformed by calling its properties and methods on `$this`, as you can see in [Example 13-16](#).

Example 13-16. Simple API resource for the Dog model

```

class Dog extends JsonResource
{
    public function toArray(Request $request): array
    {
        return [
            'id' => $this->id,
            'name' => $this->name,
            'breed' => $this->breed,
        ];
    }
}

```

To use this new resource, you'll want to update any API endpoint that returns a single Dog to wrap the response in your new resource, like in [Example 13-17](#).

Example 13-17. Using the simple Dog resource

```

use App\Dog;
use App\Http\Resources\Dog as DogResource;

Route::get('dogs/{dogId}', function ($dogId) {
    return new DogResource(Dog::find($dogId));
});

```

Resource Collections

Now, let's talk about what happens when you have more than one of your entities returning from a given API endpoint. This is possible using an API resource's `collection()` method, as you can see in [Example 13-18](#).

Example 13-18. Using the default API resource collection method

```

use App\Dog;
use App\Http\Resources\Dog as DogResource;

Route::get('dogs', function () {
    return DogResource::collection(Dog::all());
});

```

This method iterates over every entry that's passed to it, transforms it with the `DogResource` API resource, and then returns the collection.

This will likely be enough for many APIs, but if you need to customize any of the structure or add metadata to your collection responses, you'll want to instead create a custom API resource collection.

In order to do so, let's reach for the `make:resource` Artisan command again. This time we'll name it `DogCollection`, which signals to Laravel that this is an API resource collection, not just an API resource:

```
php artisan make:resource DogCollection
```

This will generate a new file very similar to the API resource file, living at `app/Http/Resources/DogCollection.php`, which again contains one method: `toArray()`. You can see what the file looks like in [Example 13-19](#).

Example 13-19. Generated API resource collection

```

<?php

namespace App\Http\Resources;

use Illuminate\Http\Resources\Json\ResourceCollection;

class DogCollection extends ResourceCollection
{
    /**
     * Transform the resource collection into an array.
     *
     * @return array<int|string, mixed>
     */
    public function toArray(Request $request): array
    {
        return parent::toArray($request);
    }
}

```

Just like with the API resource, we have access to the request and the underlying

data. But unlike with the API resource, we're dealing with a collection of items instead of just one, so we will access that (already transformed) collection as `$this->collection`. Take a look at [Example 13-20](#) for an example.

Example 13-20. A simple API resource collection for the Dog model

```
class DogCollection extends ResourceCollection
{
    public function toArray(Request $request): array
    {
        return [
            'data' => $this->collection,
            'links' => [
                'self' => route('dogs.index'),
            ],
        ];
    }
}
```

Nesting Relationships

One of the more complicated aspects of any API is how relationships are nested. The simplest way with API resources is to add a key to your returned array that's set to an API resource collection, like in [Example 13-21](#).

Example 13-21. A simple included API relationship

```
public function toArray(Request $request): array
{
    return [
        'name' => $this->name,
        'breed' => $this->breed,
        'friends' => Dog::collection($this->friends),
    ];
}
```

WARNING

If you try the code in [Example 13-21](#) and receive a 502 error, it's because you haven't loaded the "friends" relationship on your parent resource first. Keep reading to see how to get around that, but here's how to eager load that relationship when you're working with this resource, using the `with()` method:

```
return new DogResource(Dog::with('friends')->find($dogId));
```

You may also want this to be a conditional property; you can choose to only nest it if it's asked for in the request or only if it's already been eager loaded on the Eloquent object that's passed in. Take a look at [Example 13-22](#).

Example 13-22. Conditionally loading API relationship

```
public function toArray(Request $request): array
{
    return [
        'name' => $this->name,
        'breed' => $this->breed,
        // Only load this relationship if it's been eager loaded
        'bones' => BoneResource::collection($this->whenLoaded('bones')),
        // Or only load this relationship if the URL asks for it
        'bones' => $this->when(
            $request->get('include') == 'bones',
            BoneResource::collection($this->bones)
        ),
    ];
}
```

Using Pagination with API Resources

Just like you can pass a collection of Eloquent models to a resource, you can also pass a paginator instance. Take a look at [Example 13-23](#).

Example 13-23. Passing a paginator instance to an API resource collection

```
Route::get('dogs', function () {
    return new DogCollection(Dog::paginate(20));
});
```

If you pass a paginator instance, the transformed result will have additional links containing pagination information (first page, last page, prev page, and next page) and meta-information about the entire collection.

You can take a look at [Example 13-24](#) to see what this information looks like. In this example, I've set the items-per-page count to 2 by calling `Dog::paginate(2)` so you can more easily see how the links work.

Example 13-24. A sample paginated resource response with pagination links

```
{
  "data": [
    {
```

```

        "name": "Pickles",
        "breed": "Chorkie"
    },
    {
        "name": "Gandalf",
        "breed": "Golden Retriever Mix"
    }
],
"links": {
    "self": "http://gooddogbrant.com/api/dogs",
    "first": "http://gooddogbrant.com/api/dogs?page=1",
    "last": "http://gooddogbrant.com/api/dogs?page=3",
    "prev": null,
    "next": null
},
"meta": {
    "current_page": 1,
    "data": [
        {
            "name": "Pickles",
            "breed": "Chorkie",
        },
        {
            "name": "Gandalf",
            "breed": "Golden Retriever Mix",
        }
    ],
    "first_page_url": "http://gooddogbrent.com/api/dogs?page=1",
    "from": 1,
    "last_page": 3,
    "last_page_url": "http://gooddogbrent.com/api/dogs?page=3",
    "links": [
        {
            "url": null,
            "label": "&laquo; Previous",
            "active": false
        },
        {
            "url": "http://gooddogbrent.com/api/dogs?page=1",
            "label": "1",
            "active": true
        },
        {
            "url": "http://gooddogbrent.com/api/dogs?page=2",
            "label": "Next &raquo;",
            "active": false
        }
    ],
    "next_page_url": null,

```

```
"path": "http://gooddogbrent.com/api/dogs",
"per_page": 3,
"to": 3,
"total": 9
}
}
```

Conditionally Applying Attributes

You can also specify that certain attributes in your response should only be applied when a particular test is satisfied, as illustrated in [Example 13-25](#).

Example 13-25. Conditionally applying attributes

```
public function toArray(Request $request): array
{
    return [
        'name' => $this->name,
        'breed' => $this->breed,
        'rating' => $this->when(Auth::user()->canSeeRatings(), 12),
    ];
}
```

More Customizations for API Resources

The default shape of how the data property is wrapped might not be how you like it, or you may find yourself needing to add or customize metadata for the responses. Take a look at the [resources docs](#) for details on how to customize every aspect of your API responses.

API Authentication

Laravel provides two primary tools for authenticating API requests: Sanctum (most recommended) and Passport (powerful but very complex and usually overkill).

API Authentication with Sanctum

Sanctum is an API authentication system for Laravel, built for two tasks: generating simple tokens for your power users to use to interact with your API, and allowing SPAs and mobile apps to latch onto your existing authentication

system. It's not quite as configurable as OAuth 2.0, but it's *very close* and it comes with much lower cost in terms of setup and configuration.

There are several ways to use Sanctum. You can allow power users to generate tokens for your API directly in your admin panel, similar to how many developer-focused SaaS services do. You can allow users to visit a special login page to receive a token directly, which is useful for authenticating mobile apps to your API. And you can integrate with your SPA, which, using some of Sanctum's special sauce, can hook directly into Laravel's cookie-based authentication sessions, and you won't have to manage tokens at all.

Let's look at how to install Sanctum and then how to use it in each of these contexts.

Installing Sanctum

Sanctum comes preinstalled with new Laravel projects. If your project doesn't have it, you'll need to manually install it and publish its config files.

```
composer require laravel/sanctum
php artisan vendor:publish --provider="Laravel\Sanctum\SanctumServiceProvider"
php artisan migrate
```

For any routes you want to protect with Sanctum, attach the `auth:sanctum` middleware:

```
Route::get('clips', function () {
    return view('clips.index', ['clips' => Clip::all()]);
})->middleware('auth:sanctum');
```

Issuing Sanctum tokens manually

If you want to build tooling in your application to provide tokens for your users to authenticate against your API, here are the steps you'll want to take.

First, make sure your User model uses the `HasApiTokens` trait (on a new project, it'll already have it):

```
use Laravel\Sanctum\HasApiTokens;

class User extends Authenticatable
```

```
{
    use HasApiTokens, HasFactory, Notifiable;
}
```

Next, build a user interface that allows the user to generate a token. You could make a button in their settings page that says “Generate new token,” pops up a modal asking for the nickname for that token, and then posts the results to this form:

```
Route::post('tokens/create', function () {
    $token = auth()->user()->createToken(request()->token_name);

    return view('tokens.created', ['token' => $token->plainTextToken]);
});
```

You can also list out all of the tokens your user has by referencing the `tokens` property of a `user` object:

```
Route::get('tokens', function () {
    return view('tokens.index', ['tokens' => auth()->user()->tokens]);
});
```

Sanctum token abilities

A common security pattern for token-based API authentication is to allow users to generate tokens with only certain privileges, to minimize the potential damage if the token were compromised.

If you want to build a system for that, you can define (based on business logic or user preferences) which “abilities” each token has when you create it. Pass an array of strings to the `createToken()` method and each string will represent an ability that token has.

```
$token = $user->createToken(
    request()->token_name, ['list-clips', 'add-delete-clips']
);
```

Then, your code can check against the authenticated user’s tokens either directly (as in [Example 13-26](#)) or through middleware (as in [Example 13-27](#)).

Example 13-26. Manually checking a user’s access based on token abilities

```
if (request()->user()->tokenCan('list-clips')) {
    // ...
}
```

Example 13-27. Using middleware to restrict access based on token scopes

```
// routes/api.php
Route::get('clips', function () {
    // Access token has both the "list-clips" and "add-delete-clips" abilities
})->middleware(['auth:sanctum', 'abilities:list-clips,add-delete-clips']);

// or

Route::get('clips', function () {
    // Access token has at least one of the listed abilities
})->middleware(['auth:sanctum', 'ability:list-clips,add-delete-clips'])
```

NOTE

If you want to use Sanctum's middleware checks, you'll need to add the following two lines to the `middlewareAliases` property of `App\Http\Kernel`.

```
'abilities' => \Laravel\Sanctum\Http\Middleware\
    CheckAbilities::class,
'ability' => \Laravel\Sanctum\Http\Middleware\
    CheckForAnyAbility::class,
```

SPA authentication

If you plan to use Sanctum to authenticate with an SPA, there are a few steps you'll need to take first to set up your Laravel app and your SPA.

Laravel app preparation

First, uncomment the `EnsureFrontendRequestsAreStateful` class on the `api` middleware group in `app/Http/Kernel.php`.

```
'api' => [
    \Laravel\Sanctum\Http\Middleware\EnsureFrontendRequestsAreStateful::class,
    // Other API middleware here
],
```

Second, update the list of “stateful” domains in the Sanctum config. These are all the domains your SPA can make requests from. You can modify them directly in *config/sanctum.php*, or add a comma-separated list of domains to the `SANCTUM_STATEFUL_DOMAINS` key in your *.env* file.

SPA app preparation

Before allowing your users to log into your app, your SPA should request Laravel set a CSRF cookie, which most JavaScript HTTP clients like Axios will then pass with every future request.

```
axios.get('/sanctum/csrf-cookie').then(response => {  
  // Handle login  
});
```

You can then log in to your Laravel login route, either one you’ve created yourself or a route offered by an existing tool like Fortify. Future requests will remain authenticated via the session cookie Laravel sets for you.

Mobile app authentication

Here’s the workflow for allowing your mobile app users to authenticate to a Sanctum-based app: request the user’s email (or username) and their password in your mobile app. Send that, together with the name of their device (reading from the device’s name in its OS; e.g., “Matt’s iPhone”), to a route you create yourself on the backend, which will validate their login and (assuming it’s valid) create and return a token, as you can see in [Example 13-28](#), which I’ve taken directly from the docs.

Example 13-28. Route to accept mobile app logins to Sanctum-based apps

```
Route::post('sanctum/token', function (Request $request) {  
    $request->validate([  
        'email' => 'required|email',  
        'password' => 'required',  
        'device_name' => 'required',  
    ]);  
  
    $user = User::where('email', $request->email)->first();  
  
    if (! $user || ! Hash::check($request->password, $user->password)) {  
        throw ValidationException::withMessages([  
            'email' => ['The provided credentials are incorrect.'],  
        ]);  
    }  
});
```

```
    });  
}  
  
return $user->createToken($request->device_name)->plainTextToken;  
});
```

Future requests to the API should pass the token in the `Authorization` header as a Bearer type token.

Further configuration and debugging

If you're having any trouble with your Sanctum install or want to customize any of Sanctum's functionality, check out the [Sanctum docs](#) to learn more.

API Authentication with Laravel Passport

Passport (a first-party package that has to be installed, brought in via Composer) makes it easy to set up a full-featured OAuth 2.0 server in your application, complete with an API and UI components for managing clients and tokens.

A brief introduction to OAuth 2.0

OAuth is by far the most common auth system used in RESTful APIs. Unfortunately, it's far too complex a topic for us to cover here in depth. For further reading, Matt Frost has written a great book on OAuth and PHP titled *Integrating Web Services with OAuth and PHP* (php[architect]).

Here's the simplest concept behind OAuth: because APIs are stateless, we can't rely on the same session-based authentication that we do in normal browser-based viewing sessions, where the user logs in and their authenticated state is saved to the session for subsequent views. Instead, the API client needs to make a single call to an authentication endpoint and perform some form of handshake to prove itself. It then gets back a token that it must send along with every future request (via the `Authorization` header, usually) to prove its identity.

There are a few different types of OAuth "grant," which basically means that there are several different scenarios and types of interaction that can define that authentication handshake. Different projects and different sorts of end consumers will necessitate different grants.

Passport gives you everything needed to add a basic OAuth 2.0 authentication

server to your Laravel application, with a simpler and powerful API and interface.

Installing Passport

Passport is a separate package, so your first step is to install it. I'll sum up the steps here, but you can get more in-depth installation instructions in the [Passport docs](#).

First, bring it in with Composer:

```
composer require laravel/passport
```

Passport imports a series of migrations, so run those with `php artisan migrate` to create the tables necessary for OAuth clients, scopes, and tokens.

Next, run the installer with `php artisan passport:install`. This will create encryption keys for the OAuth server (`storage/oauth-private.key` and `storage/oauth-public.key`) and insert OAuth clients into the database for our personal and password grant type tokens (which we'll cover later).

You'll need to import the `Laravel\Passport\HasApiTokens` trait into your `User` model; this will add OAuth client- and token-related relationships to each `User`, as well as a few token-related helper methods.

Finally, add a new auth guard in `config/auth.php` named `api`; set the provider to `users`, and the driver to `passport`.

You now have a fully functional OAuth 2.0 server! You can create new clients with `php artisan passport:client`, and you have an API for managing your clients and tokens available under the `/oauth` route prefix.

To protect a route behind your Passport auth system, add the `auth:api` middleware to the route or route group, as shown in [Example 13-29](#).

Example 13-29. Protecting an API route with the Passport auth middleware

```
// routes/api.php
Route::get('/user', function (Request $request) {
    return $request->user();
})->middleware('auth:api');
```

In order to authenticate to these protected routes, your client apps will need to pass a token (we'll cover how to get one shortly) as a Bearer token in the Authorization header. **Example 13-30** shows what this would look like if you were making a request using the HTTP client included by Laravel.

Example 13-30. Making a sample API request with a Bearer token

```
use Illuminate\Support\Facades\Http;

$response = Http::withHeaders(['Accept' => 'application/json'])
    ->withToken($accessToken)
    ->get('http://tweeter.test/api/user');
```

Now, let's take a closer look at how it all works.

Passport's API

Passport exposes an API in your application under the `/oauth` route prefix. The API provides two primary functions: first, to authorize users with OAuth 2.0 authorization flows (`/oauth/authorize` and `/oauth/token`), and second, to allow users to manage their clients and tokens (the rest of the routes).

This is an important distinction, especially if you're unfamiliar with OAuth. Every OAuth server needs to expose the ability for consumers to authenticate with your server; that's the entire point of the service. But Passport *also* exposes an API for managing the state of your OAuth server's clients and tokens. This means you can easily build a frontend to let your users manage their information in your OAuth application. Passport actually comes with Vue-based manager components that you can either use directly or use for inspiration.

We'll cover the API routes that allow you to manage clients and tokens, and the Vue components that Passport ships with to make that easy, but first let's dig into the various ways your users can authenticate with your Passport-protected API.

Passport's available grant types

Passport makes it possible for you to authenticate users in four different ways. Two are traditional OAuth 2.0 grants (the password grant and authorization code grant) and two are convenience methods that are unique to Passport (the personal token and synchronizer token).

Password grant

The *password grant*, while less common than the authorization code grant, is much simpler. If you want users to be able to authenticate directly with your API using their username and password—for example, if you have a mobile app for your company consuming your own API—you can use the password grant.

CREATING A PASSWORD GRANT CLIENT

To use the password grant flow, you need a password grant client in your database. This is because every request to an OAuth server needs to be made by a client. Usually, the client identifies which app or site the user is authenticating against—for example, if you used Facebook to log in to a third-party website, that website would be the client.

With the password grant flow, however, there is no client coming along with the request, so you have to create one—and that’s the password grant client. One will have been added when you ran `php artisan passport:install`, but if you ever need to generate a new password grant client for any reason, you can do so as follows:

```
php artisan passport:client --password
```

```
What should we name the password grant client?
```

```
[My Application Password Grant Client]:
```

```
> Client_name
```

```
Which user provider should this client use to retrieve users? [users]:
```

```
[0] users:
```

```
> 0
```

```
Password grant client created successfully.
```

```
Client ID: 3
```

```
Client Secret: Pg1EEzt18JAnFoUIM9n38Nqewg1aekB4rvFk2Pma
```

With the password grant type, there is just one step to getting a token: sending the user’s credentials to the `/oauth/token` route, like in [Example 13-31](#).

Example 13-31. Making a request with the password grant type

```
// routes/web.php in the *consuming application*
Route::get('tweeter/password-grant-auth', function () {
    // Make call to "Tweeter," our Passport-powered OAuth server
```

```

$response = Http::post('http://tweeter.test/oauth/token', [
    'grant_type' => 'password',
    'client_id' => config('tweeter.id'),
    'client_secret' => config('tweeter.secret'),
    'username' => 'matt@mattstauffer.co',
    'password' => 'my-tweeter-password',
    'scope' => '',
]);

$thisUsersTokens = $response->json();
// Do stuff with the tokens
});

```

This route will return an `access_token`, a `refresh_token`, and two pieces of metadata: `token_type` and `expires_in` (discussed later in this chapter). You can now save those tokens to use to authenticate with the API (access token) and to request more tokens later (refresh token).

Note that the ID and secret we would use for the password grant type would be those in the `oauth_clients` database table of our Passport app in the row whose name matches that of our Passport grant client. You’ll also see entries in this table for the two clients that are generated by default when you run `passport:install`: “Laravel Personal Access Client” and “Laravel Password Grant Client.”

Authorization code grant

The most common OAuth 2.0 auth workflow is also the most complex one Passport supports. Let’s imagine we’re developing an application that’s like Twitter but for sound clips; we’ll call it Tweeter. And we’ll imagine another website, a social network for science fiction fans, called SpaceBook.

SpaceBook’s developer wants to let people embed their Tweeter data into their SpaceBook newsfeeds. We’re going to install Passport in our Tweeter app so that other apps—SpaceBook, for example—can allow their users to authenticate with their Tweeter information.

In the *authorization code grant* type, each consuming website—SpaceBook, in this example—needs to create a client in our Passport-enabled app. In most scenarios, the other sites’ admins will have user accounts at Tweeter, and we’ll build tools for them to create clients there. But for starters, we can just manually create a client for the SpaceBook admins:

```
php artisan passport:client
```

```
Which user ID should the client be assigned to?:
```

```
> 1
```

```
What should we name the client?:
```

```
> SpaceBook
```

```
Where should we redirect the request after authorization?
```

```
[http://tweeter.test/auth/callback]:
```

```
> http://spacebook.test/tweeter/callback
```

```
New client created successfully.
```

```
Client ID: 4
```

```
Client secret: 5rzqKpeCjIgz3MXpi3tjQ37HBnLLykrkWgmc18uH
```

To answer the first question, you'll need to know that every client needs to be assigned to a user in your app. Imagine user #1 is writing SpaceBook; they'll be the "owner" of this client we're creating.

Once we've run this command, we have the ID and secret for the SpaceBook client. At this point, SpaceBook can use this ID and secret to build tooling that allows an individual SpaceBook user (who is also a Tweeter user) to get an auth token from Tweeter for use when SpaceBook wants to make API calls to Tweeter on that user's behalf. [Example 13-32](#) illustrates this. (This and the following examples assume SpaceBook is a Laravel app, too; they also assume the Spacebook's developer created a file at `config/tweeter.php` that returns the ID and secret we just created.)

Example 13-32. A consumer app redirecting a user to our OAuth server

```
// In SpaceBook's routes/web.php:
Route::get('tweeter/redirect', function () {
    $query = http_build_query([
        'client_id' => config('tweeter.id'),
        'redirect_uri' => url('tweeter/callback'),
        'response_type' => 'code',
        'scope' => '',
    ]);

    // Builds a string like:
    // client_id={client_id}&redirect_uri={redirect_uri}&response_type=code

    return redirect('http://tweeter.test/oauth/authorize?' . $query);
});
```

When users hit that route in SpaceBook, they'll now be redirected to the

/oauth/authorize Passport route in our Tweeter app. At this point they'll see a confirmation page—you can use the default Passport confirmation page by running this command:

```
php artisan vendor:publish --tag=passport-views
```

This will publish the view to `resources/views/vendor/passport/authorize.blade.php`, and your users will see the page shown in [Figure 13-1](#).

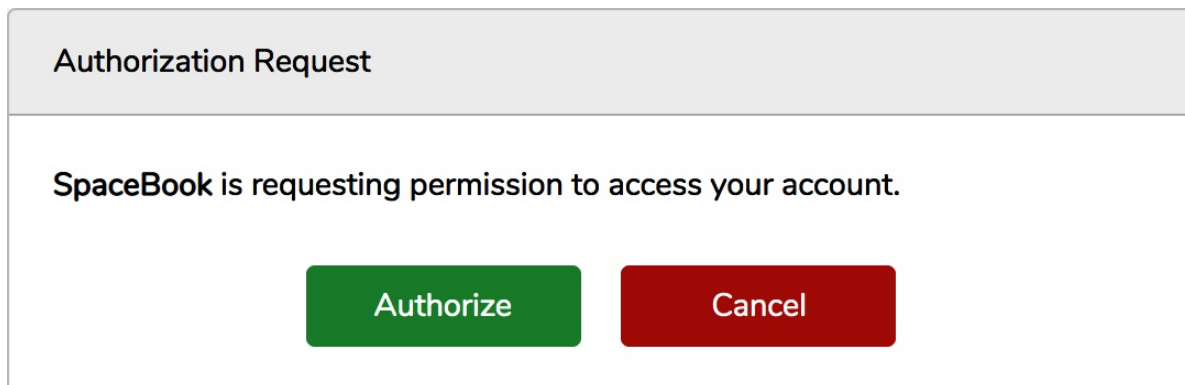


Figure 13-1. OAuth authorization code approval page

Once a user chooses to accept or reject the authorization, Passport will redirect that user back to the provided `redirect_uri`. In [Example 13-32](#) we set a `redirect_uri` of `url('tweeter/callback')`, so the user will be redirected back to `http://spacebook.test/tweeter/callback`.

An approval request will contain a code that our consumer app's callback route can now use to get a token back from our Passport-enabled app, Tweeter. A rejection request will contain an error. SpaceBook's callback route might look something like [Example 13-33](#).

Example 13-33. The authorization callback route in the sample consuming app

```
// In SpaceBook's routes/web.php:
Route::get('tweeter/callback', function (Request $request) {
    if ($request->has('error')) {
        // Handle error condition
    }

    $response = Http::post('http://tweeter.test/oauth/token', [
        'grant_type' => 'authorization_code',
        'client_id' => config('tweeter.id'),
    ]
```

```

        'client_secret' => config('tweeter.secret'),
        'redirect_uri' => url('tweeter/callback'),
        'code' => $request->code,
    ]);

    $thisUsersTokens = $response->json();
    // Do stuff with the tokens
});

```

What the SpaceBook developer has done here is build an HTTP request, using the Laravel HTTP client, to the `/oauth/token` Passport route on Tweeter. They then send a POST request containing the authorization code they received when the user approved access, and Tweeter will return a JSON response containing a few keys:

`access_token`

The token SpaceBook will want to save for this user. This token is what the user will use to authenticate in future requests to Tweeter (using the `Authorization` header).

`refresh_token`

A token SpaceBook will need *if* you decide to set your tokens to expire. By default, Passport's access tokens last for one year.

`expires_in`

The number of seconds until an `access_token` expires (needs to be refreshed).

`token_type`

The type of token you're getting back, which will be `Bearer`; this means you pass a header with all future requests with the name of `Authorization` and the value of `Bearer YOURTOKENHERE`.

USING REFRESH TOKENS

If you'd like to force users to reauthenticate more often, you need to set a shorter refresh time on the tokens, and then you can use `refresh_token` to

request a new `access_token` when needed—most likely whenever you receive a 401 (Unauthorized) response from an API call.

Example 13-34 illustrates how to set a shorter refresh time.

Example 13-34. Defining token refresh times

```
// AuthServiceProvider's boot() method
public function boot(): void
{
    Passport::routes();

    // How long a token lasts before needing refreshing
    Passport::tokensExpireIn(
        now()->addDays(15)
    );

    // How long a refresh token will last before re-auth
    Passport::refreshTokensExpireIn(
        now()->addDays(30)
    );
}
```

To request a new token using a refresh token, the consuming application will need to have first saved `refresh_token` from the initial auth response in **Example 13-33**. Once it's time to refresh, it will make a call similar to that example, but modified slightly as shown in **Example 13-35**.

Example 13-35. Requesting a new token using a refresh token

```
// In SpaceBook's routes/web.php:
use Illuminate\Support\Facades\Http;

Route::get('tweeter/request-refresh', function (Request $request) {
    $response = Http::post('http://tweeter.test/oauth/token', [
        'grant_type' => 'refresh_token',
        'client_id' => config('tweeter.id'),
        'client_secret' => config('tweeter.secret'),
        'redirect_uri' => url('tweeter/callback'),
        'refresh_token' => $theTokenYouSavedEarlier,
        'scope' => '',
    ]);

    $thisUsersTokens = $response->json();

    // Do stuff with the tokens
});
```

In the response, the consuming app will receive a fresh set of tokens to save

to its user.

You now have all the tools you need to perform basic authorization code flows. We'll cover how to build an admin panel for your clients and tokens later, but first, let's take a quick look at the other grant types.

Personal access tokens

The authorization code grant is great for your users' apps, and the password code grant is great for your own apps, but what if your users want to create tokens for themselves to test out your API or to use when they're developing their apps? That's what personal tokens are for.

CREATING A PERSONAL ACCESS CLIENT

To create personal tokens, you need a personal access client in your database. Running `php artisan passport:install` will have added one already, but if you ever need to generate a new personal access client for any reason, you can run `php artisan passport:client --personal`:

```
php artisan passport:client --personal
```

```
What should we name the personal access client?
```

```
[My Application Personal Access Client]:
```

```
> My Application Personal Access Client
```

```
Personal access client created successfully.
```

Personal access tokens are not quite a “grant” type; there's no OAuth-prescribed flow here. Rather, they're a convenience method that Passport adds to make it easy to have a single client registered in your system that exists solely to facilitate the creation of convenience tokens for your users who are developers.

For example, maybe you have a user who's developing a competitor to SpaceBook named RaceBook (it's for marathon runners), and they want to toy around with the Tweeter API a bit to figure out how it works *before* starting to code. Does this developer have the facility to create tokens using the authorization code flow? Not yet—they haven't even written any code yet!

That's what personal access tokens are for.

You can create personal access tokens through the JSON API, which we'll cover shortly, but you can also create one for your user directly in code:

```
// Creating a token without scopes
$token = $user->createToken('Token Name')->accessToken;

// Creating a token with scopes
$token = $user->createToken('My Token', ['place-orders'])->accessToken;
```

Your users can use these tokens just as if they were tokens created with the authorization code grant flow. We'll talk more about scopes in “**Passport scopes**”.

Tokens from Laravel session authentication (synchronizer tokens)

There's one final way for your users to get tokens to access your API, and it's another convenience method that Passport adds but that normal OAuth servers don't provide. This method is for when your users are already authenticated because they've logged in to your Laravel app like normal, and you want your app's JavaScript to be able to access the API. It'd be a pain to have to reauthenticate the users with the authorization code or password grant flow, so Laravel provides a helper for that.

If you add the `Laravel\Passport\Http\Middleware\CreateFreshApiToken` middleware to your web middleware group (in `app/Http/Kernel.php`), every response Laravel sends to your authenticated users will have a cookie named `laravel_token` attached to it. This cookie is a JSON Web Token (JWT) that contains encoded information about the CSRF token. Now, if you send the normal CSRF token with your JavaScript requests in the `X-CSRF-TOKEN` header and also send the `X-Requested-With` header with any API requests you make, the API will compare your CSRF token with this cookie and this will authenticate your users to the API just like any other token.

JSON WEB TOKENS (JWT)

JWT is a relatively new format for “representing claims securely between two parties” that has gained prominence over the last few years. A JSON

Web Token is a JSON object containing all of the information necessary to determine a user's authentication state and access permissions. This JSON object is digitally signed using a keyed-hash message authentication code (HMAC) or RSA, which is what makes it trustworthy.

The token is usually encoded and then delivered via URL or POST request, or in a header. Once a user authenticates with the system somehow, every HTTP request after that will contain the token, describing the user's identity and authorization.

JSON Web Tokens consist of three Base64-encoded strings separated by dots (.); something like *xxx.yyy.zzz*. The first section is a Base64-encoded JSON object containing information about which hashing algorithm is being used; the second section is a series of "claims" about the user's authorization and identity; and the third is the signature, or the first and second sections encrypted and signed using the algorithm specified in the first section.

To learn more about JWT, check out [JWT.IO](#) or the [jwt-auth Laravel package](#).

The default JavaScript bootstrap setup that Laravel comes bundled with sets up this header for you, but if you're using a different framework, you'll need to set it up manually. [Example 13-36](#) shows how to do it with jQuery.

Example 13-36. Setting jQuery to pass Laravel's CSRF tokens and the X-Requested-With header with all Ajax requests

```
$.ajaxSetup({
  headers: {
    'X-CSRF-TOKEN': "{{ csrf_token() }}",
    'X-Requested-With': 'XMLHttpRequest'
  }
});
```

If you add the `CreateFreshApiToken` middleware to your web middleware group and pass those headers with every JavaScript request, your JavaScript requests will be able to hit your Passport-protected API routes without worrying about any of the complexity of the authorization code or password grants.

Passport scopes

If you're familiar with OAuth, you've probably noticed we haven't talked much about scopes yet. Everything we've covered so far can be customized by scope—but before we get into that, let's first quickly cover what scopes are.

In OAuth, *scopes* are defined sets of privileges that are something other than “can do everything.” If you've ever gotten a GitHub API token before, for example, you might've noticed that some apps want access just to your name and email address, some want access to all of your repos, and some want access to your gists. Each of these is a “scope,” which allows both the user and the consumer app to define what access the consumer app needs to perform its job.

As shown in [Example 13-37](#), you can define the scopes for your application in the `boot()` method of your `AuthServiceProvider`.

Example 13-37. Defining Passport scopes

```
// AuthServiceProvider
use Laravel\Passport\Passport;
...
public function boot(): void
{
    ...

    Passport::tokensCan([
        'list-clips' => 'List sound clips',
        'add-delete-clips' => 'Add new and delete old sound clips',
        'admin-account' => 'Administer account details',
    ]);
}
```

Once you have your scopes defined, the consumer app can define which scopes it's asking for access to. Just add a space-separated list of tokens in the `scope` field in the initial redirect, as shown in [Example 13-38](#).

Example 13-38. Requesting authorization to access specific scopes

```
// In SpaceBook's routes/web.php:
Route::get('tweeter/redirect', function () {
    $query = http_build_query([
        'client_id' => config('tweeter.id'),
        'redirect_uri' => url('tweeter/callback'),
        'response_type' => 'code',
        'scope' => 'list-clips add-delete-clips',
    ]);

    return redirect('http://tweeter.test/oauth/authorize?' . $query);
});
```

```
});
```

When the user tries to authorize with this app, it'll present the list of requested scopes. This way, the user will know whether "SpaceBook is requesting to see your email address" or "SpaceBook is requesting access to post as you and delete your posts and message your friends."

You can check for scope using middleware or on the User instance. **Example 13-39** shows how to check on the User.

Example 13-39. Checking whether the token a user authenticated with can perform a given action

```
Route::get('/events', function () {
    if (auth()->user()->tokenCan('add-delete-clips')) {
        //
    }
});
```

There are two middleware you can use for this too, `scope` and `scopes`. To use these in your app, add them to `$middlewareAliases` in your `app/Http/Kernel.php` file:

```
'scopes' => \Laravel\Passport\Http\Middleware\CheckScopes::class,
'scope' => \Laravel\Passport\Http\Middleware\CheckForAnyScope::class,
```

You can now use the middleware as illustrated in **Example 13-40**. `scopes` requires *all* of the defined scopes to be on the user's token in order for the user to access the route, while `scope` requires *at least one* of the defined scopes to be on the user's token.

Example 13-40. Using middleware to restrict access based on token scopes

```
// routes/api.php
Route::get('clips', function () {
    // Access token has both the "list-clips" and "add-delete-clips" scopes
})->middleware('scopes:list-clips,add-delete-clips');

// or

Route::get('clips', function () {
    // Access token has at least one of the listed scopes
})->middleware('scope:list-clips,add-delete-clips')
```

If you haven't defined any scopes, the app will just work as if they don't exist.

The moment you use scopes, however, your consumer apps will have to explicitly define which scopes they’re requesting access with. The one exception to this rule is that if you’re using the password grant type, your consumer app can request the `*` scope, which gives the token access to everything.

Deploying Passport

The first time you deploy your Passport-powered app, the Passport API won’t function until you generate keys for the app. This can be accomplished by running `php artisan passport:keys` on your production server, which will generate the encryption keys Passport uses to generate tokens.

Customizing 404 Responses

Laravel offers customizable error-message pages for normal HTML views, but you can also customize the default 404 fallback response for calls with a JSON content type. To do so, add a `Route::fallback()` call to your API, as shown in [Example 13-41](#).

Example 13-41. Defining a fallback route

```
// routes/api.php
Route::fallback(function () {
    return response()->json(['message' => 'Route Not Found'], 404);
})->name('api.fallback.404');
```

Triggering the Fallback Route

If you want to customize which route is returned when Laravel catches “not found” exceptions, you can update the exception handler using the `respondWithRoute()` method, as illustrated in [Example 13-42](#).

Example 13-42. Calling the fallback route when “not found” exceptions are caught

```
// App\Exceptions\Handler
use Illuminate\Support\Facades\Route;
use Symfony\Component\HttpKernel\Exception\NotFoundHttpException;
use Illuminate\Http\Request;

public function register(): void
{
```

```

$this->renderable(function (NotFoundException $e, Request $request) {
    if ($request->isJson()) {
        return Route::respondWithRoute('api.fallback.404');
    }
});
}

```

Testing

Fortunately, testing APIs is actually simpler than testing almost anything else in Laravel.

We cover this in more depth in [Chapter 12](#), but there are a series of methods for making assertions against JSON. Combine that capability with the simplicity of full-stack application tests, and you can put together your API tests quickly and easily. Take a look at the common API testing pattern in [Example 13-43](#).

Example 13-43. A common API testing pattern

```

...
class DogsApiTest extends TestCase
{
    use WithoutMiddleware, RefreshDatabase;

    public function test_it_gets_all_dogs()
    {
        $dog1 = Dog::factory()->create();
        $dog2 = Dog::factory()->create();

        $response = $this->getJson('api/dogs');

        $response->assertJsonFragment(['name' => $dog1->name]);
        $response->assertJsonFragment(['name' => $dog2->name]);
    }
}

```

Note that we're using `WithoutMiddleware` to avoid worrying about authentication. You'll want to test that separately, if at all (for more on authentication, see [Chapter 9](#)).

In this test we insert two `Dogs` into the database, then visit the API route for listing all `Dogs` and make sure both are present in the output.

You can cover all of your API routes simply and easily here, including

modifying actions like POST and PATCH.

Testing Passport

You can use the `actingAs()` method on the `Passport` facade to test your scopes. Take a look at [Example 13-44](#) to see a common pattern for testing scopes in Passport.

Example 13-44. Testing scoped access

```
public function test_it_lists_all_clips_for_those_with_list_clips_scope()
{
    Passport::actingAs(
        User::factory()->create(),
        ['list-clips']
    );

    $response = $this->getJson('api/clips');
    $response->assertStatus(200);
}
```

TL;DR

Laravel is geared toward building APIs and makes it simple to work with JSON and RESTful APIs. There are some conventions, like for pagination, but much of the definition of exactly how your API will be sorted, authenticated, or whatever else is up to you.

Laravel provides tools for authentication and testing, easy manipulation and reading of headers, and working with JSON, even automatically encoding all Eloquent results to JSON if they're returned directly from a route.

Laravel Passport is a separate package that makes it simple to create and manage an OAuth server in your Laravel apps.

Chapter 14. Storage and Retrieval

We looked at how to store data in relational databases in [Chapter 5](#), but there's a lot more that can be stored, both locally and remotely. In this chapter we'll cover filesystem and in-memory storage, file uploads and manipulation, nonrelational data stores, sessions, the cache, logging, cookies, and full-text search.

Local and Cloud File Managers

Laravel provides a series of file manipulation tools through the Storage facade and a few helper functions.

Laravel's filesystem access tools can connect to the local filesystem as well as to S3, Rackspace, and FTP. The S3 and Rackspace file drivers are provided by [Flysystem](#), and it's simple to add additional Flysystem providers, such as Dropbox or WebDAV, to your Laravel app.

Configuring File Access

The definitions for Laravel's file manager live in *config/filesystems.php*. Each connection is called a “disk,” and [Example 14-1](#) lists the disks that are available out of the box.

Example 14-1. Default available storage disks

```
...
'disks' => [
    'local' => [
        'driver' => 'local',
        'root' => storage_path('app'),
        'throw' => false,
    ],
    'public' => [
        'driver' => 'local',
        'root' => storage_path('app/public'),
        'url' => env('APP_URL').'/storage',
        'visibility' => 'public',
        'throw' => false,
    ],
],
```

```

's3' => [
    'driver' => 's3',
    'key' => env('AWS_ACCESS_KEY_ID'),
    'secret' => env('AWS_SECRET_ACCESS_KEY'),
    'region' => env('AWS_DEFAULT_REGION'),
    'bucket' => env('AWS_BUCKET'),
    'url' => env('AWS_URL'),
    'endpoint' => env('AWS_ENDPOINT'),
    'use_path_style_endpoint' => env('AWS_USE_PATH_STYLE_ENDPOINT', false),
    'throw' => false,
],
],

```

THE STORAGE_PATH() HELPER

The `storage_path()` helper used in [Example 14-1](#) links to Laravel’s configured storage directory, *storage/*. Anything you pass to it is added to the end of the directory name, so `storage_path('public')` will return the string *storage/public*.

The `local` disk connects to your local storage system and presumes it will be interacting with the *app* directory of the storage path, which is *storage/app*.

The `public` disk is also a local disk (although you can change it if you’d like), which is intended for use with any files you intend to be served by your application. It defaults to the *storage/app/public* directory, and if you want to use this directory to serve files to the public, you’ll need to add a symbolic link (*symlink*) to somewhere within the *public/* directory. Thankfully, there’s an Artisan command that maps *public/storage* to serve the files from *storage/app/public*:

```
php artisan storage:link
```

The `s3` disk shows how Laravel connects to cloud-based file storage systems. If you’ve ever connected to S3 or any other cloud storage provider, this will be familiar; pass it your key and secret and some information defining the “folder” you’re working with, which in S3 is the region and the bucket.

S3, FTP, OR SFTP DRIVERS PACKAGES REQUIRED

To use the S3, FTP, or SFTP drivers, you first need to install a Composer package for the desired driver.

For S3:

```
composer require -W league/flysystem-aws-s3-v3 "^3.0"
```

For FTP:

```
composer require league/flysystem-ftp "^3.0"
```

For SFTP:

```
composer require league/flysystem-sftp-v3 "^3.0"
```

Using the Storage Facade

In *config/filesystem.php* you can set the default disk, which is what will be used any time you call the Storage facade without specifying a disk. To specify a disk, call `disk('diskname')` on the facade:

```
Storage::disk('s3')->get('file.jpg');
```

The filesystems all provide the following methods:

`get('file.jpg')`

Retrieves the file at *file.jpg*

`json('file.json', $flags)`

Retrieves the file at *file.json* and decodes its JSON contents

`put('file.jpg', $contentsOrStream)`

Puts the given file contents to *file.jpg*

`putFile('myDir', $file)`

Puts the contents of a provided file (in the form of an instance of either `Illuminate\Http\File` or `Illuminate\Http\UploadedFile`) to the *myDir* directory, but with Laravel managing the entire streaming process and naming the file

`exists('file.jpg')`

Returns a Boolean indicating whether *file.jpg* exists

`getVisibility('myPath')`

Gets the visibility for the given path (“public” or “private”)

`setVisibility('myPath')`

Sets the visibility for the given path (“public” or “private”)

`copy('file.jpg', 'newfile.jpg')`

Copies *file.jpg* to *newfile.jpg*

`move('file.jpg', 'newfile.jpg')`

Moves *file.jpg* to *newfile.jpg*

`prepend('my.log', 'log text')`

Adds the *log text* content at the beginning of *my.log*

`append('my.log', 'log text')`

Adds the *log text* content to the end of *my.log*

`delete('file.jpg')`

Deletes *file.jpg*

`size('file.jpg')`

Returns the size in bytes of *file.jpg*

`lastModified('file.jpg')`

Returns the Unix timestamp when *file.jpg* was last modified

`files('myDir')`

Returns an array of filenames in the directory *myDir*

`allFiles('myDir')`

Returns an array of filenames in the directory *myDir* and all its subdirectories

`directories('myDir')`

Returns an array of directory names in the directory *myDir*

`allDirectories('myDir')`

Returns an array of directory names in the directory *myDir* and all its subdirectories

`makeDirectory('myDir')`

Creates a new directory

`deleteDirectory('myDir')`

Deletes *myDir*

`readStream('my.log')`

Gets a resource to read *my.log*

`writeStream('my.log', $resource)`

Writes a new file (*my.log*) using a stream

INJECTING AN INSTANCE

If you'd prefer injecting an instance instead of using the File facade, typehint or inject `Illuminate\Filesystem\Filesystem` and you'll have all the same methods available to you.

Adding Additional Flysystem Providers

If you want to add an additional Flysystem provider, you'll need to “extend” Laravel's native storage system. In a service provider somewhere—it could be the `boot()` method of `AppServiceProvider`, but it'd be more appropriate to create a unique service provider for each new binding—use the Storage facade to add new storage systems, as seen in [Example 14-2](#).

Example 14-2. Adding additional Flysystem providers

```
// Some service provider
public function boot(): void
{
    Storage::extend('dropbox', function ($app, $config) {
        $client = new DropboxClient(
            $config['accessToken'], $config['clientIdentifier']
        );

        return new Filesystem(new DropboxAdapter($client));
    });
}
```

Basic File Uploads and Manipulation

One of the more common usages for the Storage facade is accepting file uploads from your application's users. Let's look at a common workflow for that in [Example 14-3](#).

Example 14-3. Common user upload workflow

```
...
class DogController
{
    public function updatePicture(Request $request, Dog $dog)
    {
        Storage::put(
            "dogs/{$dog->id}",
            file_get_contents($request->file('picture')->getRealPath())
        );
    }
}
```

We `put()` to a file named *dogs/id*, and we grab our contents from the uploaded file. Every uploaded file is a descendant of the `SplFileInfo` class, which provides a `getRealPath()` method that returns the path to the file's location. So,

we get the temporary upload path for the user's uploaded file, read it with `file_get_contents()`, and pass it into `Storage::put()`.

Since we have this file available to us here, we can do anything we want to the file before we store it—use an image manipulation package to resize it if it's an image, validate it and reject it if it doesn't meet our criteria, or whatever else we like.

If we want to upload this same file to S3 and we have our credentials stored in `config/filesystems.php`, we can simply adjust [Example 14-3](#) to call `Storage::disk('s3')->put()`; we'll now be uploading to S3. Take a look at [Example 14-4](#) to see a more complex upload example.

Example 14-4. A more complex example of file uploads, using Intervention

```
...
class DogController
{
    public function updatePicture(Request $request, Dog $dog)
    {
        $original = $request->file('picture');

        // Resize image to max width 150
        $image = Image::make($original)->resize(150, null, function ($constraint) {
            $constraint->aspectRatio();
        })->encode('jpg', 75);

        Storage::put(
            "dogs/thumbs/{$dog->id}",
            $image->getEncoded()
        );
    }
}
```

I used an image library called [Intervention](#) in [Example 14-4](#) just as an example; you can use any library you want. The important point is that you have the freedom to manipulate the files however you want before you store them.

USING STORE() AND STOREAS() ON THE UPLOADED FILE

You can also store an uploaded file using the file itself. Learn more in [Example 7-18](#).

Simple File Downloads

Just like `Storage` makes it easy to accept uploads from users, it also simplifies the task of returning files to them. Take a look at [Example 14-5](#) for the simplest example.

Example 14-5. Simple file downloads

```
public function downloadMyFile()
{
    return Storage::download('my-file.pdf');
}
```

Sessions

Session storage is the primary tool we use in web applications to store state between page requests. Laravel's session manager supports session drivers using files, cookies, a database, Memcached or Redis, DynamoDB, or in-memory arrays (which expire after the page request and are only good for tests).

You can configure all of your session settings and drivers in *config/session.php*. You can choose whether or not to encrypt your session data, select which driver to use (`file` is the default), and specify more connection-specific details like the length of session storage and which files or database tables to use. Take a look at [the session docs](#) to learn about specific dependencies and settings you need to prepare for whichever driver you choose to use.

The general API of the session tools allows you to save and retrieve data based on individual keys: `session()->put('user_id')` and `session()->get('user_id')`, for example. Make sure to avoid saving anything to a flash session key, since Laravel uses that internally for flash (only available for the next page request) session storage.

Accessing the Session

The most common way to access the session is using the `Session` facade:

```
Session::get('user_id');
```


But you can also use the `session()` method on any given Illuminate Request object, as in [Example 14-6](#).

Example 14-6. Using the `session()` method on a Request object

```
Route::get('dashboard', function (Request $request) {  
    $request->session()->get('user_id');  
});
```

Or you can inject an instance of `Illuminate\Session\Store`, as in [Example 14-7](#).

Example 14-7. Injecting the backing class for sessions

```
Route::get('dashboard', function (Illuminate\Session\Store $session) {  
    return $session->get('user_id');  
});
```

Finally, you can use the global `session()` helper. Use it with no parameters to get a session instance, with a single string parameter to “get” from the session, or with an array to “put” to the session, as demonstrated in [Example 14-8](#).

Example 14-8. Using the global `session()` helper

```
// Get  
$value = session()->get('key');  
$value = session('key');  
// Put  
session()->put('key', 'value');  
session(['key', 'value']);
```

If you’re new to Laravel and not sure which to use, I’d recommend using the global helper.

Methods Available on Session Instances

The two most common methods are `get()` and `put()`, but let’s take a look at each of the available methods and their parameters:

`session()->get($key, $fallbackValue)`

`get()` pulls the value of the provided key out of the session. If there is no value attached to that key, it will return the fallback value instead (and if you don’t provide a fallback, it will return `null`). The fallback value can be a simple value or a closure, as you can see in the following examples:

```
$points = session()->get('points');
```

```
$points = session()->get('points', 0);
```

```
$points = session()->get('points', function () {  
    return (new PointGetterService)->getPoints();  
});
```

`session()->put($key, $value)`

`put()` stores the provided value in the session at the provided key:

```
session()->put('points', 45);
```

```
$points = session()->get('points');
```

`session()->push($key, $value)`

If any of your session values are arrays, you can use `push()` to add a value to the array:

```
session()->put('friends', ['Saúl', 'Quang', 'Mechteld']);
```

```
session()->push('friends', 'Javier');
```

`session()->has($key)`

`has()` checks whether there's a value set at the provided key:

```
if (session()->has('points')) {  
    // Do something
```

```
}
```

You can also pass an array of keys, and it only returns `true` if all of the keys exist.

SESSION()->HAS() AND NULL VALUES

If a session value is set but the value is `null`, `session()->has()` will return `false`.

`session()->exists($key)`

`exists()` checks whether there's a value set at the provided key, like `has()`, but unlike `has()`, it will return `true` even if the set value is `null`:

```
if (session()->exists('points')) {  
    // returns true even if 'points' is set to null  
}
```

`session()->all()`

`all()` returns an array of everything that's in the session, including those values set by the framework. You'll likely see values under keys like `_token` (CSRF tokens), `_previous` (previous page, for `back()` redirects), and `flash` (for flash storage).

`session()->only()`

`only()` returns an array of only the specified values in the session.

`session()->forget($key)`, `session()->flush()`

`forget()` removes a previously set session value. `flush()` removes every session value, even those set by the framework:

```
session()->put('a', 'awesome');  
  
session()->put('b', 'bodacious');  
  
session()->forget('a');  
  
// a is no longer set; b is still set  
  
session()->flush();  
  
// Session is now empty
```

`session()->pull($key, $fallbackValue)`

`pull()` is the same as `get()`, except that it deletes the value from the session after pulling it.

`session()->regenerate()`

It's not common, but if you need to regenerate your session ID, `regenerate()` is there for you.

Flash Session Storage

There are three more methods we haven't covered yet, and they all have to do with something called *flash session storage*.

One very common pattern for session storage is to set a value that you only want available for the next page load. For example, you might want to store a message like "Updated post successfully." You could manually get that message and then wipe it on the next page load, but if you use this pattern a lot it can get wasteful. Enter flash session storage: keys that are expected to only last for a single page request.

Laravel handles the work for you, and all you need to do is use `flash()` instead of `put()`. These are the useful methods here:

`session()->flash($key, $value)`

`flash()` sets the session key to the provided value for just the next page request.

`session()->reflash()`, `session()->keep($key)`

If you need the previous page's flash session data to stick around for one more request, you can use `reflash()` to restore all of it for the next request or `keep($key)` to just restore a single flash value for the next request. `keep()` can also accept an array of keys to reflash.

Cache

Caches are structured very similarly to sessions. You provide a key and Laravel stores it for you. The biggest difference is that the data in a cache is cached per application, and the data in a session is cached per user. That means caches are more commonly used for storing results from database queries, API calls, or other slow queries that can stand to get a little bit “stale.”

The cache configuration settings are available at *config/cache.php*. Just like with a session, you can set the specific configuration details for any of your drivers and also choose which will be your default. Laravel uses the `file` cache driver by default, but you can also use Memcached or Redis, APC, DynamoDB, or a database, or write your own cache driver. Take a look at [the cache docs](#) to learn about specific dependencies and settings you need to prepare for whichever driver you choose to use.

Accessing the Cache

Just like with sessions, there are several ways to access a cache. You can use the facade:

```
$users = Cache::get('users');
```

Or you can get an instance from the container, as in [Example 14-9](#).

Example 14-9. Injecting an instance of the cache

```
Route::get('users', function (Illuminate\Contracts\Cache\Repository $cache) {  
    return $cache->get('users');  
});
```

You can also use the global `cache()` helper, as in [Example 14-10](#).

Example 14-10. Using the global `cache()` helper

```
// Get from cache
$users = cache('key', 'default value');
$users = cache()->get('key', 'default value');
// Put for $seconds duration
$users = cache(['key' => 'value'], $seconds);
$users = cache()->put('key', 'value', $seconds);
```

If you're new to Laravel and not sure which to use, I'd recommend using the global helper.

Methods Available on Cache Instances

Let's take a look at the methods you can call on a Cache instance:

```
cache()->get($key, $fallbackValue),
cache()->pull($key, $fallbackValue)
```

`get()` makes it easy to retrieve the value for any given key. `pull()` is the same as `get()` except it removes the cached value after retrieving it.

```
cache()->put($key, $value, $secondsOrExpiration)
```

`put()` sets the value of the specified key for a given number of seconds. If you'd prefer setting an expiration date/time instead of a number of seconds, you can pass a Carbon object as the third parameter:

```
cache()->put('key', 'value', now()->addDay());
```

```
cache()->add($key, $value)
```

`add()` is similar to `put()`, except if the value already exists, `add()` won't set it. Also, the method returns a Boolean indicating whether or not the value was actually added:

```
$someDate = now();

cache()->add('someDate', $someDate); // returns true

$someOtherDate = now()->addHour();
```

```
cache()->add('someDate', $someOtherDate); // returns false
```

cache()->forever(\$key, \$value)

forever() saves a value to the cache for a specific key; it's the same as put(), except the values will never expire (until they're removed with forget()).

cache()->has(\$key)

has() returns a Boolean indicating whether or not there's a value at the provided key.

cache()->remember(\$key, \$seconds, \$closure),

cache()->rememberForever(\$key, \$closure)

remember() provides a single method to handle a very common flow: look up whether a value exists in the cache for a certain key, and if it doesn't, get that value somehow, save it to the cache, and return it.

remember() lets you provide a key to look up, the number of seconds it should be saved for, and a closure to define how to look it up, in case the key has no value set. rememberForever() is the same, except it doesn't need you to set the number of seconds it should be saved for. Take a look at the following example to see a common user scenario for remember():

```
// Either returns the value cached at "users" or gets "User::all()",  
// caches it at "users", and returns it  
$users = cache()->remember('users', 7200, function () {  
    return User::all();  
});
```

cache()->increment(\$key, \$amount), cache()->decrement(\$key, \$amount)

`increment()` and `decrement()` allow you to increment and decrement integer values in the cache. If there is no value at the given key, it'll be treated as if it were 0, and if you pass a second parameter to increment or decrement, it'll increment or decrement by that amount instead of by 1.

`cache()->forget($key)`, `cache()->flush()`

`forget()` works just like `Session`'s `forget()` method: pass it a key and it'll wipe that key's value. `flush()` wipes the entire cache.

Cookies

You might expect cookies to work the same as sessions and the cache. A facade and a global helper are available for these too, and our mental models of all three are similar: you can get or set their values in the same way.

But because cookies are inherently attached to the requests and responses, you'll need to interact with cookies differently. Let's look really briefly at what makes cookies different.

Cookies in Laravel

Cookies can exist in three places in Laravel. They can come in via the request, which means the user had the cookie when they visited the page. You can read that with the `Cookie` facade, or you can read it off of the request object.

They can also be sent out with a response, which means the response will instruct the user's browser to save the cookie for future visits. You can do this by adding the cookie to your response object before returning it.

And last, a cookie can be *queued*. If you use the `Cookie` facade to set a cookie, you have put it into a "CookieJar" queue, and it will be removed and added to the response object by the `AddQueuedCookiesToResponse` middleware.

Accessing the Cookie Tools

You can get and set cookies in three places: the `Cookie` facade, the `cookie()` global helper, and the request and response objects.

The cookie facade

The `Cookie` facade is the most full-featured option, allowing you to not only read and make cookies, but also to queue them to be added to the response. It provides the following methods:

`Cookie::get($key)`

To pull the value of a cookie that came in with the request, you can just run `Cookie::get('cookie-name')`. This is the simplest option.

`Cookie::has($key)`

You can check whether a cookie came in with the request using `Cookie::has('cookie-name')`, which returns a Boolean.

`Cookie::make(...$params)`

If you want to *make* a cookie without queueing it anywhere, you can use `Cookie::make()`. The most likely use for this would be to make a cookie and then manually attach it to the response object, which we'll cover in a bit.

Here are the parameters for `make()`, in order:

- `$name` is the name of the cookie.
- `$value` is the content of the cookie.
- `$minutes` specifies how many minutes the cookie should live.
- `$path` is the path under which your cookie should be valid.
- `$domain` lists the domains for which your cookie should work.
- `$secure` indicates whether the cookie should only be transmitted over a secure (HTTPS) connection.
- `$httpOnly` indicates whether the cookie will be made accessible only through the HTTP protocol.
- `$raw` indicates whether the cookie should be sent without URL encoding.
- `$sameSite` indicates whether the cookie should be available for cross-

site requests; options are `lax`, `strict`, or `null`.

`Cookie::make()`

Returns an instance of `Symfony\Component\HttpFoundation\Cookie`.

DEFAULT SETTINGS FOR COOKIES

The `CookieJar` used by the `Cookie` facade instance reads its defaults from the session config. So, if you change any of the configuration values for the session cookie in *config/session.php*, those same defaults will be applied to all of your cookies that you create using the `Cookie` facade.

`Cookie::queue(Cookie // params)`

If you use `Cookie::make()`, you'll still need to attach the cookie to your response, which we'll cover shortly. `Cookie::queue()` has the same syntax as `Cookie::make()`, but it enqueues the created cookie to be automatically attached to the response by middleware.

If you'd like, you can also just pass a cookie you've created yourself into `Cookie::queue()`.

Here's the simplest possible way to add a cookie to the response in Laravel:

```
Cookie::queue('dismissed-popup', true, 15);
```

WHEN YOUR QUEUED COOKIES WON'T GET SET

Cookies can only be returned as part of a response. So, if you enqueue cookies with the `Cookie` facade and then your response isn't returned correctly—for example, if you use PHP's `exit()` or something halts the execution of your script—your cookies won't be set.

The `cookie()` global helper

The `cookie()` global helper will return a `CookieJar` instance if you call it with no parameters. However, two of the most convenient methods on the `Cookie` facade—`has()` and `get()`—exist *only* on the facade, not on the `CookieJar`. So, in this context, I think the global helper is actually less useful than the other options.

The one task for which the `cookie()` global helper is useful is creating a cookie. If you pass parameters to `cookie()`, they'll be passed directly to the equivalent of `Cookie::make()`, so this is the fastest way to create a cookie:

```
$cookie = cookie('dismissed-popup', true, 15);
```

INJECTING AN INSTANCE

You can also inject an instance of `Illuminate\Cookie\CookieJar` anywhere in the app, but you'll have the same limitations discussed here.

Cookies on Request and Response objects

Since cookies come in as a part of the request and are set as a part of the response, those `Illuminate` objects are the places they actually live. The `Cookie` facade's `get()`, `has()`, and `queue()` methods are just proxies to interact with the `Request` and `Response` objects.

So, the simplest way to interact with cookies is to pull cookies from the request and set them on the response.

Reading cookies from Request objects

Once you have a copy of your `Request` object—if you don't know how to get one, just try `app('request')`—you can use the `Request` object's `cookie()` method to read its cookies, as shown in [Example 14-11](#).

Example 14-11. Reading a cookie from a Request object

```
Route::get('dashboard', function (Illuminate\Http\Request $request) {  
    $userDismissedPopup = $request->cookie('dismissed-popup', false);  
});
```

As you can see in this example, the `cookie()` method has two parameters: the

cookie's name and, optionally, the fallback value.

Setting cookies on Response objects

When you have your Response object ready, you can use the `cookie()` method on it to add a cookie to the response, like in [Example 14-12](#).

Example 14-12. Setting a cookie on a Response object

```
Route::get('dashboard', function () {
    $cookie = cookie('saw-dashboard', true);

    return Response::view('dashboard')
        ->cookie($cookie);
});
```

If you're new to Laravel and not sure which option to use, I'd recommend setting cookies on the Request and Response objects. It's a bit more work, but will lead to fewer surprises if future developers don't understand the CookieJar queue.

Logging

We've seen a few really brief examples of logging so far in this book when we were talking about other concepts like the container and facades, so let's briefly look at what options you have with logging beyond just

```
Log::info('Message').
```

The purpose of logs is to increase *discoverability*, or your ability to understand what's going on at any given moment in your application.

Logs are short messages, sometimes with some data embedded in a human-readable form, that your code generates for the sake of understanding what's happening during the execution of an app. Each log must be captured at a specific *level*, which can vary from **emergency** (something very bad happened) to **debug** (something of almost no significance happened).

Without any modifications, your app will write any log statements to a file located at `storage/logs/laravel.log`, and each log statement will look a little bit like this:

```
[2018-09-22 21:34:38] local.ERROR: Something went wrong.
```

You can see we have the date, time, environment, error level, and message all on one line. However, Laravel also (by default) logs any uncaught exceptions, and in that case you'll see the entire stack trace inline.

We'll cover how to log, why to log, and how to log elsewhere (for example, in Slack) in this following section.

When and Why to Use Logs

The most common use case for logs is to act as a semidisposable record of things that have happened that you *may* care about later, but to which you definitively don't need programmatic access. The logs are more about learning what's going on in the app and less about creating structured data your app can consume.

For example, if you want to have code that consumes a record of every user login and does something interesting with it, that's a use case for a *logins* database table. However, if you have a casual interest in those logins but you're not entirely certain whether you care or whether you need that information programmatically, you may just throw a `debug-` or `info-`level log on it and forget about it.

Logs are also common when you need to see the value of something at the moment it goes wrong, or at a certain time of day, or something else that means you want the data at a time when you're not around. Throw a log statement in the code, get the data you need out of the logs, and either keep it in the code for later usage or just delete it again.

Writing to the Logs

The simplest way to write a log entry in Laravel is to use the `Log` facade and use the method on that facade that matches the severity level you'd like to record. The levels are the same as those defined in [RFC 5424](#):

```
Log::emergency($message);  
Log::alert($message);  
Log::critical($message);  
Log::error($message);
```

```
Log::warning($message);
Log::notice($message);
Log::info($message);
Log::debug($message);
```

You can also, optionally, pass a second parameter that's an array of connected data:

```
Log::error('Failed to upload user image.', ['user' => $user]);
```

This additional information may be captured differently by different log destinations, but here's how this looks in the default local log (although it will be just a single line in the log):

```
[2018-09-27 20:53:31] local.ERROR: Failed to upload user image. {
  "user": "[object] (App\\User: {
    \"id\": 1,
    \"name\": \"Matt\",
    \"email\": \"matt@tighten.co\",
    \"email_verified_at\": null,
    \"api_token\": \"long-token-here\",
    \"created_at\": \"2018-09-22 21:39:55\",
    \"updated_at\": \"2018-09-22 21:40:08\"
  })"
}
```

Log Channels

Like many other aspects of Laravel (file storage, database, mail, etc.), you can configure your logs to use one or more predefined log types, which you define in the config file. Using each type involves passing various configuration details to a specific log driver.

These log types are called *channels*, and out of the box you'll have options for `stack`, `single`, `daily`, `slack`, `stderr`, `syslog`, and `errorlog`. Each channel is connected to a single driver; the available drivers are `stack`, `single`, `daily`, `slack`, `syslog`, `errorlog`, `monolog`, and `custom`.

We'll cover the most common channels here: `single`, `daily`, `slack`, and `stack`. To learn more about the drivers and the full list of channels available, take a look at the [logging docs](#).

The single channel

The `single` channel writes every log entry to a single file, which you'll define in the `path` key. You can see its default configuration in [Example 14-13](#):

Example 14-13. Default configuration for the single channel

```
'single' => [
    'driver' => 'single',
    'path' => storage_path('logs/laravel.log'),
    'level' => env('LOG_LEVEL', 'debug'),
],
```

This means it'll only log events at the `debug` level or higher, and it will write them all to a single file, `storage/logs/laravel.log`.

The daily channel

The `daily` channel splits out a new file for each day. You can see its default config in [Example 14-14](#).

Example 14-14. Default configuration for the daily channel

```
'daily' => [
    'driver' => 'daily',
    'path' => storage_path('logs/laravel.log'),
    'level' => env('LOG_LEVEL', 'debug'),
    'days' => 14,
],
```

It's similar to `single`, but we now can set how many days of logs to keep before they're cleaned up, and the date will be appended to the filename we specify. For example, the preceding config will generate a file named `storage/logs/laravel-<yyyy-mm-dd>.log`.

The Slack channel

The `slack` channel makes it easy to send your logs (or, more likely, only certain logs) over to Slack.

It also illustrates that you're not limited to just the handlers that come out of the box with Laravel. We'll cover this in a second, but this isn't a custom Slack implementation; it's just Laravel building a log driver that connects to the Monolog Slack handler, and if you can use any Monolog handler, you have a *lot*

of options available to you.

The default configuration for this channel is shown in [Example 14-15](#).

Example 14-15. Default configuration for the slack channel

```
'slack' => [
    'driver' => 'slack',
    'url' => env('LOG_SLACK_WEBHOOK_URL'),
    'username' => 'Laravel Log',
    'emoji' => ':boom:',
    'level' => env('LOG_LEVEL', 'critical'),
],
```

The stack channel

The stack channel is the channel that's enabled by default on your application. Its default configuration is shown in [Example 14-16](#).

Example 14-16. Default configuration for the stack channel

```
'stack' => [
    'driver' => 'stack',
    'channels' => ['single'],
    'ignore_exceptions' => false,
],
```

The stack channel allows you to send all your logs to more than one channel (listed in the `channels` array). So, while this is the channel that's configured by default on your Laravel apps, because its `channels` array is set to `single` by default, in reality your app is just using the `single` log channel.

But what if you wanted everything of the level `info` and above to go to the daily files, but you wanted `critical` and higher log messages to go to Slack? It's easy with the stack driver, as [Example 14-17](#) demonstrates.

Example 14-17. Customizing the stack driver

```
'channels' => [
    'stack' => [
        'driver' => 'stack',
        'channels' => ['daily', 'slack'],
    ],

    'daily' => [
        'driver' => 'daily',
        'path' => storage_path('logs/laravel.log'),
    ],
],
```



```

        'level' => 'info',
        'days' => 14,
    ],

    'slack' => [
        'driver' => 'slack',
        'url' => env('LOG_SLACK_WEBHOOK_URL'),
        'username' => 'Laravel Log',
        'emoji' => ':boom:',
        'level' => 'critical',
    ],
]

```

Writing to specific log channels

There may also be times when you want to control exactly which log messages go where. You can do that by specifying the channel when you call the Log facade:

```
Log::channel('slack')->info("This message will go to Slack.");
```

ADVANCED LOG CONFIGURATION

If you'd like to customize how each log is sent to each channel, or implement custom Monolog handlers, check out the [logging docs](#) to learn more.

Full-Text Search with Laravel Scout

Laravel Scout is a separate package that you can bring into your Laravel apps to add full-text search to your Eloquent models. Scout makes it easy to index and search the contents of your Eloquent models; it ships with drivers for Algolia, Meilisearch, and databases (MySQL/PostgreSQL), but there are also community packages for other providers. I'll assume you're using Algolia.

Installing Scout

First, pull in the package in any Laravel app:

```
composer require laravel/scout
```

Next you'll want to set up your Scout configuration. Run this command:

```
php artisan vendor:publish --provider="Laravel\Scout\ScoutServiceProvider"
```

and paste your Algolia credentials in *config/scout.php*.

Finally, install the Algolia SDK:

```
composer require algolia/algoliasearch-client-php
```

Marking Your Model for Indexing

In your model (we'll use *Review*, for a book review, for this example), import the `Laravel\Scout\Searchable` trait.

You can define which properties are searchable using the `toSearchableArray()` method (it defaults to mirroring `toArray()`), and define the name of the model's index using the `searchableAs()` method (it defaults to the table name).

Scout subscribes to the create/delete/update events on your marked models. When you create, update, or delete any rows, Scout will sync those changes up to Algolia. It'll either make those changes synchronously with your updates or, if you configure Scout to use a queue, queue the updates.

Searching Your Index

Scout's syntax is simple. For example, to find any *Review* with the word *Llew* in it:

```
Review::search('Llew')->get();
```

You can also modify your queries as you would with regular Eloquent calls:

```
// Get all records from the Review that match the term "Llew",  
// limited to 20 per page and reading the page query parameter,  
// just like Eloquent pagination  
Review::search('Llew')->paginate(20);  
  
// Get all records from the Review that match the term "Llew"
```

```
// and have the account_id field set to 2
Review::search('Llew')->where('account_id', 2)->get();
```

What comes back from these searches? A collection of Eloquent models, rehydrated from your database. The IDs are stored in Algolia, which returns a list of matched IDs; Scout then pulls the database records for those and returns them as Eloquent objects.

You don't have full access to the complexity of SQL WHERE commands, but it provides a basic framework for comparison checks like you can see in the code samples here.

Queues and Scout

At this point your app will be making HTTP requests to Algolia on every request that modifies any database records. This can slow down your application quickly, which is why Scout makes it easy to push all of its actions onto a queue.

In *config/scout.php*, set `queue` to `true` so that these updates are indexed asynchronously. Your full-text index is now operating under “eventual consistency”; your database records will receive the updates immediately, and the updates to your search indexes will be queued and updated as fast as your queue worker allows.

Performing Operations Without Indexing

If you need to perform a set of operations and avoid triggering the indexing in response, wrap the operations in the `withoutSyncingToSearch()` method on your model:

```
Review::withoutSyncingToSearch(function () {
    // Make a bunch of reviews, e.g.
    Review::factory()->count(10)->create();
});
```

Conditionally Indexing Models

Sometimes you might only want to index records if they meet a certain condition. You may use the `shouldBeSearchable()` method on the model class

to achieve this:

```
public function shouldBeSearchable()
{
    return $this->isApproved();
}
```

Manually Triggering Indexing via Code

If you want to manually trigger indexing your model, you can do it using code in your app or via the command line.

To manually trigger indexing from your code, add `searchable()` to the end of any Eloquent query and it will index all of the records that were found in that query:

```
Review::all()->searchable();
```

You can also choose to scope the query to only those records you want to index. However, Scout is smart enough to insert new records and update old records, so you may choose to just reindex the entire contents of the model's database table.

You can also run `searchable()` on relationship methods:

```
$user->reviews()->searchable();
```

If you want to unindex any records with the same sort of query chaining, just use `unsearchable()` instead:

```
Review::where('sucky', true)->unsearchable();
```

Manually Triggering Indexing via the CLI

You can also trigger indexing with an Artisan command:

```
php artisan scout:import "App\Review"
```

This will chunk all of the Review models and index them all.

The HTTP Client

Laravel's HTTP client isn't quite a storage mechanism, but it is a retrieval mechanism, and to be honest, I'm not sure where else it fits in this book. Let's get to it!

The HTTP client makes it possible for your Laravel app to make calls—POST, GET, whatever—to external web services and APIs with a simple, clean interface.

If you've ever worked with Guzzle, you understand what it can do, and you can also likely understand why a simple interface is worth mentioning: Guzzle is incredibly powerful, but also incredibly complex, and it's become increasingly so over the years.

Using the HTTP Facade

Most of the time, if you're working with the HTTP client, you'll rely on its facade, calling methods like `get()` and `post()` directly on the facade. Take a look at [Example 14-18](#) for an example.

Example 14-18. Basic usage examples of the HTTP facade

```
use Illuminate\Support\Facades\Http;

$response = Http::get('http://my-api.com/posts');

$response = Http::post('http://my-api.com/posts/2/comments', [
    'title' => 'I loved this post!',
]);
```

The `$response` you receive back from a call on the HTTP facade is an instance of `Illuminate\Http\Client\Response`, which gives you a suite of methods to inspect the response. You can take a look at [the docs](#) for a full list, but you can also see a few common methods in [Example 14-19](#).

Example 14-19. Commonly used methods on the HTTP Client Response object

```
$response = Http::get('http://my-api.com/posts');

$response->body(); // string
$response->json(); // array
$response->json('key', 'default') // string
$response->successful(); // bool
```

As you can see from [Example 14-18](#), you can send data along with POST requests, but there are many other ways you can send data along with your requests.

Once again, here are a few common examples, and you can see more in the docs:

```
$response = Http::withHeaders([
    'X-Custom-Header' => 'header value here'
])->post(/* ... */);

$response = Http::withToken($authToken)->post(/* ... */);

$response = Http::accept('application/json')->get('http://my-api.com/users');
```

Handling Errors and Timeouts and Checking Statuses

By default, the HTTP client will wait 30 seconds before failing out of a request, and not retry it at all. But you can customize many aspects of how the client responds to unexpected situations.

To define the timeout, chain `timeout()` and pass the number of seconds it should wait:

```
$response = Http::timeout(120)->get(/* ... */);
```

If you expect there to be failures with your attempts, you can define that the client should retry each request a given number of times, using the `retry()` chained method:

```
$response = Http::retry($retries, $millisecondsBetweenRetries)->post(/* ... */);
```

Some of the other methods on the response object allow us to check whether the request worked and what HTTP status we got back; here are a few:

```
$response->successful(); // 200 or 300
$response->failed(); // 400 or 500 errors
$response->clientError(); // 400 errors
$response->serverError(); // 500 errors

// A few of the specific checks we can run for given status codes
$response->ok(); // 200 OK
$response->movedPermanently(); // 301 Moved Permanently
```

```
$response->unauthorized(); // 401 Unauthorized
$response->serverError(); // 500 Internal Server Error
```

You can also define a callback to be run any time there's an error:

```
$response->onError(function (Response $response) {
    // handle error
});
```

Testing

Testing most of these features is as simple as using them in your tests; no need to mock or stub. The default configuration will already work—for example, take a look at *phpunit.xml* to see that your session driver and cache driver have been set to values appropriate for tests.

However, there are a few convenience methods and a few gotchas that you should know about before you attempt to test them all.

File Storage

Testing file uploads can be a bit of a pain, but follow these steps and it will be clear.

Uploading fake files

First, let's look at how to manually create an `Illuminate\Http\UploadedFile` object for use in our application testing ([Example 14-20](#)).

Example 14-20. Creating a fake `UploadedFile` object for testing

```
public function test_file_should_be_stored()
{
    Storage::fake('public');

    $file = UploadedFile::fake()->image('avatar.jpg');

    $response = $this->postJson('/avatar', [
        'avatar' => $file,
    ]);

    // Assert the file was stored
    Storage::disk('public')->assertExists("avatars/{$file->hashName()}");
}
```

```

    // Assert a file does not exist
    Storage::disk('public')->assertMissing('missing.jpg');
}

```

We've created a new instance of `UploadedFile` that refers to our testing file, and we can now use it to test our routes.

Returning fake files

If your route is expecting a real file to exist, sometimes the best way to make it testable is to make that real file actually exist. Let's say every user must have a profile picture.

First, let's set up the model factory for the user to use Faker to make a copy of the picture, as in [Example 14-21](#).

Example 14-21. Returning fake files with Faker

```

public function definition ()
{
    return [
        'picture' => fake()->file(
            base_path('tests/stubs/images'), // Source directory
            storage_path('app'), // Target directory
            false, // Return just filename, not full path
        ),
        'name' => fake()->name(),
    ];
};

```

Faker's `file()` method picks a random file from the source directory, copies it to the target directory, and then returns the filename. So, we've just picked a random file from the `tests/stubs/images` directory, copied it to the `storage/app` directory, and set its filename as the `picture` property on our `User`. At this point we can use a `User` in tests on routes that expect the `User` to have a picture, as seen in [Example 14-22](#).

Example 14-22. Asserting that an image's URL is echoed

```

public function test_user_profile_picture_echoes_correctly()
{
    $user = User::factory()->create();

    $response = $this->get(route('users.show', $user->id));
}

```



```
$response->assertSee($user->picture);  
}
```

Of course, in many contexts you can just generate a random string there without even copying a file. But if your routes check for the file's existence or run any operations on the file, this is your best option.

Session

If you need to assert something has been set in the session, you can use some convenience methods Laravel makes available in every test. All of these methods are available in your tests on the `Illuminate\Testing\TestResponse` object:

`assertSessionHas($key, $value = null)`

Asserts that the session has a value for a particular key, and, if the second parameter is passed, that that key is a particular value:

```
public function test_some_thing()  
{  
    // Do stuff that ends up with a $response object...  
    $response->assertSessionHas('key', 'value');  
}
```

`assertSessionHasAll(array $bindings)`

If passed an array of key/value pairs, asserts that all of the keys are equal to all of the values. If one or more of the array entries is just a value (with PHP's default numeric key), it will just be checked for existence in the session:

```
$check = [  
    'has',  
    'hasWithThisValue' => 'thisValue',  
];
```

```
];
```

```
$response->assertSessionHasAll($check);
```

`assertSessionMissing($key)`

Asserts that the session does *not* have a value for a particular key.

`assertSessionHasErrors($bindings = [], $format = null)`

Asserts that the session has an `errors` value. This is the key Laravel uses to send errors back from validation failures.

If the array contains just keys, it will check that errors are set with those keys:

```
$response = $this->post('test-route', ['failing' => 'data']);  
$response->assertSessionHasErrors(['name', 'email']);
```

You can also pass values for those keys, and, optionally, a *\$format*, to check that the messages for those errors came back the way you expected:

```
$response = $this->post('test-route', ['failing' => 'data']);  
$response->assertSessionHasErrors([  
    'email' => '<strong>The email field is required.</strong>',  
], '<strong>:message</strong>');
```

Cache

There's nothing special about testing your features that use the cache—just do it:

```
Cache::put('key', 'value', 900);  
$this->assertEquals('value', Cache::get('key'));
```

Laravel uses the array cache driver by default in your testing environment, which just stores your cache values in memory.

Cookies

What if you need to set a cookie before testing a route in your application tests? You can set cookies on a request using the `withCookies()` method. To learn more, check out [Chapter 12](#).

EXCLUDING YOUR COOKIE FROM ENCRYPTION DURING TESTING

Your cookies won't work in your tests unless you exclude them from Laravel's cookie encryption middleware. You can do this by teaching the `EncryptCookies` middleware to temporarily disable itself for those cookies:

```
use Illuminate\Cookie\Middleware\EncryptCookies;
...

$this->app->resolving(
    EncryptCookies::class,
    function ($object) {
        $object->disableFor('cookie-name');
    }
);

// ...run test
```

That means you can set a cookie and check against it with something like [Example 14-23](#).

Example 14-23. Running unit tests against cookies

```
public function test_cookie()
{
    $this->app->resolving(EncryptCookies::class, function ($object) {
        $object->disableFor('my-cookie');
    });

    $response = $this->call(
        'get',
```

```

        'route-echoing-my-cookie-value',
        [],
        ['my-cookie' => 'baz']
    );
    $response->assertSee('baz');
}

```

If you want to test that a response has a cookie set, you can use `assertCookie()` to test for the cookie:

```

$response = $this->get('cookie-setting-route');
$response->assertCookie('cookie-name');

```

Or you could use `assertPlainCookie()` to test for the cookie and to assert that it's not encrypted.

Log

The simplest way to test that a certain log was written is by making assertions against the Log facade (learn more in “[Faking Other Facades](#)”). [Example 14-24](#) shows how this works.

Example 14-24. Making assertions against the Log facade

```

// Test file
public function test_new_accounts_generate_log_entries()
{
    Log::shouldReceive('info')
        ->once()
        ->with('New account created!');

    // Create a new account
    $this->post(route('accounts.store'), ['email' => 'matt@mattstauffer.com']);
}

// AccountController
public function store()
{
    // Create account

    Log::info('New account created!');
}

```

There's also a package called [Log Fake](#) that expands on what you can do with the facade testing shown here and allows you to write more customized

assertions against your logs.

Scout

If you need to test code that uses Scout data, you're probably not going to want your tests triggering indexing actions or reading from Scout. Simply add an environment variable to your *phpunit.xml* to disable Scout's connection to Algolia:

```
<env name="SCOUT_DRIVER" value="null"/>
```

HTTP Client

One incredible benefit of using Laravel's HTTP client is that it makes it possible to fake responses in your tests with minimal configuration.

The simplest option is to run `Http::fake()`, which will return an empty successful response from every call you make.

However, you can also customize the specific responses you want to come back from your HTTP client calls, as you can see in [Example 14-25](#).

Example 14-25. Customizing responses to HTTP clients by URL

```
Http::fake([
    // Return a JSON response for a particular API
    'my-api.com/*' => Http::response(['key' => 'value'], 200, $headersArray),

    // Return a string response for all other endpoints
    '*' => Http::response('This is a fake API response', 200, $headersArray),
]);
```

If, instead, you need to define that requests made to a given endpoint (or that match a given endpoint pattern) follow a particular sequence, you can also define that, as you can see in [Example 14-26](#).

Example 14-26. Defining a sequence of responses to a given endpoint

```
Http::fake([
    // Return a sequence of responses for consecutive calls to this API
    'my-api.com/*' => Http::sequence()
        ->push('Initial string response', 200)
        ->push(['secondary' => 'response'], 200)
        ->pushStatus(404),
]);
```

You can also make assertions against the data your application is sending to particular endpoints, as in [Example 14-27](#).

Example 14-27. Asserting against the calls your application makes

```
Http::fake();

Http::assertSent(function (Request $request) {
    return $request->hasHeader('X-Custom-Header', 'certain-value') &&
        $request->url() == 'http://my-api.com/users/2/comments' &&
        $request['name'] == 'New User';
});
```

TL;DR

Laravel provides simple interfaces to many common storage operations: filesystem access, sessions, cookies, the cache, and search. Each of these APIs is the same regardless of which provider you use, which Laravel enables by allowing multiple “drivers” to serve the same public interface. This makes it simple to switch providers depending on the environment or as the needs of the application change.

Chapter 15. Mail and Notifications

Sending an application's users notifications via email, Slack, SMS, or another notification system is a common but surprisingly complex requirement.

Laravel's mail and notification features provide consistent APIs that abstract away the need to pay too close attention to any particular provider. Just like in [Chapter 14](#), you'll write your code once and choose at the configuration level which provider you'll use to send your email or notifications.

Mail

Laravel's mail functionality is a convenience layer on top of [Symfony Mailer](#). Out of the box, Laravel comes with drivers for SMTP, Mailgun, Postmark, Amazon SES, and Sendmail.

For all of the cloud services, you'll set your authentication information in *config/services.php*. However, if you take a look, you'll see there are already keys there—and in *config/mail.php*—that allow you to customize your application's mail functionality in *.env* using variables like `MAIL_MAILER` and `MAILGUN_SECRET`.

CLOUD-BASED API DRIVER DEPENDENCIES

If you're using any of the cloud-based API drivers, you may need to bring in external dependencies to support them.

If you're using Mailgun, you'll want to bring in Symfony's Mailgun Mailer and its HTTP client:

```
composer require symfony/mailgun-mailer \
    symfony/http-client
```

If you're using Postmark, you'll pull in Symfony's Postmark Mailer and its HTTP client:

```
composer require symfony/postmark-mailer \
    symfony/http-client
```

And if you use the SES driver, you'll want to pull in the AWS SDK:

```
composer require aws/aws-sdk-php
```

Basic “Mailable” Mail Usage

Every mail message you'll send in a modern Laravel app will be an instance of a specific PHP class, created to represent each email, called a *mailable*.

To make a mailable, use the `make:mail` Artisan command:

```
php artisan make:mail AssignmentCreated
```

Example 15-1 shows what that class looks like.

Example 15-1. An autogenerated mailable PHP class

```
<?php
```

```
namespace App\Mail;
```

```
use Illuminate\Bus\Queueable;
use Illuminate\Contracts\Queue\ShouldQueue;
use Illuminate\Mail\Mailable;
use Illuminate\Mail\Mailables\Content;
use Illuminate\Mail\Mailables\Envelope;
use Illuminate\Queue\SerializesModels;
```

```
class AssignmentCreated extends Mailable
```

```
{
```

```
    use Queueable, SerializesModels;
```

```
    /**
```

```
     * Create a new message instance.
```

```
     */
```

```
    public function __construct()
```

```
    {
```

```
        //
```

```
    }
```

```
    /**
```



```

    * Get the message envelope.
    */
    public function envelope(): Envelope
    {
        return new Envelope(
            subject: 'Assignment Created',
        );
    }

    /**
     * Get the message content definition.
     */
    public function content(): Content
    {
        return new Content(
            view: 'view.name',
        );
    }

    /**
     * Get the attachments for the message.
     *
     * @return array<int, \Illuminate\Mail\Mailables\Attachment>
     */
    public function attachments(): array
    {
        return [];
    }
}

```

You may notice a few similarities between mailables and jobs; this class even imports the `Queueable` trait for queuing your mail and the `SerializesModels` trait so any Eloquent models you pass to the constructor will be serialized correctly.

So, how does this work? The class constructor is the place where you'll pass in any data, and any properties you set as public on your mailable class will be available to the template.

In the `envelope()` method, you'll set configuration details about the mail—sender, subject, metadata.

In the `content()` method, you'll define the content—which view you're using to render, any Markdown contents, and text parameters.

And if you want to attach any files to the mail, you'll use the `attachments()`

method.

Take a look at [Example 15-2](#) to see how we might update the autogenerated mailable for our assignment example.

Example 15-2. A sample mailable

```
<?php

namespace App\Mail;

use Illuminate\Bus\Queueable;
use Illuminate\Contracts\Queue\ShouldQueue;
use Illuminate\Mail\Mailable;
use Illuminate\Mail\Mailables\Address;
use Illuminate\Mail\Mailables\Content;
use Illuminate\Mail\Mailables\Envelope;
use Illuminate\Queue\SerializesModels;

class AssignmentCreated extends Mailable
{
    use Queueable, SerializesModels;

    public function __construct(public $trainer, public $trainee) {}

    public function envelope(): Envelope
    {
        return new Envelope(
            subject: 'New assignment from ' . $this->trainer->name,
            from: new Address($this->trainer->email, $this->trainer->name),
        );
    }

    public function content(): Content
    {
        return new Content(
            view: 'emails.assignment-created'
        );
    }

    public function attachments(): array
    {
        return [];
    }
}
```

Once you've created the mailable class, it's time to send it. First, you create an instance of the mailable class, passing in the appropriate data; then, you chain

`Mail::to($user)->send($mailable)` to send the mail. You can also customize some other details of your mail, like the CC and BCC, as part of the inline call chain. Take a look at [Example 15-3](#) to see a few examples.

Example 15-3. How to send mailables

```
$mail = new AssignmentCreated($trainer, $trainee);

// Simple
Mail::to($user)->send($mail);

// With CC/BCC/etc.
Mail::to($user1)
    ->cc($user2)
    ->bcc($user3)
    ->send($mail);

// With string email address and collections
Mail::to('me@app.com')
    ->bcc(User::all())
    ->send($mail)
```

Mail Templates

Mail templates are just like any other template. They can extend other templates, use sections, parse variables, contain conditional or looping directives, and do anything else you can do in a normal Blade view.

Take a look at [Example 15-4](#) to see a possible `emails.assignment-created` template for [Example 15-2](#).

Example 15-4. Sample assignment-created email template

```
<!-- resources/views/emails/assignment-created.blade.php -->
<p>Hey {{ $trainee->name }}!</p>

<p>You have received a new training assignment from <b>{{ $trainer->name }}</b>.
Check out your <a href="{{ route('training-dashboard') }}">training
dashboard</a> now!</p>
```

In [Example 15-2](#), both `$trainer` and `$trainee` are public properties on your mailable, which makes them available to the template. Had one of them been private, it would not have been available.

If you want to explicitly define which variables are passed to the template, you can use the `with` parameter on the `Content` of your mailable, as in [Example 15-](#)

5.

Example 15-5. Customizing the template variables

```
use Illuminate\Mail\Mailables\Content;

public function content(): Content
{
    return new Content(
        view: 'emails.assignment-created',
        with: ['assignment' => $this->event->name],
    );
}
```

HTML VERSUS PLAIN-TEXT EMAILS

So far we've used the `view` parameter on the new `Content()` instantiation. This expects the template we're referencing to pass back HTML. If you'd like to pass a plain-text version, use the `text` parameter to define your plain-text view:

```
public function content(): Content
{
    return new Content(
        html: 'emails.assignment-created',
        text: 'emails.assignment-created-text',
    );
}
```

Methods Available in `envelope()`

We've already looked at how we can customize the subject and the “from” address using the `envelope()` method. Note that the way we're customizing these is by passing different named parameters to the constructor of the `Envelope` class:

```
public function envelope(): Envelope
{
    return new Envelope(
        subject: 'New assignment from ' . $this->trainer->name,
        from: new Address($this->trainer->email, $this->trainer->name),
    );
}
```

This isn't an exhaustive list, but here is a short list of parameters the `envelope()` method allows us to pass to the `Envelope` class to customize our email. Any parameter that can accept an `Address` can also accept a string email address or an array containing a mixture of `Address` objects and/or strings.

`from:` *Address*

Sets the “from” name and address—represents the author

`subject:` *string*

Sets the email subject

`cc:` *Address*

Sets the CC

`bcc:` *Address*

Sets the BCC

`replyTo:` *Address*

Sets the “reply to”

`tags:` *array*

Sets the tags, if relevant for your email sender

`metadata:` *array*

Sets the metadata, if relevant for your email sender

Finally, if you want to perform any manual modifications on the underlying Symfony message, you can do that in the `using` parameter, as shown in [Example 15-6](#).

Example 15-6. Modifying the underlying `SymfonyMessage` object

```
public function envelope(): Envelope
{
    return new Envelope(
        subject: 'Howdy!',
        view: 'emails.howdy',
```

```

        using: [
            function (Email $message) {
                $message->setReplyTo('noreply@email.com');
            },
        ],
    );
}

```

Attaching Files and Inlining Images

To attach a file to your mail, return an array (in which each entry is an Attachment) from the `attachments()` method, as shown in [Example 15-7](#).

Example 15-7. Attaching files or data to mailables

```

use Illuminate\Mail\Mailables\Attachment;

// Attach a file using the local filename
public function attachments(): array
{
    return [
        Attachment::fromPath('/absolute/path/to/file'),
    ];
}

// Attach a file using storage disks
public function attachments(): array
{
    return [
        // Attach from default disk
        Attachment::fromStorage('/path/to/file'),
        // Attach from custom disk
        Attachment::fromStorageDisk('s3', '/path/to/file'),
    ];
}

// Attach a file passing the raw data
public function attachments(): array
{
    return [
        Attachment::fromData(fn () => file_get_contents($this->pdf), 'whitepaper.pdf')
            ->withMime('application/pdf'),
    ];
}

```

Attachable mail objects

If you have a PHP class that can be represented as an attachment to an email, or

if you'd like to build a PHP class full of logic around the objects you attach to your emails, you'll want to try Laravel's attachable objects.

Each of these objects need only be a PHP class that implements the `Illuminate\Contracts\Mail\Attachable` interface, which requires a `toMailAttachment()` method, which returns an instance of `Illuminate\Mail\Attachment`.

One common example would be if you want to make one of your Eloquent models attachable. In our example, we've been emailing our clients about a new assignment from their trainer, so let's try making `Assignment` attachable. Take a look at [Example 15-8](#).

Example 15-8. Making an Eloquent model attachable

```
<?php

namespace App\Models;

use Illuminate\Contracts\Mail\Attachable;
use Illuminate\Database\Eloquent\Model;
use Illuminate\Mail\Attachment;

class Assignment extends Model implements Attachable
{
    /**
     * Get the attachable representation of the model.
     */
    public function toMailAttachment(): Attachment
    {
        return Attachment::fromPath($this->pdf_path);
    }
}
```

If a class implements `Attachable`, you can use any instances of that class as entries in the array you return from `attachments()`:

```
public function attachments(): array
{
    return [$this->assignment];
}
```

Inlining images

If you want to attach images inline, directly into your email, Laravel provides a

feature for that as well, as you can see in [Example 15-9](#).

Example 15-9. Inlining images in email

```
<!-- emails/image.blade.php -->
```

Here is an image:

```

```

Or, the same image embedding the data:

```

```

Markdown Mailables

Markdown mailables allow you to write your email content in Markdown, after which it will be converted into full HTML (and plain-text) emails with Laravel's built-in, responsive HTML templates. You can also tweak these templates to make a customized email template that's simple for your developers and nondevelopers to create content for.

First, run the `make:mail` Artisan command with the `markdown` flag:

```
php artisan make:mail AssignmentCreated --markdown=emails.assignment-created
```

You can see an example of what the generated mail file looks like in [Example 15-10](#).

Example 15-10. Generated Markdown mailable

```
class AssignmentCreated extends Mailable
{
    // ...

    public function content(): Content
    {
        return new Content(
            markdown: 'emails.assignment-created',
        );
    }
}
```

As you can see, this is almost exactly the same as a normal mailable file in Laravel. The main difference is that you're passing your template into the

markdown parameter instead of the view parameter. Also note that the template you're referencing should represent a Markdown template, not a normal Blade template.

What's a *Markdown template*? Unlike a normal Blade email template, which is expected—with the use of includes and inheritance like any Blade file—to generate a full HTML email, Markdown templates simply pass Markdown content to a few predefined components.

These components will look like `<x-mail::component-name-here>`, and, as such, the main body of your Markdown email should be passed into a component named `<x-mail::message>`. Take a look at [Example 15-11](#) to see an example of a simple Markdown mail template.

Example 15-11. Simple assignment Markdown email

```
{{-- resources/views/emails/assignment-created.blade.php --}}
<x-mail::message>
# Hey {{ $trainee->name }}!

You have received a new training assignment from **{{ $trainer->name }}**

<x-mail::button :url="route('training-dashboard')">
View Your Assignment
</x-mail::button>

Thanks,<br>
{{ config('app.name') }}
</x-mail::message>
```

As you can see in [Example 15-11](#), there's a parent `mail::message` component to which you pass the body of your email, but you're also provided with other, smaller components you can sprinkle into your emails. We used the `mail::button` component here, which takes the content (“View Your Assignment”) but also requires the `url` attribute to be passed.

There are three types of components available:

Button

Generates a centered button link. The button component requires a `url` attribute and allows an optional `color` attribute, to which you can pass `primary`, `success`, or `error`.

Panel

Renders the provided text with a slightly lighter background than the rest of the message.

Table

Converts the content passed into it via the Markdown table syntax.

CUSTOMIZING THE COMPONENTS

These Markdown components are built into the core of the Laravel framework, but if you need to customize how they work, you can publish their files and edit them:

```
php artisan vendor:publish --tag=laravel-mail
```

You can learn more about customizing these files and their themes in the [Laravel docs](#).

Rendering Mailables to the Browser

When you're developing emails in your applications, it's helpful to be able to preview how they'll render. You can rely on a tool like Mailtrap for this, and that is a useful tool, but it can also be helpful to render the mails directly in your browser and see your changes made immediately.

Take a look at [Example 15-12](#) to see a sample route you can add to your application to render a given mailable.

Example 15-12. Rendering a mailable to a route

```
Route::get('preview-assignment-created-mailable', function () {  
    $trainer = Trainer::first();  
    $trainee = Trainee::first();  
  
    return new \App\Mail\AssignmentCreated($trainer, $trainee);  
});
```

Laravel also provides a way to quickly preview a notification in the browser:

```
Route::get('preview-notification', function () {  
    $trainer = Trainer::first();  
    $trainee = Trainee::first();  
  
    return (new App\Notifications\AssignmentCreated($trainer, $trainee))  
        ->toMail($trainee);  
});
```

Queues

Sending email is a time-consuming task that can cause applications to slow down, so it's common to move it to a background queue. It's so common, in fact, that Laravel has a set of built-in tools to make it easier to queue your messages without writing queue jobs for each email:

queue()

To queue a mail object instead of sending it immediately, simply pass your mailable object to `Mail::queue()` instead of `Mail::send()`:

```
Mail::to($user)->queue(new AssignmentCreated($trainer, $trainee));
```

later()

`Mail::later()` works the same as `Mail::queue()`, but it allows you to add a delay—either in minutes or by specifying a specific time by passing an instance of `DateTime` or `Carbon`—specifying when the email will be pulled from the queue and sent:

```
$when = now()->addMinutes(30);  
Mail::to($user)->later($when, new AssignmentCreated($trainer, $trainee));
```

CONFIGURING QUEUES

Your queues must be configured correctly for these methods to work. Take a look at [Chapter 16](#) to learn more about how queues work and how to get them running in your application.

For both `queue()` and `later()`, if you'd like to specify which queue or queue connection your mail is added to, use the `onConnection()` and `onQueue()` methods on your mailable object:

```
$message = (new AssignmentCreated($trainer, $trainee))
    ->onConnection('sqs')
    ->onQueue('emails');

Mail::to($user)->queue($message);
```

If you'd like to direct that a given mailable should always be queued, you can make the mailable implement the `Illuminate\Contracts\Queue\ShouldQueue` interface.

Local Development

This is all well and good for sending mail in your production environments. But how do you test it all out? There are two primary tools you'll want to consider: Laravel's log driver and fake inboxes for testing, such as Mailtrap.

The log driver

Laravel provides a log driver that logs every email you try to send to your local *laravel.log* file (which is, by default, in *storage/logs*).

To use this, edit *.env* and set `MAIL_MAILER` to `log`. Now open up or tail *storage/logs/laravel.log* and send an email from your app. You'll see something like this:

```
Message-ID: <04ee2e97289c68f0c9191f4b04fc0de1@localhost>
Date: Tue, 17 May 2016 02:52:46 +0000
Subject: Welcome to our app!
From: Matt Stauffer <matt@mattstauffer.com>
To: freja@jensen.no
MIME-Version: 1.0
Content-Type: text/html; charset=utf-8
Content-Transfer-Encoding: quoted-printable

Welcome to our app!
```

You can optionally specify that logged mail gets sent to a different log channel

than the rest of your logs. Either modify *config/mail.php* or set the `MAIL_LOG_CHANNEL` variable in your *.env* file to the name of any existing log channel.

Fake inboxes

If you want to see your test emails as they'll look in a real inbox, you can use one of several services that allow you to send your emails to them and show you your emails in a full fake inbox.

The two most common such services are Mailtrap, a paid SaaS that requires no setup and allows you to share your inbox with coworkers and clients, and Mailpit, a service you can run on your local machine through Docker.

Mailtrap

Mailtrap is a service for capturing and inspecting emails in development environments. You send your mail to the Mailtrap servers via SMTP, but instead of sending those emails off to the intended recipients, Mailtrap captures them all and provides you with a web-based email client for inspecting them, regardless of which email address is in the to field.

To set up Mailtrap, sign up for a free account and visit the base dashboard for your demo. Copy your username and password from the SMTP column.

Then edit your app's *.env* file and set the following values in the `mail` section:

```
MAIL_MAILER=smtp
MAIL_HOST=mailtrap.io
MAIL_PORT=2525
MAIL_USERNAME=your_username_from_mailtrap_here
MAIL_PASSWORD=your_password_from_mailtrap_here
MAIL_ENCRYPTION=null
```

Now, any email you send from your app will show up in your Mailtrap inbox.

Mailpit

If you like the idea of Mailtrap, but you'd like to run the application locally (and for free), you can use **Mailpit**, a Mailtrap alternative you can spin up in a local Docker container.

Notifications

The purpose of most of the mail that's sent from web apps is to notify users that a particular action has happened or needs to happen. As users' communication preferences grow more and more diverse, we gather ever more—and more disparate—packages to communicate via Slack, SMS, and other means.

To support these preferences, Laravel introduced a concept called, fittingly, *notifications*. Just like a mailable, a notification is a PHP class that represents a single communication that you might want to send to your users. For now, let's imagine we're notifying the users of our physical training app that they have a new workout available.

Each class represents all of the information necessary to send notifications to your users *using one or many notification channels*. A single notification could send an email, send an SMS via Vonage, send a WebSocket ping, add a record to a database, send a message to a Slack channel, and much more.

So, let's create our notification:

```
php artisan make:notification WorkoutAvailable
```

Example 15-13 shows what that gives us.

Example 15-13. An autogenerated notification class

```
<?php

namespace App\Notifications;

use Illuminate\Bus\Queueable;
use Illuminate\Notifications\Notification;
use Illuminate\Contracts\Queue\ShouldQueue;
use Illuminate\Notifications\Messages\MailMessage;

class WorkoutAvailable extends Notification
{
    use Queueable;

    /**
     * Create a new notification instance.
     */
    public function __construct()
    {
        //
    }
}
```

```

}

/**
 * Get the notification's delivery channels.
 *
 * @return array<int, string>
 */
public function via(object $notifiable): array
{
    return ['mail'];
}

/**
 * Get the mail representation of the notification.
 */
public function toMail(object $notifiable): MailMessage
{
    return (new MailMessage)
        ->line('The introduction to the notification.')
        ->action('Notification Action', url('/'))
        ->line('Thank you for using our application!');
}

/**
 * Get the array representation of the notification.
 *
 * @return array<string, mixed>
 */
public function toArray(object $notifiable): array
{
    return [
        //
    ];
}
}

```

We can learn a few things here. First, we're going to pass relevant data into the constructor. Second, there's a `via()` method that allows us to define, for a given user, which notification channels to use (`$notifiable` represents whatever entities you want to notify in your system; for most apps, it'll be a user, but that's not always the case). And third, there are individual methods for each notification channel that allow us to specifically define how to send one of these notifications through that channel.

WHEN WOULD A \$NOTIFIABLE NOT BE A USER?

Although the most common notification targets will be users, it's possible you may want to notify something else. This may simply be because your application has multiple user types—so, you might want to be able to notify both trainers and trainees. But you also might find yourself wanting to notify a group, a company, or a server.

So, let's modify this class for our `WorkoutAvailable` example. Take a look at [Example 15-14](#).

Example 15-14. Our `WorkoutAvailable` notification class

```
...
class WorkoutAvailable extends Notification
{
    use Queueable;

    public function __construct(public $workout) {}

    public function via(object $notifiable): array
    {
        // This method doesn't exist on the User... we're going to make it up
        return $notifiable->preferredNotificationChannels();
    }

    public function toMail(object $notifiable): MailMessage
    {
        return (new MailMessage)
            ->line('You have a new workout available!')
            ->action('Check it out now', route('workout.show', [$this->workout]))
            ->line('Thank you for training with us!');
    }

    public function toArray(object $notifiable): array
    {
        return [];
    }
}
```

Defining the `via()` Method for Your Notifiables

As you can see in [Example 15-14](#), we're responsible for deciding, for each notification and each notifiable, which notification channels we're going to use.

You could just send everything as mail or just send everything as an SMS ([Example 15-15](#)).

Example 15-15. Simplest possible `via()` method

```
public function via(object $notifiable): array
{
    return 'vonage';
}
```

You could also let each user choose a preferred method and save that on the User itself ([Example 15-16](#)).

Example 15-16. Customizing the via() method per user

```
public function via(object $notifiable): array
{
    return $notifiable->preferred_notification_channel;
}
```

Or, as we imagined in [Example 15-14](#), you could create a method on each notifiable that allows for some complex notification logic. For example, you could notify the user over certain channels during work hours and other channels in the evening. What is important is that via() is a PHP class method, so you can do whatever complex logic you want there.

Sending Notifications

There are two ways to send a notification: using the Notification facade, or adding the Notifiable trait to an Eloquent class (likely your User class).

Sending notifications using the notification facade

The Notification facade is the clumsier of the two methods, since you have to pass both the notifiable and the notification. However, it's helpful because you can choose to pass more than one notifiable at the same time, as shown in [Example 15-17](#).

Example 15-17. Sending a notification using the Notification facade

```
use App\Notifications\WorkoutAvailable;
...
Notification::send($users, new WorkoutAvailable($workout));
```

Sending notifications using the notifiable trait

Any model that imports the Laravel\Notifications\Notifiable trait (which the App\User class does by default) has a notify() method that can be passed a

notification, which will look like [Example 15-18](#).

Example 15-18. Sending a notification using the Notifiable trait

```
use App\Notifications\WorkoutAvailable;
...
$user->notify(new WorkoutAvailable($workout));
```

Queueing Notifications

Most notification drivers need to use HTTP requests to send notifications, which could slow down your user experience. To deal with this you'll probably want to queue your notifications. All notifications import the `Queueable` trait by default, so all you need to do is add `implements ShouldQueue` to your notification, and Laravel will instantly move it to a queue.

As with any other queued features, you'll need to make sure you have your queue settings configured correctly and a queue worker running.

If you'd like to delay the delivery of a notification, you can run the `delay()` method on the notification:

```
$delayUntil = now()->addMinutes(15);

$user->notify((new WorkoutAvailable($workout))->delay($delayUntil));
```

Out-of-the-Box Notification Types

Out of the box, Laravel comes with notification drivers for email, database, broadcast, Vonage SMS, and Slack. I'll cover each briefly, but I'd recommend referring to the [notifications docs](#) for more thorough introductions to each.

It's also easy to create your own notification drivers, and dozens of people already have; you can find them at the [Laravel Notification Channels website](#).

Email notifications

Let's take a look at how the email from our earlier example, [Example 15-14](#), is built:

```
public function toMail(object $notifiable): MailMessage
{
```

```
return (new MailMessage)
    ->line('You have a new workout available!')
    ->action('Check it out now', route('workouts.show', [$this->workout]))
    ->line('Thank you for training with us!');
}
```

The result is shown in **Figure 15-1**. The email notification system puts your application’s name in the header of the email; you can customize that app name in the `name` key of *config/app.php*.

This email is automatically sent to the `email` property on the notifiable, but you can customize this behavior by adding a method to your notifiable class named `routeNotificationForMail()` that returns the email address you’d like email notifications sent to.

The email’s subject is set by parsing the notification class name and converting it to words. So, our `WorkoutAvailable` notification would have the default subject of “Workout Available.” You can also customize this by chaining the `subject()` method on the `MailMessage` in the `toMail()` method.

If you want to modify the templates, publish them and edit to your heart’s content:

```
php artisan vendor:publish --tag=laravel-notifications
```



Hello!

You have a new workout available!

Check it out now

Thank you for training with us!

Regards,
Laravel

If you're having trouble clicking the "Check it out now" button, copy and paste the URL below into your web browser: <https://trainremote.co/workouts/12345>

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Figure 15-1. An email sent with the default notification template

MARKDOWN MAIL NOTIFICATIONS

If you like working with Markdown emails (see “[Markdown Mailables](#)”), you can also use the same `markdown()` method in your notifications, as shown in [Example 15-19](#):

Example 15-19. Using the `markdown()` method with notifications

```
public function toMail(object $notifiable): MailMessage
{
    return (new MailMessage)
        ->subject('Workout Available')
        ->markdown('emails.workout-available', ['workout' => $this->workout]);
}
```

You can change the style of the default template to be an “error” message, which uses slightly different language and changes the primary button color to red. To do this, just add a call to the `error()` method to your

MailMessage call chain in the toMail() method.

Database notifications

You can send notifications to a database table using the database notification channel. First, create your table with `php artisan notifications:table`. Next, create a `toDatabase()` method on your notification and return an array of data there. This data will be encoded as JSON and stored in the database table's data column.

The `Notifiable` trait adds a `notifications` relationship to any model it's imported into, allowing you to easily access records in the notifications table. So if you're using database notifications, you could do something like [Example 15-20](#):

Example 15-20. Iterating over a user's database notifications

```
User::first()->notifications->each(function ($notification) {  
    // Do something  
});
```

The database notification channel also has the concept of whether or not a notification is “read.” You can scope to only the “unread” notifications as shown in [Example 15-21](#):

Example 15-21. Iterating over a user's unread database notifications

```
User::first()->unreadNotifications->each(function ($notification) {  
    // Do something  
});
```

And you can mark one or all notifications as read, as [Example 15-22](#) demonstrates.

Example 15-22. Marking database notifications as read

```
// Individual  
User::first()->unreadNotifications->each(function ($notification) {  
    if ($condition) {  
        $notification->markAsRead();  
    }  
});  
  
// All  
User::first()->unreadNotifications->markAsRead();
```

Broadcast notifications

The broadcast channel sends notifications using Laravel’s event broadcasting features, which are powered by WebSockets (we’ll learn more about these in [“Broadcasting Events Over WebSockets, and Laravel Echo”](#)).

Create a `toBroadcast()` method on your notification and return an array of data. If your app is correctly configured for event broadcasting, that data will be broadcast on a private channel named `notifiable.id`. The `id` will be the ID of the notifiable, and `notifiable` will be the notifiable’s fully qualified class name, with the slashes replaced by periods—for example, the private channel for the `App\User` with the ID of 1 will be `App.User.1`.

SMS notifications

SMS notifications are sent via [Vonage](#), so if you want to send SMS notifications, sign up for a Vonage account and follow the instructions in the [notifications docs](#). Like with the other channels, you’ll be setting up a `toVonage()` method and customizing the SMS message there.

SMS NOTIFICATION PACKAGE INSTALL

In Laravel, the SMS notification channel is a first-party package. If you want to use Vonage SMS notifications, simply require this package with Composer:

```
composer require laravel/vonage-notification-channel \
    guzzlehttp/guzzle
```

Slack notifications

The slack notification channel allows you to customize the appearance of your notifications and even attach files to them. As with the other channels, you’ll set up a `toSlack()` method and customize the message there.

SLACK NOTIFICATION PACKAGE INSTALL

Laravel’s Slack notification channel is a first-party package. If you want to use Slack

notifications, simply require this package with Composer:

```
composer require laravel/slack-notification-channel
```

Other notifications

Looking to send your notifications through other channels than those that come out of the box? There's a robust community effort to provide an incredible variety of notification channels; check out what's on offer at the [Laravel Notifications Channels website](#).

Testing

Let's take a look at how to test mail and notifications.

Mail

There are two aspects of our mail we can write assertions against: the mail's content and attributes, and the fact that it was actually triggered to be sent. Let's start with assertions against the mail's content.

Asserting against the mail

First, we can run assertions against the `envelope()` type data, as you can see in [Example 15-23](#).

Example 15-23. Asserting against a mailable's envelope data

```
$mailable = new AssignmentCreated($trainer, $trainee);

$mailable->assertFrom('noreply@mytrainingapp.com');
$mailable->assertTo('user@gmail.com');
$mailable->assertHasCc('trainer@mytrainingapp.com');
$mailable->assertHasBcc('records@mytrainingapp.com');
$mailable->assertHasReplyTo('trainer@mytrainingapp.com');
$mailable->assertHasSubject('New assignment from Faith Elizabeth');
$mailable->assertHasTag('assignments');
$mailable->assertHasMetadata('clientId', 4);
```

Next, we can run assertions against the contents of the message, as you can see

in [Example 15-24](#).

Example 15-24. Asserting against a mailable's contents

```
$mailable->assertSeeInHtml($trainee->name);
$mailable->assertSeeInHtml('You have received a new training assignment');
$mailable->assertSeeInOrderInHtml(['Hey', 'You have received']);

$mailable->assertSeeInText($trainee->name);
$mailable->assertSeeInOrderInText(['Hey', 'You have received']);
```

We can assert against attachments, as you can see in [Example 15-25](#).

Example 15-25. Asserting against mailable attachments

```
$mailable->assertHasAttachment('/pdfs/assignment-24.pdf');
$mailable->assertHasAttachment(Attachment::fromPath('/pdfs/assignment-24.pdf'));
$mailable->assertHasAttachedData($pdfData, 'assignment-24.pdf', [
    'mime' => 'application/pdf',
]);
$mailable->assertHasAttachmentFromStorage(
    '/pdfs/assignment-24.pdf',
    'assignment-24.pdf',
    ['mime' => 'application/pdf']
);
$mailable->assertHasAttachmentFromStorageDisk(
    's3',
    '/pdfs/assignment-24.pdf',
    'assignment-24.pdf',
    ['mime' => 'application/pdf']
);
```

Asserting whether the mail was sent

To test what mail was (or wasn't) sent, we'll first want to run `Mail::fake()` to capture the mail actions for inspection. Then, we can run our various assertions, as you can see in [Example 15-26](#).

Example 15-26. Asserting against whether or not mail was sent

```
Mail::fake();

// Call the code that sends the email

// Assert that no mailables were sent
Mail::assertNothingSent();

// Assert that a mailable was sent
Mail::assertSent(AssignmentCreated::class);
```



```
// Assert a mailable was sent a certain number of times
Mail::assertSent(AssignmentCreated::class, 4);

// Assert a mailable was not sent
Mail::assertNotSent(AssignmentCreated::class);

// Assertions for queued emails
Mail::assertQueued(AssignmentCreated::class);
Mail::assertNotQueued(AssignmentCreated::class);
Mail::assertNothingQueued();
```

Laravel also allows us to pass a closure as the second parameter to these assertions, inspecting the emails to make sure they fit what we're expecting. Take a look at [Example 15-27](#).

Example 15-27. Examining an email's properties in assertions

```
Mail::assertSent(
    AssignmentCreated::class,
    function (AssignmentCreated $mail) use ($trainer, $trainee) {
        return $mail->hasTo($trainee->email) &&
            $mail->hasSubject('New assignment from ' . $trainer->name);
    }
);
```

You can also use `hasCc()`, `hasBcc()`, `hasReplyTo()`, and `hasFrom()`.

Notifications

Laravel provides a built-in set of assertions for testing your notifications. [Example 15-28](#) demonstrates.

Example 15-28. Asserting notifications were sent

```
public function test_new_signups_triggers_admin_notification()
{
    Notification::fake();

    Notification::assertSentTo($user, NewUsersSignedup::class,
        function ($notification, $channels) {
            return $notification->user->email == 'user-who-signed-up@gmail.com'
                && $channels == ['mail'];
        });

    // Assert that the email was sent to a given user
    Notification::assertSentTo(
        [$user],
        NewUsersSignedup::class
    );
}
```

```
);  
  
// You can also use assertNotSentTo()  
Notification::assertNotSentTo(  
    [$userDidntSignUp], NewUsersSignedup::class  
);  
}
```

TL;DR

Laravel's mail and notification features provide simple, consistent interfaces to a variety of messaging systems. Laravel's mail system uses mailables, PHP classes that represent emails, to provide a consistent syntax to different mail drivers. The notification system makes it easy to build a single notification that can be delivered in many different media—from emails to SMS messages to physical postcards.

Chapter 16. Queues, Jobs, Events, Broadcasting, and the Scheduler

So far we've covered some of the most common structures that power web applications: databases, mail, filesystems, and more. All of these are common across a majority of applications and frameworks.

Laravel also provides facilities for some less common architecture patterns and application structures. In this chapter we'll cover Laravel's tools for implementing queues, queued jobs, events, and WebSocket event publishing. We'll also cover Laravel's scheduler, which makes manually edited cron schedules a thing of the past.

Queues

To understand what a queue is, just think about the idea of “queueing up” in a line at the bank. Even if there are multiple lines—queues—only one person is being served at a time from each queue, and each person will eventually reach the front and be served. In some banks, it's a strict first-in-first-out sort of policy, but in other banks, there's not an exact guarantee that someone won't cut ahead of you in line at some point. Essentially, someone can get added to the queue, be removed from the queue prematurely, or be successfully “processed” and then removed. Someone might even hit the front of the queue, not be able to be served correctly, return to the queue for a time, and then be processed again.

Queues in programming are very similar. Your application adds a “job” to a queue, which is a chunk of code that tells the application how to perform a particular behavior. Then some other separate application structure, usually a “queue worker,” takes the responsibility for pulling jobs off of the queue one at a time and performing the appropriate behavior. Queue workers can delete the jobs, return them to the queue with a delay, or mark them as successfully

processed.

Laravel makes it easy to serve your queues using Redis, *beanstalkd*, Amazon Simple Queue Service (SQS), or a database table. You can also choose the `sync` driver to have the jobs run right in your application without actually being queued, or the `null` driver for jobs to just be discarded; these two are usually used in local development or testing environments.

Why Queues?

Queues make it easy to remove a costly or slow process from any synchronous call. The most common example is sending mail—doing so can be slow, and you don’t want your users to have to wait for mail to send in response to their actions. Instead, you can trigger a “send mail” queued job and let the users get on with their day. And sometimes you may not be worried about saving your users time, but you might have a process like a cron job or a webhook that has a lot of work to get through; rather than letting it all run at once (and potentially time out), you may choose to queue its individual pieces and let the queue worker process them one at a time.

Additionally, if you have some heavy processing that’s more than your server can handle, you can spin up more than one queue worker to work through your queue faster than your normal application server could on its own.

Basic Queue Configuration

Like many other Laravel features that abstract multiple providers, queues have their own dedicated config file (*config/queue.php*) that allows you to set up multiple drivers and define which will be the default. This is also where you’ll store your SQS, Redis, or *beanstalkd* authentication information.

SIMPLE REDIS QUEUES ON LARAVEL FORGE

Laravel Forge is a hosting management service provided by Taylor Otwell, the creator of Laravel, which makes serving queues with Redis a breeze. Every server you create has Redis configured automatically, so if you visit any site’s Forge console, you can just go to the Queue tab and hit Start Worker and you’re ready to use Redis as your queue driver; you can leave all the default settings, and no other work is necessary.

Queued Jobs

Remember our bank analogy? Each person in the bank's *queue* (line) is, in programming terms, a *job*. Queued jobs can, depending on the environment, take many shapes, like arrays of data or simple strings. In Laravel, each job is a collection of information containing the job name, the data payload, the number of attempts that have been made so far to process this job, and some other simple metadata.

But you don't need to worry about any of that in your interactions with Laravel. Laravel provides a structure called `Job`, which is intended to encapsulate a single task—a behavior that your application can be commanded to do—and allow it to be added to and pulled from a queue. There are also simple helpers to make it easy to queue Artisan commands and mail.

Let's start with an example in which every time a user changes their plan with your SaaS app, you want to rerun some calculations about your overall profit.

Creating a job

As always, there's an Artisan command for that:

```
php artisan make:job CrunchReports
```

Take a look at [Example 16-1](#) to see what you'll get.

Example 16-1. The default template for jobs in Laravel

```
<?php

namespace App\Jobs;

use Illuminate\Bus\Queueable;
use Illuminate\Contracts\Queue\ShouldBeUnique;
use Illuminate\Contracts\Queue\ShouldQueue;
use Illuminate\Foundation\Bus\Dispatchable;
use Illuminate\Queue\InteractsWithQueue;
use Illuminate\Queue\SerializesModels;

class CrunchReports implements ShouldQueue
{
    use Dispatchable, InteractsWithQueue, Queueable, SerializesModels;

    /**
```

```

        * Create a new job instance.
        */
    public function __construct()
    {
        //
    }

    /**
     * Execute the job.
     */
    public function handle(): void
    {
        //
    }
}

```

As you can see, this template imports the `Dispatchable`, `InteractsWithQueue`, `Queueable`, and `SerializesModels` traits, and implements the `ShouldQueue` interface.

We also get two methods from this template: the constructor, which you'll use to attach data to the job, and the `handle()` method, which is where the job's logic should reside (and is also the method signature you'll use to inject dependencies).

The traits and interface provide the class with the ability to be added to, and interact with, the queue. `Dispatchable` gives it methods to dispatch itself; `InteractsWithQueue` allows each job, while being handled, to control its relationship with the queue, including deleting or requeueing itself; `Queueable` allows you to specify how Laravel should push this job to the queue; and `SerializesModels` gives the job the ability to serialize and deserialize Eloquent models.

SERIALIZING MODELS

The `SerializesModels` trait gives jobs the ability to *serialize* (convert to a flatter format that can be stored in a data store like a database or queue system) injected models so that your job's `handle()` method will have access to them. However, because it's too difficult to reliably serialize an entire Eloquent object, the trait ensures that just the primary keys of any attached Eloquent objects are serialized when the job is pushed onto the queue. When the job is deserialized and handled, the trait pulls those Eloquent models fresh from the database by their primary key. This means that when your job runs, it will be pulling a fresh instance of this model, not whatever state it was in when you queued the job.

Let's fill out the methods for our sample class, as in [Example 16-2](#).

Example 16-2. An example job

```
...
use App\ReportGenerator;

class CrunchReports implements ShouldQueue
{
    use Dispatchable, InteractsWithQueue, Queueable, SerializesModels;

    protected $user;

    public function __construct($user)
    {
        $this->user = $user;
    }

    public function handle(ReportGenerator $generator): void
    {
        $generator->generateReportsForUser($this->user);

        Log::info('Generated reports.');
```

We're expecting the User instance to be injected when we create the job, and then when it's handled, we're typehinting a ReportGenerator class (which we presumably wrote). Laravel will read the typehint and inject that dependency automatically.

Pushing a job onto a queue

There are multiple methods by which you can dispatch a job, including some methods available to every controller and a global `dispatch()` helper. But the simpler and preferred method is calling the `dispatch()` method on the job itself, so that's what we'll do for the rest of the chapter.

To dispatch your job, you can just create an instance of it and then call its `dispatch()` method, passing any necessary data directly into that method. Take a look at [Example 16-3](#) for an example.

Example 16-3. Dispatching jobs

```
$user = auth()->user();  
$daysToCrunch = 7;  
\App\Jobs\CrunchReports::dispatch($user, $daysToCrunch);
```

There are three settings you can control to customize exactly how you dispatch a job: the connection, the queue, and the delay.

Customizing the connection

If you ever have multiple queue connections in place at once, you can customize the connection by chaining `onConnection()` after the `dispatch()` method:

```
DoThingJob::dispatch()->onConnection('redis');
```

Customizing the queue

Within queue servers, you can specify which named queue you’re pushing a job onto. For example, you may differentiate your queues based on their importance, naming one `low` and one `high`.

You can customize which queue you’re pushing a job onto with the `onQueue()` method:

```
DoThingJob::dispatch()->onQueue('high');
```

Customizing the delay

You can customize the amount of time your queue workers should wait before processing a job with the `delay()` method, which accepts either an integer representing the number of seconds to delay a job or a *DateTime/Carbon* instance:

```
// Delays five minutes before releasing the job to queue workers  
$delay = now()->addMinutes(5);  
DoThingJob::dispatch()->delay($delay);
```

Note that Amazon SQS doesn’t allow delays longer than 15 minutes.

Job chaining

If you need a series of jobs to run in order one after the other, you can “chain”

them together. Each job will wait to run until the previous job completes, and if one job fails, the rest after it won't run.

```
$user = auth()->user();
$daysToCrunch = 7;

Bus::chain([
    new CrunchReports($user, $daysToCrunch),
    new SendReport($user),
])->dispatch();
```

When one of the chained jobs fails, you can execute with the `catch()` method:

```
$user = auth()->user();
$daysToCrunch = 7;

Bus::chain([
    new CrunchReports($user, $daysToCrunch),
    new NotifyNewReportsDone($user)
])->catch(function (Throwable $e) {
    new ReportsNotCrunchedNotification($user)
})->dispatch($user);
```

Job batching

Job batching makes it possible to push a group of jobs onto the queue at the same time, inspect the status of the batch, and take an action after the batch is complete.

This feature requires a database table to keep track of the jobs; as you might expect, there's an Artisan command to create it:

```
php artisan queue:batches-table
php artisan migrate
```

To mark a job as batchable, include the `Illuminate\Bus\Batchable` trait. This trait adds a `batch()` method to your job, which will allow you to retrieve information about the current batch your job is running in.

Take a look at [Example 16-4](#) to see how this works. In this example, you can see that one of the most important steps to take on a batchable job is to make sure it doesn't take any action if its batch was canceled.

Example 16-4. Batchable job in Laravel

```
...
class SampleBatchableJob implements ShouldQueue
{
    use Batchable, Dispatchable, InteractsWithQueue, Queueable, SerializesModels;

    public function handle(): void
    {
        // Don't run if this batch is canceled
        if ($this->batch()->cancelled()) {
            return;
        }

        // Otherwise, run like normal
        // ...
    }
}
```

Dispatching batchable jobs

The Bus facade offers a method `batch()`, which allows you to dispatch a batch of jobs. You can also define actions to take after the batch succeeds or fails, using the `then()` (succeeds), `catch()` (fails), or `finally()` (succeeds or fails) methods.

You can see how these can be called in [Example 16-5](#).

Example 16-5. Dispatching batchable jobs

```
use App\Jobs\CrunchReports;
use Illuminate\Support\Facades\Bus;

$user = auth()->user();
$admin = User::admin()->first();
$supervisor = User::supervisor()->first();

$daysToCrunch = 7;

Bus::batch([
    new CrunchReports::dispatch($user, $daysToCrunch),
    new CrunchReports::dispatch($admin, $daysToCrunch),
    new CrunchReports::dispatch($supervisor, $daysToCrunch)
])->then(function (Batch $batch) {
    // Run when the batch is completed successfully
})->catch(function (Batch $batch, Throwable $e) {
    // Run when any job fails
})->finally(function (Batch $batch) {
    // Run when the batch is complete
})
```

```
})->dispatch();
```

Adding jobs to batches from a job

If the jobs in your batch have a responsibility of adding jobs to a batch—for example, if you initially dispatch a few job-dispatcher type jobs—they can use the `add()` method on the `Batch` object returned by `batch()`:

```
public function handle(): void
{
    if ($this->batch()->cancelled()) {
        return;
    }

    $this->batch()->add([
        new \App\Jobs\ImportContacts,
        new \App\Jobs\ImportContacts,
        new \App\Jobs\ImportContacts,
    ]);
}
```

Cancelling a batch

If a job has a reason to cancel its batch, it can:

```
public function handle(): void
{
    if (/* This batch should be canceled for whatever reason */) {
        return $this->batch()->cancel();
    }

    // ...
}
```

Batch failures

By default, if a single job in a batch fails, the batch will be marked as “canceled.” If you want to define a different behavior, you can chain `allowFailures()` when you dispatch the batch:

```
$batch = Bus::batch([
    // ...
])->allowFailures()->dispatch();
```

Cleaning up the batches table

The batches table isn't self-pruning, so you'll want to schedule your app to "prune" that table:

```
$schedule->command('queue:prune-batches')->daily();
```

Running a Queue Worker

So what is a queue worker, and how does it work? In Laravel, it's an Artisan command that runs forever (until it's stopped manually) and takes the responsibility for pulling down jobs from your queue and running them:

```
php artisan queue:work
```

This command starts a daemon "listening" to your queue; every time there are jobs on the queue, it will pull down the first job, handle it, delete it, and move on to the next. If at any point there are no jobs, it "sleeps" for a configurable amount of time before checking again to see if there are any more jobs.

You can define how many seconds a job should be allowed to run before the queue listener stops it (`--timeout`), how many seconds the listener should "sleep" when there are no jobs left (`--sleep`), how many tries each job should be allowed before being deleted (`--tries`), which connection the worker should listen to (the first parameter after `queue:work`), and which queues it should listen to (`--queue=`):

```
php artisan queue:work redis --timeout=60 --sleep=15 --tries=3  
--queue=high,medium
```

You can also process just a single job with `php artisan queue:work`.

Handling Errors

So, what happens when something goes wrong with a job that's in the middle of processing?

Exceptions in handling

If an exception is thrown, the queue listener will release that job back onto the queue. The job will be rereleased to be processed again and again until it is able to finish successfully or until it has reached the maximum number of attempts allowed by your queue listener.

Limiting the number of tries

The maximum number of tries is defined by the `--tries` switch passed to the `queue:listen` or `queue:work` Artisan command.

THE DANGER OF INFINITE RETRIES

If you don't set `--tries`, or if you set it to `0`, the queue listener will allow infinite retries. That means if there are any circumstances in which a job can just *never* be completed—for example, if it relies on a tweet that has since been deleted—your app will slowly crawl to a halt as it retries forever.

The [documentation](#) and Laravel Forge both show 3 as the recommended starting point for the maximum number of retries. So, in case of confusion, start there and adjust:

```
php artisan queue:work --tries=3
```

If at any point you'd like to check how many times a job has been attempted already, use the `attempts()` method on the job itself, as in [Example 16-6](#).

Example 16-6. Checking how many times a job has already been tried

```
public function handle(): void
{
    ...
    if ($this->attempts() > 3) {
        //
    }
}
```

You can also specify the maximum number of times a given job can be retried on the job class itself by defining a `$tries` property. When specified, this value will take precedence over the value set with the `--tries` switch:

```
public $tries = 3;
```

You can set the `$maxExceptions` property in a job class to specify how many times the job can throw an exception (and therefore be retried) before it should be considered failed:

```
// Can attempt this job 10 times.
public $tries = 10;

// If the job fails 3 times because an exception was thrown,
// stop attempting the job and fail it.
public $maxExceptions = 3;
```

You can also specify when a job should time out, instructing the framework to attempt the job any number of times within a specified timeframe. You can specify a `retryUntil()` method on a job and from that return a `DateTime/Carbon` instance:

```
public function retryUntil()
{
    return now()->addSeconds(30);
}
```

Job-based retry delay

We can specify how long to wait before retrying a failed job by setting a `$retryAfter` property on the job, equivalent to the minutes to wait. For more complex calculations, we can instead define a `retryAfter` method, which should also return the minutes to wait:

```
public $retryAfter = 10;

public function retryAfter() {...}
```

Job middleware

We can run jobs through middleware, just like we run our HTTP requests through middleware. This is a great opportunity to extract logic that guards or validates your jobs or the conditions they'll run in:

```
<?php

namespace App\Jobs\Middleware;
```

```

use Illuminate\Http\Response;

class MyMiddleware
{
    public function handle($job, $next): Response
    {
        if ($something) {
            $next($job);
        } else {
            $job->release(5);
        }
    }
}

```

To assign a middleware to a job, specify a `middleware()` method in the job class:

```

...
use App\Jobs\Middleware\MyMiddleware;

...
public function middleware()
{
    return [new MyMiddleware];
}

```

You can also specify a middleware when dispatching jobs using the `through` method:

```

DoThingJob::dispatch()->through([new MyMiddleware]);

```

Rate limiting middleware for jobs

Laravel comes out of the box with a rate limiting job middleware. To use it, define a rate limiter using `RateLimiter::for()` in the `boot()` method of a service provider, as shown in [Example 16-7](#).

Example 16-7. A sample job rate limiting middleware

```

// In a service provider
public function boot(): void
{
    RateLimiter::for('imageConversions', function (object $job) {
        return $job->user->paidForPriorityConversions()
    });
}

```

```

        ? Limit::none()
        : Limit::perHour(1)->by($job->user->id);
    });
}

```

The syntax of the job rate limiting middleware is the same as the route rate limiting middleware (“**Rate Limiting**”).

Handling failed jobs

Once a job has exceeded its allowable number of retries, it’s considered a “failed” job. Before you do anything else—even if all you want to do is limit the number of times a job can be tried—you’ll need to create a “failed jobs” database table.

There’s an Artisan command to create the migration (and you’ll then want to migrate):

```

php artisan queue:failed-table
php artisan migrate

```

Any job that has surpassed its maximum number of allowed attempts will be dumped there. But there are quite a few things you can do with your failed jobs.

First, you can define a `failed()` method on the job itself, which will run when that job fails (see **Example 16-8**).

Example 16-8. Defining a method to run when a job fails

```

...
class CrunchReports implements ShouldQueue
{
    ...

    public function failed()
    {
        // Do whatever you want, like notify an admin
    }
}

```

Next, you can register a global handler for failed jobs. Somewhere in the application’s bootstrap—if you don’t know where to put it, just put it in the `boot()` method of `AppServiceProvider`—place the code in **Example 16-9** to define a listener.

Example 16-9. Registering a global handler to handle failed jobs

```
// Some service provider
use Illuminate\Support\Facades\Queue;
use Illuminate\Queue\Events\JobFailed;
// ...
public function boot(): void
{
    Queue::failing(function (JobFailed $event) {
        // $event->connectionName
        // $event->job
        // $event->exception
    });
}
```

There is also a suite of Artisan tools for interacting with the failed jobs table.

`queue:failed` shows you a list of your failed jobs:

```
php artisan queue:failed
```

The list will look something like this:

ID	Connection	Queue	Class	Failed At
9	database	default	App\Jobs\AlwaysFails	2018-08-26 03:42:55

From there, you can grab the ID of any individual failed job and retry it with `queue:retry`:

```
php artisan queue:retry 9
```

If you'd rather retry all of the jobs, pass `all` instead of an ID:

```
php artisan queue:retry all
```

You can delete an individual failed job with `queue:forget`:

```
php artisan queue:forget 5
```

You can delete all of your failed jobs over a certain age (by default, it's 24 hours,

but you can also pass a custom number of hours using `--hours=48`):

```
php artisan queue:prune-failed
```

And you can delete all of your failed jobs with `queue:flush`:

```
php artisan queue:flush
```

Controlling the Queue

Sometimes, from within the handling of a job, you'll want to add conditions that will potentially either release the job to be restarted later or delete the job forever.

To release a job back onto the queue, use the `release()` method, as in [Example 16-10](#).

Example 16-10. Releasing a job back onto the queue

```
public function handle()
{
    ...
    if (condition) {
        $this->release($numberOfSecondsToDelayBeforeRetrying);
    }
}
```

If you want to delete a job during its handling, you can just `return` at any point, as seen in [Example 16-11](#); that's the signal to the queue that the job was handled appropriately and should not be returned to the queue.

Example 16-11. Deleting a job

```
public function handle(): void
{
    // ...
    if ($jobShouldBeDeleted) {
        return;
    }
}
```

Queues Supporting Other Functions

The primary use for queues is to push jobs onto them, but you can also queue

mail using the `Mail::queue` functionality. You can learn more about this in “Queues”. You can also queue Artisan commands, which we covered in Chapter 8.

Laravel Horizon

Laravel Horizon, like some of the other tools we’ve covered (Scout, Passport, etc.), is a tool provided by Laravel that doesn’t come bundled with the core.

Horizon provides insight into the status of your Redis queued jobs. You can see which jobs have failed, how many are queued, and how fast they’re working, and you can even get notifications when any of your queues are overloaded or failing. The Horizon dashboard is shown in Figure 16-1.

Installing and running Horizon is relatively straightforward, and the documentation is thorough, so if you’re interested, take a look at the [Horizon docs](#) to learn how to install, configure, and deploy it.

Please note that you will need to have your queue connection set to `redis`, in your `.env` or the `config/queue.php` config file, in order to run Horizon.

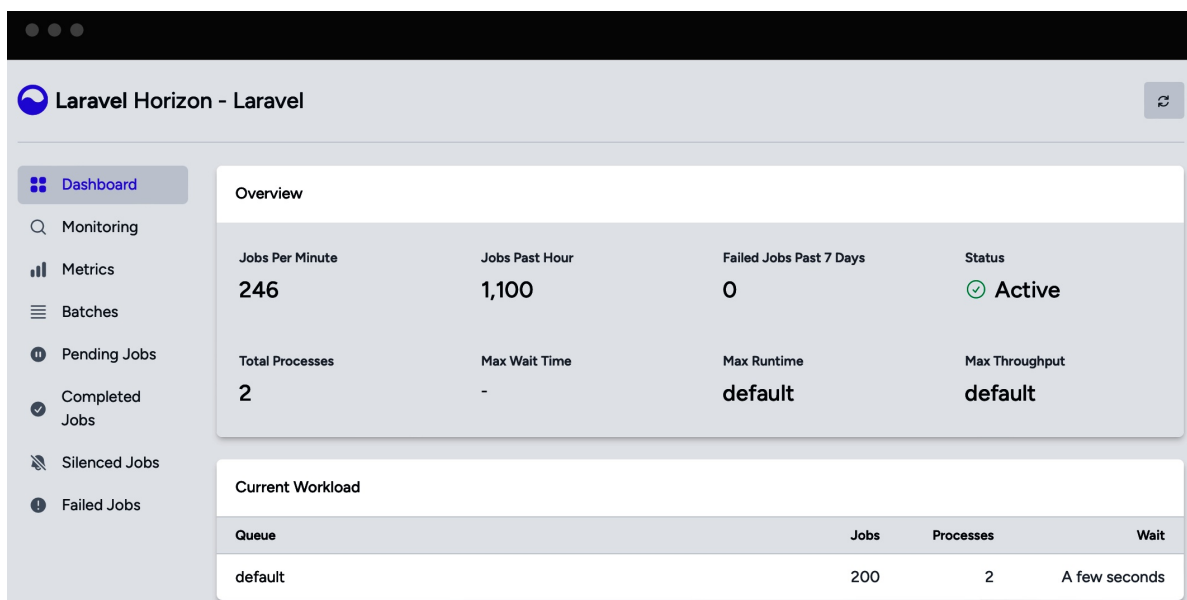


Figure 16-1. The Horizon dashboard

Events

With jobs, the calling code informs the application that it should *do something*: `CrunchReports` or `NotifyAdminOfNewSignup`.

With an event, the calling code instead informs the application that *something happened*: `UserSubscribed`, `UserSignedUp`, or `ContactWasAdded`. *Events* are notifications that something has taken place.

Some of these events may be “fired” by the framework itself. For example, Eloquent models fire events when they are saved, created, or deleted. But some events can also be manually triggered by the application’s code.

An event being fired doesn’t do anything on its own. However, you can bind *event listeners*, whose sole purpose is to listen for the broadcasting of specific events and to act in response. Any event can have anywhere from zero to many event listeners.

Laravel’s events are structured like the observer, or “pub/sub,” pattern. Many events are fired out into the application; some may never be listened for, and others may have a dozen listeners. The events don’t know or care.

Firing an Event

There are three ways to fire an event. You can use the `Event` facade, inject the `Dispatcher`, or use the `event()` global helper, as illustrated in [Example 16-12](#).

Example 16-12. Three ways to fire an event

```
Event::fire(new UserSubscribed($user, $plan));  
// or  
$dispatcher = app(Illuminate\Contracts\Events\Dispatcher::class);  
$dispatcher->fire(new UserSubscribed($user, $plan));  
// or  
event(new UserSubscribed($user, $plan));
```

If in doubt, I’d recommend using the global helper function.

To create an event to fire, use the `make:event` Artisan command:

```
php artisan make:event UserSubscribed
```

That’ll make a file that looks something like [Example 16-13](#).

Example 16-13. The default template for a Laravel event

```

<?php

namespace App\Events;

use Illuminate\Broadcasting\Channel;
use Illuminate\Broadcasting\InteractsWithSockets;
use Illuminate\Broadcasting\PresenceChannel;
use Illuminate\Broadcasting\PrivateChannel;
use Illuminate\Contracts\Broadcasting\ShouldBroadcast;
use Illuminate\Foundation\Events\Dispatchable;
use Illuminate\Queue\SerializesModels;

class UserSubscribed
{
    use Dispatchable, InteractsWithSockets, SerializesModels;

    /**
     * Create a new event instance.
     */
    public function __construct()
    {
        //
    }

    /**
     * Get the channels the event should broadcast on.
     *
     * @return array<int, \Illuminate\Broadcasting\Channel>
     */
    public function broadcastOn(): array
    {
        return [
            new PrivateChannel('channel-name'),
        ];
    }
}

```

Let's take a look at what we get here. `SerializesModels` works just like with jobs; it allows you to accept Eloquent models as parameters.

`InteractsWithSockets`, `ShouldBroadcast`, and the `broadcastOn()` method provide the backing functionality for broadcasting events using WebSockets, which we'll cover in a bit.

It might seem strange that there's no `handle()` or `fire()` method here. But remember, this object exists not to determine a particular action, but just to encapsulate some data. The first piece of data is its name; `UserSubscribed` tells

us that a particular event happened (a user subscribed). The rest of the data is any data we pass into the constructor and associate with this entity.

Example 16-14 shows what we might want to do with our `UserSubscribed` event.

Example 16-14. Injecting data into an event

```
...
class UserSubscribed
{
    use InteractsWithSockets, SerializesModels;

    public $user;
    public $plan;

    public function __construct($user, $plan)
    {
        $this->user = $user;
        $this->plan = $plan;
    }
}
```

Now we have an object that appropriately represents the event that happened: `$event->user` subscribed to the `$event->plan` plan. Remember, firing this event is as simple as `event(new UserSubscribed($user, $plan))`.

Listening for an Event

We have an event and the ability to fire it. Now let's look at how to listen for it.

First, we'll create an event listener. Let's say we want to email the app's owner every time a new user subscribes:

```
php artisan make:listener EmailOwnerAboutSubscription --event=UserSubscribed
```

That gives us the file in **Example 16-15**.

Example 16-15. The default template for a Laravel event listener

```
<?php

namespace App\Listeners;

use App\Events\UserSubscribed;
use Illuminate\Contracts\Queue\ShouldQueue;
```

```

use Illuminate\Queue\InteractsWithQueue;

class EmailOwnerAboutSubscription
{
    /**
     * Create the event listener.
     */
    public function __construct()
    {
        //
    }

    /**
     * Handle the event.
     */
    public function handle(UserSubscribed $event): void
    {
        //
    }
}

```

This is where the action happens—where the `handle()` method lives. This method expects to be passed an event of type `UserSubscribed` and act in response to it.

So, let's make it send an email ([Example 16-16](#)).

Example 16-16. A sample event listener

```

...
use App\Mail\UserSubscribed as UserSubscribedMessage;

class EmailOwnerAboutSubscription
{
    public function handle(UserSubscribed $event): void
    {
        Log::info('Emailed owner about new user: ' . $event->user->email);

        Mail::to(config('app.owner-email'))
            ->send(new UserSubscribedMessage($event->user, $event->plan));
    }
}

```

Now, one last task: we need to set this listener to listen to the `UserSubscribed` event. We'll do that in the `$listen` property of the `EventServiceProvider` class (see [Example 16-17](#)).

Example 16-17. Binding listeners to events in EventServiceProvider

```
class EventServiceProvider extends ServiceProvider
{
    protected $listen = [
        \App\Events\UserSubscribed::class => [
            \App\Listeners\EmailOwnerAboutSubscription::class,
        ],
    ];
}
```

As you can see, the key of each array entry is the class name of the event, and the value is an array of listener class names. We can add as many class names as we want under the `UserSubscribed` key, and they will all listen and respond to each `UserSubscribed` event.

Automatic event discovery

You can also instruct Laravel to automatically connect events and their matching listeners, without having to manually bind them in `EventServiceProvider`. This feature, called *automatic event discovery*, is disabled by default, but can be enabled by setting the `shouldDiscoverEvents()` method to return `true` in the `EventServiceProvider`:

```
/**
 * Determine if events and listeners should be automatically discovered.
 */
public function shouldDiscoverEvents(): bool
{
    return true;
}
```

If this feature is enabled, Laravel will map events to their matching listeners based on the typehints in the listeners. It'll have to match them on every request, which will introduce a small lag to your app, but like many slow features, you can cache these lookups using `php artisan event:cache` and clear the cache with `php artisan event:clear`.

Event subscribers

There's one more structure you can use to define the relationship between your events and their listeners. Laravel has a concept called an *event subscriber*, which is a class that contains a collection of methods that act as separate listeners to unique events, and also contains the mapping of which method

should handle which event. In this case it's easier to show than to tell, so take a look at [Example 16-18](#). Note that event subscribers are not a particularly commonly used tool.

Example 16-18. A sample event subscriber

```
<?php

namespace App\Listeners;

class UserEventSubscriber
{
    public function onUserSubscription($event)
    {
        // Handles the UserSubscribed event
    }

    public function onUserCancellation($event)
    {
        // Handles the UserCanceled event
    }

    public function subscribe($events)
    {
        $events->listen(
            \App\Events\UserSubscribed::class,
            'App\Listeners\UserEventSubscriber@onUserSubscription'
        );

        $events->listen(
            \App\Events\UserCanceled::class,
            'App\Listeners\UserEventSubscriber@onUserCancellation'
        );
    }
}
```

Subscribers need to define a `subscribe()` method, which is passed an instance of the event dispatcher. We'll use that to pair events with their listeners, but in this case, those are methods on this class, instead of entire classes.

As a refresher, any time you see an `@` inline like this, it means the class name is to the left of the `@` and the method name is to the right. So, in [Example 16-18](#), we're defining that the `onUserSubscription()` method of this subscriber will listen to any `UserSubscribed` events.

There's one last thing we need to do: in

App\Providers\EventServiceProvider, we need to add our subscriber’s class name to the \$subscribe property, as seen in [Example 16-19](#).

Example 16-19. Registering an event subscriber

```
...
class EventServiceProvider extends ServiceProvider
{
    ...
    protected $subscribe = [
        \App\Listeners\UserEventSubscriber::class
    ];
}
```

Broadcasting Events Over WebSockets, and Laravel Echo

WebSocket (often called WebSockets) is a protocol, popularized by Pusher (a hosted WebSocket SaaS), that makes it simple to provide near real-time communication between web devices. Rather than relying on information passing via HTTP requests, WebSockets libraries open a direct connection between the client and the server. WebSockets are behind tools like the chat boxes in Gmail and Facebook, where you don’t have to wait for the page to reload or for Ajax requests to receive or send data; instead, data is both sent and received in real time.

WebSockets work best with small pieces of data passed in a pub/sub structure—just like Laravel’s events. Laravel has a built-in set of tools that makes it easy to define that one or more of your events should be broadcast to a WebSocket server; it’s straightforward, for example, to have a MessageWasReceived event that is published to the notifications box of a certain user or set of users the instant a message arrives at your application.

LARAVEL ECHO

Laravel also has a more powerful tool designed for more complex event broadcasting. If you need presence notification or want to keep your rich frontend data model in sync with your Laravel app, check out Laravel Echo, which we’ll cover in [“Advanced Broadcasting Tools”](#). Much of what

comprises Echo is built into the Laravel core, but some of it requires pulling in the external JavaScript Echo library, which we cover in “[Laravel Echo \(the JavaScript Side\)](#)”.

Configuration and Setup

Take a look at `config/broadcasting.php` to find the configuration settings for your event broadcasting. Laravel supports three drivers for broadcasting: Pusher, a paid SaaS offering; Redis, for locally running WebSocket servers; and `log`, for local development and debugging.

QUEUE LISTENERS

In order for event broadcasting to move quickly, Laravel pushes the instruction to broadcast events onto a queue. That means you’ll need to have a queue worker running (or use the sync queue driver for local development). See “[Running a Queue Worker](#)” to learn how to run a queue worker.

Laravel suggests a default delay of three seconds before the queue worker looks for new jobs. However, with event broadcasting, you may notice some events take a second or two to broadcast. To speed this up, update your queue settings to wait only one second before looking for new jobs.

Broadcasting an Event

To broadcast an event, you need to mark that event as a broadcast event by having it implement the

`Illuminate\Contracts\Broadcasting\ShouldBroadcast` interface. This interface requires you to add the `broadcastOn()` method, which will return an array of either strings or `Channel` objects, each representing a WebSocket channel.

THE STRUCTURE OF WEBSOCKET EVENTS

Every event you send with WebSockets can have three primary characteristics: the name, the channel, and the data.

The *name* of an event might be something like `user-was-subscribed`, but

Laravel's default is to use the fully qualified class name of the event; that is something like `App\Events\UserSubscribed`. You can customize this by passing the name to the optional `broadcastAs()` method in your event class.

The *channel* is the way of describing which clients should receive this message. It's a very common pattern to have a channel for each user (e.g., `users.1`, `users.2`, etc.), and possibly a channel for all users (e.g., `users`), and maybe one for just users who are members of a certain account (`accounts.1`).

If the channel you're targeting is a private channel, preface the channel name with `private-`, and if it's a presence channel, preface the channel name with `presence-`. So, a private Pusher channel named `groups.5` should instead be named `private-groups.5`. If you use Laravel's `PrivateChannel` and `PresenceChannel` objects in your `broadcastOn()` method, they'll take care of adding those prefaces to your channel names for you.

If you're not familiar with public, private, and presence channels, see the note in [“The broadcast service provider”](#).

The *data* is a payload, usually JSON, of information relevant to the event—the message, maybe, or information about the user or plan that can be acted upon by the consuming JavaScript.

Example 16-20 shows our `UserSubscribed` event, modified to broadcast on two channels: one for the user (to confirm the user's subscription) and one for admins (to notify them of a new subscription).

Example 16-20. An event broadcasting on multiple channels

```
...
use Illuminate\Contracts\Broadcasting\ShouldBroadcast;

class UserSubscribed implements ShouldBroadcast
{
    use Dispatchable, InteractsWithSockets, SerializesModels;

    public $user;
    public $plan;

    public function __construct($user, $plan)
    {
```

```

        $this->user = $user;
        $this->plan = $plan;
    }

    public function broadcastOn(): array
    {
        // String syntax
        return [
            'users.' . $this->user->id,
            'admins'
        ];

        // Channel object syntax
        return [
            new Channel('users.' . $this->user->id),
            new Channel('admins'),
            // If it were a private channel: new PrivateChannel('admins'),
            // If it were a presence channel: new PresenceChannel('admins'),
        ];
    }
}

```

By default, any public properties of your event will be serialized as JSON and sent along as the data of your broadcast event. That means the data of one of our broadcast `UserSubscribed` events might look like [Example 16-21](#).

Example 16-21. Sample broadcast event data

```

{
    'user': {
        'id': 5,
        'name': 'Fred McFeely',
        ...
    },
    'plan': 'silver'
}

```

You can override this by returning an array of data from the `broadcastWith()` method on your event, as in [Example 16-22](#).

Example 16-22. Customizing the broadcast event data

```

public function broadcastWith()
{
    return [
        'userId' => $this->user->id,
        'plan' => $this->plan
    ];
}

```

You can customize which queue your event is pushed onto by setting the `$broadcastQueue` property on the event class:

```
public $broadcastQueue = 'websockets-for-faster-processing';
```

You may choose to do this so you can keep other queue items from slowing down your event broadcast; real-time WebSockets aren't much fun if a long-running job that's higher in the queue keeps the events from going out in time.

You can also force a given event to skip the queue entirely (using the “sync” queue driver, which is processed by the current PHP thread), by having it implement the `ShouldBroadcastNow` contract ([Example 16-23](#)).

Example 16-23. Forcing an event to skip the broadcast queue

```
use Illuminate\Contracts\Broadcasting\ShouldBroadcastNow;

class UserSubscribed implements ShouldBroadcastNow
{
    //
}
```

And, finally, you can choose to customize whether a given event should be broadcast at all by giving it a `broadcastWhen()` methods as in [Example 16-24](#):

Example 16-24. Conditionally determining whether an event should be broadcast

```
public function broadcastWhen()
{
    // Notify me only when users sign up from the White House
    return Str::contains($this->user->email, 'whitehouse.gov');
}
```

Receiving the Message

As of this book's publication, the most common solution Laravel developers use is [Pusher](#). Plans over a certain size cost money, but there's a generous free plan. Pusher makes it incredibly easy to set up a simple WebSocket server, and its JavaScript SDK handles all of the authentication and channel management with almost no work on your part. SDKs are available for iOS, Android, and many more platforms, languages, and frameworks.

If you'd like to host your own Pusher-compatible WebSockets server, there are two great options. First, you can try a Laravel-based tool called **Laravel WebSockets**. You can install the package into your current Laravel app (the same app you're broadcasting from) or into a separate microservice.

Second, if you're using Docker (including Sail) you can install **Soketi**, a free Pusher replacement developed in TypeScript.

If you choose to work with a server other than Pusher, you'll follow all of the directions in this book as if you were working with Pusher, but your configuration settings will be a bit different.

It's helpful to understand how to listen to Laravel's broadcast events without Echo even if you choose to use Echo in the end. But because much of the code here is not necessary if you use Echo, I'd recommend reading this section, and then reading "**Laravel Echo (the JavaScript Side)**" before you start implementing any of it; you can decide which way you prefer and then write your code from there.

To get started, pull in Pusher's library, get an API key from your Pusher account, and subscribe to any events on any channels with code like that in **Example 16-25**.

Example 16-25. Basic usage of Pusher

```
...
<script src="https://js.pusher.com/4.3/pusher.min.js"></script>
<script>
// Enable Pusher logging - don't include this in production
Pusher.logToConsole = true;

// Globally, perhaps; just a sample of how to get data in
var App = {
  'userId': {{ auth()->id() }},
  'pusherKey': '{{ config('broadcasting.connections.pusher.key') }}'
};

// Locally
var pusher = new Pusher(App.pusherKey, {
  cluster: '{{ config('broadcasting.connections.pusher.options.cluster') }}',
  encrypted: {{ config('broadcasting.connections.pusher.options.encrypted') }}
});

var pusherChannel = pusher.subscribe('users.' + App.userId);
```

```
pusherChannel.bind('App\\Events\\UserSubscribed', (data) => {  
    console.log(data.user, data.plan);  
});  
</script>
```

ESCAPING BACKSLASHES IN JAVASCRIPT

Since `\` is a control character in JavaScript, you need to write `\\` to represent a backslash in your strings, which is why there are two backslashes between each namespace segment in [Example 16-25](#).

To publish to Pusher from Laravel, get your Pusher key, secret, cluster, and app ID from your Pusher account dashboard, and then set them in your `.env` file under the keys `PUSHER_KEY`, `PUSHER_SECRET`, `PUSHER_APP_CLUSTER`, and `PUSHER_APP_ID`, respectively.

If you serve your app, visit a page with the JavaScript from [Example 16-25](#) embedded in it in one window, push a broadcast event in another window or from your terminal, have a queue listener running or are using the sync driver, and all of your authentication information is set up correctly, you should see event logs popping up in your JavaScript window's console in near real time.

With this power, it's now easy for you to keep your users up to date with what's happening with their data any time they're in your app. You can notify users of the actions of other users, of long-running processes that have just finished, or of your application's responses to external actions like incoming emails or webhooks. The possibilities are endless.

REQUIREMENTS

If you want to broadcast with Pusher or Redis, you'll need to bring in these dependencies:

- Pusher: `pusher/pusher-php-server` "~3.0"
- Redis: `redis/redis`

Advanced Broadcasting Tools

Laravel has a few more tools to make it possible to perform more complex interactions in event broadcasting. These tools, a combination of framework features and a JavaScript library, are called *Laravel Echo*.

These framework features work best when you use Laravel Echo in your JavaScript frontend (which we'll cover in “[Laravel Echo \(the JavaScript Side\)](#)”), but you can still enjoy some of the benefits of Echo without using the JavaScript components. Echo will work with both Pusher and Redis, but I'm going to use Pusher for any examples.

Excluding the current user from broadcast events

Every connection to Pusher is assigned a unique “socket ID” identifying that socket connection. And it's easy to define that any given socket (user) should be excluded from receiving a specified broadcast event.

This feature makes it possible to define that certain events should not be broadcast to the user who fired them. Let's say every user in a team gets notified when other users create a task; would you want to be notified of a task you just created? No, and that's why we have the `toOthers()` method.

To implement this, there are two steps to follow. First, you need to set up your JavaScript to send a certain POST to `/broadcasting/socket` when your WebSocket connection is initialized. This attaches your `socket_id` to your Laravel session. Echo does this for you, but you can also do it manually—take a look at the [Echo source](#) to see how it works.

Next, you'll want to update every request that your JavaScript makes to have an `X-Socket-ID` header that contains that `socket_id`. [Example 16-26](#) shows how to do that with Axios or in jQuery. Note that your event must use the `Illuminate\Broadcasting\InteractsWithSockets` trait in order to call the `toOthers()` method.

Example 16-26. Sending the socket ID along with each Ajax request with Axios or in jQuery

```
// Run this right after you initialize Echo
// With Axios
window.axios.defaults.headers.common['X-Socket-Id'] = Echo.socketId();

// With jQuery
```

```
$.ajaxSetup({
  headers: {
    'X-Socket-Id': Echo.socketId()
  }
});
```

Once you've handled this, you can exclude any event from being broadcast to the user who triggered it by using the `broadcast()` global helper, instead of the `event()` global helper, and then chaining `toOthers()` after it:

```
broadcast(new UserSubscribed($user, $plan))->toOthers();
```

The broadcast service provider

All of the other features that Echo provides require your JavaScript to authenticate with the server. Take a look at `App\Providers\BroadcastServiceProvider`, where you'll define how to authorize users' access to your private and presence channels.

The two primary actions you can take are to define the middleware that will be used on your broadcasting auth routes, and to define the authorization settings for your channels.

If you're going to use these features, you'll need to uncomment the `App\Providers\BroadcastServiceProvider::class` line in `config/app.php`.

And if you'll be using these features *without* Laravel Echo, you'll either need to manually handle sending CSRF tokens along with your authentication requests, or exclude `/broadcasting/auth` and `/broadcasting/socket` from CSRF protection by adding them to the `$except` property of the `VerifyCsrfToken` middleware.

Binding authorization definitions for WebSocket channels

Private and presence WebSocket channels need to be able to ping your application to learn whether the current user is authorized for that channel. You'll use the `Broadcast::channel()` method to define the rules for this authorization in your `routes/channels.php` file.

PUBLIC, PRIVATE, AND PRESENCE CHANNELS

There are three types of channels in WebSockets: public, private, and presence:

Public channels

Can be subscribed to by any user, authenticated or not.

Private channels

Require the end user's JavaScript to authenticate against the application to prove that the user is both authenticated and authorized to join this channel.

Presence channels

A type of private channel, but instead of allowing for message passing, they simply keep track of which users join and leave the channel, and make this information available to the application's frontend.

`Broadcast::channel()` takes two parameters: first, a string representing the channel(s) you want it to match, and second, a closure that defines how to authorize users for any channel matching that string. The closure will be passed an Eloquent model of the current user as its first parameter, and any matched *variableNameHere* segments as additional parameters. For example, a channel authorization definition with a string of `teams.teamId`, when matched against the channel `teams.5`, will pass its closure `$user` as the first parameter and `5` as the second parameter.

If you're defining the rules for a private channel, your `Broadcast::channel()` closure will need to return a Boolean: is this user authorized for this channel or not? If you're defining the rules for a presence channel, your closure should return an array of data you want available to the presence channel for any users that you want to show up in the channel. **Example 16-27** illustrates defining rules for both kinds of channel.

Example 16-27. Defining authorization rules for private and presence WebSocket channels

```
...  
// routes/channels.php  
  
// Define how to authenticate a private channel  
Broadcast::channel('teams.{teamId}', function ($user, $teamId) {  
    return (int) $user->team_id === (int) $teamId;  
});
```

```

});

// Define how to authenticate a presence channel; return any data
// you want the app to have about the user in the channel
Broadcast::channel('rooms.{roomId}', function ($user, $roomId) {
    if ($user->rooms->contains($roomId)) {
        return [
            'name' => $user->name
        ];
    }
});

```

You might be wondering how this information gets from your Laravel application to your JavaScript frontend. Pusher’s JavaScript library sends a POST to your application; by default it will hit `/pusher/auth`, but you can customize that (and Echo customizes it for you) to hit Laravel’s authentication route, `/broadcasting/auth`:

```

var pusher = new Pusher(App.pusherKey, {
    authEndpoint: '/broadcasting/auth'
});

```

Example 16-28 shows how we can tweak **Example 16-25** for private and presence channels, *without* Echo’s frontend components.

Example 16-28. Basic use of Pusher for private and presence channels

```

...
<script src="https://js.pusher.com/4.3/pusher.min.js"></script>
<script>
    // Enable Pusher logging - don't include this in production
    Pusher.logToConsole = true;

    // Globally, perhaps; just a sample of how to get data in
    var App = {
        'userId': {{ auth()->id() }},
        'pusherKey': '{{ config('broadcasting.connections.pusher.key') }}'
    };

    // Locally
    var pusher = new Pusher(App.pusherKey, {
        cluster: '{{ config('broadcasting.connections.pusher.options.cluster') }}',
        encrypted: {{ config('broadcasting.connections.pusher.options.encrypted') }},
        authEndpoint: '/broadcasting/auth'
    });

    // Private channel

```

```

var privateChannel = pusher.subscribe('private-teams.1');

privateChannel.bind('App\\Events\\UserSubscribed', (data) => {
    console.log(data.user, data.plan);
});

// Presence channel
var presenceChannel = pusher.subscribe('presence-rooms.5');

console.log(presenceChannel.members);
</script>

```

We now have the ability to send WebSocket messages to users depending on whether they pass a given channel's authorization rules. We can also keep track of which users are active in a particular group or section of the site and display relevant information to each user about other users in the same group.

Laravel Echo (the JavaScript Side)

Laravel Echo comprises two pieces: the advanced framework features we just covered and a JavaScript package that takes advantage of those features and drastically reduces the amount of boilerplate code you need to write powerful WebSocket-based frontends. The Echo JavaScript package makes it easy to handle authentication, authorization, and subscribing to private and presence channels. Echo can be used with the SDKs for either Pusher (for Pusher or a custom Pusher-compatible server) or `socket.io` (for Redis).

Bringing Echo into your project

To use Echo in your project's JavaScript, add it to *package.json* using `npm install --save` (be sure to bring in the appropriate Pusher or `socket.io` SDK as well):

```
npm install pusher-js laravel-echo --save
```

Let's assume you have a basic Vite file compiling your *app.js*, like in Laravel's default installation setup.

Laravel's default *resources/js/app.js* structure has a great example of how best to initialize your Echo install. Take a look at [Example 16-29](#) to see how that works between that file and *resources/js/bootstrap.js*.

Example 16-29. Initializing Echo in *app.js* and *bootstrap.js*

```
// app.js
require('./bootstrap');

// ... lots of Vue stuff ...

// Add your Echo bindings here
// bootstrap.js
import Echo from "laravel-echo";

window.Echo = new Echo({
  broadcaster: 'pusher',
  key: process.env.MIX_PUSHER_APP_KEY,
  cluster: process.env.MIX_PUSHER_APP_CLUSTER
});
```

For CSRF protection, you'll also need to add a csrf-token `<meta>` tag to your HTML template:

```
<meta name="csrf-token" content="{{ csrf_token() }}">
```

And, of course, remember to link to your compiled *app.js* in your HTML template:

```
<script src="{{ asset('js/app.js') }}"></script>
```

Now we're ready to get started.

CHANGES TO THE CONFIGURATION WHEN USING THE LARAVEL WEBSOCKETS SERVER PACKAGE

If you're working with a Laravel WebSockets server (using the package discussed earlier in ["Receiving the Message"](#)), the configuration details in [Example 16-29](#) will be a little bit different. See the [Laravel WebSockets package docs](#) for more info.

Using Echo for basic event broadcasting

This is nothing different from what we've already used Pusher for, but [Example 16-30](#) is a simple code sample to show how to use Echo to listen to public channels for basic event information.

Example 16-30. Listening to a public channel with Echo

```
var currentTeamId = 5; // Likely set elsewhere

Echo.channel(`teams.${currentTeamId}`)
  .listen('UserSubscribed', (data) => {
    console.log(data);
  });
```

Echo provides a few methods for subscribing to various types of channels; `channel()` will subscribe you to a public channel. Note that when you listen to an event with Echo, you can ignore the full event namespace and just listen for the unique class name of this event.

We now have access to the public data that's passed along with our event, represented in the `data` object. We can also chain `listen()` handlers, as in [Example 16-31](#).

Example 16-31. Chaining event listeners in Echo

```
Echo.channel(`teams.${currentTeamId}`)
  .listen('UserSubscribed', (data) => {
    console.log(data);
  })
  .listen('UserCanceled', (data) => {
    console.log(data);
  });
```

REMEMBER TO COMPILE AND INCLUDE!

Did you try these code samples and not see anything change in your browser? Make sure to run `npm run dev` (if you're running it locally) or `npm run build` (to build it once) to compile your code. And, if you haven't yet, be sure to actually include *app.js* in your template somewhere.

Private channels and basic authentication

Echo also has a method for subscribing to private channels: `private()`. It works the same as `channel()`, but requires you to have set up channel authorization definitions in *routes/channel.php*, like we covered earlier. Additionally, unlike with the SDKs, you don't need to put `private-` in front of your channel name.

[Example 16-32](#) shows what it looks like to listen to a private channel named

private-teams.5.

Example 16-32. Listening to a private channel with Echo

```
var currentTeamId = 5; // Likely set elsewhere

Echo.private(`teams.${currentTeamId}`)
  .listen('UserSubscribed', (data) => {
    console.log(data);
  });
```

Presence channels

Echo makes it much simpler to join and listen to events in presence channels. This time you'll want to use the `join()` method to bind to the channel, as in [Example 16-33](#).

Example 16-33. Joining a presence channel

```
var currentTeamId = 5; // Likely set elsewhere

Echo.join(`teams.${currentTeamId}`)
  .here((members) => {
    console.log(members);
  });
```

`join()` subscribes to the presence channel, and `here()` allows you to define the behavior when the user joins and also when any other users join or leave the presence channel.

You can think of a presence channel like a “who’s online” sidebar in a chat room. When you first join a presence channel, your `here()` callback will be called and provided a list of all the members at that time. And any time any members join or leave, that callback will be called again with the updated list. There’s no messaging happening here, but you can play sounds, update the on-page list of members, or do whatever else you want in response to these actions.

There are also specific methods for individual events, which you can use individually or chained (see [Example 16-34](#)).

Example 16-34. Listening for specific presence events

```
var currentTeamId = 5; // Likely set elsewhere

Echo.join('teams.' + currentTeamId)
  .here((members) => {
    // Runs when you join
```



```

        console.table(members);
    })
    .joining((joiningMember, members) => {
        // Runs when another member joins
        console.table(joiningMember);
    })
    .leaving((leavingMember, members) => {
        // Runs when another member leaves
        console.table(leavingMember);
    });

```

Excluding the current user

We covered this previously in the chapter, but if you want to exclude the current user, you can use the `broadcast()` global helper instead of the `event()` global helper and then chain the `toOthers()` method after your broadcast call. But with Echo, the JavaScript side of this is already handled for you. It'll just work.

As you can see, the Echo JavaScript library doesn't do anything you couldn't do on your own—but it makes a lot of common tasks much simpler and provides a cleaner, more expressive syntax for common WebSocket tasks.

Subscribing to notifications with Echo

Laravel's notifications come with a broadcast driver out of the box that pushes notifications out as broadcast events. You can subscribe to these notifications with Echo using `Echo.notification()`, as in [Example 16-35](#).

Example 16-35. Subscribing to a notification with Echo

```

Echo.private(`App.User.${userId}`)
    .notification((notification) => {
        console.log(notification.type);
    });

```

Client events

If you'd like to send quick, performant messages between your users without the messages even hitting your Laravel application—for example, to send “typing...” notifications—you can use Echo's `whisper()` method, as shown in [Example 16-36](#).

Example 16-36. Bypassing the Laravel server using Echo's whisper() method

```

Echo.private('room')

```

```
.whisper('typing', {  
    name: this.user.name  
});
```

And then use `listenForWhisper()` to listen, as in [Example 16-37](#).

Example 16-37. Listening for whisper events with Echo

```
Echo.private('room')  
    .listenForWhisper('typing', (e) => {  
        console.log(e.name);  
    });
```

Scheduler

If you've ever written a cron job before, you likely already wish for a better tool. Not only is the syntax onerous and frustratingly difficult to remember, but it's one significant aspect of your application that can't be stored in version control.

Laravel's scheduler makes handling scheduled tasks simple. You'll write your scheduled tasks in code, and then point one cron job at your app: once per minute, run `php artisan schedule:run`. Every time this Artisan command is run, Laravel checks your schedule definitions to find out if any scheduled tasks should run.

Here's the cron job to define that command:

```
* * * * * cd /home/myapp.com && php artisan schedule:run >> /dev/null 2>&1
```

There are many task types you can schedule and many time frames you can use to schedule them.

`app/Console/Kernel.php` has a method named `schedule()`, which is where you'll define any tasks you'd like to schedule.

Available Task Types

First, let's take a look at the simplest option: a closure, run every minute ([Example 16-38](#)). Every time the cron job hits the `schedule:run` command, it will call this closure.

Example 16-38. Scheduling a closure to run once every minute

```
// app/Console/Kernel.php
public function schedule(Schedule $schedule): void
{
    $schedule->call(function () {
        CalculateTotals::dispatch();
    })->everyMinute();
}
```

There are two other types of tasks you can schedule: Artisan and shell commands.

You can schedule Artisan commands by passing their syntax exactly as you would call them from the command line:

```
$schedule->command('scores:tally --reset-cache')->everyMinute();
```

And you can run any shell commands that you can run with PHP's `exec()` method:

```
$schedule->exec('/home/myapp.com/bin/build.sh')->everyMinute();
```

Available Time Frames

The beauty of the scheduler isn't just that you can define your tasks in code; it's that you can schedule them in code, too. Laravel keeps track of time passing and evaluates whether it's time for any given task to run. That's easy with `everyMinute()` because the answer is always simple: run the task. But Laravel keeps the rest simple for you, too, even for the most complex of requests.

Let's take a look at your options by starting with a monstrous definition that's simple in Laravel:

```
$schedule->call(function () {
    // Runs once a week on Sunday at 23:50
})->weekly()->sundays()->at('23:50');
```

Notice that we can chain times together: we can define frequency and specify the day of the week and the time, and of course we can do much more.

Table 16-1 shows a list of potential date/time modifiers for use when scheduling a job.

Table 16-1. Date/time modifiers for use with the scheduler

Command	Description
<code>->timezone('America/Detroit')</code>	Set the time zone for schedules
<code>->cron('* * * * *')</code>	Define the schedule using the traditional cron notation
<code>->everyMinute()</code>	Run every minute
<code>->everyTwoMinutes()</code>	Run every 2 minutes
<code>->everyThreeMinutes()</code>	Run every 3 minutes
<code>->everyFourMinutes()</code>	Run every 4 minutes
<code>->everyFiveMinutes()</code>	Run every 5 minutes
<code>->everyTenMinutes()</code>	Run every 10 minutes
<code>->everyFifteenMinutes())</code>	Run every 15 minutes
<code>->everyThirtyMinutes()</code>	Run every 30 minutes
<code>->hourly()</code>	Run every hour
<code>->hourlyAt(14)</code>	Run every hour at 14 minutes past
<code>->everyTwoHours()</code>	Run every 2 hours
<code>->everyThreeHours()</code>	Run every 3 hours

<code>->everyFourHours()</code>	Run every 4 hours
<code>->everySixHours()</code>	Run every 6 hours
<code>->daily()</code>	Run every day at midnight
<code>->dailyAt('14:00')</code>	Run every day at 14:00
<code>->twiceDaily(1, 14)</code>	Run every day at 1:00 and 14:00
<code>->twiceDailyAt(1, 14, 6)</code>	Run every day at 1:06 and 14:06 (the third argument is the minutes to start)
<code>->weekly()</code>	Run every week (midnight on Sunday)
<code>->weeklyOn(5, '10:00')</code>	Run every week on Friday at 10:00
<code>->monthly()</code>	Run every month (midnight on the 1st)
<code>->monthlyOn(15, '23:00')</code>	Run every month on the 15th at 23:00
<code>->quarterly()</code>	Run every quarter (midnight on the 1st of January, April, July, and October)
<code>->yearly()</code>	Run every year (midnight on the 1st of January)
<code>->yearlyOn(6)</code>	Run every year (midnight on the 1st of June)
<code>->when(closure)</code>	Limit the task to when the closure returns <code>true</code>
<code>->skip(closure)</code>	Limit the task to when the closure returns <code>false</code>

<code>->between('8:00', '12:00')</code>	Limit the task to between the given times
<code>->unlessBetween('8:00', '12:00')</code>	Limit the task to any time except between the given times
<code>->weekdays()</code>	Limit to weekdays
<code>->sundays()</code>	Limit to Sundays
<code>->mondays()</code>	Limit to Mondays
<code>->tuesdays()</code>	Limit to Tuesdays
<code>->wednesdays()</code>	Limit to Wednesdays
<code>->thursdays()</code>	Limit to Thursdays
<code>->fridays()</code>	Limit to Fridays
<code>->saturdays()</code>	Limit to Saturdays
<code>->days([1,2])</code>	Limit to Sundays and Mondays
<code>->environments(<i>staging</i>)</code>	Limit to only the staging environment

Most of these can be chained one after another, but of course, any combinations that don't make sense chained can't be chained.

Example 16-39 shows a few combinations you could consider.

Example 16-39. Some sample scheduled events

```
// Both run weekly on Sunday at 23:50
$schedule->command('do:thing')->weeklyOn(0, '23:50');
$schedule->command('do:thing')->weekly()->sundays()->at('23:50');
```

```
// Run once per hour, weekdays, 8am-5pm
$schedule->command('do:thing')->weekdays()->hourly()->when(function () {
    return date('H') >= 8 && date('H') <= 17;
});

// Run once per hour, weekdays, 8am-5pm using the "between" method
$schedule->command('do:thing')->weekdays()->hourly()->between('8:00', '17:00');

// Run every 30 minutes except when directed not to by the SkipDetector
$schedule->command('do:thing')->everyThirtyMinutes()->skip(function () {
    return app('SkipDetector')->shouldSkip();
});
```

Defining Time Zones for Scheduled Commands

You can define the time zone on a specific scheduled command using the `timezone()` method:

```
$schedule->command('do:it')->weeklyOn(0, '23:50')->timezone('America/Chicago');
```

You can also set a default time zone (separate from the application time zone) that all of your scheduled times will be defined in, by defining the `scheduleTimezone()` method in `App\Console\Kernel`:

```
protected function scheduleTimezone()
{
    return 'America/Chicago';
}
```

Blocking and Overlap

If you want to avoid having your tasks overlap each other—for example, if you have a task running every minute that may sometimes take longer than a minute to run—end the schedule chain with the `withoutOverlapping()` method. This method skips a task if the previous instance of that task is still running:

```
$schedule->command('do:thing')->everyMinute()->withoutOverlapping();
```

Handling Task Output

Sometimes the output from your scheduled task is important, whether for

logging, notifications, or just ensuring that the task ran.

If you want to write the returned output of a task to a file (potentially overwriting what is already in the file), use `sendOutputTo()`:

```
$schedule->command('do:thing')->daily()->sendOutputTo($filePath);
```

If you want to append it to a file instead, use `appendOutputTo()`:

```
$schedule->command('do:thing')->daily()->appendOutputTo($filePath);
```

And if you want to email the output to a designated recipient, write it to a file first and then add `emailOutputTo()`:

```
$schedule->command('do:thing')
->daily()
->sendOutputTo($filePath)
->emailOutputTo('me@myapp.com');
```

Make sure that your email settings are configured correctly in Laravel's basic email configuration.

CLOSURE SCHEDULED EVENTS CAN'T SEND OUTPUT

The `sendOutputTo()`, `appendOutputTo()`, and `emailOutputTo()` methods only work for `command()`-scheduled tasks. You can't use them for closures, unfortunately.

You may also want to send some output to a webhook to verify that your tasks ran correctly. There are a few services that provide this sort of uptime monitoring, most significantly **Laravel Envoyer**, a zero-downtime deployment service that also provides cron uptime monitoring, and **Dead Man's Snitch**, a tool designed purely for monitoring cron job uptime.

These services don't expect something to be emailed to them, but rather expect an HTTP "ping," so Laravel makes that easy with `pingBefore()` and `thenPing()`:

```
$schedule->command('do:thing')
```



```
->daily()  
->pingBefore($beforeUrl)  
->thenPing($afterUrl);
```

If you want to use the ping features, you'll need to pull in Guzzle using Composer:

```
composer require guzzlehttp/guzzle
```

Task Hooks

Speaking of running something *before* and *after* your task, there are hooks for that, with `before()` and `after()`:

```
$schedule->command('do_thing')  
->daily()  
->before(function () {  
    // Prepare  
})  
->after(function () {  
    // Cleanup  
});
```

Running the Scheduler in Local Development

Since the scheduler depends on cron, it's simpler to set up on a server than on your local machine. If you'd like to have the scheduler running locally, run the `schedule:work` Artisan command, which will invoke the scheduler every minute, just like a cron job:

```
php artisan schedule:work
```

Testing

Testing queued jobs (or anything else in the queue) is easy. In `phpunit.xml`, which is the configuration file for your tests, the `QUEUE_DRIVER` environment variable is set to `sync` by default. That means your tests will run your jobs or other queued tasks synchronously, directly in your code, without relying on a queue system of any sort. You can test them just like any other code.

However, you can assert against the specific job itself, as in [Example 16-40](#).

Example 16-40. Using a closure to verify that a dispatched job meets given criteria

```
use Illuminate\Support\Facades\Bus;
...
public function test_changing_subscriptions_triggers_crunch_job()
{
    // ...

    Bus::fake();

    Bus::assertDispatched(CrunchReports::class, function ($job) {
        return $job->subscriptions->contains(5);
    });

    // Also can use assertNotDispatched()
}
```

There are also the `assertPushedWithChain()` and `assertPushedWithoutChain()` methods.

```
Bus::fake();

Bus::assertPushedWithChain(
    CrunchReports::class,
    [ChainedJob::class],
    function ($job) {
        return $job->subscriptions->contains(5);
    }
);

// Also can use assertPushedWithoutChain()
Bus::assertPushedWithChain(CrunchReports::class, function ($job) {
    return $job->subscriptions->contains(5);
});
```

To test that an event fired, you have two options. First, you can just test that the behavior you expected happened, without concerning yourself with the event itself.

Second, you can run a test against the event that was fired, as in [Example 16-41](#).

Example 16-41. Using a closure to verify that a fired event meets given criteria

```
use Illuminate\Support\Facades\Event;
...
```

```

public function test_usersubscribed_event_fires()
{
    Event::fake();

    // ...

    Event::assertDispatched(UserSubscribed::class, function ($e) {
        return $e->user->email = 'user-who-subscribed@mail.com';
    });

    // Also can use assertNotDispatched()
}

```

Another common scenario is that you’re testing code that incidentally fires events, and you want to disable the event listeners during that test. You can disable the event system with the `withoutEvents()` method, as in [Example 16-42](#).

Example 16-42. Disabling event listeners during a test

```

public function test_something_subscription_related()
{
    $this->withoutEvents();

    // ...
}

```

TL;DR

Queues allow you to separate chunks of your application’s code from the synchronous flow of user interactions out to a list of commands to be processed by a “queue worker.” This allows your users to resume interactions with your application while slower processes are handled asynchronously in the background.

Jobs are classes that are structured with the intention of encapsulating a chunk of application behavior so that it can be pushed onto a queue.

Laravel’s event system follows the pub/sub or observer pattern, allowing you to send out notifications of an event from one part of your application, and elsewhere bind listeners to those notifications to define what behavior should happen in response to them. Using WebSockets, events can also be broadcast to frontend clients.

Laravel's scheduler simplifies scheduling tasks. Point an every-minute cron job to `php artisan schedule:run` and then schedule your tasks with even the most complex of time requirements using the scheduler, and Laravel will handle all the timings for you.

Chapter 17. Helpers and Collections

We’ve already covered many global functions throughout the book: these are little helpers that make it easier to perform common tasks, like `dispatch()` for jobs, `event()` for events, and `app()` for dependency resolution. We also talked a bit about Laravel’s collections, or arrays on steroids, in [Chapter 5](#).

In this chapter we’ll cover some of the more common and powerful helpers and some of the basics of programming with collections. Many of the “helpers” in this section that once were global functions are now calls on facades; `array_first()`, the global function, has been replaced by `Arr::first()`, the auth call. So while these aren’t all *technically* helpers, as they’re not all global functions anymore, they still have the same position in our toolbox.

Helpers

You can find a full list of the helpers Laravel offers in the [helpers docs](#), but we’re going to cover a few of the most useful functions here.

Arrays

PHP’s native array manipulation functions give us a lot of power, but sometimes there are standard manipulations we want to make that require unwieldy loops and logic checks. Laravel’s array helpers make a few common array manipulations much simpler:

```
Arr::first($array, $callback, $default = null)
```

Returns the first array value that passes a test, defined in a callback closure. You can optionally set the default value as the third parameter. Here’s an example:

```

$people = [
    [
        'email' => 'm@me.com',
        'name' => 'Malcolm Me'
    ],
    [
        'email' => 'j@jo.com',
        'name' => 'James Jo'
    ],
];

$value = Arr::first($people, function ($person, $key) {
    return $person['email'] == 'j@jo.com';
});

```

Arr::get(\$array, \$key, \$default = null)

Makes it easy to get values out of an array, with two added benefits: it won't throw an error if you ask for a key that doesn't exist (and you can provide defaults with the third parameter), and you can use dot notation to traverse nested arrays. For example:

```

$array = ['owner' => ['address' => ['line1' => '123 Main St.']]];

$line1 = Arr::get($array, 'owner.address.line1', 'No address');
$line2 = Arr::get($array, 'owner.address.line2');

```

Arr::has(\$array, \$keys)

Makes it easy to check whether an array has a particular value set using dot

notation for traversing nested arrays. The `$keys` parameter can be a single entry or an array of entries, which will check whether every entry in the array exists:

```
$array = ['owner' => ['address' => ['line1' => '123 Main St.']]];

if (Arr::has($array, 'owner.address.line2')) {
    // Do stuff
}
```

`Arr::hasAny($array, $keys)`

Makes it easy to check whether an array has any of the specified keys using dot notation for traversing nested arrays. The `$keys` parameter can be a single key or an array of keys, which will check whether any of the keys exists in the array:

```
$array = ['owner' => ['address' => ['line1' => '123 Main St.']]];

if (Arr::hasAny($array, ['owner.address', 'default.address'])) {
    // Do stuff
}
```

`Arr::pluck($array, $value, $key = null)`

Returns an array of the values corresponding to the provided key:

```
$array = [
    ['owner' => ['id' => 4, 'name' => 'Tricia']],
    ['owner' => ['id' => 7, 'name' => 'Kimberly']],
];
```

```
$array = Arr::pluck($array, 'owner.name');
```

```
// Returns ['Tricia', 'Kimberly'];
```

If you want the returned array to be keyed by another value from the source array, you can pass that value's dot-notated reference as the third parameter:

```
$array = Arr::pluck($array, 'owner.name', 'owner.id');
```

```
// Returns [4 => 'Tricia', 7 => 'Kimberly'];
```

Arr::random(\$array, \$num = null)

Returns a random item from the provided array. If you provide a \$num parameter, it will pull an array of that many results, randomly selected:

```
$array = [  
    ['owner' => ['id' => 4, 'name' => 'Tricia']],  
    ['owner' => ['id' => 7, 'name' => 'Kimberly']],  
];
```

```
$randomOwner = Arr::random($array);
```

Arr::join(\$array, \$glue, \$finalGlue = '')

Joins the items from \$array into a string, adding \$glue between them. If \$finalGlue is provided, it is added before the last element of the array, instead of \$glue:

```
$array = ['Malcolm', 'James', 'Tricia', 'Kimberly'];
```



```
Arr::join($array, ', ');  
  
// Malcolm, James, Tricia, Kimberly  
  
Arr::join($array, ', ', ', ', and');  
  
// Malcolm, James, Tricia, and Kimberly
```

Strings

Just like with arrays, there are some string manipulations and checks that are possible with native PHP functions, but that can be cumbersome. Laravel's helpers make a few common string operations faster and simpler:

`e($string)`

An alias to `htmlentities()`; prepares a (often user-provided) string for safe echoing on an HTML page. For example:

```
e('<script>do something nefarious</script>');  
  
// Returns &lt;script&gt;do something nefarious&lt;/script&gt;
```

`str($string)`

Used for casting stringables; is an alias for `Str::of($string)`:

```
str('http') === Str::of('http');  
  
// true
```

`Str::startsWith($haystack, $needle)`, `Str::endsWith($haystack, $needle)`, `Str::contains($haystack, $needle, $ignoreCase)`

Returns a Boolean indicating whether the provided `$haystack` string starts

with, ends with, or contains the provided `$needle` string:

```
if (Str::startsWith($url, 'https')) {  
    // Do something  
}  
  
if (Str::endsWith($abstract, '...')) {  
    // Do something  
}  
  
if (Str::contains($description, '1337 h4x0r')) {  
    // Run away  
}
```

`Str::limit($value, $limit = 100, $end = '...')`

Limits a string to the provided number of characters. If the string's length is less than the limit, just returns the string; if it's greater, trims to the number of characters provided and then appends either `...` or the provided `$end` string. For example:

```
$abstract = Str::limit($loremIpsum, 30);  
// Returns "Lorem ipsum dolor sit amet, co..."  
  
$abstract = Str::limit($loremIpsum, 30, "&hellip;");  
// Returns "Lorem ipsum dolor sit amet, co&hellip;"
```

`Str::words($value, $words = 100, $end = '...')`

Limits a string to the provided number of words. If the string's length is less than the number of words, just returns the string; if it's greater, trims to the

number of words provided and then appends either ... or the provided \$end string. For example:

```
$abstract = Str::words($loremIpsum, 3);  
// Returns "Lorem ipsum dolor..."  
  
$abstract = Str::words($loremIpsum, 5, " &hellip;");  
// Returns "Lorem ipsum dolor sit amet, &hellip;"
```

`Str::before($subject, $search)`, `Str::after($subject, $search)`,
`Str::beforeLast($subject, $search)`, `Str::afterLast($subject, $search)`

Returns the subsections of a string before or after another string, or the last instance of another string. For example:

```
Str::before('Nice to meet you!', 'meet you');  
// Returns "Nice to "  
  
Str::after('Nice to meet you!', 'Nice');  
// Returns " to meet you!"  
  
Str::beforeLast('App\Notifications>WelcomeNotification', '\\');  
// Returns "App\Notifications"  
  
Str::afterLast('App\Notifications>WelcomeNotification', '\\');  
// Returns "WelcomeNotification"
```

`Str::is($pattern, $value)`

Returns a Boolean indicating whether or not a given string matches a given

pattern. The pattern can be a regex pattern, or you can use asterisks to indicate wildcard positions:

```
Str::is('*.dev', 'myapp.dev');      // true
Str::is('*.dev', 'myapp.dev.co.uk'); // false
Str::is('*dev*', 'myapp.dev');      // true
Str::is('*myapp*', 'www.myapp.dev'); // true
Str::is('my*app', 'myfantasticapp'); // true
Str::is('my*app', 'myapp');         // true
```

HOW TO PASS A REGEX TO STR::IS()

If you're curious about what regex patterns are acceptable to pass to `Str::is()`, check out the method definition here (shortened for space) to see how it works:

```
public function is($pattern, $value)
{
    if ($pattern == $value) return true;

    $pattern = preg_quote($pattern, '#');
    $pattern = Str::replace('\*', '.*', $pattern);
    if (preg_match('#^'.$pattern.'\z#u', $value) === 1) {
        return true;
    }

    return false;
}
```

Str::isUuid(\$value)

Determines whether the value is a valid UUID:

```
Str::isUuid('33f6115c-1c98-49f3-9158-a4a4376dfbe1'); // Returns true
Str::isUuid('laravel-up-and-running'); // Returns false
```

`Str::random($length = n)`

Returns a random string of alphanumeric mixed-case characters of the length specified:

```
$hash = Str::random(64);  
  
// Sample: J40uNWAvY60wE4BPEWxu7BZFQEmxEHmGiLmQncj0ThMGJK705Kfgptyb9uL wspmh
```

`Str::slug($title, $separator = '-', $language = 'en')`

Returns a URL-friendly slug from a string—often used for creating a URL segment for a name or title:

```
Str::slug('How to Win Friends and Influence People');  
  
// Returns 'how-to-win-friends-and-influence-people'
```

`Str::plural($value, $count = n)`

Converts a string to its plural form. This function currently only supports the English language:

```
Str::plural('book');  
  
// Returns books  
  
Str::plural('person');  
  
// Returns people  
  
Str::plural('person', 1);  
  
// Returns person
```

`__( $key, $replace = [], $locale = null)`

Translates the given translation string or translation key using your

localization files:

```
echo __('Welcome to your dashboard');
```

```
echo __('messages.welcome');
```

FLUENT STRING OPERATIONS

While the Str helpers are hugely powerful tools, one downside is that you often find yourself in the middle of four nested calls (Str::trim(Str::replace)), etc.).

You can now also use the Str helper methods fluently by chaining the calls on the response of the Str::of method. Take a look:

```
return (string) Str::of(' Go to town!! ')
    ->trim()
    ->replace('town', 'bed')
    ->slug(); // Returns "go-to-bed"
```

Application Paths

When you're dealing with the filesystem, it can often be tedious to make links to certain directories for getting and saving files. These helpers give you quick access to find the fully qualified paths to some of the most important directories in your app.

Note that each of these can be called with no parameters, but if a parameter is passed, it will be appended to the end of the normal directory string and returned as a whole:

```
app_path($append = '')
```

Returns the path for the *app* directory:

```
app_path();
```

```
// Returns /home/forge/myapp.com/app
```

`base_path($path = '')`

Returns the path for the root directory of your app:

```
base_path();
```

```
// Returns /home/forge/myapp.com
```

`config_path($path = '')`

Returns the path for configuration files in your app:

```
config_path();
```

```
// Returns /home/forge/myapp.com/config
```

`database_path($path = '')`

Returns the path for database files in your app:

```
database_path();
```

```
// Returns /home/forge/myapp.com/database
```

`storage_path($path = '')`

Returns the path for the *storage* directory in your app:

```
storage_path();
```

```
// Returns /home/forge/myapp.com/storage
```

`lang_path($path = '')`

Returns the path for the *lang* directory in your app:

```
lang_path();  
  
// Returns /home/forged/myapp.com/resources/lang
```

URLs

Some frontend file paths are consistent but at times annoying to type—for example, paths to assets—and it's helpful to have convenient shortcuts to them, which we'll cover here. But some can actually vary as route definitions move, so some of these helpers are vital in making sure all of your links and assets work correctly:

`action($action, $parameters = [], $absolute = true)`

Assuming a controller method has a single URL mapped to it, returns the correct URL given a controller and method name pair (separated by @) or using tuple notation:

```
<a href="{{ action('PersonController@index') }}">See all People</a>  
  
// Or, using tuple notation:  
  
<a href="{{ action(  
    [App\Http\Controllers\PersonController::class, 'index']  
    ) }}">  
    See all People  
</a>  
  
// Returns <a href="http://myapp.com/people">See all People</a>
```

If the controller method requires parameters, you can pass them in as the second parameter (as an array, if there's more than one required parameter). You can key them if you want for clarity, but what matters is just that they're in the right order:

```
<a href="{{ action(  
    'PersonController@create',  
    ['name' => 'John', 'email' => 'john@example.com']  
    ) }}">
```



```

        'PersonController@show',

        ['id => 3]

    ) }}">See Person #3</a>

// or

<a href="{ action(

    'PersonController@show',

    [3]

) }}">See Person #3</a>

// Returns <a href="http://myapp.com/people/3">See Person #3</a>

```

If you pass `false` to the third parameter, your links will generate as relative (`/people/3`) instead of absolute (`http://myapp.com/people/3`).

`route($name, $parameters = [], $absolute = true)`

If a route has a name, returns the URL for that route:

```

// routes/web.php

Route::get('people', [PersonController::class, 'index'])
    ->name('people.index');

// A view somewhere

<a href="{ route('people.index') }}">See all People</a>

// Returns <a href="http://myapp.com/people">See all People</a>

```

If the route definition requires parameters, you can pass them in as the second parameter (as an array if more than one parameter is required). Again, you can key them if you want for clarity, but what matters is just that

they're in the right order:

```
<a href="{ route('people.show', ['id' => 3]) }" >See Person #3</a>

// or

<a href="{ route('people.show', [3]) }" >See Person #3</a>

// Returns <a href="http://myapp.com/people/3" >See Person #3</a>
```

If you pass `false` to the third parameter, your links will generate as relative instead of absolute.

`url($string)` and `secure_url($string)`

Given any path string, converts to a fully qualified URL. (`secure_url()` is the same as `url()` but forces HTTPS):

```
url('people/3');

// Returns http://myapp.com/people/3
```

If no parameters are passed, this instead gives an instance of `Illuminate\Routing\UrlGenerator`, which makes method chaining possible:

```
url()->current();

// Returns http://myapp.com/abc

url()->full();

// Returns http://myapp.com/abc?order=reverse

url()->previous();
```

```
// Returns http://myapp.com/login
```

```
// And many more methods available on the UrlGenerator...
```

Miscellaneous

There are a few other global helpers that I'd recommend getting familiar with. Of course, you should check out the [whole list](#), but the ones mentioned here are definitely worth taking a look at:

`abort($code, $message, $headers)`, `abort_unless($boolean, $code, $message, $headers)`, `abort_if($boolean, $code, $message, $headers)`

Throws HTTP exceptions. `abort()` throws the exception defined, `abort_unless()` throws it if the first parameter is false, and `abort_if()` throws it if the first parameter is true:

```
public function controllerMethod(Request $request)
{
    abort(403, 'You shall not pass');

    abort_unless(request()->filled('magicToken'), 403);

    abort_if(request()->user()->isBanned, 403);
}
```

auth()

Returns an instance of the Laravel authenticator. Like the Auth facade, you can use this to get the current user, to check for login state, and more:

```
$user = auth()->user();

$userId = auth()->id();
```

```
if (auth()->check()) {
```

```
    // Do something
}
```

back()

Generates a “redirect back” response, sending the user to the previous location:

```
Route::get('post', function () {
    // ...

    if ($condition) {
        return back();
    }
});
```

collect(\$array)

Takes an array and returns the same data, converted to a collection:

```
$collection = collect(['Rachel', 'Hototo']);
```

We’ll cover collections in just a bit.

config(\$key)

Returns the value for any dot-notated configuration item:

```
$defaultDbConnection = config('database.default');
```

csrf_field(), csrf_token()

Returns a full HTML hidden input field (csrf_field()) or just the

appropriate token value (`csrf_token()`) for adding CSRF verification to your form submission:

```
<form>

    {{ csrf_field() }}

</form>
```

// or

```
<form>

    <input type="hidden" name="_token" value="{{ csrf_token() }}">

</form>
```

`dump($variable), dd($variable...)`

Runs an output similar to `var_dump()` on all provided parameters; `dd()` also runs `exit()` to quit the application (this is used for debugging):

```
// ...

dump($var1, $var2); // Check the output...

// ...

dd($var1, $var2, $state); // Why is this not working???
```

`env($key, $default = null)`

Returns the environment variable for the given key:

```
$key = env('API_KEY', '');
```

Remember, do not ever use `env()` outside of config files.

`dispatch($job)`

Dispatches a job:

```
dispatch(new EmailAdminAboutNewUser($user));
```

`event($event)`

Fires an event:

```
event(new ContactAdded($contact));
```

`old($key = null, $default = null)`

Returns the old value (from the last user form submission) for this form key, if it exists:

```
<input name="name" value="{{ old('value', 'Your name here') }}"
```

`redirect($path)`

Returns a redirect response to the given path:

```
Route::get('post', function () {  
    // ...  
  
    return redirect('home');  
});
```

Without parameters, this generates an instance of the `Illuminate\Routing\Redirector` class.

`response($content, $status = 200, $headers)`

If passed with parameters, returns a prebuilt instance of Response. If passed with no parameters, it returns an instance of the Response factory:

```
return response('OK', 200, ['X-Header-Greatness' => 'Super great']);
```

```
return response()->json(['status' => 'success']);
```

`tap($value, $callback = null)`

Calls the closure (the second argument), passing it the first argument, and then returns the first argument (instead of the output of the closure):

```
return tap(Contact::first(), function ($contact) {  
    $contact->name = 'Aheahe';  
    $contact->save();  
});
```

`view($viewPath)`

Returns a view instance:

```
Route::get('home', function () {  
    return view('home'); // Gets /resources/views/home.blade.php  
});
```

`fake()`

Returns an instance of Faker:

```
@for($i = 0; $i <= 4; $i++)  
    <td>Purchased by {{ fake()->unique()->name() }}</td>  
@endfor
```

Collections

Collections are one of the most powerful yet underappreciated tools Laravel provides. We covered them a bit in “[Eloquent Collections](#)”, but here’s a quick recap.

Collections are essentially arrays with superpowers. The array-traversing methods you normally have to pass arrays into (`array_walk()`, `array_map()`, `array_reduce()`, etc.), all of which have confusingly inconsistent method signatures, are available as consistent, clean, chainable methods on every collection. You can get a taste of functional programming and map, reduce, and filter your way to cleaner code.

We’ll cover some of the basics of Laravel’s collections and collection pipeline programming here, but for a much deeper overview, check out Adam Wathan’s book *Refactoring to Collections* (Gumroad).

The Basics

Collections are not a new idea within Laravel. Many languages make collection-style programming available on arrays out of the box, but with PHP we’re not quite so lucky.

Using PHP’s `array*()` functions, we can take the monstrosity shown in [Example 17-1](#) and turn it into the slightly less monstrous monstrosity shown in [Example 17-2](#).

Example 17-1. A common, but ugly, foreach loop

```
$users = [...];

$admins = [];

foreach ($users as $user) {
    if ($user['status'] == 'admin') {
        $user['name'] = $user['first'] . ' ' . $user['last'];
        $admins[] = $user;
    }
}

return $admins;
```

Example 17-2. Refactoring the foreach loop with native PHP functions

```

$users = [...];

return array_map(function ($user) {
    $user['name'] = $user['first'] . ' ' . $user['last'];
    return $user;
}, array_filter($users, function ($user) {
    return $user['status'] == 'admin';
}));

```

Here, we've gotten rid of a temporary variable (\$admins) and converted one confusing foreach loop into two distinct actions: map and filter.

The problem is, PHP's array manipulation functions are awful and confusing. Just look at this example; `array_map()` takes the closure first and the array second, but `array_filter()` takes the array first and the closure second. In addition, if we added any complexity to this, we'd have functions wrapping functions wrapping functions. It's a mess.

Laravel's collections take the power of PHP's array manipulation methods and give them a clean, fluent syntax—and they add many methods that don't even exist in PHP's array manipulation toolbox. Using the `collect()` helper method that turns an array into a Laravel collection, we can do what's shown in [Example 17-3](#).

Example 17-3. Refactoring the foreach loop with Laravel's collections

```

$users = collect([...]);

return $users->filter(function ($user) {
    return $user['status'] == 'admin';
})->map(function ($user) {
    $user['name'] = $user['first'] . ' ' . $user['last'];
    return $user;
});

```

This isn't the most extreme of examples. There are plenty of others where the reduction in lines of code and the increased simplicity would make an even stronger case. But this right here is *so common*.

Look at the original example and how muddy it is. It's not entirely clear until you understand the entire code sample what any given piece is there for.

The biggest benefit collections provide, over anything else, is breaking the actions you're taking to manipulate an array into simple, discrete,

understandable tasks. You can now do something like this:

```
$users = [...]  
$countAdmins = collect($users)->filter(function ($user) {  
    return $user['status'] == 'admin';  
})->count();
```

or something like this:

```
$users = [...];  
$greenTeamPoints = collect($users)->filter(function ($user) {  
    return $user['team'] == 'green';  
})->sum('points');
```

Many of the examples we'll look at in the rest of this chapter operate on this mythical `$users` collection we've started imagining here. Each entry in the `$users` array will represent a single human; they'll likely all be array-accessible. The specific properties each user will have may vary a bit depending on the example. But any time you see this `$users` variable, know that that's what we're working with.

A Few Collection Operations

There's much more you can do than what we've covered so far. I recommend you take a look at the [Laravel collections docs](#) to learn more about all the methods you can use, but to get you started, here are just a few of the core methods:

`all()`, `toArray()`

If you'd like to convert your collection to an array, you can do so with either `all()` or `toArray()`. `toArray()` flattens to arrays not just the collection but also any Eloquent objects underneath it. `all()` converts *only* the collection to an array; any Eloquent objects contained within the collection will be preserved as Eloquent objects. Here are a few examples:

```
$users = User::all();
```

```
$users->toArray();
```

```
/* Returns
```

```
[
    ['id' => '1', 'name' => 'Agouhanna'],
    ...
]
```

```
*/
```

```
$users->all();
```

```
/* Returns
```

```
[
    Eloquent object { id : 1, name: 'Agouhanna' },
    ...
]
```

```
*/
```

filter(), reject()

When you want to get a subset of your original collection by checking each item against a closure, you'll use `filter()` (which keeps an item if the closure returns `true`) or `reject()` (which keeps an item if the closure returns `false`):

```
$users = collect([...]);
```

```
$admins = $users->filter(function ($user) {
```

```
    return $user->isAdmin;
```

```
});
```

```
$paidUsers = $user->reject(function ($user) {  
    return $user->isTrial;  
});
```

where()

`where()` makes it easy to provide a subset of your original collection where a given key is equal to a given value. Anything you can do with `where()` you can also do with `filter()`, but it's a shortcut for a common scenario:

```
$users = collect([...]);  
$admins = $users->where('role', 'admin');
```

whereNull(), whereNotNull()

`whereNull()` makes it easy to provide a subset of your original collection where a given key is equal to null; `whereNotNull()` is the inverse:

```
$users = collect([...]);  
$active = $users->whereNull('deleted_at');  
$deleted = $users->whereNotNull('deleted_at');
```

first(), last()

If you want just a single item from your collection, you can use `first()` to pull from the beginning of the list or `last()` to pull from the end.

If you call `first()` or `last()` with no parameters, they'll just give you the first or last item in the collection, respectively. But if you pass either a closure, they'll instead give you the first or last item in the collection *that returns true when passed to that closure*.

Sometimes you'll do this because you want the actual first or last item. But

sometimes it's the easiest way to get one item even if you only expect there to be one:

```
$users = collect([...]);  
  
$owner = $users->first(function ($user) {  
    return $user->isOwner;  
});  
  
$firstUser = $users->first();  
  
$lastUser = $users->last();
```

You can also pass a second parameter to each method, which is the default value and will be provided as a fallback if the closure doesn't provide any results.

each()

If you'd like to do something with each item of a collection, but it doesn't include modifying the items or the collection itself, you can use `each()`:

```
$users = collect([...]);  
  
$users->each(function ($user) {  
    EmailUserAThing::dispatch($user);  
});
```

map()

If you'd like to iterate over all the items in a collection, make changes to them, and return a new collection with all of your changes, you'll want to use `map()`:

```
$users = collect([...]);
```

```

$users = $users->map(function ($user) {
    return [
        'name' => $user['first'] . ' ' . $user['last'],
        'email' => $user['email'],
    ];
});

```

reduce()

If you'd like to get a single result from your collection, like a count or a string, you'll probably want to use `reduce()`. This method works by taking an initial value (called the *carry*) and then allowing each item in the collection to change that value somehow. You can define an initial value for the carry, and a closure that accepts the current state of the carry, and then each item as parameters:

```

$users = collect([...]);

$points = $users->reduce(function ($carry, $user) {
    return $carry + $user['points'];
}, 0); // Start with a carry of 0

```

pluck()

If you want to pull out just the values for a given key under each item in a collection, you can use `pluck()`:

```

$users = collect([...]);

$emails = $users->pluck('email')->toArray();

```

chunk(), take()

`chunk()` makes it easy to split your collection into groups of a predefined size, and `take()` pulls just the provided number of items:

```
$users = collect([...]);
```

```
$rowsOfUsers = $users->chunk(3); // Separates into groups of 3
```

```
$topThree = $users->take(3); // Pulls the first 3
```

takeUntil(), takeWhile()

`takeUntil()` returns all items in the collection until the callback returns `true`. `takeWhile()` returns all items in the collection until the callback returns `false`. If the callback passed to `takeUntil()` never returns `true`, or if the callback passed to `takeWhile()` never returns `false`, the entire collection is returned:

```
$items = collect([1, 2, 3, 4, 5, 6, 7, 8, 9]);
```

```
$subset = $items->takeUntil(function ($item) {
```

```
    return $item >= 5;
```

```
})->toArray();
```

```
// [1, 2, 3, 4]
```

```
$subset = $items->takeWhile(function ($item) {
```

```
    return $item < 4;
```

```
})->toArray();
```

```
// [1, 2, 3]
```

groupBy()

If you want to group all of the items in your collection by the value of one of their properties, you can use `groupBy()`:

```
$users = collect([...]);

$usersByRole = $users->groupBy('role');

/* Returns:

[
    'member' => [...],
    'admin' => [...],
]

*/
```

You can also pass a closure, and whatever you return from the closure will be what's used to group the records:

```
$heroes = collect([...]);

$heroesByAbilityType = $heroes->groupBy(function ($hero) {

    if ($hero->canFly() && $hero->isInvulnerable()) {

        return 'Kryptonian';

    }

    if ($hero->bitByARadioactiveSpider()) {

        return 'Spidermanesque';

    }

});
```



```

    if ($hero->color === 'green' && $hero->likesSmashing()) {
        return 'Hulk-like';
    }

    return 'Generic';
});

```

reverse(), shuffle()

`reverse()` reverses the order of the items in your collection, and `shuffle()` randomizes them:

```

$numbers = collect([1, 2, 3]);

$numbers->reverse()->toArray(); // [3, 2, 1]
$numbers->shuffle()->toArray(); // [2, 3, 1]

```

skip()

`skip()` returns a new collection without the given number of items:

```

$numbers = collect([1, 2, 3, 4, 5]);

$numbers->skip(3)->values(); // [4, 5]

```

skipUntil()

`skipUntil()` skips items until the callback returns `true`. You can also pass in a value to `skipUntil`, and it will skip all values until the given value is found. If the value is never found or the callback never returns `true`, an

empty collection is returned:

```
$numbers = collect([1, 2, 3, 4, 5]);

$numbers->skipUntil(function ($item) {
    return $item > 3;
})->values();
// [4, 5]

$numbers->skipUntil(3)->values();
// [3, 4, 5]
```

skipWhile()

`skipWhile()` skips items while the callback returns true. If the callback never returns false, an empty collection is returned:

```
$numbers = collect([1, 2, 3, 4, 5]);

$numbers->skipWhile(function ($item) {
    return $item <= 3;
})->toArray();
// [4, 5]
```

sort(), sortBy(), sortByDesc()

If your items are simple strings or integers, you can use `sort()` to sort them:

```
$sortedNumbers = collect([1, 7, 6])->sort()->toArray(); // [1, 6, 7]
```

If they're more complex, you can pass a string (representing the property) or a closure to `sortBy()` or `sortByDesc()` to define your sorting behavior:

```
$users = collect([...]);

// Sort an array of users by their 'email' property
$users->sort('email');

// Sort an array of users by their 'email' property
$users->sortBy(function ($user, $key) {
    return $user['email'];
});
```

countBy()

`countBy` counts every occurrence of each value in a collection:

```
$collection = collect([10, 10, 20, 20, 20, 30]);

$collection->countBy()->all();

// [10 => 2, 20 => 3, 30 => 1]
```

Each key in the resulting collection is one of the original values; its paired value is the number of times that value occurred in the original collection.

The `countBy` method also accepts a callback, which customizes the value that's used to count each item in the collection:

```
$collection = collect(['laravel.com', 'tighten.co']);
```

```
$collection->countBy(function ($address) {  
    return Str::after($address, '.');  
})->all();
```

```
// all: ["com" => 1, "co" => 1]
```

count(), isEmpty(), isEmpty()

You can see how many items there are in your collection using `count()`, `isEmpty()`, or `isEmpty()` tell you whether there are items:

```
$numbers = collect([1, 2, 3]);
```

```
$numbers->count(); // 3
```

```
$numbers->isEmpty(); // false
```

```
$numbers->isEmpty() // true
```

avg(), sum()

If you're working with a collection of numbers, `avg()` and `sum()` do what their method names say and don't require any parameters:

```
collect([1, 2, 3])->sum(); // 6
```

```
collect([1, 2, 3])->avg(); // 2
```

But if you're working with arrays, you can pass the key of the property you'd like to pull from each array to operate on:

```
$users = collect([...]);
```

```
$sumPoints = $users->sum('points');
```

```
$avgPoints = $users->avg('points');
```

join

`join()` joins the collection values into a single output string, joining each with a provided string—like PHP’s `join()` method. You can also (optionally) customize the final concatenation operator:

```
$collection = collect(['a', 'b', 'c', 'd', 'e']);  
$collection->join(', ', ' and ');
```

```
// 'a, b, c, d, and e'
```

USING COLLECTIONS OUTSIDE OF LARAVEL

Have you fallen in love with collections, and do you want to use them on your non-Laravel projects?

Just use the `composer require illuminate/collections` command and you’ll have the `Illuminate\Support\Collection` class ready to use in your code—along with the `collect()` helper.

TL;DR

Laravel provides a suite of global helper functions that simplify all sorts of tasks. They make it easier to manipulate and inspect arrays and strings, they facilitate generating paths and URLs, and they provide simple access to some consistent and vital functionality.

Laravel’s collections are powerful tools that bring the possibility of collection pipelines to PHP.

Chapter 18. The Laravel Ecosystem

As Laravel has grown, the Laravel team has built a suite of tools to support and simplify the lives and workflows of Laravel developers. Much of the new work has gone straight into the core, but there are quite a few packages and SaaS offerings that aren't part of the core but are still very much a part of the Laravel experience.

We've already covered quite a few of them, and for those I'll provide pointers to where to go in the book for more information. For the tools we haven't covered, I'll give each a quick description and a link to the relevant website.

Tools Covered in This Book

We've already taken a look at these, but here are some brief reminders of what they are and links to where you can find the relevant sources in the book.

Valet

Valet is a local development server (for Mac, but with forks for Windows and Linux) that makes it quick and easy to serve all of your projects to your browser with almost no effort. You'll install Valet globally on your local development machine via Composer.

With a few commands you can have Nginx, MySQL, Redis, and more serving every Laravel app on your machine at a `.test` domain.

Valet is covered in “[Laravel Valet](#)”.

Homestead

Homestead is a configuration layer on top of Vagrant that makes it simple to serve multiple Laravel applications from a Laravel-friendly Vagrant setup.

Homestead was introduced briefly in [“Laravel Homestead”](#).

Herd

Herd is a native macOS app that wraps Valet and its dependencies up into a single app you can install without wrangling Docker, Homebrew, or any other dependency managers.

Herd is covered in [“Laravel Herd”](#).

The Laravel Installer

The Laravel installer is a package installed globally on your local development machine (via Composer) that makes it easy and quick to set up a new Laravel project.

The installer is covered in [“Installing Laravel with the Laravel Installer Tool”](#).

Dusk

Dusk is a frontend testing framework built for testing your entire application, JavaScript and all. It’s a powerful package you can pull into your application via Composer and that drives actual browsers with ChromeDriver.

Dusk is covered in [“Testing with Dusk”](#).

Passport

Passport is a powerful, simple-to-set-up OAuth 2.0 server for authenticating clients to your APIs. You’ll install it in each application as a Composer package, and with very little work you can have a full OAuth 2.0 flow accessible to your users.

Passport is covered in [“API Authentication with Laravel Passport”](#).

Sail

Sail is Laravel’s default local development environment, powered by Docker.

Sail is covered in [“Laravel Sail”](#).

Sanctum

Sanctum is an authentication system to power mobile apps, SPAs, and simple token-based APIs. It's a simpler but still quite powerful alternative to the much more complex OAuth.

Sanctum is covered in [“API Authentication with Sanctum”](#).

Fortify

Fortify is a headless authentication system. It provides the routes and controllers for all the authentication features Laravel requires, from login and registration to password reset and more, to be consumed by whichever frontend you choose.

Fortify is covered in [“Fortify”](#).

Breeze

Breeze is a minimal set of routes and controllers for all of Laravel's required authentication features, paired together with frontend templates for each. Breeze can be served through Blade, Vue, React, or Inertia.

Breeze is covered in [“Laravel Breeze”](#).

Jetstream

Jetstream is a robust application starter kit, providing all of the authentication features Breeze offers, along with email verification, two-factor authentication, session management, API authentication, and team management features. Unlike Breeze, Jetstream is only available in two frontend tooling flavors: Livewire and Inertia/Vue.

Jetstream is covered in [“Laravel Jetstream”](#).

Horizon

Horizon is a queue monitoring package you can install into each application via Composer. It exposes a full user interface for monitoring the health, performance, failures, and history of your Redis queued jobs.

Horizon is introduced briefly in “[Laravel Horizon](#)”.

Echo

Echo is a JavaScript library (introduced along with a series of improvements to Laravel’s notification system) that makes it simple to subscribe to events and channels broadcast from your Laravel app via WebSockets.

Echo is covered in “[Laravel Echo \(the JavaScript Side\)](#)”.

Tools Not Covered in This Book

These are a few tools that I did not cover because they are beyond the scope of this book. Some of these are just for use in special circumstances (Cashier for taking payments, Socialite for social login, etc.), but some I use every day (Forge, especially).

Here’s a brief introduction, beginning with the ones you’re most likely to encounter in your work. Note that this list is not exhaustive!

Forge

Forge is a paid SaaS tool for creating and managing virtual servers on hosts like DigitalOcean, Linode, AWS, and more. It provisions Laravel-ready servers (and individual sites on those servers) with all the tools you need to run them, from queues and queue workers to Let’s Encrypt SSL certificates. It can also set up simple shell scripts to autodeploy your sites when you push up new code to GitHub or Bitbucket.

Forge is incredibly useful for spinning up sites quickly and easily, but it’s not so minimal that you can’t also run your apps on it in the longer term or at larger scale. You can scale up your server sizes, add load balancers, and manage private networking between your servers, all within Forge.

Vapor

Vapor is a paid SaaS tool for deploying Laravel applications to AWSs Lambda, using the “serverless” hosting pattern. It manages cache, queues, database, asset

building, domain direction, autoscaling, content delivery networks, environment management, and most of anything else you'll need to move to serverless deployment for your Laravel apps.

Envoyer

Envoyer is a paid SaaS tool that's branded as offering "zero downtime PHP deployment." Unlike Forge, Envoyer doesn't spin up your servers or manage them. Its primary job is to listen to triggers—usually when you push new code, but you can also manually trigger deployments or trigger them with webhooks—and perform your deployment steps in response.

There are three ways that Envoyer does this much better than Forge's push-to-deploy tool and most other push-to-deploy solutions:

- It has a robust toolset for building out your deployment pipeline as a simple but powerful multistage process.
- It deploys your app using Capistrano-style zero-downtime deploys; each new deployment is built into its own folder, and only once the build process has completed successfully is that deploy folder symlinked to your actual web root. Because of this, there's no moment when your server is broken while Composer installs or NPM builds.
- Because of this folder-based system, it's easy and quick to roll back any breaking changes to a previous release; Envoyer just updates the symlink back to a previous deploy folder and it's immediately serving an older build.

You can also set up regular health checks (pings against your servers that report errors to you if the pings don't get back a 200 HTTP response), expectations that your cron jobs will ping Envoyer on a regular schedule, and chat-based notifications of any significant events.

Envoyer is more of a niche tool than Forge. I don't know many Laravel developers who don't use Forge, but those who pay for Envoyer are more likely to have websites that will suffer if they can't immediately roll back a problematic commit, or websites that get enough traffic (or important enough traffic) that 10 seconds of downtime here and there can be a big issue. If your

site is in that category, Envoyer will feel like magic.

Cashier

Cashier is a free package that provides a simple interface in front of Stripe's subscription billing offerings. Cashier handles much of the basic functionality of subscribing users, changing their plans, giving them access to invoices, handling webhook callbacks from the billing service, managing cancellation grace periods, and more.

If you want to allow your users to sign up for subscriptions using Stripe, Cashier will make your life a lot easier.

Socialite

Socialite is a free package that makes it incredibly simple to add social login (for example, via GitHub or Facebook) to your apps.

Nova

Nova is a paid package for building admin panels. If you imagine your average complex Laravel app, it may have several parts: the public-facing website or customer view, the administration section for making changes to the core data or customer list, and maybe an API.

Nova drastically simplifies the process of building the admin panel part of the site using Vue and a Laravel API. It makes it easy to generate CRUD (create, read, update, delete) pages for all of your resources, together with more complex custom views for your data, custom actions and relationships on each of your resources, and even custom tools for adding non-CRUD tooling to the same general admin space.

Spark

Spark is a paid package for generating a SaaS that accepts payments and makes it easy to manage users, teams, and subscriptions. It provides Stripe and Paddle integration, invoices, per-seat or team-based billing, and a full billing portal that stands alone from the rest of your app so you're not stuck using Spark's default

tech stack.

Envoy

Envoy is a local task runner that makes it easy to define common tasks that will run on your remote servers, commit those tasks' definitions to version control, and run them simply and predictably.

Take a look at **Example 18-1** to get a sense of what a common Envoy task looks like.

Example 18-1. A common Envoy task

```
@servers(['web-1' => '192.168.1.1', 'web-2' => '192.168.1.2'])

@task('deploy', ['on' => ['web-1', 'web-2']])
    cd mysite.com
    git pull origin {{ $branch }}
    php artisan migrate
    php artisan route:cache
@endtask
```

To run **Example 18-1**, you'd run the following command from your local terminal:

```
envoy run deploy --branch=master
```

Telescope

Telescope is a free debugging tool, installable as a package, for Laravel applications. It generates a dashboard where you can dig into the current status of jobs, queue workers, HTTP requests, database queries, and much more.

Octane

Octane is a free tool that makes it possible for you to serve your Laravel applications using asynchronous, concurrent PHP web servers designed for speed and power. There are three such tools at the writing of this book: Swoole, Open Swoole, and RoadRunner. With Octane, these tools load your application into memory once and then facilitate serving your application each request in the most performant way, taking advantage of language and system-level tooling for

concurrency.

Pennant

Pennant is a Laravel-native implementation of “feature flags,” a pattern that allows you to easily define whether each request should see a certain feature in your app—often because of the user who’s requesting and what they have access to. Pennant allows you to define once which metrics should be used to define whether the request should or shouldn’t be served a given feature, providing a syntax very similar to Laravel’s access control list layer.

Folio

Folio is a Laravel package that allows you to build your app’s routing based on the arrangement of template files inside of folders. Similar to Next and Nuxt, Folio allows you to create individual templates (e.g., `/index.blade.php` displays at `mysite.com/`, `/about.blade.php` at `mysite.com/about`, and `/users/index.blade.php` at `mysite.com/users`) or define templates that should match placeholders in the URL (e.g. `/users/[id].blade.php` at `mysite.com/users/14`).

Volt

Volt augments Livewire with the ability to write single-file, functional components. Volt also offers a directive, `@volt`, that allows you to assign a subsection of a template to be managed by the Livewire component definition, while the rest of the template is still normal Blade.

Pint

Pint is a code style tool built to enforce Laravel’s default code style in your apps. It’s built on top of PHP-CS-Fixer, providing a few tooling improvements and also a pre-configured Laravel-specific set of code rules.

Other Resources

I've mentioned many of these already, but here's a nonexhaustive list of resources folks often turn to to learn Laravel:

- [Laravel Bootcamp](#)
- [Laravel News](#)
- [Laracasts](#)
- [@TaylorOtwell](#) and [@LaravelPHP](#) on Twitter
- [Adam Wathan's courses](#)
- [Chris Fidao's courses](#)
- [Laravel Daily](#)
- [DevDojo](#)
- [CodeCourse](#)
- [The Laravel Podcast](#)

There are many blogs (I have one at [mattstauffer.com](#) and Tighten has one at [tighten.com](#), and there are plenty of others that are incredibly useful), many excellent Twitter-ers, many superb package authors, and far too many Laravel practitioners who I respect to fit into a list here. This is a rich, diverse, and giving community, full of developers who love to share everything they're learning; the hard part is not finding good content but finding the time to consume it all.

I can't list every person or resource you should look to in your journey as a Laravel developer, but if you start with the resources and folks listed here, you will be off to a great start in getting up and running with Laravel.

Glossary

accessor

A method defined on an Eloquent model that customizes how a given property will be returned. Accessors make it possible to define that getting a given property from a model will return a different (or, more likely, differently formatted) value than what is stored in the database for that property.

ActiveRecord

A common database ORM pattern, and also the pattern that Laravel's Eloquent uses. In ActiveRecord, the same model class defines both how to retrieve and persist database records *and* how to represent them. Additionally, each database record is represented by a single entity in the application, and each entity in the application is mapped to a single database record.

API

Technically *application programming interface*, but most commonly used to refer to a series of endpoints (and instructions on how to use them) that can be used to make HTTP-based calls to read and modify data from outside of a system. Sometimes, the term API is also used to describe the set of interfaces, or affordances, any given package or library or class exposes to its consumers.

application test

Often called acceptance or functional tests, application tests test the entire behavior of the application, usually at an outer boundary, by employing something like a DOM crawler—which is exactly what Laravel's application test suite offers.

argument (Artisan)

Arguments are parameters that can be passed to Artisan console commands. Arguments aren't prefaced with `--` or followed by `=`, but instead just accept a single value.

Artisan

The tool that makes it possible to interact with Laravel applications from the command line.

assertion

In testing, an assertion is the core of the test: you are *asserting* that something should be equal to (or less than or greater than) something else, or that it should have a given count, or whatever else you like. Assertions are the things that can either pass or fail.

authentication

Correctly identifying oneself as a member/user of an application is the act of authentication. Authentication doesn't define *what* you may do, but simply *who* you are (or aren't).

authorization

Assuming you've either succeeded or failed at authenticating yourself, authorization defines what you're *allowed* to do given your particular identification. Authorization is about access and control.

autowiring

When a dependency injection container will inject an instance of a resolvable class without a developer having explicitly taught it how to resolve that class, that's called autowiring. With a container that doesn't have autowiring, you can't even inject a plain PHP object with no dependencies until you have explicitly bound it to the container. With autowiring, you only have to explicitly bind something to the container if its dependencies are too complex or vague for the container to figure out on its own.

beanstalkd

Beanstalk is a work queue. It's simple and excels at running multiple asynchronous tasks—which makes it a common driver for Laravel's queues. *beanstalkd* is its daemon.

Blade

Laravel's templating engine.

Carbon

A PHP package that makes working with dates much easier and more expressive.

Cashier

A Laravel package that makes billing with Stripe or Braintree, especially in subscription contexts, easier and more consistent and powerful.

closure

Closures are PHP's version of anonymous functions. A closure is a function that you can pass around as an object, assign to a variable, pass as a parameter to other functions and methods, or even serialize.

CodeIgniter

An older PHP framework that Laravel was inspired by.

collection

The name of a development pattern and also Laravel's tool that implements it. Like arrays on steroids, collections provide map, reduce, filter, and many other powerful operations that PHP's native arrays don't.

command

The name for a custom Artisan console task.

Composer

PHP's dependency manager. Like RubyGems or NPM.

container

Somewhat of a catchall word, in Laravel “container” refers to the application container that’s responsible for dependency injection. Accessible via `app()` and also responsible for resolving calls to controllers, events, jobs, and commands, the container is the glue that holds each Laravel app together.

contract

Another name for an interface.

controller

A class that is responsible for routing user requests through to the application’s services and data, and returning some form of useful response back to the user.

CSRF (cross-site request forgery)

A malicious attack where an external site makes requests against your application by hijacking your users’ browsers (with JavaScript, likely) while they’re still logged in to your site. Protected against by adding a token (and a check for that token on the POST side) to every form on the site.

dependency injection

A development pattern where dependencies are injected in from the outside—usually through the constructor—instead of being instantiated in the class.

directive

Blade syntax options like `@if`, `@unless`, etc.

dot notation

Navigating down inheritance trees using `.` to reference a jump down to a new level. If you have an array like `['owner' => ['address' => ['line1' => '123 Main St.' ]]]`, you have three levels of nesting. Using dot notation, you would represent “123 Main St.” as `"owner.address.line1"`.

Dusk

Laravel's frontend testing package that can test JavaScript (primarily Vue) and DOM interactions by spinning up ChromeDriver to run the tests.

eager loading

Avoiding $N+1$ problems by adding a second smart query to your first query to get a set of related items. Usually you have a first query that gets a collection of thing A. But each A has many Bs, and so every time you get the Bs from an A, you need a new query. Eager loading means doing two queries: first you get all the As, and then you get *all* the Bs related to all those As, in a single query. Two queries, and you're done.

Echo

A Laravel product that makes WebSocket authentication and syncing of data simple.

Eloquent

Laravel's ActiveRecord ORM. The tool you'll use to define and query something like a User model.

environment variables

Variables that are defined in an `.env` file that is expected to be excluded from version control. This means that they don't sync between environments and that they're also kept safe.

Envoy

A Laravel package for writing scripts to run common tasks on remote servers. Envoy provides a syntax for defining tasks and servers and a command-line utility for running the tasks.

Envoyer

A Laravel SaaS product for zero-down-time deployment, multiserver deployments, and server and cron health checks.

event

Laravel's tool for implementing a pub/sub or observer pattern. Each event represents that an event happened: the name of the event describes what happened (e.g., `UserSubscribed`) and the payload allows for attaching relevant information. Designed to be “fired” and then “listened” for (or published and subscribed, if you prefer the pub/sub concept).

facade

A tool in Laravel for simplifying access to complex tools. Facades provide static access to core services in Laravel. Since every facade is backed by a class in the container, you can replace any call to something like `Cache::put()`; with a two-line call to something like `$cache = app('cache'); $cache->put();`.

Faker

A PHP package that makes it easy to generate random data. You can request data in different categories, like names, addresses, and timestamps.

flag

A parameter anywhere that is on or off (Boolean).

fluent

Methods that can be chained one after another are said to be fluent. To provide a fluent syntax, each method must return the instance, preparing it to be chained again. This allows for something like `People::where('age', '>', 14)->orderBy('name')->get()`.

Flysystem

The package that Laravel uses to facilitate its local and cloud file access.

Forge

A Laravel product that makes it easy to spin up and manage virtual servers on major cloud providers like DigitalOcean and AWS.

Fortify

Headless backend authentication system providing the routes and controllers for all of Laravel's important authentication systems.

FQCN (fully qualified class name)

The full namespaced name of any given class, trait, or interface. `Controller` is the class name; `Illuminate\Routing\Controller` is the FQCN.

helper

A globally accessible PHP function (or, in Laravel, sometimes a call on a globally accessible facade) that makes some other functionality easier.

Homestead

A Laravel tool that wraps Vagrant and makes it easier to spin up Forge-parallel virtual servers for local Laravel development.

Horizon

A Laravel package that provides tooling for managing queues with greater nuance than Laravel's defaults, and also provides insight into the current and historic operating state of the queue workers and their jobs.

HTTP client

An HTTP client built into Laravel that provides the ability to make outgoing requests to other web applications.

Illuminate

The top-level namespace of all Laravel components.

integration test

Integration tests test the way individual units work together and pass messages.

IoC (inversion of control)

The concept of giving “control” over how to make a concrete instance of an interface to the higher-level code of the package instead of the lower-level code. Without IoC, each individual controller and class might decide what instance of `Mailer` it wanted to create. IoC makes it so that the low-level code—those controllers and classes—just get to ask for a `Mailer`, and some high-level configuration code defines *once* per application which instance should be provided to satisfy that request.

job

A class that intends to encapsulate a single task. Jobs are intended to be able to be pushed onto a queue and run asynchronously.

JSON (JavaScript Object Notation)

A syntax for data representation.

JWT (JSON Web Token)

A JSON object containing all of the information necessary to determine a user’s authentication state and access permissions. This JSON object is digitally signed, which is what makes it trustworthy, using HMAC or RSA. Usually delivered in the header.

mailable

An architectural pattern designed to encompass the functionality of sending mail into a single “sendable” class.

Markdown

A formatting language designed for formatting plain text and outputting to multiple formats. Commonly used for formatting text that has a good chance of being processed by a script or read by humans in its raw form—for example, Git READMEs.

mass assignment

The ability to pass many parameters at once to create or update an Eloquent model, using a keyed array.

Memcached

An in-memory data store designed to provide simple but fast data storage. Memcached only supports basic key/value storage.

middleware

A series of wrappers around an application that filter and decorate its inputs and outputs.

migration

A manipulation to the state of the database, stored in and run from code.

Mockery

A library included with Laravel that makes it easy to mock PHP classes in your tests.

model

A class used to represent a given database table in your system. In ActiveRecord ORMs like Laravel's Eloquent, this class is used both to represent a single record from the system and to interact with the database table.

model factory

A tool for defining how the application can generate an instance of your model if needed for testing or seeding. Usually paired with a fake data generator like Faker.

multitenancy

A single app serving multiple clients, each of which has its customers. Multitenancy often suggests that each client of your application gets its own theming and domain name with which to differentiate its service to its customers vis-à-vis your other clients' potential services.

mutator

A tool in Eloquent that allows you to manipulate the data being saved to a

model property before it is saved to the database.

Nginx

A web server similar to Apache.

notification

A Laravel framework tool allowing a single message to be sent via myriad notification channels (e.g., email, Slack, SMS) to one or more recipients.

Nova

A paid Laravel package for building admin panels for your Laravel apps.

NPM (Node Package Manager)

A central web-based repository for Node packages, at npmjs.org; also a utility used on your local machine to install a project's frontend dependencies into the *node_modules* directory based on the specifications of *package.json*.

OAuth

The most common authentication framework for APIs. OAuth has multiple grant types, each of which describes a different flow of how consumers retrieve, use, and refresh the “tokens” that identify them after the initial authentication handshake.

option (Artisan)

Like arguments, options are parameters that can be passed to Artisan commands. They're prefaced with `--` and can be used as a flag (`--force`) or to provide data (`--userId=5`).

ORM (object-relational mapper)

A design pattern that is centered around using objects in a programming language to represent data, and its relationships, in a relational database.

Passport

A Laravel package that can be used to easily add an OAuth authentication server to your Laravel app.

PHPSpec

A PHP testing framework.

PHPUnit

A PHP testing framework. The most common and connected to the most of Laravel's custom testing code.

polymorphic

In database terms, able to interact with multiple database tables with similar characteristics. A polymorphic relationship will allow entities of multiple models to be attached in the same way.

preprocessor

A build tool that takes in a special form of a language (for CSS, one special form is LESS) and generates code with just the normal language (CSS). Preprocessors build in tools and features that are not in the core language.

primary key

Most database tables have a single column that is intended to represent each row. This is called the primary key and is commonly named `id`.

queue

A stack onto which jobs can be added. Usually associated with a queue worker, which pulls jobs one at a time from a queue, works on them, and then discards them.

React

A JavaScript framework. Created and maintained by Facebook.

real-time facades

Similar to facades, but without requiring a separate class. Real-time facades

can be used to make any class's methods callable as static methods by importing that class with `Facades\` in front of its namespace.

Redis

Like Memcached, a data store simpler than most relational databases but powerful and fast. Redis supports a very limited set of structures and data types but makes up for it in speed and scalability.

REST (Representational State Transfer)

The most common format for APIs these days. Usually suggests that interactions with an API should each authenticate separately and should be “stateless”; also usually suggests that the HTTP verbs are used for basic differentiation of requests.

route

A definition of a way or ways the user might visit a web application. A route is a pattern definition; it can be something like `/users/5`, or `/users`, or `/users/id`.

S3 (Simple Storage Service)

Amazon's “object storage” service, which makes it easy to use AWS's incredible computing power to store and serve files.

SaaS (Software as a Service)

Web-based applications that you pay money to use.

Sanctum

API token authentication system for single-page applications, mobile applications, and simple token-based APIs.

scope

In Eloquent, a tool for defining how to consistently and simply narrow down a query.

Scout

A Laravel package for full-text search on Eloquent models.

serialization

The process of converting more complex data (usually an Eloquent model) to something simpler (in Laravel, usually an array or JSON).

service provider

A structure in Laravel that registers and boots classes and container bindings.

Socialite

A Laravel package making it simple to add social authentication (e.g., login via Facebook) to Laravel apps.

soft delete

Marking a database row as “deleted” without actually deleting it; usually paired with an ORM that by default hides all “deleted” rows.

Spark

A Laravel tool that makes it easy to spin up a new subscription-based SaaS app.

Symfony

A PHP framework that focuses on building excellent components and making them accessible to others. Symfony’s HTTPFoundation is at the core of Laravel and every other modern PHP framework.

Telescope

A Laravel package for adding a debugging assistant to Laravel apps.

Tinker

Laravel’s REPL, or read–evaluate–print loop. It’s a tool that allows you to perform complex PHP operations within the full context of your app from the

command line.

TL;DR

Too long; didn't read. "Summary."

typehinting

Prefacing a variable name in a method signature with a class or interface name. Tells PHP (and Laravel, and other developers) that the only thing that's allowed to be passed in that parameter is an object with the given class or interface.

unit test

Unit tests target small, relatively isolated units—a class or method, usually.

Vagrant

A command-line tool that makes it easy to build virtual machines on your local computer using predefined images.

Valet

A Laravel package (for macOS users, but there are forks for Linux and Windows) that makes it easy to serve your applications from your development folder of choice, without worrying about Vagrant or virtual machines.

validation

Ensuring that user input matches expected patterns.

view

An individual file that takes data from the backend system or framework and converts it into HTML.

view composer

A tool that defines that, every time a given view is loaded, it will be provided a certain set of data.

Vue

A JavaScript framework. Preferred by Laravel. Written by Evan You.

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Colophon

The animal on the cover of *Laravel: Up & Running* is a gemsbok (*Oryx gazella*). This large antelope is native to the deserts of South Africa, Botswana, Namibia (where it is featured on the country's coat of arms), and Zimbabwe.

Gemsbok measure about 5 feet 7 inches tall at the shoulder and can weigh from 250 to 390 pounds. They are typically pale gray or brown, with black and white facial markings and long black tails. A black stripe extends from the chin to the lower edge of the neck. The gemsbok's impressive straight horns, used in defensive maneuvers, average 33 inches in length and are regarded as charms in many cultures. In medieval England, they were often marketed as unicorn horns.

Although these horns make the gemsbok a highly sought trophy animal, the population remains stable throughout Southern Africa. In 1969, gemsbok were introduced to southern New Mexico, where their current population is around 3,000.

Gemsbok are well-suited to such desert environments, with the ability to survive without drinking water for most of the year. To achieve this, they do not pant or sweat, allowing their body temperature to rise several degrees above normal on hot days. Their lifespan is approximately 18 years in the wild.

Many of the animals on O'Reilly covers are endangered; all of them are important to the world.

The cover illustration is by Karen Montgomery, based on a black and white engraving from *Riverside Natural History*. The cover fonts are Gilroy Semibold and Guardian Sans. The text font is Adobe Minion Pro; the heading font is Adobe Myriad Condensed; and the code font is Dalton Maag's Ubuntu Mono.